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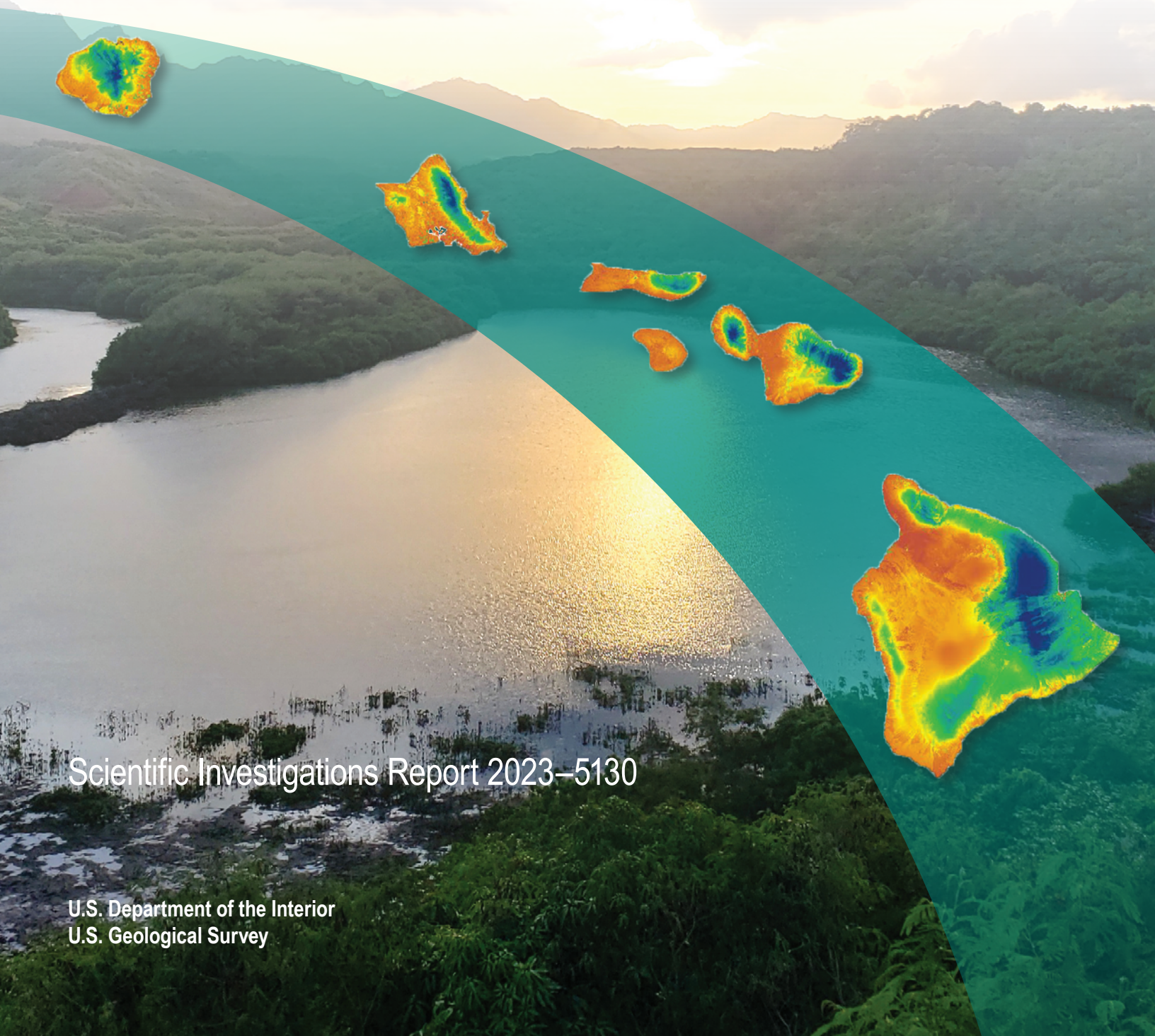
Estimated Groundwater Recharge for Mid-Century and End-of-Century Climate Projections, Kaua‘i, O‘ahu, Moloka‘i, Lāna‘i, Maui, and the Island of Hawai‘i



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Cover. Mean annual groundwater recharge maps of Kauaʻi, Oʻahu, Molokaʻi, Lānaʻi, Maui, and the Island of Hawaiʻi for the 1978–2007 scenario and 2020 land-cover conditions. Background photograph of view looking west from Menehune Fishpond Overlook, Kauaiʻi, Hawaiʻi. U.S. Geological Survey photograph by Heidi Kane, October 1, 2019.

Estimated Groundwater Recharge for Mid-Century and End-of-Century Climate Projections, Kauaʻi, Oʻahu, Molokaʻi, Lānaʻi, Maui, and the Island of Hawaiʻi

By Heidi L. Kāne, Alan Mair, Adam G. Johnson, Kolja Rotzoll, James Mifflin, and Delwyn S. Oki

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Conversion Factors

U.S. customary units to International System of Units

Multiply	By	To obtain
	Length	
inch (in.)	2.54	centimeter (cm)
inch (in.)	25.4	millimeter (mm)
foot (ft)	0.3048	meter (m)
mile (mi)	1.609	kilometer (km)
	Area	
acre	4,047	square meter (m^2)
acre	0.4047	hectare (ha)
square mile (mi^2)	259.0	hectare (ha)
square mile (mi^2)	2.590	square kilometer (km^2)
	Volume	
gallon (gal)	3.785	liter (L)
gallon (gal)	0.003785	cubic meter (m^3)
million gallons (Mgal)	3,785	cubic meter (m^3)
	Flow rate	
million gallons per day (Mgal/d)	0.04381	cubic meter per second (m^3/s)

Temperature in degrees Celsius (°C) may be converted to degrees Fahrenheit (°F) as follows: °F = (1.8 × °C) + 32.

Temperature in degrees Fahrenheit (°F) may be converted to degrees Celsius (°C) as follows: °C = (°F – 32) / 1.8.

Datum

Vertical coordinate information is referenced to local mean sea level.

Horizontal coordinate information is referenced to the North American Datum of 1983 (NAD 83).

Altitude, as used in this report, refers to distance above the vertical datum.

Supplemental Information

In this study, future projections of climate variables, such as rainfall, temperature, and reference evapotranspiration, are often described as one of two types of anomalies: (1) absolute anomaly or (2) relative anomaly. An absolute anomaly is calculated as the difference between the projected future value and the value from a reference condition, expressed in the units of the climate variable. A relative anomaly is calculated as the ratio of the absolute anomaly to the value from the reference condition.

Abbreviations

CMIP	Coupled Model Intercomparison Project
CWRM	State of Hawai‘i Commission on Water Resource Management
DEM	digital elevation model
DLNR	State of Hawai‘i Department of Land and Natural Resources
ET	evapotranspiration
FAO-PM	Food and Agriculture Organization of the United Nations Penman-Monteith
GCM	general circulation model
GIS	geographic information system
HRCM	Hawai‘i Regional Climate Model
NHDPlus HR	National Hydrography Dataset Plus High Resolution
OSDS	onsite sewage disposal systems
RCP	representative concentration pathway
RCM	regional climate model
SD	statistical downscaling
TMK	tax map key
USGS	U.S. Geological Survey
WATRMod	Water-budget Accounting for Tropical Regions Model

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By Heidi L. Kāne, Alan Mair, Adam G. Johnson, Kolja Rotzoll, James Mifflin, and Delwyn S. Oki

Abstract

Demand for freshwater in the State of Hawaiʻi is expected to increase by roughly 13 percent from 2020 to 2035. Groundwater availability in Hawaiʻi is affected by a number of factors, including land cover, rainfall, runoff, evapotranspiration, and climate change. To evaluate the availability of fresh groundwater under projected future-climate conditions, estimates of groundwater recharge are needed. A water-budget model with a daily computation interval was used to estimate the spatial distribution of groundwater recharge for Kauaʻi, Oʻahu, Molokaʻi, Lānaʻi, Maui, and the Island of Hawaiʻi for recent climate conditions and three future-climate scenarios. Climate conditions from 1978 to 2007 were used as the reference period for recent climate conditions on each island. The three future-climate scenarios were developed using available high-resolution downscaled climate projections that include (1) a mid-century scenario using projected rainfall conditions for the Representative Concentration Pathway (RCP) scenario during 2041–71 with a total radiative forcing of 8.5 watts per square meter by the year 2100 (RCP8.5 2041–71 scenario), (2) a dry-climate scenario using projected rainfall conditions for the RCP8.5 scenario during 2071–99, and (3) a wet-climate scenario using projected rainfall conditions for the "Special Report on Emissions Scenarios" A1B scenario during 2080–99 for Maui, the RCP4.5 scenario during 2080–99 for Kauaʻi, Lānaʻi, and the Island of Hawaiʻi, and the RCP8.5 scenario during 2080–99 for Oʻahu and Molokaʻi. An additional drought scenario was added for Lānaʻi to assess the effect of extreme drought conditions during 2008–12 on groundwater recharge. All scenarios used 2020 land cover.

Mean annual groundwater recharge is estimated to decrease between 5 and 55 percent on all six islands in this study for the mid-century and dry-climate scenarios relative to the reference-period recharge. Recharge is estimated to increase for Kauaʻi, Oʻahu, Molokaʻi, Lānaʻi, and Maui between 2 and 43 percent and decrease for the Island of Hawaiʻi by about 4 percent for the wet-climate scenario. Comparing the mid-century and dry-climate scenarios, all 110 aquifer systems (management areas defined by the State of Hawaiʻi Commission on Water Resource Management) from all six islands show similar direction in drying (104 aquifer systems) or wetting (6 aquifer systems) changes for recharge. However, among the three future scenarios, only 35 of 110 aquifer systems show similar direction in drying (30 aquifer systems) or wetting (5 aquifer systems) changes for recharge.

Introduction

Reliable estimates of groundwater recharge are critical for evaluating the availability of groundwater in the Hawaiian Islands. Accurate recharge estimates that include the best available information regarding the effects of climate change are important for understanding Hawaiʻi's future water resources. Groundwater recharge is water derived from precipitation and other sources, such as irrigation and leakage from surface reservoirs, that moves through or bypasses the plant-soil system and replenishes aquifers. Groundwater provides 99 percent of Hawaiʻi's drinking water and about 50 percent of all freshwater used in the State (Gingerich and Oki, 2000; Izuka and others, 2018). Groundwater availability in Hawaiʻi is affected by changes in precipitation and evapotranspiration (ET), saltwater intrusion related to withdrawals, and contamination from anthropogenic sources (Hunt, 2004; Gingerich, 2008; Tribble, 2008; Izuka and others, 2018; Frazier and others, 2023). Climate change can affect groundwater availability in Hawaiʻi mainly by influencing groundwater recharge rates. Recent studies have indicated that observed climate change has already affected groundwater resources in the Hawaiian Islands with long-term downward trends of rainfall across more than 90 percent of the State (Frazier and Giambelluca, 2017) and upward trends of surface temperature, particularly at higher altitudes (Giambelluca and others, 2008; Diaz and others, 2011; McKenzie and others, 2019), generally coinciding with observed long-term downward trends of stream base flow (Oki, 2004; Bassiouni and Oki, 2013; Clilverd and others, 2019). These historical changes imply a decreasing trend in groundwater recharge and storage and an overall decline in groundwater availability. Furthermore, published mid-century and end-of-century climate projections for the main Hawaiian Islands (Elison Timm and others, 2015; Zhang and others, 2016b; Zhang and Wang, 2017) indicate that rainfall is projected to decrease across large sections of each island, which has important implications for future groundwater availability.

Two reports by the Intergovernmental Panel on Climate Change, the Sixth Assessment Report and the Special Report

on Global Warming of 1.5 °C (Intergovernmental Panel on Climate Change, 2018, 2021), indicate that continued emissions of greenhouse gases will cause further warming and changes in all components of the Earth's climate system. In response to the warming, General Circulation Model (GCM) projections indicate that changes in the water cycle will not be uniform and that contrasts in precipitation between wet and dry regions and wet and dry seasons will increase over the 21st century. In the Hawaiian Islands, the interaction of prevailing northeasterly trade winds with the island chain's topography gives rise to a diversity of climates on each island where the mean annual rainfall for recent climate conditions can vary by a factor of 15 or more across a distance of six miles (mi) or less (Schroeder, 1993; Giambelluca and others, 2013). Many GCMs have a horizontal grid-size resolution of about 60 mi or greater (Taylor and others, 2012). Consequently, topographic features, such as the Hawaiian Islands, are below the typical horizontal and vertical resolution of many GCMs and are not accurately represented in the GCMs (Timm and Diaz, 2009). High-resolution climate information is needed to incorporate geographical variations at horizontal grid-size resolutions of 1 kilometer (km) or less and resolve their influence on Hawai'i's climate. To address this need, dynamical and statistical downscaling methods have recently been used to post-process GCM results into mid- to late-21st century high-resolution climate-projection datasets for the Hawaiian Islands (Elison Timm and others, 2015; Zhang and others, 2016a, b; Elison Timm, 2017; Zhang and Wang, 2017).

With projected water demand in the Hawaiian Islands expected to increase by roughly 13 percent from 2020 to 2035, from about 239 to 269 million gallons per day (Mgal/d; Townscape, Inc., 2019), estimates of the spatial distribution of groundwater recharge are needed for the State of Hawai'i to evaluate the effects of projected changes in climate and increased water demand on future groundwater availability. Hydrologic models representing the diverse topography and climate of several of the Hawaiian Islands have been developed to produce spatially distributed estimates of recharge rates for the islands (Engott, 2011; Engott and others, 2017; Izuka and others, 2018; Johnson and others, 2018; Mair and others, 2019; Oki and others, 2020). Estimates of recharge are used by the State of Hawai'i Commission on Water Resource Management (CWRM) in the calculation of sustainable-yield values for aquifer systems on each of the main Hawaiian Islands (Townscape, Inc., 2019). Spatially distributed estimates of recharge are also a critical input to numerical groundwater models that assess groundwater availability under different recharge and demand scenarios. Numerical groundwater models have helped inform decision-making by water-resource managers in Hawai'i (Gingerich, 2008; Gingerich and Engott, 2012; Oki and others, 2020; Izuka and others, 2021; Rotzoll and others, 2022).

For this study, the Water-budget Accounting for Tropical Regions Model (WATRMod; Oki, 2022) was used to estimate recharge for recent and future-climate conditions on Kaua'i, O'ahu, Moloka'i, Lāna'i, Maui, and the Island of Hawai'i (fig. 1), assuming 2020 land-cover conditions. Three high-resolution,

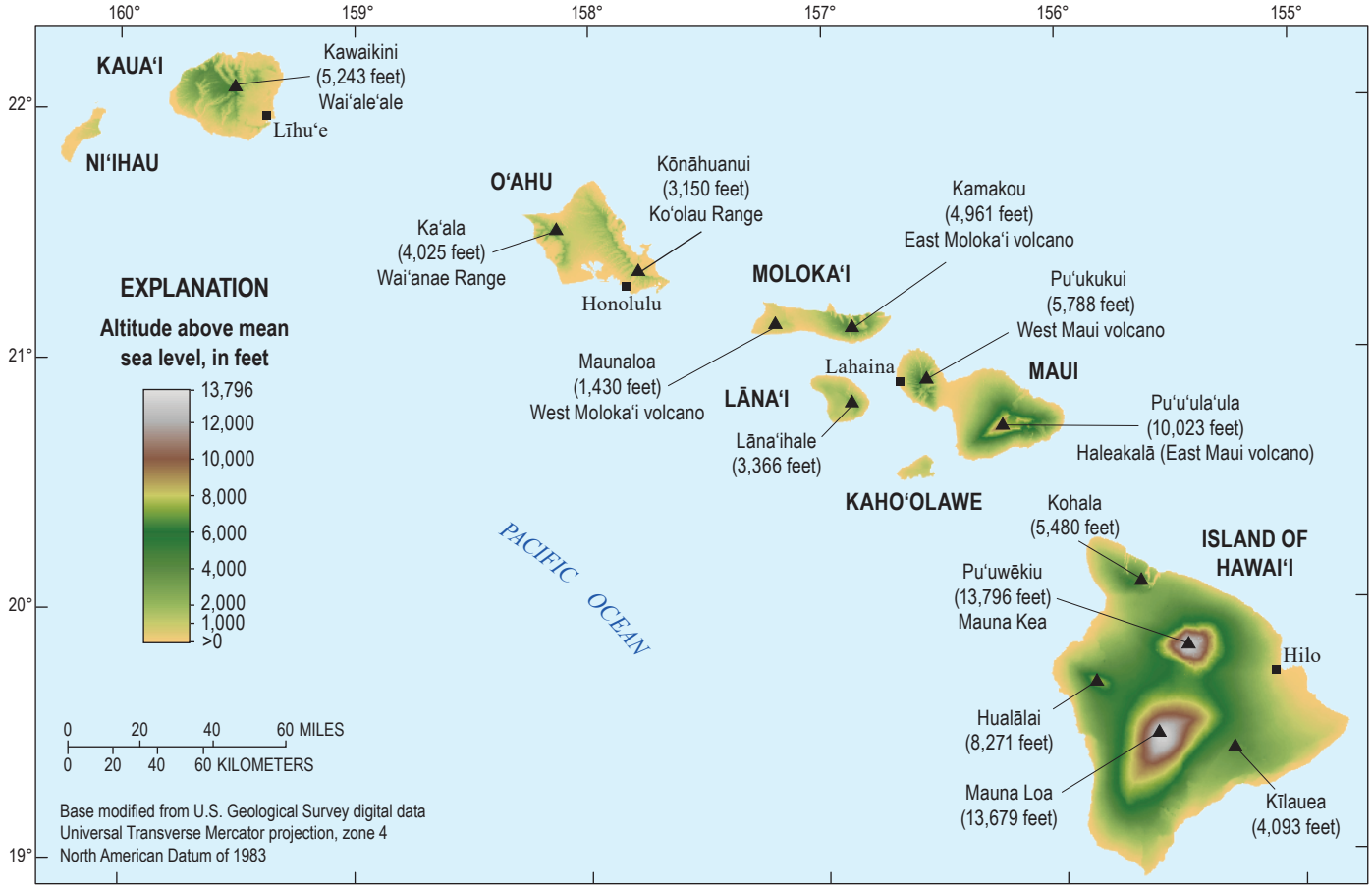

Figure 1. Map of the main Hawaiian Islands showing general topography and high points of individual islands.

downscaled climate-projection datasets were selected to develop inputs for WATRMod that capture the available range of projected changes in island-wide rainfall for each island. For Lānaʻi, an additional drought scenario was selected to assess the effect of episodic dry periods on recharge estimates. The recharge estimates from the climate-change scenarios in this study can be used in numerical groundwater models to evaluate the additional effects of groundwater withdrawals on groundwater levels, freshwater volume in the aquifer, streamflow, coastal discharge, and salinity of water withdrawn from production wells. This study was conducted in cooperation with the Pacific Islands Climate Adaptation Science Center and CWRM and in collaboration with Pūlama Lānaʻi.

Purpose and Scope

This report describes the spatial distribution of mean annual groundwater recharge on Kauaʻi, Oʻahu, Molokaʻi, Lānaʻi, Maui, and the Island of Hawaiʻi for recent climate conditions and three future-climate scenarios. An additional drought scenario was included for Lānaʻi to assess the effects of episodic dry periods on groundwater recharge. Recharge was computed with WATRMod using a daily computation interval (Oki, 2022). Hydrological processes and physical conditions that affect groundwater recharge for recent- and future-climate conditions were simulated in WATRMod using the most recent datasets available—including maps of 2020 land-cover conditions, rainfall for each month during 1978–2007, daily rainfall during 1990–2014, and mean monthly reference grass ET (Giambelluca and others, 2014; Frazier and others, 2016; Longman and others, 2019). Published high-resolution downscaled climate projections (Elison Timm and others, 2015; Zhang and others, 2016a, b; Zhang and Wang, 2017) were used to estimate future monthly rainfall and seasonal runoff for simulating the effects of future-climate conditions on groundwater recharge and other water-budget components.

The water-budget model and input datasets for recent and future-climate conditions were used to estimate the spatial distribution of mean annual groundwater recharge for four water-budget model scenarios for each island: (1) recent reference-period conditions during 1978–2007, (2) a mid-century scenario using projected rainfall conditions for a Representative Concentration Pathway (RCP) scenario during 2041–71 with a total radiative forcing of 8.5 watts per square meter (W/m^2) by the year 2100 (RCP8.5 2041–71 scenario), (3) a dry-climate scenario using projected rainfall conditions for a RCP8.5 scenario during 2071–99 (RCP8.5 2071–99 scenario), and (4) a wet-climate scenario using projected rainfall conditions for a "Special Report on Emissions Scenarios" A1B scenario during 2080–99 (A1B 2080–99 scenario) for Maui, a RCP scenario during 2080–99 with a total radiative forcing of 4.5 W/m^2 by the year 2100 (RCP4.5 2080–99 scenario) for the Kauaʻi, Lānaʻi, and the Island of Hawaiʻi, and a RCP8.5 scenario during 2080–99 (RCP8.5 2080–99 scenario) for Oʻahu and Molokaʻi. The A1B and RCP scenarios represent possible future greenhouse gas emission and concentration trajectories, as described by Nakićenović and others (2000) and the Intergovernmental Panel on Climate Change (2014). Projected rainfall conditions for mid-century and dry-climate scenarios are presented in Elison Timm and others (2015); wet-climate scenarios are presented in Zhang and others (2016a, b) for Maui, and Zhang and Wang (2017) for the other five islands. A fifth water-budget scenario was added as a historical drought scenario for Lānaʻi to assess the effect of dry conditions on groundwater recharge by using rainfall conditions for a consecutive 5-year period with the lowest mean annual island-wide rainfall during 1920–2012. The RCP8.5 2041–71 scenario represents the projected rainfall for mid-century conditions, whereas the RCP8.5 2071–99 scenario represents the driest climate scenario among the available set of future-climate projections for each island. The RCP4.5 2080–99, RCP8.5 2080–99, and the A1B 2080–99 scenarios represent the wettest climate scenario among the available set of future-climate projections for each island. This study also included the development of 2020 land-cover maps of each island for use in each water-budget model scenario. WATRMod simulation results were used to quantify regional hydrologic changes between the 1978–2007 reference-period scenario and future-climate scenarios.

The climate-projection datasets developed by Zhang and others (2016a, b) and Zhang and Wang (2017), identified in this study as the wet-climate scenarios, provide additional information for modifying input datasets related to daily rainfall frequency, mean monthly reference ET, and forest-canopy evaporation. The input datasets of mean monthly reference ET were further modified on the basis of projected increases in mean atmospheric carbon dioxide (CO_2) concentrations associated with each climate-projection dataset. The effects of selectively changing inputs of monthly rainfall, daily rainfall frequency, seasonal runoff, mean monthly reference ET, and forest-canopy evaporation on recharge estimates were evaluated for the wet-climate scenario for each island considered in this study. Incomplete sets of similar information in the climate-projection datasets from Elison Timm and others (2015) precluded similar analyses from being conducted for the mid-century and dry-climate scenarios. Investigating the causes of differences in climate conditions among the published climate projections was beyond the scope of this study.

Previous Studies

As of 2021, estimates of island-wide groundwater recharge for Kauaʻi, Oʻahu, Molokaʻi, Lānaʻi, Maui, and the Island of Hawaiʻi were developed from water-budget studies that used a range of climate and land-cover conditions (table 1). These studies estimated recharge for drought or recent-climate conditions, and pre-development or recent land-cover conditions (Hardy, 1996; Engott, 2011; Engott and others, 2017; Izuka and others, 2018; Johnson and others, 2018; Mair and others, 2019; Brewington and others, 2019; Oki and others, 2020; Rotzoll and Johnson, 2023). Four of these studies also estimated recharge for projected future-climate and hypothetical land-cover conditions for Oʻahu, Maui, and the Island of Hawaiʻi (Engott, 2011; Brewington and others, 2019; Mair and others, 2019; Rotzoll and Johnson, 2023). McLean and others (2022) developed a web-based tool that allows users to quantify the effects of changing land cover and

Table 1. Selected island-wide water-budget studies for Kauaʻi, Oʻahu, Molokaʻi, Lānaʻi, Maui, and the Island of Hawaiʻi.

[SD, statistical downscaling; HRCM1, Hawaiʻi Regional Climate Model version 1 from Zhang and others (2016a, b); RCP8.5, Representative Concentration Pathway with total radiative forcing of 8.5 Watts per square meter by the year 2100 from Intergovernmental Panel on Climate Change (2014); A1B, A1B emission scenario from Nakićenović and others (2000); Mgal/d, million gallons per day]

Island	Reference	Land-cover scenario	Climate scenario	Island-wide recharge (Mgal/d)
Kauaʻi	Izuka and others (2018)	1870	1978–2007	871
		2010	1978–2007	875
Oʻahu	Engott and others (2017)	2010	1978–2007	660
		2010	1998–2002	417
	Izuka and others (2018)	1870	1978–2007	675
	Rotzoll and Johnson (2023)	2010	SD RCP8.5 2041–71	538
Molokaʻi	Oki and others (2020)	2000s	1978–2007	227
		2000s	1998–2002	140
Lānaʻi	Hardy (1996)	Not specified	1916–83	62
Maui	Johnson and others (2018)	2010	1978–2007	1,308
		2010	1998–2002	1,010
	Izuka and others (2018)	1870	1978–2007	1,279
	Mair and others (2019)	2017	1978–2007	1,232
		2017	SD RCP8.5 2071–99	1,059
		2017	HRCM1 A1B 2080–99	1,376
	Brewington and others (2019)	Four hypothetical scenarios	SD RCP8.5 2071–99	1,059 to 1,190
		Four hypothetical scenarios	HRCM1 A1B 2080–99	1,385 to 1,520
Island of Hawaiʻi	Engott (2011)	2008	1916–83	6,595
		2008	Lowest 5-year average[a]	4,271
		Three hypothetical scenarios	1916–83	6,604 to 6,705
		One hypothetical scenario	SD A1B 2071–99[b]	5,861 to 8,289
	Izuka and others (2018)	1870	1978–2007	5,291

[a] Selected as period with lowest 5-year average rainfall during 1916–83 in each of six rainfall-variability zones

[b] Values from mean, and upper and lower confidence interval of rainfall projections from Timm and others (2009)

future-climate conditions on groundwater recharge on Oʻahu. To the authors' knowledge, no study has been published that documents climate-change effects on island-wide groundwater recharge for Kauaʻi, Molokaʻi, and Lānaʻi.

Engott (2011) used a water-budget model to estimate groundwater recharge on the Island of Hawaiʻi for average climate conditions during 1916–1983 and three future-climate scenarios, and four land-cover scenarios. He estimated rainfall-change factors on the basis of the means and lower and upper margins of 95-percent statistical confidence intervals of rainfall anomalies for the late 21st-century as reported by Timm and Diaz (2009). Estimated changes in island-wide recharge for the Island of Hawaiʻi ranged from an 11-percent decrease to a 26-percent increase when compared to the 1916–1983 climate scenario and 2008 land-cover conditions (table 1).

Mair and others (2019) utilized a water-budget model developed by Johnson and others (2018) to estimate the spatial distribution of recharge on Maui for average climate conditions during 1978–2007 and two future-climate scenarios, for 2017 land-cover conditions. The two future-climate scenarios, including one drier and one wetter than the 1978–2007 scenario, were developed using high-resolution climate projections of late 21st-century rainfall by Elison Timm and others (2015) and Zhang and others (2016a, b). Estimated changes in island-wide recharge for Maui ranged from a 14-percent decrease to a 12-percent increase when compared to the 1978–2007 scenario (table 1). Brewington and others (2019) utilized a water-budget model developed by Johnson and others (2018) to estimate the spatial distribution of recharge on Maui for four hypothetical land-cover scenarios and the two future-climate scenarios used by Mair and others (2019). For the dry future-climate scenario, island-wide recharge for the four land-cover scenarios increased by as much as 12 percent relative to 2017 land-cover conditions (table 1). For the wet future-climate, island-wide recharge for the four land-cover scenarios increased by as much as 10 percent relative to 2017 land-cover conditions.

Rotzoll and Johnson (2023) utilized a water-budget model developed by Engott and others (2017) to estimate the spatial distribution of recharge on O‘ahu for one future-climate scenario and land-cover conditions in 2010. The future-climate scenario was developed using high-resolution climate projections of mid-21st-century rainfall by Elison Timm and others (2015). Island-wide recharge for the future-climate scenario decreased by 18 percent relative to average climate conditions during 1978–2007 and 2010 land-cover conditions as reported by Engott and others (2017).

Other studies have quantified how climate change affects groundwater recharge and other water-budget components on parts of O‘ahu and the Island of Hawai‘i using water-budget or watershed models and inputs derived from regional downscaled projections of rainfall and temperature and high-resolution downscaled projections of rainfall (Safeeq and Fares, 2012; Leta and others, 2016, 2017, 2018; Povak and others, 2017; Strauch and others, 2017; Ghazal and others, 2019). Three of these studies quantified estimated effects on groundwater recharge (Leta and others, 2016, 2017; Ghazal and others, 2019).

Leta and others (2016) used a watershed model to simulate changes in groundwater recharge and other water-budget components in an eastern O’ahu watershed for a set of three climate scenarios that incorporated projections of atmospheric CO_2 concentrations, temperature, and rainfall (Intergovernmental Panel on Climate Change, 2007; Timm and Diaz, 2009; Diaz and others, 2011; Elison Timm and others, 2011). In their first scenario, they found that a projected 10-percent decrease in wet season rainfall and a projected 5-percent increase in dry-season rainfall resulted in a 7-percent decrease in mean annual recharge. In a second scenario, projected changes in rainfall coupled with a 1.1 °C increase in temperature resulted in a 10-percent decrease in mean annual recharge. In their third scenario, projected changes in rainfall and temperature coupled with an increase of 220 parts per million in atmospheric CO_2 concentration resulted in about a 1-percent decrease in mean annual recharge.

Leta and others (2017) used a watershed model to simulate changes in groundwater recharge and other water-budget components in a southeastern O‘ahu watershed for a set of 48 climate scenarios that incorporated projections of solar radiation, temperature, and rainfall (Intergovernmental Panel on Climate Change, 2014; Elison Timm and others, 2015). For the scenarios incorporating only projected changes in rainfall, results from Leta and others (2017) ranged from a 44-percent decrease to a 14-percent increase in mean monthly recharge. Scenarios that coupled projected changes in rainfall with projected changes in temperature resulted in mean monthly recharge changes ranging from a 51-percent decrease to an 8-percent increase. Finally, scenarios that coupled projected changes in rainfall and temperature with projected changes in solar radiation resulted in mean monthly recharge changes ranging from a 57-percent decrease to a 12-percent increase.

Ghazal and others (2019) used a watershed model to simulate changes in groundwater recharge and other water-budget components in a wetland located in an eastern O‘ahu watershed on O‘ahu for four future-climate scenarios (Elison Timm and others, 2015) and one hypothetical land-cover scenario. For scenarios that only incorporated projected changes in rainfall, their results ranged from a 12-percent decrease in wet-season recharge to an 85-percent decrease in dry-season recharge. Scenarios that coupled projected changes in rainfall with hypothetical changes in land cover resulted in changes in seasonal recharge ranging from a 51-percent decrease during the wet season to a 92-percent decrease during the dry season.

Study Area

The Hawaiian archipelago, located in the central basin of the Pacific Ocean, is geographically isolated from continental landmasses. The eight main islands—Ni‘ihau, Kaua‘i, O‘ahu, Moloka‘i, Lāna‘i, Kaho‘olawe, Maui, and the Island of Hawai‘i (fig. 1)—are the emergent tops of basaltic shield volcanoes that rise from the floor of the Pacific Ocean and are hereinafter referred to as the Hawaiian Islands. The older islands are in the northwestern part of the chain and the younger islands are in the southeast. Ni‘ihau and Kaua‘i are the oldest islands at over 5.1 million years old, whereas the Island of Hawai‘i is the youngest island with active volcanism (Macdonald and others, 1983). Each island is made up of one or more shield volcanoes formed by thousands of lava flows emanating from vents in rift zones and, in most cases, a central caldera. Lava flows on the flanks of the volcanoes form productive aquifers containing freshwater-lens systems. These aquifers are a primary source of freshwater needed by the 1.5 million residents (U.S. Census Bureau, 2021), agricultural pursuits, industries, and the U.S. military. Fresh groundwater in coastal aquifers is underlain by saltwater from the ocean and can potentially be affected by short- and long-term climatic changes and human activity (Izuka and others, 2018). The six islands considered in this study are Kaua‘i, O‘ahu, Moloka‘i, Lāna‘i, Maui, and the Island of Hawai‘i, which have a combined land area of 6,305 square miles (mi^2). For management purposes, CWRM divides each island into aquifer systems, with boundaries delineated on the basis of geologic, topographic, and geographic considerations (fig. 2; State of Hawai‘i, 2014). CWRM estimates groundwater sustainable-yield values for each aquifer system using an analytical model that requires an estimate of recharge as input. To address management and community needs, the groundwater-recharge results for this study are summarized by aquifer system.

Physical Setting

The islands considered for this study are the six largest of the Hawaiian Islands and constitute 98 percent of the total land area in the archipelago. The islands range in size from 141 (Lāna‘i) to 4,029 mi^2 (Island of Hawai‘i), with peak altitudes ranging from 3,366 (Lāna‘i) to 13,796 feet (ft; Island of Hawai‘i). O‘ahu is the most densely populated of the islands, with about 1 million residents or 70 percent of Hawai‘i’s population (U.S. Census Bureau, 2021).

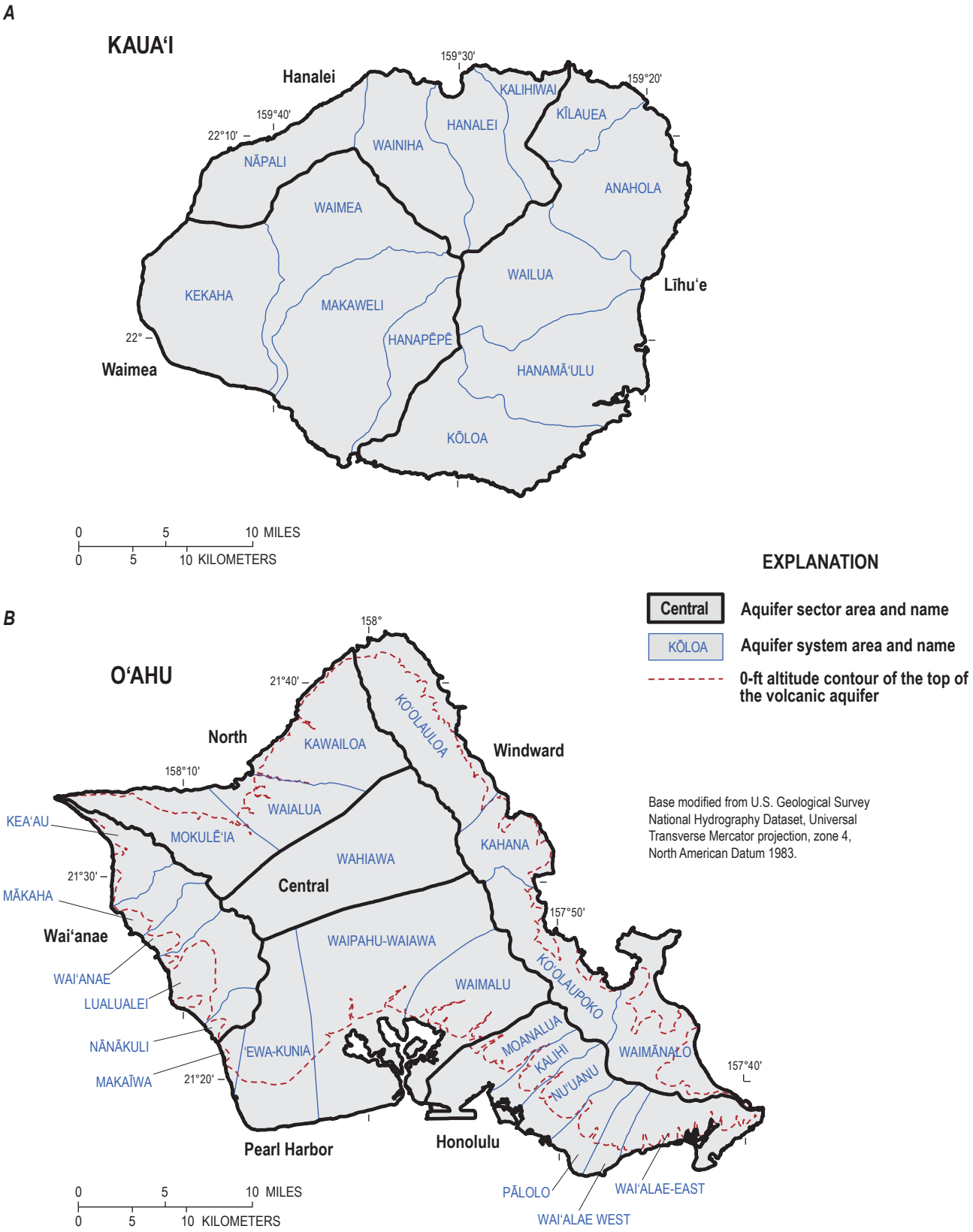

Figure 2. Maps of aquifer systems and sectors for (*A*) Kaua'i, (*B*) O'ahu, (*C*) Moloka'i, (*D*) Lāna'i, (*E*) Maui, and (*F*) the Island of Hawai'i (modified from State of Hawai'i, 2014) and the 0-foot altitude contour of the top of the volcanic aquifer for (*B*) O'ahu and (*E*) Maui (modified from Sherrod and others, 2007; Izuka and others, 2018). ft, feet.

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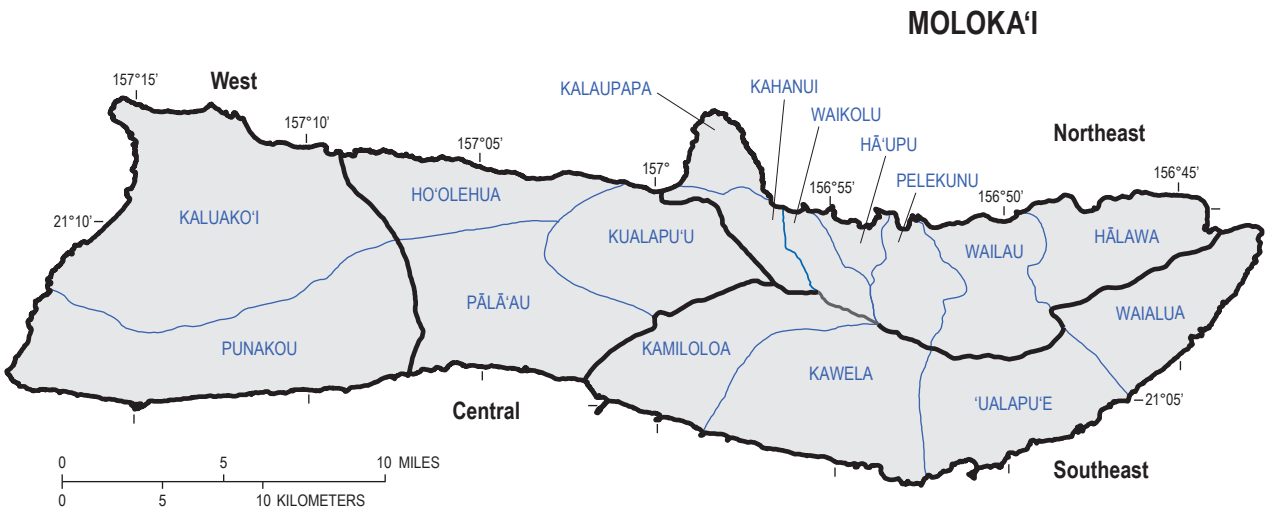

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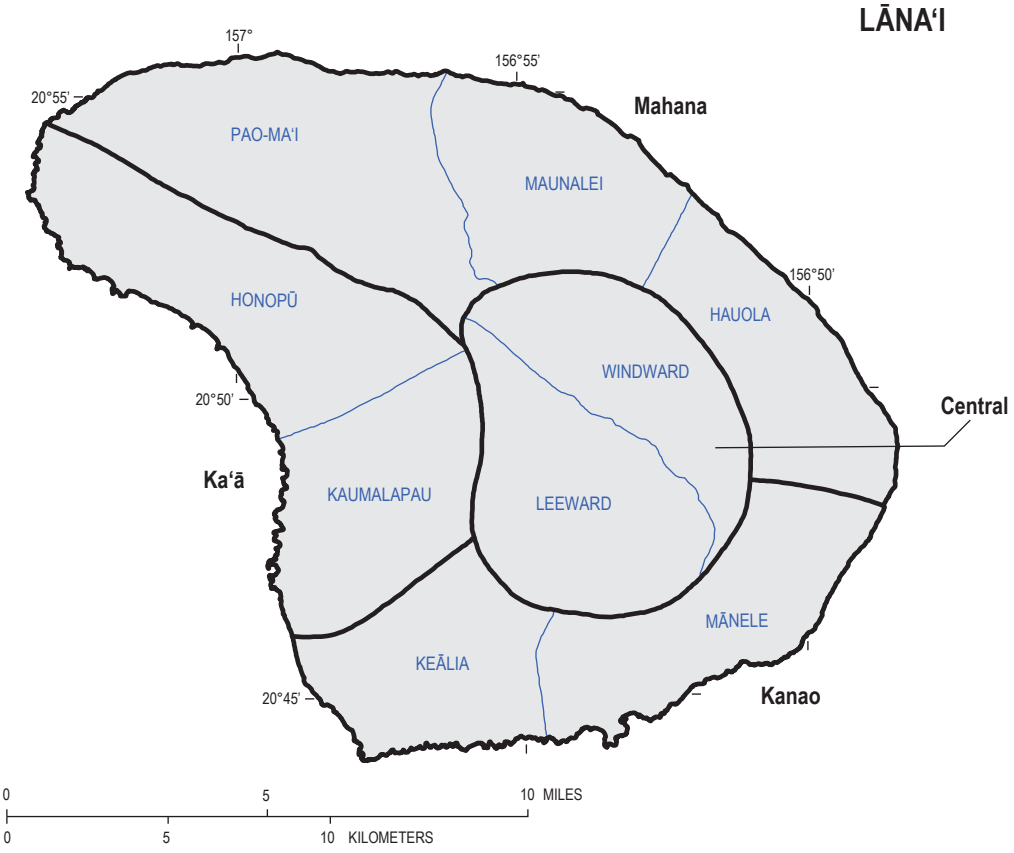

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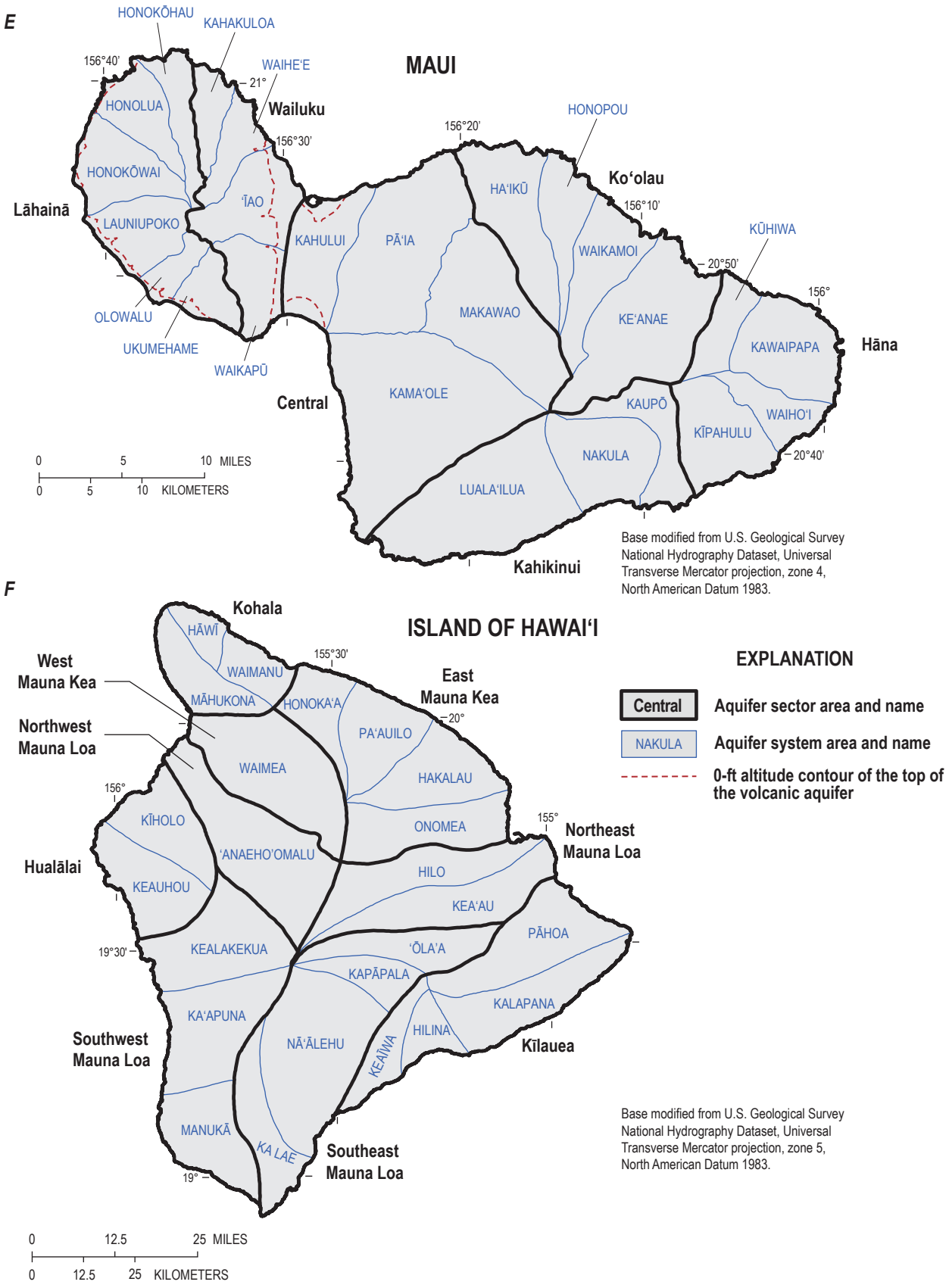

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Kaua‘i, the fourth largest island, is located near the northwestern end of the archipelago and covers an area of 554 mi^2. Kaua‘i is essentially a large, broad domed remanent of a shield volcano carved with large canyons and a broad sloping summit plateau. The highest point on Kaua‘i is Kawaikini on Wai‘ale‘ale at 5,243 ft (fig. 1) and receives 393 inches of rain a year (fig. 3; Giambelluca and others, 2013).

O‘ahu, the third largest island, has an area of 595 mi^2 and is formed by two elongated shield volcanoes, the eroded remnants of which are the Wai‘anae Range in the west and the Ko‘olau Range in the east (fig. 1). Weathering and erosion have greatly modified the volcanoes, carving out large amphitheater valleys and steep inter-fluvial ridges in the island’s interior. O‘ahu has extensive coastal-plain deposits that partly overlie the shield volcano lava flows and almost completely encircle the island. The coastal-plain sediments and rejuvenated-stage volcanic rocks on O‘ahu have been referred to as “caprock” because they act as a semiconfining unit over parts of the high-permeability lava-flow aquifers (Sherrod and others, 2007; Izuka and others, 2018). O‘ahu is the center of commerce, industry, and government in Hawai‘i and the site of the State capital, Honolulu.

Moloka‘i is the fifth largest of the Hawaiian Islands and occupies an area of 260 mi^2. The island primarily consists of two shield volcanoes, the older and smaller West Moloka‘i volcano rising to an altitude of 1,430 ft at Maunaloa, and the younger East Moloka‘i volcano which reaches an altitude of 4,961 ft at Kamakou peak (Stearns and Macdonald, 1947; Sherrod and others, 2021). Between the summit areas of the volcanoes is a relatively flat central saddle region.

Lāna‘i is the sixth largest of the Hawaiian Islands and has an area of 141 mi^2, and is the smallest of the islands examined in this study. Lāna‘i was formed by a single shield volcano and, except where modified by erosion or faulting, is generally gently sloped, rising from sea level at the coast to 3,366 ft near the central part of the island. About 94 percent of the land surface of Lāna‘i is below 2,000 ft, the base of the cloud zone (fig. 4*D*). Groundwater is the main source of water on Lāna‘i owing to the lack of reliable surface-water resources.

Maui is the second largest of the Hawaiian Islands and has an area of 728 mi^2. The island is formed by two shield volcanoes, the West Maui volcano and the East Maui volcano known as Haleakalā. An intervening saddle of broad, gently sloping land sits between the two volcanoes (fig. 1). Erosion of West Maui volcano has carved deep valleys resulting in sharp-crested ridges. On Haleakalā, the valleys on the rainy northeastern (windward) slope are separated by broad areas and ridges, whereas the drier southwestern (leeward) slope is less incised and retains the broad, shield shape of the volcano. Wedges of low-permeability sedimentary caprock impede the

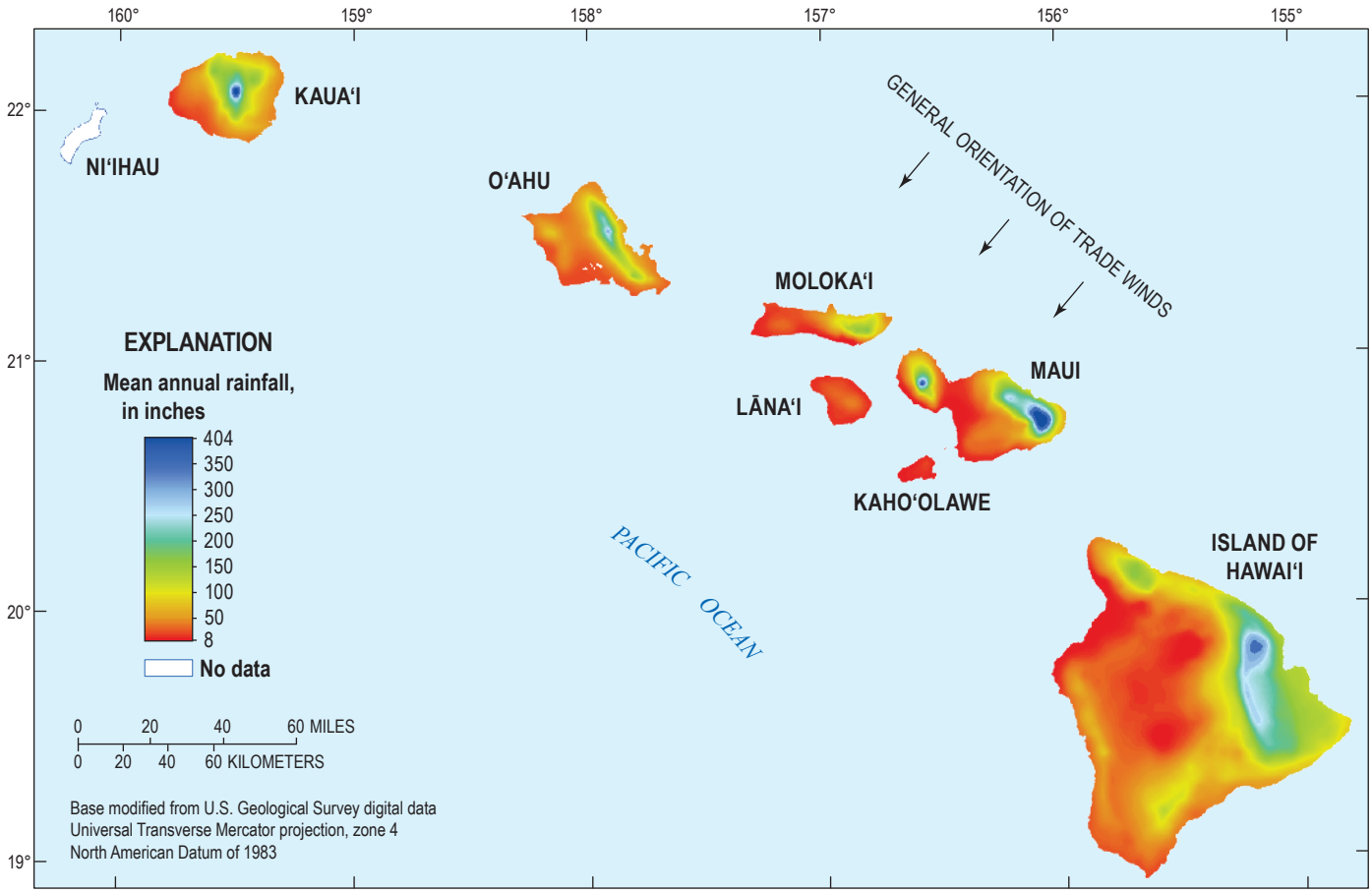


Figure 3. Map showing mean annual rainfall in the main Hawaiian Islands, excluding Ni‘ihau (modified from Giambelluca and others, 2013).

seaward flow of fresh groundwater in freshwater-lens systems along parts of the northeastern and southwestern flanks of West Maui volcano (Sherrod and others, 2007; Izuka and others, 2018). Wedges of caprock between West Maui volcano and Haleakalā also impede the flow of fresh groundwater between West Maui volcano and the saddle between volcanoes.

The Island of Hawai‘i lies at the southeastern end of the Hawaiian archipelago (fig. 1). With an area of 4,029 mi^2, it is nearly twice as large as the combined areas of the other Hawaiian Islands (Macdonald and others, 1983). The island consists of five volcanoes: Kohala, Mauna Kea, Hualālai, Mauna Loa, and Kīlauea. Mauna Kea is the highest of these volcanoes, rising to an altitude of 13,796 ft, and Mauna Loa is the largest by volume. With the exception of the northeastern slopes of Kohala and Mauna Kea, the geologically youthful land surface generally retains a broad, shield-shaped morphology from the volcanic summits to the ocean, except where modified by faulting, volcanic vents, and (or) erosion. On the northeastern sides of Kohala and Mauna Kea, large streams resulting from high rainfall have carved deep valleys in the slopes of these relatively older volcanoes.

Climate

The Hawaiian Islands are located in the tropics and in the path of the trade-wind belt of the North Pacific anticyclone. The climate for most of the islands is characterized by mild temperatures, moderate humidity, and prevailing northeasterly trade winds (Giambelluca and Schroeder, 1998). Hawai‘i’s climate is diverse, however, and includes deserts, tropical rain forests, and snow-capped mountains. Climate distribution is influenced by the prevailing northeasterly trade winds and their interaction with the islands’ volcanic mountains. Rain is generated by the orographic effect—trade winds blow against the mountain slopes and air carrying moisture from the ocean is forced to rise and cool adiabatically. The cooling causes condensation of water vapor and results in high rainfall amounts over mountain crests and on northeast-facing (windward) slopes and low amounts in leeward areas (fig. 3). The contrast between dry leeward areas and wet windward slopes is substantial—mean annual rainfall ranges from less than 10 inches on some leeward coasts to greater than 400 inches on the windward slopes—and can vary by more than an order of magnitude within 5 horizontal miles (Giambelluca and Schroeder, 1998; Giambelluca and others, 2013). The interaction of trade winds and mountains provides a voluminous and persistent influx of freshwater to Hawai‘i. Without the orographic effect, rainfall would be less than 30 inches per year (in/yr), similar to that of the open ocean in this region of the Pacific (Giambelluca and Schroeder, 1998). The orographic effect and the rainfall it generates are limited to altitudes at or below about 8,200 ft by the trade-wind temperature inversion, thus the climate at the peaks of the highest mountains—Haleakalā on Maui, and Mauna Kea and Mauna Loa on the Island of Hawai‘i—is dry and mean annual rainfall values are among the lowest in the Hawaiian Islands (figs. 1, 3). The southwest-facing slopes of the Island of Hawai‘i, though sheltered from the trade winds, receive air flow from trade winds that wrap around Mauna Loa and sea breezes generated by diurnal heating and cooling of the large land mass of the Island of Hawai‘i. Combined, these winds produce a band of higher precipitation on the middle-altitude slopes (Giambelluca and Schroeder, 1998; Giambelluca and others, 2013). Hawai‘i also receives rain from occasional migratory storm systems unrelated to the orographic effect. These storms can bring heavy rain to any part of an island and are the main source of rainfall in the dry leeward areas and on mountain tops above the trade-wind inversion (Giambelluca and others, 2013).

Rainfall seasonality across much of the Hawaiian Islands is characterized by a dry and a wet season. The dry season (May–September) is characterized by warm temperatures and steady trade winds, whereas the wet season (October–April) has cooler temperatures and less persistent trade winds (Blumenstock and Price, 1967; Sanderson, 1993). In drier leeward areas, where most rainfall comes from storms, mean monthly rainfall is highest in December and January and lowest in June and July (Giambelluca and others, 2013; Frazier and others, 2016). In wet windward areas that receive abundant orographic rainfall, mean monthly rainfall tends to be more evenly distributed throughout the year. On the southwest-facing slopes of the Island of Hawai‘i, rainfall tends to be highest from May to September when the sea breezes generated by the diurnal heating and cooling of the large land mass are strongest. Precipitation in the Hawaiian Islands also varies with multi-year cycles linked to oceanic and atmospheric climate cycles, such as El Nino and La Nina, which repeats roughly every 2–7 years with atmospheric events that typically last about 9–12 months, the Pacific Decadal Oscillation (PDO), a recurring pattern of widespread variations in ocean temperature and atmospheric climate in the Pacific basin, and the Pacific North American pattern (PNA), a pattern of air pressure anomalies over the Pacific Ocean and North America. The El Nino, La Nina, PDO, and PNA cycles cause some years to be wetter or drier than the long-term average (Chu and Chen, 2005; Frazier and others, 2018). Rainfall in Hawai‘i is typically lower during El Nino events (Giambelluca and others, 2013).

Land Cover

The diversity of land cover in the Hawaiian Islands reflects the influence of island age, steep climate gradients, agricultural practices, and land development (fig. 4). The islands are dominated by native and non-native forests, shrub and grasslands, and varying amounts of agriculture and developed regions that are distributed at and around the main metropolitan areas. Large tracts of forest, generally dominated by the native canopy species, exist in the wetter areas of the island, with shrubland and grassland in the drier areas (fig. 4). Sparsely vegetated land generally exists on lava flows that have not been sufficiently weathered to allow for plant colonization or along dry ridges and coastlines. Small wetlands exist on some of the larger islands in some of the

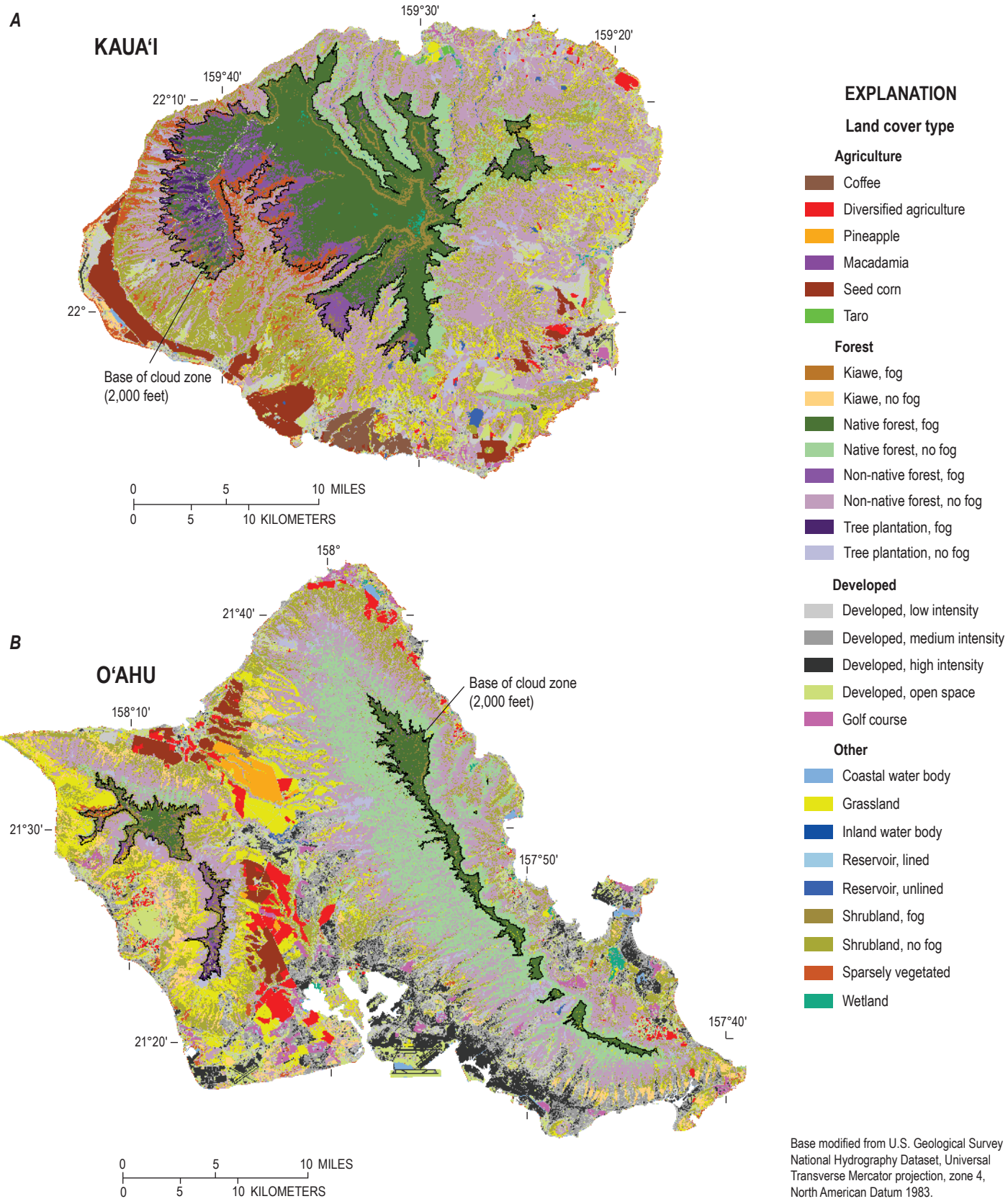
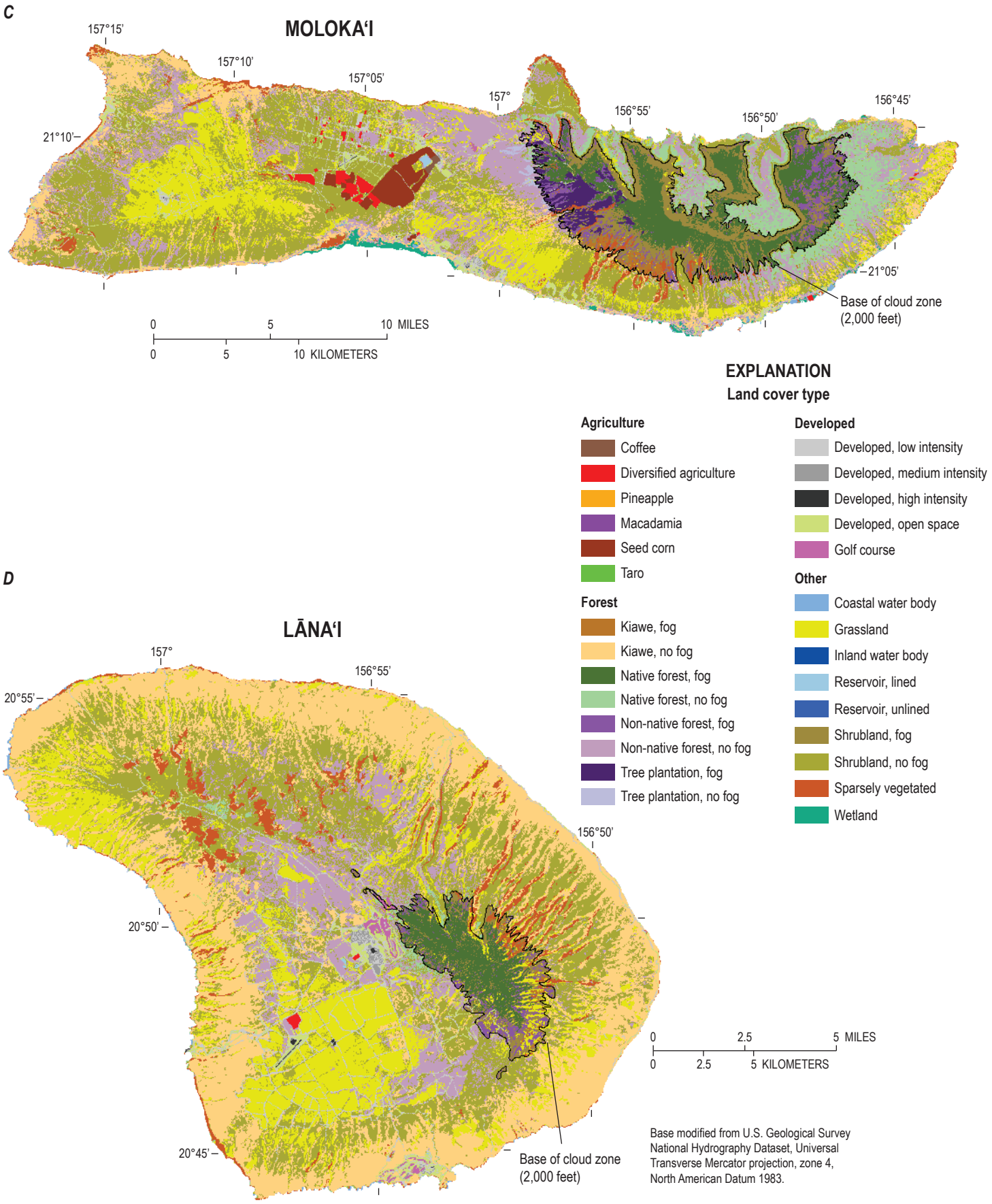


Figure 4. Maps of land cover in 2020 for (*A*) Kaua'i, (*B*) O'ahu, (*C*) Moloka'i, (*D*) Lāna'i, (*E*) Maui, and (*F*) the Island of Hawai'i. Note that reservoir, taro, and developed land cover types (excluding golf courses) are difficult to discern at map scale.



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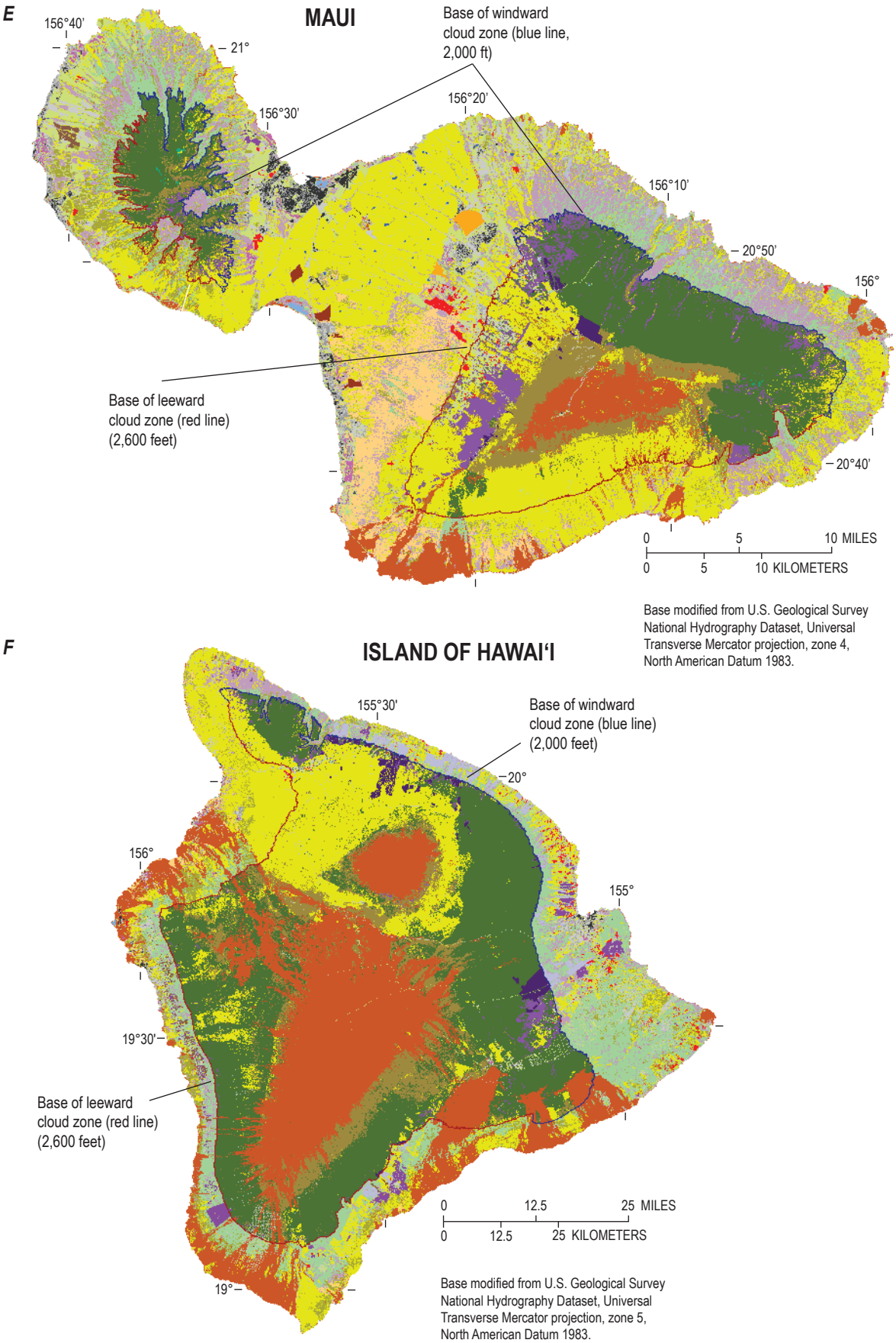


Figure 4.—Continued

flat summits of old volcanic craters, such as Waiʻaleʻale on Kauaʻi, Kaʻala on Oʻahu, Puʻukukui on West Maui volcano, and upland areas of eastern Haleakalā (figs. 1, 4).

On Kauaʻi, non-native forest and shrubland dominate the wet upland areas of eastern Kauaʻi within the cloud zone with wetland areas occurring near Kawaikini and Waiʻaleʻale region (figs. 1, 4*A*). The wet upland western slope of Waiʻaleʻale is dominated by native forest and non-native forest within and below the cloud zone. The mid-altitude areas on the windward side are partially developed but are mainly comprised of grasslands, shrublands, and non-native forest areas. The southern to southwestern coastal lands have large agricultural areas of seed corn and coffee with patches of kiawe near the coast (see Appendix 1 for a description of kiawe). The main developed regions are near the northern, eastern, and southern coastlines with the highest intensity occurring near the city of Līhuʻe.

Oʻahu is the most developed island with the coastal lands of southern Oʻahu and the Honolulu region having the most high-intensity developed areas (fig. 4*B*). The region between the Koʻolau and Waiʻanae Ranges is covered in grassland and agriculture fields (seed corn, coffee, pineapple, and other diversified agriculture). The wet upland areas of the Waiʻanae Range have native forest surrounded by non-native forest and shrubland. Shrub and grasslands dominate the lower altitudes near the southwestern coastline. The wet upland regions of the Koʻolau Range are dominated by native forest with non-native forest becoming more prevalent on the mid-altitude slopes. The leeward coastal areas have mixed non-native forest, shrubland, and grassland with developed areas closest to the coast.

On Molokaʻi, the wet upland areas of the East Molokaʻi volcano are dominated by native and non-native forests and tree plantations (fig. 4*C*). The mid-to-low altitude areas along the slopes of the East Molokaʻi volcano are dominated by shrub and grasslands with patches of sparsely vegetated areas occurring in the southern region. Low altitude areas for both eastern and western Molokaʻi are predominantly covered by shrub and grasslands, with large patches of kiawe growing near the coast, the largest of which occurs along the northwestern coastline. The central saddle area of Molokaʻi has patches of diversified agriculture and seed corn.

The dominant land-cover feature on Lānaʻi is the kiawe that encircles the entire coastline (fig. 4*D*). The small wet summit region of Lānaʻihale is covered in native and non-native forests. The mid-altitude regions of the island are dominated by shrubland, non-native forest, and areas that are sparsely vegetated. The remaining regions are comprised mostly of shrub and grasslands intermixed with small sections that are minimally developed.

On West Maui volcano, native forest and shrubland dominate the wet upland areas near Puʻukukui, whereas non-native forest, grassland, urban development, and agricultural fields (macadamia and coffee) cover the drier lowland areas (figs. 1, 4*E*). Mid-altitude areas on northwestern Haleakalā are developed with patches of pineapple and diversified agriculture. In the high-altitude area of Haleakalā, the peak, Puʻuʻulaʻula, is sparsely vegetated barren rock surrounded by a region of shrubland. Mostly native forest covers the mid-altitude wet northeastern slopes below the shrubland region and transitions to coastal regions dominated by non-native forest and grassland. Much of the arid southwestern slopes of Haleakalā are covered with non-native forest and grassland mixed with patches of development, native forest, and diversified agriculture. Coastal areas between West Maui volcano and Haleakalā are developed with grassland in between (fig. 4*E*).

On the Island of Hawaiʻi, the active shield volcanoes of Mauna Loa and Mauna Kea rise above the base of the trade-wind inversion. Areas above the base of the trade-wind inversion consist of sparsely vegetated barren rock that transition into shrubland and native forest (fig. 4*F*). The shrubland and native forest areas on Mauna Kea transition into grassland that extends northwest across the leeward slope of Mauna Kea down to the coast. The mid-altitude region of Mauna Loa and the eastern flank of Mauna Kea are mainly native and non-native forest with areas of tree plantations and macadamia groves. Areas near the west, south, and southeast coasts are mostly grassland and sparsely vegetated with small areas of developed land. The eastern, coastal areas of the island are covered by native and non-native forest with developed areas occurring around Hilo. The northern region of Kohala is covered by native and non-native forest on the windward side, but mostly grass and shrubland on the leeward side.

The maps of land cover developed for this study (fig. 4) reflect the latest complete sets of land-cover information for all six islands and differ from land-cover maps used in previous water-budget studies. For example, the land-cover maps in this study reflect the complete cessation of sugarcane cultivation and the expansion of diversified agriculture in the Hawaiian Islands. Other changes include the distinction of forest and shrubland land cover above and below the base of the cloud zone, and the delineation of kiawe land cover on each island. Lastly, owing to updated information on reservoirs in Hawaiʻi (State of Hawaiʻi, 2021), lined and unlined reservoirs are distinguished in the land-cover map for each island. See the "Model Input" section for discussion on the development of the land-cover maps used in this report.

Climate Scenarios

The effect of projected future-climate conditions on recharge was evaluated using climate projections developed for a set of global emission and warming scenarios adopted by Nakićenović and others (2000) and the Intergovernmental Panel on Climate Change (2014), hereinafter IPCC scenarios. The IPCC scenarios describe different climate futures, all of which are considered possible depending on the volume of greenhouse gases emitted between now and 2100. The climate projections used in this study are based on downscaled products developed for the three IPCC global emission and warming scenarios and are described in the following sections.

Special Report on Emissions Scenarios—A1B Scenario

The "Special Report on Emissions Scenarios" (SRES) A1B scenario is a global emission scenario characterized by very rapid global economic growth, global population that peaks in mid-century, rapid introduction of more efficient technologies, and a balanced dependence on fossil and non-fossil intensive energy sources (Nakićenović and others, 2000). The A1B scenario ranks as a mid-level projection among the SRES emission scenarios in terms of projected global carbon dioxide (CO_2) emissions for 2100.

Representative Concentration Pathway—RCP4.5 and RCP8.5 Scenarios

In 2014, the IPCC adopted a new approach for defining future scenarios to support research on potential policy responses to climate change, which resulted in the Representative Concentration Pathway (RCP) set of greenhouse gas concentration scenarios (Intergovernmental Panel on Climate Change, 2014). The RCP scenarios describe different pathways of greenhouse gas emissions and atmospheric greenhouse gas concentrations, air pollutant emissions, and land use that result in different levels of radiative forcing. The RCP4.5 scenario is an intermediate scenario characterized by greenhouse gas concentration levels that start declining by about 2050 and result in total radiative forcing of 4.5 watts per square meter (W/m^2) by the year 2100. The RCP8.5 scenario is characterized by increasing greenhouse gas emissions over time that lead to high greenhouse gas concentration levels and result in total radiative forcing of 8.5 W/m^2 by the year 2100. Among the total set of RCP scenarios, the RCP8.5 corresponds to the pathway with the highest greenhouse gas emissions.

Downscaled Climate Projections for the Hawaiian Islands

Simulations of the Earth's climate using GCMs for the A1B emission scenario, and the RCP4.5 and RCP8.5 warming scenarios were used by researchers from the University of Hawai'i at Mānoa International Pacific Research Center to develop a set of downscaled climate projections for the Hawaiian Islands as described below. Downscaling is a method used to translate large-scale GCM data into smaller spatial scales that can be better utilized by regional and local stakeholders. Recent downscaling of phases 3 and 5 of the Coupled Model Intercomparison Project (CMIP3 and CMIP5) experiments using dynamical and statistical downscaling approaches have produced mid- and late-21st century high-resolution climate projection datasets for the Hawaiian Islands (tables 2, 3; Elison Timm and others, 2015; Zhang and others, 2016a, b; Elison Timm, 2017; Zhang and Wang, 2017).

Elison Timm and others (2015) used a statistical downscaling (SD) approach to produce wet (November–April) and dry (May–October) season rainfall anomalies for two time periods (2041–71 and 2071–99) at a 250-meter (m) grid resolution for each island in their study, for two CMIP5 RCP4.5 and RCP8.5 scenarios (Intergovernmental Panel on Climate Change, 2014), hereinafter SD RCP4.5 2041–71, SD RCP4.5 2071–99, SD RCP8.5 2041–71, and SD RCP8.5 2071–99 scenarios. The seasonal rainfall

Table 2. Summary of projected mean annual rainfall anomalies for available high-resolution downscaled climate projections for Kaua'i, O'ahu, Moloka'i, Lāna'i, Maui, and the Island of Hawai'i.

[SD, statistical downscaling; RCP4.5 and RCP8.5, Representative Concentration Pathway with total radiative forcing of 4.5 and 8.5 Watts per square meter, respectively, by the year 2100 from Intergovernmental Panel on Climate Change (2014); HRCM1, Hawai'i Regional Climate Model version 1 from Zhang and others (2016a, b); A1B, A1B emission scenario from Nakićenović and others (2000); HRCM2, Hawai'i Regional Climate Model version 2 from Zhang and Wang (2017); --, not applicable]

Source	Future climate scenario and projection period	Reference period	Dataset grid resolution, in meters	Island-wide rainfall anomalies[c]					
				Kaua'i	O'ahu	Moloka'i	Lāna'i	Maui	Island of Hawai'i
Elison Timm and others (2015)	SD RCP4.5 2041–71	1978–2007	250	–0.141	–0.135	–0.127	–0.226	–0.091[d]	–0.098
	SD RCP8.5 2041–71	1978–2007	250	–0.155	–0.158	–0.131	–0.252	–0.083[d]	–0.109
	SD RCP4.5 2071–99	1978–2007	250	–0.159	–0.151	–0.148	–0.246	–0.099[d]	–0.107
	SD RCP8.5 2071–99	1978–2007	250	–0.233	–0.222	–0.207[d]	–0.353[d]	–0.128[d]	–0.163[d]
Zhang and others (2016a, b)	HRCM1 A1B 2080–99	1978–2007[a]	1,000[b]	--	--	--	--	0.098	--
Zhang and Wang (2017)	HRCM2 RCP4.5 2080–99	1978–2007[a]	800	0.019	0.037	0.010	0.072	–0.024	–0.036
	HRCM2 RCP8.5 2080–99	1978–2007[a]	800	–0.051	0.040	0.125	–0.020	–0.031	–0.064

[a]Rainfall anomalies for the HRCM1 and HRCM2 scenarios were adjusted to reflect a 1978–2007 reference period.

[b]Climate projection datasets for the HRCM1 scenario are available at 1-kilometer resolution for Maui only.

[c]For mean annual rainfall, computed as the area-weighted average of the product of the reference period mean and the relative mean annual anomaly at each cell of the rainfall atlas map by Giambelluca and others (2013) and then divided by the island-wide reference period mean.

[d]Negative values were present in the datasets of seasonal rainfall anomalies from Elison Timm and others (2015). All negative values were set to zero before computing the island-wide rainfall anomaly.

Table 3. Summary of projected mean annual temperature and mean atmospheric carbon dioxide concentration anomalies for available high-resolution downscaled climate projections for Kaua'i, O'ahu, Moloka'i, Lāna'i, Maui, and the Island of Hawai'i.

[SD, statistical downscaling; RCP4.5 and RCP8.5, Representative Concentration Pathway with total radiative forcing of 4.5 and 8.5 Watts per square meter, respectively, by the year 2100 from Intergovernmental Panel on Climate Change (2014); HRCM1, Hawai'i Regional Climate Model version 1 from Zhang and others (2016a, b); A1B, A1B emission scenario from Nakićenović and others (2000); HRCM2, Hawai'i Regional Climate Model version 2 from Zhang and Wang (2017); Temperature, air temperature at land surface; CO_2, carbon dioxide; ppm, parts per million; --, not applicable]

Source	Future climate scenario and projection period	Reference period	Dataset grid resolution, in meters	Range of temperature anomalies, in degrees Celsius						Mean temperature anomaly, in degrees Celsius	Mean atmospheric CO_2 concentration anomaly[b], in ppm
				Kaua'i	O'ahu	Moloka'i	Lāna'i	Maui	Island of Hawai'i	All islands	
Elison Timm (2017)	SD RCP4.5 2040–69	1976–2005	100	1.3–1.4	1.3–1.4	1.3–1.4	1.3	1.3–1.7	1.3–1.9	1.3	--
	SD RCP8.5 2040–69	1976–2005	100	1.8–2.0	1.8–2.0	1.8–2.0	1.8–1.9	1.8–2.4	1.8–2.8	1.9	--
	SD RCP4.5 2070–99	1976–2005	100	1.6–1.8	1.6–1.8	1.6–1.8	1.6–1.7	1.6–2.2	1.6–2.5	1.7	--
	SD RCP8.5 2070–99	1976–2005	100	3.1–3.5	3.1–3.4	3.1–3.5	3.1–3.3	3.1–4.2	3.1–4.8	3.3	--
Zhang and others (2016a, b)	HRCM1 A1B 2080–99	1990–2009	1,000[a]	--	--	--	--	2.1–3.0	--	--	315
Zhang and Wang (2017)	HRCM2 RCP4.5 2080–99	1990–2009	800	1.6–1.9	1.7–1.9	1.8–2.1	1.8–2.1	1.8–2.6	1.9–3.0	2.0	164
	HRCM2 RCP8.5 2080–99	1990–2009	800	3.0–3.6	3.1–3.5	3.1–3.6	3.1–3.6	3.1–4.6	3.1–5.2	3.6	472

[a]Climate projection datasets for the HRCM1 scenario are available at 1,000-meter resolution for Maui only.

[b]Mean atmospheric CO_2 concentration anomalies were calculated for the HRCM1 and HRCM2 scenarios as the difference between mean atmospheric CO_2 concentrations during the 2080–99 projection period and the 1990–2009 reference period. The mean atmospheric CO_2 concentration during the 2080–99 projection period and 1990–2009 reference period was estimated as the mean of the minimum and maximum CO_2 concentrations used by Zhang and others (2016a, b) and Zhang and Wang (2017). Mean atmospheric CO_2 concentration anomalies for the SD set of scenarios were not determined for this study.

anomalies computed by Elison Timm and others (2015) represent departures from the mean seasonal rainfall amounts during 1978–2007 that were estimated by Giambelluca and others (2013). Elison Timm (2017) used a statistical downscaling approach to develop projections of ambient surface air temperature anomalies for two time periods (2040–69 and 2070–99) at a 250-m grid resolution for RCP4.5 and RCP8.5 scenarios, hereinafter SD RCP4.5 2040–69, SD RCP4.5 2070–99, SD RCP8.5 2040–69, and SD RCP8.5 2070–99 scenarios. Ambient surface temperature anomalies for the RCP scenarios were computed relative to mean values for the 1976–2005 reference period.

Zhang and others (2012) used a dynamical downscaling approach by configuring a Weather Research Forecasting (WRF) regional climate model for the Hawaiian Islands and surrounding ocean region, hereinafter the Hawai'i Regional Climate Model version 1 (HRCM1), to conduct a one-year retrospective simulation (November 2005–October 2006; Skamarock and others, 2008). Zhang and others (2012) used the HRCM1 to simulate the interaction of sea-surface temperature, cloud-base height, trade-wind strength and direction, and trade-wind inversion height and frequency with Hawai'i's complex topography at a 3-kilometer (km) grid spacing. The HRCM1 produced simulated meteorological parameters including rainfall, surface temperature, relative humidity, and wind speed and direction. Lauer and others (2013) noted limitations in the simulated rainfall of the HRCM1 over areas with steep topography, particularly on Maui and O'ahu. In response to these limitations, Zhang and others (2016a) modified the HRCM1 to include an additional 1-km resolution domain nested inside the 3-km grid around Maui to further refine and improve the ability of the HRCM1 to simulate rainfall across steep and complex topography. Using the modified HRCM1, Zhang and others (2016a) produced climate datasets for a 20-year retrospective simulation of Maui during 1990–2009, hereinafter the HRCM1 1990–2009 simulation. Zhang and others (2016b) used the modified HRCM1 to produce climate datasets of Maui for a CMIP3 SRES A1B emission scenario (Nakićenović and others, 2000) during 2080–99, hereinafter the HRCM1 A1B 2080–99 scenario.

Zhang and Wang (2017) also used a dynamical downscaling approach by configuring a regional climate model for the western North and South Pacific region, including the Hawaiian Islands, hereinafter the Hawai'i Regional Climate Model version 2 (HRCM2). The model domain for the HRCM2 extends across a substantially larger part of the Pacific Ocean when compared to the domain of the HRCM1. In the region surrounding the Hawaiian Islands, they added a nested domain at an 800-m grid spacing to improve the ability of the HRCM2 to simulate rainfall across the complex topography of the islands. Similar to the approach used for the HRCM1, they produced climate datasets for a 20-year retrospective simulation of the Hawaiian Islands during 1990–2009, hereinafter the HRCM2 1990–2009 simulation. They also used the HRCM2 to produce climate datasets for the

Hawaiian Islands during 2080–99 for CMIP5 RCP4.5 and RCP8.5 scenarios (Intergovernmental Panel on Climate Change, 2014), hereinafter HRCM2 RCP4.5 2080–99 and HRCM2 RCP8.5 2080–99 scenarios, respectively. See "Water-Budget Scenarios" section of this report for further discussion of the scenarios used for each island.

Selection of Climate Scenarios

For this study, the high-resolution downscaled climate projection datasets for each island represent three future-climate scenarios for the water-budget analyses. Projected mean annual rainfall and temperature anomalies were computed and compared to determine the range of projected changes in rainfall and temperature. On the basis of projected changes in mean annual rainfall, two climate projections were chosen to capture a range of future-climate conditions, a dry-climate condition and a wet-climate condition. The timeframe indicated by each downscaled climate projection was not considered in the identification of the dry-climate condition and wet-climate condition. A third climate projection was selected from the available mid-century climate projection datasets. Finally, a drought scenario based on historical rainfall was added for Lānaʻi to quantify the effects of episodic dry conditions on groundwater recharge.

Mean Annual Rainfall Anomalies

The projected wet- and dry-season rainfall anomalies for the RCP scenarios were transformed into mean annual rainfall anomalies to compare future-climate projections across annual hydrologic cycles for each island. The seasonal rainfall anomalies developed by Elison Timm and others (2015) are available as geospatial datasets: (1) relative seasonal rainfall anomalies expressed as a fraction of a reference mean, (2) absolute seasonal rainfall anomalies expressed as millimeters (mm) above or below the reference mean, and (3) absolute seasonal rainfall anomalies expressed as inches above or below the reference mean. These datasets were obtained from the State University of New York at Albany (see data product labeled SDSRA-HI-V2 at http://www.atmos.albany.edu/facstaff/timm/products_data.html). First, absolute mean annual rainfall anomalies, expressed in millimeters greater or less than the reference mean, were computed as the sum of wet- and dry-season rainfall anomalies at each grid cell. Next, relative mean annual rainfall anomalies, expressed as a fraction of the reference mean, were computed by dividing the absolute mean annual rainfall anomalies by the mean annual rainfall grid values developed by Giambelluca and others (2013) for 1978–2007. Mean annual rainfall during 1978–2007 (Giambelluca and others, 2013) was used as the reference mean because the future wet- and dry-season rainfall anomalies by Elison Timm and others (2015) were computed relative to the same 1978–2007 period.

The HRCM1 and HRCM2 retrospective simulations and climate projections developed by Zhang and others (2016a, b) and Zhang and Wang (2017) were also transformed into mean annual rainfall anomalies. Gridded, geospatial datasets of daily rainfall produced by Zhang and others (2016a, b) and Zhang and Wang (2017) were provided by the University of Hawai'i at Mānoa (HRCM1 monthly rainfall data were obtained from http://apdrc.soest.hawaii.edu/projects/HRCM/; HRCM2 hourly rainfall data were obtained from Zhang [2016, 2017]). To transform the climate projections into mean annual rainfall anomalies, mean annual rainfall values were computed for each grid cell of the retrospective simulations and climate projections. Next, absolute mean annual rainfall anomalies for each climate projection were computed as the difference between the mean annual rainfall of the climate projection and the retrospective simulation for each grid cell. The relative mean annual rainfall anomalies for each climate projection, expressed as a fraction of the 1990–2009 mean, were then computed by dividing the absolute mean rainfall anomalies by the retrospective simulation mean at each grid cell.

The wet- and dry-season rainfall anomalies for the SD RCP4.5 2041–71, SD RCP4.5 2071–99, SD RCP8.5 2041–71, and SD RCP8.5 2071–99 scenarios (Elison Timm and others, 2015) were developed relative to mean rainfall during 1978–2007 (Giambelluca and others, 2013). However, the climate projections for the HRCM1 and HRCM2 scenarios (Zhang and others, 2016b; Zhang and Wang, 2017) were developed relative to simulated climate conditions during 1990–2009 (Zhang and others, 2016a; Zhang and Wang, 2017). For this study, the approach developed by Mair and others (2019) was used to establish a common reference period for the rainfall projections for the SD, HRCM1, and HRCM2 scenarios (see "Future Monthly Rainfall" section). The mean annual rainfall anomalies developed using a common reference period for the SD, HRCM1, and HRCM2 scenarios indicate a range of projected changes in rainfall across each of the islands (table 2).

Mean Annual Surface Air Temperature Anomalies

Datasets of the mean annual surface air temperature anomalies developed by Elison Timm (2017) for the RCP scenarios were obtained from the State University of New York at Albany (see data product labeled Version 1.1 at http://www.atmos.albany.edu/facstaff/timm/products_data.html). The HRCM1 and HRCM2 1990–2009 retrospective simulations and climate projections developed by Zhang and others (2016a, b) and Zhang and Wang (2017) were transformed into mean annual surface air temperature anomalies to compare with the surface air temperature anomalies developed by Elison Timm (2017) for the RCP scenarios. First, mean annual surface air temperatures for the HRCM1 and HRCM2 projections were computed for each grid cell using simulated hourly data of surface air temperature (2 meters above the land surface) developed by Zhang and others (2016a, b) and Zhang and Wang (2017) and provided by the University of Hawai'i at Mānoa (see HRCM1 data products at http://apdrc.soest.hawaii.edu/projects/HRCM/ and HRCM2 data products available at Zhang [2016, 2017]). Next, mean annual temperature anomalies for the HRCM1 and HRCM2 projections were computed as the difference between the future mean and the 1990–2009 mean for each grid cell. The range of projected warming across all islands varies from

a mean of 1.3 degrees Celsius (°C) for a SD RCP4.5 2040–69 scenario to 3.6 °C for a HRCM2 RCP8.5 2080–99 scenario (table 3). The greatest rates of warming on each island occur in the SD RCP8.5 2070–99 and HRCM2 RCP8.5 2080–99 scenarios. All scenarios project greater rates of warming at higher altitudes.

Selected Future-Climate Scenarios

Three future-climate scenarios were selected for each island to quantify possible changes in groundwater recharge using a water-budget model (see "Water-Budget Model" section). The SD RCP8.5 2041–71 scenario was chosen as the mid-21st-century projection for each island on the basis of input from CWRM and Pūlama Lānaʻi. The island-wide rainfall anomalies for the remaining projections were analyzed to identify two additional scenarios that capture the range of available projected changes in island-wide rainfall for each island, namely, a dry-climate and wet-climate scenario. The timeframe indicated by the downscaled climate projections was not considered in the selection of the dry-climate and wet-climate scenarios. The SD RCP8.5 2071–99 scenario, the driest scenario among the available set of rainfall projections (table 2), was chosen as the dry-climate scenario for each island. The wettest climate projections available for each island were chosen as the wet-climate scenario: HRCM2 RCP4.5 2080–99 for Kauaʻi, Lānaʻi, and the Island of Hawaiʻi; HRCM2 RCP8.5 2080–99 for Oʻahu and Molokaʻi; HRCM1 A1B 2080–99 for Maui. The projected rates of warming across each island (table 3) were not considered in the selection of the future-climate scenarios.

Selected Drought Scenario for Lānaʻi

The 93-year gridded dataset of monthly rainfall maps during 1920–2012 (Frazier and others, 2016) was used to identify the 5-year period with the lowest mean annual island-wide rainfall. Mean annual island-wide rainfall on Lānaʻi was lowest during 2008–12 and was therefore selected for the Lānaʻi drought scenario.

Water-Budget Model

Spatially distributed mean annual groundwater recharge was estimated for Kauaʻi, Oʻahu, Molokaʻi, Lānaʻi, Maui, and the Island of Hawaiʻi using a threshold-type water-budget model approach. A water budget of the plant-soil system accounts for water inputs (rainfall, fog interception, irrigation, septic-system discharge, and other inputs), water outputs (runoff, ET, and recharge), and changes in stored water during a specified time interval. Recharge was assumed to occur when water in the soil root zone exceeded the soil's moisture-storage capacity.

The water-budget code used in this study, a Water-budget Accounting for Tropical Regions Model (WATRMod), computes a daily water budget for each hydrologically independent subarea in the study area (Oki, 2022). Each subarea is defined by its climatic, soil, land-cover, and human-related characteristics (for example, irrigation; see "Land Cover" section). WATRMod simulates the daily hydrologic processes and physical conditions that affect recharge on the islands. Hydrologic processes simulated by the model include rainfall, fog interception, irrigation, septic-system discharge, direct recharge that bypasses the plant-soil system, runoff, canopy evaporation in forested areas, and ET. The model represents physical conditions like soil moisture-storage capacity, vegetation, and land cover using specified input parameters, depending on the characteristic of the subarea. For each model subarea, the water-budget model calculates recharge and other water-budget components for each day during the assigned simulation period.

WATRMod was used to estimate spatially variable daily water-budget components, including groundwater recharge, for each of the six islands in this study (Oki, 2022). The WATRMod code is a modified version of the code used in earlier studies to estimate groundwater recharge for Pacific islands (Engott, 2011; Engott and others, 2017; Izuka and others, 2018; Johnson and others, 2018; Mair and others, 2019; Brewington and others, 2019; Oki and others, 2020; Rotzoll and Johnson, 2023). Detailed descriptions of WATRMod, calculations performed by WATRMod, and exclusions and limitations are provided in Oki (2022).

Water-Budget Scenarios

In this study, we selected four water-budget scenarios to span a range of potential future-climate conditions for each island: (1) recent conditions during the 1978–2007 reference period, hereinafter the 1978–2007 scenario, (2) a mid-century scenario using projected rainfall conditions for an SD RCP8.5 2041–71 scenario, (3) a dry-climate scenario using projected rainfall conditions for an SD RCP8.5 2071–99 scenario, and (4) a wet-climate scenario using projected rainfall conditions for an HRCM1 A1B 2080–99 scenario for Maui, HRCM2 RCP4.5 2080–99 scenario for Kauaʻi, Lānaʻi, and the Island of Hawaiʻi, and a HRCM2 RCP8.5 2080–99 scenario for Oʻahu and Molokaʻi (table 4). A 30-year simulation period was used for each of the four water-budget scenarios. A fifth water-budget scenario was added as a drought scenario for Lānaʻi to assess the effect of dry conditions for a 5-year simulation period during 2008–12. Mean annual recharge for each island and each individual aquifer system was calculated by summing the recharge values of the model subareas located within their respective boundaries.

Model Input

This study incorporates separate model input datasets for each water-budget scenario to characterize rainfall and direct runoff. Common model input datasets were used for all water-budget scenario parameters, which included land-cover conditions, reference-ET rates, soil moisture-storage capacities,

Table 4. Summary of water-budget scenarios for Kauaʻi, Oʻahu, Molokaʻi, Lānaʻi, Maui, and the Island of Hawaiʻi.

[SD, statistical downscaling; RCP4.5 and RCP8.5, Representative Concentration Pathway with total radiative forcing of 4.5 and 8.5 Watts per square meter, respectively, by the year 2100 from Intergovernmental Panel on Climate Change (2014); HRCM1, Hawaiʻi Regional Climate Model version 1 from Zhang and others (2016a, b); A1B, A1B emission scenario from Nakićenović and others (2000); HRCM2, Hawaiʻi Regional Climate Model version 2 from Zhang and Wang (2017); A, applicable; --, not applicable]

Water-budget scenario	Study-defined climate condition	Kauaʻi	Oʻahu	Molokaʻi	Lānaʻi	Maui	Island of Hawaiʻi
1978–2007	Reference climate	A	A	A	A	A	A
SD RCP8.5 2041–71	Mid-century scenario	A	A	A	A	A	A
SD RCP8.5 2071–99	Dry-climate scenario	A	A	A	A	A	A
HRCM1 A1B 2080–99	Wet-climate scenario	--	--	--	--	A	--
HRCM2 RCP4.5 2080–99	Wet-climate scenario	A	--	--	A	--	A
HRCM2 RCP8.5 2080–99	Wet-climate scenario	--	A	A	--	--	--
2008–12	Drought scenario	--	--	--	A	--	--

fog-interception rates, irrigation methods, septic effluent discharge rates, and storm-drain capture conditions.

Land Cover

Land-cover maps representative of land-cover conditions for 2020 were developed for this study, hereinafter the 2020 land-cover maps (fig. 4). The 2020 land-cover maps identify 28 types of land covers and were generated by modifying the LANDFIRE Existing Vegetation Type map for each island (U.S. Geological Survey, 2016), hereinafter referred to as the Landfire maps. The Landfire maps were generalized by combining similar land-cover types. Additional land-cover types were added to distinguish golf courses, kiawe, selected agricultural crops, coastal water bodies, inland water bodies, and lined and unlined reservoirs. The 2020 land-cover maps were further modified to distinguish between (1) forested and shrubland areas within the fog-interception zone, and (2) forested and shrubland areas below the fog-interception zone. Detailed descriptions of the development of the 2020 land-cover map for each island are described in the metadata of Kāne and others (2024).

Similar Landfire groups were combined into more general classes for the 2020 land-cover maps. Grassland Landfire groups "Hawaiʻi Lowland Dry Grassland," "Hawaiian Lowland Mesic Grassland," "Hawaiʻi Montane-Subalpine Dry Grassland," "Hawaiʻi Montane-Subalpine Mesic Grassland," and "Introduced Perennial Grassland" were combined into the "Grassland" land-cover type (fig. 4). Shrubland Landfire groups "Hawaiʻi Wet Cliff and Ridge Crest Shrubland," "Hawaiʻi Lowland Dry Shrubland," "Hawaiʻi Lowland Mesic Shrubland," "Hawaiʻi Montane-Subalpine Dry Shrubland," "Hawaiʻi Alpine Dwarf-Shrubland," "Hawaiʻi Subalpine Mesic Shrubland," "Hawaiʻi Introduced Deciduous Shrubland," and "Hawaiʻi Introduced Evergreen Shrubland" were combined into the "Shrubland" land-cover type. Forest Landfire groups "Hawaiʻi Lowland Rainforest," "Hawaiʻi Montane Cloud Forest," "Hawaiʻi Montane Rainforest," "Hawaiʻi Lowland Dry Forest," "Hawaiʻi Montane-Subalpine Dry Forest and Woodland," and "Hawaiʻi Montane-Subalpine Mesic Forest" were combined into "Native forest" land-cover type. "Hawaiʻi Introduced Dry Forest" and "Hawaiʻi Introduced Wet-Mesic Forest" were combined into the "Non-native forest" land-cover type. Forest land-cover types were further differentiated as Native forest (fog and no fog), Non-native forest (fog and no fog), and Tree plantation (fog and no fog) types. Shrubland land covers were also further differentiated as Shrubland (fog and no fog) land-cover types. Wetland Landfire groups "Hawaiʻi Bog," "Hawaiʻi Islands Introduced Wetland Vegetation-Tree," "Hawaiʻi Introduced Wetland Vegetation-Shrub," "Hawaiʻi Introduced Wetland Vegetation-Herbaceous," "Hawaiʻi Introduced Coastal Wetland Vegetation-Tree," "Hawaiʻi Introduced Coastal Wetland Vegetation-Shrub," and "Hawaiʻi Introduced Coastal Wetland Vegetation-Herbaceous" areas were designated as "Wetland" land-cover type. Landfire groups "Hawaiʻi Dry Cliff," "Hawaiʻi Dry Coastal Strand," "Hawaiʻi Wet-Mesic Coastal Strand," and "Barren" were combined into the "Sparsely vegetated" land-cover type. "Water" Landfire areas located in the tidal zone (fishponds) or adjacent to the coastline were designated as "Coastal water body"; all other Landfire "Water" land cover areas were designated as "Inland water body," unless designated as a reservoir land cover.

Locations of reservoirs, golf courses, and selected agricultural crops were delineated in the 2020 land-cover maps using other sources of information. The boundaries of most reservoirs were delineated using the National Hydrography Dataset Waterbodies (U.S. Geological Survey, 2019). The remaining reservoirs were digitized using publicly available satellite imagery (Google Earth, https://earth.google.com/web/) and Geographic Information System (GIS) software. Reservoirs were designated as "Reservoir, lined" if one or both of the following conditions were met: (1) a reservoir-lining system was described in an online database of dams (State of Hawaiʻi, 2021) or (2) a review of satellite imagery indicated the presence

of a clay, concrete, geotextile, or other synthetic liner. All other reservoirs were classified as "Reservoir, unlined." Irrigated areas of golf courses, such as fairways and putting greens, were delineated using a combination of sources including a 2014 golf-course map (State of Hawai'i, 2017), delineations of golf courses from previous water-budget studies (Engott, 2011; Izuka and others, 2018; Oki and others, 2020), and satellite imagery and GIS software.

Agricultural crop designations and boundary delineations in the Landfire maps were updated to reflect (1) the cessation of sugarcane cultivation on Maui by Hawaiian Commercial & Sugar Company in December 2016, and (2) the agricultural land-cover information described in the 2015 Hawai'i Statewide Agricultural Land Use Baseline map (Melrose and others 2016). Similar agricultural land-cover groups were combined into more general classes for the 2020 land-cover maps. Agricultural land-use groups "Banana," "Flowers/Foliage/Landscape," "Diversified Crop," and "Tropical Fruits" were combined into "Diversified agriculture" for the 2020 land-cover maps. The "Seed production" land-cover group was assigned "Seed corn" because seed corn was reported as being the primary seed crop grown in Hawai'i (National Agricultural Statistics Service, 2021). The "Commercial forestry" and "Pasture" land-cover groups were assigned "Tree plantation" and "Grassland," respectively. Agricultural land-use information for selected islands was updated in a 2020 Hawai'i Statewide Agricultural Land Use map (Perroy and Collier, 2021). However, a complete set of updated maps for all islands was not available when analyses for this study were conducted. The boundaries of kiawe land cover on each island were delineated using maps of *Prosopis pallida*, one of the dominant species of kiawe found in Hawai'i (U.S. Geological Survey Gap Analysis Project, 2016).

The 2020 land-cover map also distinguishes between forested areas within the fog-interception zone and forested areas below the fog-interception zone, which is assumed to be (1) at altitudes of 2,000 ft and higher on Kaua'i, O'ahu, Moloka'i, Lāna'i, and (2) at varying altitudes for Maui and the Island of Hawai'i (fig. 4). The 2020 land-cover map for each island was used in the computation of recharge for all water-budget simulation scenarios for each island (table 4).

The water-budget subareas were developed by merging multiple spatial datasets that characterize the spatial distribution of rainfall, cloud-water (or fog) interception, irrigation, reference ET, direct runoff, soil type, cesspool seepage and septic effluent rates, aquifer-system boundaries, caprock (O'ahu and Maui only), and the 2020 land cover. The number of subareas for each island ranges from 61,416 for Lāna'i to 1,130,963 for the Island of Hawai'i (Kāne and others, 2024).

Impervious Surfaces

Impervious surfaces include paved surfaces and buildings. In the water budget, the amount of excess water from impervious surfaces of a subarea that is distributed to the pervious surfaces of the subarea depends on the impervious fraction of the subarea. The impervious fraction of each subarea was computed from a map of impervious surfaces for each island (Office for Coastal Management, 2020a, b, c, d, e, f). Subareas with a water body, reservoir, or taro land-cover classification were assigned an impervious fraction value of zero. All remaining subareas were reassigned an impervious fraction value of 0.99 whenever the computed value exceeded 0.99.

Rainfall

1978–2007 Monthly Rainfall

Monthly spatial and temporal distributions of rainfall in the water-budget calculations of the 1978–2007 scenario were defined using gridded monthly rainfall maps for the 30-year timeframe. These maps are a subset of the 1,116 1920–2012 monthly rainfall maps for the Hawaiian Islands that Frazier and others (2016) generated for the Rainfall Atlas of Hawai'i (Giambelluca and others, 2013). Monthly rainfall for each subarea was computed in WATRMod as the product of the monthly rainfall value from Frazier and others (2016) and a mean monthly adjustment factor. Each rainfall grid cell was assigned 12 mean monthly adjustment factors to account for slight differences in the Frazier and others (2016) and Giambelluca and others (2013) datasets. The mean monthly rainfall adjustment factors for each subarea were determined from the following equation:

$$\tau(i) = \frac{\bar{P}_{Atlas}(i)}{\bar{P}_{1978-2007}(i)} \quad (1)$$

where

$\tau(i)$ is the mean monthly rainfall adjustment factor for month [dimensionless],

$\bar{P}_{Atlas}(i)$ is the mean rainfall for month i during 1978–2007 from Giambelluca and others (2013) [inches],

$\bar{P}_{1978-2007}(i)$ is the mean rainfall for month i during 1978–2007 from Frazier and others (2016) [inches], and

i is the variable designating month in a calendar year and ranges from a value of 1 to 12.

The final adjusted monthly rainfall maps were used as the rainfall dataset for the 1978–2007 scenario.

Future Monthly Rainfall

High-resolution regional climate model (RCM) data are used in hydrologic studies at the regional and watershed scale. However, systematic biases are often evident in RCM data when comparing simulated meteorological variables with observations of a reference climate. These biases can affect the output of hydrologic models (Graham and others, 2007; van Roosmalen and others, 2010, 2011); however, bias correction can be applied before using RCM data in hydrologic models (Wilby and others, 2000). A common bias-correction approach is the change-factor method (Hay and others, 2000), which applies relative change factors, derived by comparing simulated reference climate and projected future-climate RCM data (van Roosmalen and others,

2007, 2009, 2010, 2011; Navarro-Racines and others, 2020) to a reference climate series. The main underlying assumptions of the change-factor method are that (1) the biases for the reference and future-climate simulations are equivalent, and (2) the RCMs simulate relative changes better than absolute values.

In this study, the change-factor method was used to develop datasets of future monthly rainfall by applying relative change factors to datasets of monthly rainfall (Frazier and others, 2016) during a selected reference period. Three sets of change factors for monthly rainfall, one for each of the selected future scenarios, were prepared for each island. Two sets of change factors were calculated for each island from the wet- and dry-season rainfall anomalies of the SD RCP8.5 2041–71 and SD RCP8.5 2071–99 scenarios (Elison Timm and others, 2015). The third set of change factors for Kauaʻi, Lānaʻi, and the Island of Hawaiʻi was calculated from the differences in monthly rainfall means between the HRCM2 1990–2009 simulation and the HRCM2 RCP4.5 2080–99 scenario (Zhang and Wang, 2017). The third set of change factors for Oʻahu and Molokaʻi was calculated from the differences in monthly rainfall means between the HRCM2 1990–2009 simulation and the HRCM2 RCP8.5 2080–99 scenario (Zhang and Wang, 2017). For Maui, the third set of change factors was calculated from the differences in monthly rainfall means between the HRCM1 1990–2009 simulation and the HRCM1 A1B 2080–99 scenario (Zhang and others, 2016a, b).

The wet- and dry-season rainfall anomalies of the SD RCP8.5 2041–71 and SD RCP8.5 2071–99 scenarios (Elison Timm and others, 2015) were developed relative to mean wet- and dry-season rainfall during 1978–2007 (Giambelluca and others, 2013). However, the rainfall anomalies for the climate projections of the HRCM1 A1B 2080–99, HRCM2 RCP4.5 2080–99, and HRCM2 RCP8.5 2080–99 scenarios (Zhang and others, 2016b; Zhang and Wang, 2017) were developed relative to simulated climate conditions during a 1990–2009 reference period (Zhang and others, 2016a; Zhang and Wang, 2017). Zhang and others (2016b) and Zhang and Wang (2017) used a pseudo-global-warming approach that imposed the large-scale GCM-based climate change signal from the 2080–99 time period on the boundary conditions used by HRCM1 and HRCM2 for the 1990–2009 reference period. Hence, the HRCM1 and HRCM2 climate projections reflect changes in climate relative to a 1990–2009 reference period. For this study, the 1978–2007 reference period was selected to develop change factors for each of the future-climate scenarios to establish a common reference period among different water-budget model scenarios. The three sets of change factors were then used to adjust monthly rainfall during 1978–2007 to produce three future scenario rainfall datasets per island for WATRMod simulations.

The monthly rainfall input for each subarea of the future scenarios was determined with the equation:

$$P_{\delta}(i,j) = F_P(i) \times P_{1978-2007}(i,j) \times \tau(i) \quad (2)$$

where

$P_{\delta}(i,j)$ is the rainfall input for month i and year j in the future scenario [inches],

$F_P(i)$ is the relative change factor for month i in a calendar year [dimensionless],

$P_{1978-2007}(i,j)$ is the gridded monthly rainfall for month i and year j during 1978–2007 from Frazier and others (2016) [inches],

i is the variable designating month in a calendar year and ranges from a value of 1 to 12.

j is the variable used to denote the year counter and ranges from 1 to 30 in the 1978–2007 and future scenarios, and

$\tau(i)$ is the mean monthly rainfall adjustment factor for month [dimensionless].

For the SD RCP8.5 2041–71 and SD RCP8.5 2071–99 scenarios, the relative change factor, $F_P(i)$, for rainfall at each subarea was calculated with the following equation:

$$F_P(i) = (1 + A_{SD}) \quad (3)$$

where

A_{SD} is the wet (November to April) or dry (May to October) season rainfall anomaly for the SD RCP8.5 2041–71 or SD RCP8.5 2071–99 scenario (Elison Timm and others, 2015), expressed as a fraction of 1978–2007 mean rainfall [dimensionless].

For the HRCM1 A1B 2080–99, HRCM2 RCP4.5 2080–99, and HRCM2 RCP8.5 2080–99 scenarios, the relative change factor, $F_P(i)$, for rainfall at each subarea was calculated with the following equation:

$$F_P(i) = (1 + A_{HRCM}(i)) \quad (4)$$

where

$A_{HRCM}(i)$ is the rainfall anomaly for month i determined from the HRCM1 1990–2009 simulation and HRCM1 A1B 2080–99 scenario (Zhang and others, 2016a, b), or the HRCM2 1990–2009 simulation and HRCM2 future scenarios, expressed as a fraction of the 1978–2007 mean (Zhang and Wang, 2017) [dimensionless].

The rainfall anomaly for month i, $A_{HRCM}(i)$ at each subarea was calculated with the following equation:

$$A_{HRCM}(i) = \frac{\left((1+\delta_{HRCM}(i)) \times \bar{P}_{1990-2009}(i)\right) - \bar{P}_{1978-2007}(i)}{\bar{P}_{1978-2007}(i)} \quad (5)$$

where

$\delta_{HRCM}(i)$ is the mean rainfall anomaly for month i for the HRCM1 or HRCM2 future scenario, expressed as a fraction of the mean rainfall from the HRCM1 or HRCM2 1990–2009 simulation [dimensionless],

$\bar{P}_{1990-2009}(i)$ is the mean rainfall for month i during 1990–2009 from Frazier and others (2016) [inches].

The mean rainfall anomaly for month i, $\delta_{\mathrm{HRCM}}(i)$, was computed with the following equation:

$$\delta_{\mathrm{HRCM}}(i) = \left[\frac{\bar{P}_{fut}(i) - \bar{P}_{ret}(i)}{\bar{P}_{ret}(i)} \right] \quad (6)$$

where

$\bar{P}_{fut}(i)$ is the mean rainfall for month i during the HRCM1 or HRCM2 future scenario [inches],

$\bar{P}_{ret}(i)$ is the mean rainfall for month i during the HRCM1 or HRCM2 1990–2009 simulation [inches].

1978–2007 Daily Rainfall

The water-budget model generated daily rainfall data for the 1978–2007 scenario by disaggregating the monthly values of the 1978–2007 rainfall distribution maps using the method of fragments (for example, Oki, 2002). High-resolution, 250-m grid size maps of daily rainfall during 1990–2014 for the Hawaiian Islands, developed by Longman and others (2019), were used to disaggregate monthly rainfall into daily rainfall fragments for the water-budget model. For each grid cell, daily rainfall fragments were computed by dividing daily rainfall values during a particular month by the total rainfall at the cell for that month. Daily rainfall fragment computations resulted in a set of fragments for which the total number of fragments in each set was equal to the number of days in the corresponding month. Fragment sets were compiled for every cell for each month during the 1990–2014 period, except for grid cells for which the total rainfall computed from the daily maps was zero over an entire calendar month. For grid cells with no rainfall during a month, the number of fragment sets at that cell was reduced by the number of months with no rainfall.

Fragment sets were grouped by grid cell and month of the year during 1990–2014 for each island, hereinafter the daily rainfall fragment dataset for the 1978–2007 reference period, and then used in the model simulations for the 1978–2007 scenario. In the water-budget calculation, the fragment set used for a given grid cell for a given month (for example, March, 1995) was selected randomly from all the possible fragments sets for that month (or all March fragment sets across 1990–2014 in this example). Daily rainfall data for a given month was synthesized by multiplying the corresponding total monthly rainfall by each fragment in the set. The lack of fragment sets consisting entirely of zero values does not preclude simulation of zero rainfall on each day of a month because months with zero rainfall will invariably have zero rainfall on each day regardless of the fragment set selected.

The simulated daily rainfall fragment datasets from the HRCM1 and HRCM2 1990–2009 simulations, and the daily rainfall fragment dataset for the 1978–2007 reference period represent different reference conditions in terms of daily rainfall frequency and could have dissimilar effects on estimated recharge using the water-budget model. Therefore, only the daily rainfall fragment dataset for the 1978–2007 reference period was used in the model simulations for the 1978–2007 scenario (table 4).

Future Daily Rainfall

The daily rainfall fragment sets developed for the 1978–2007 scenario were also used for each of the future-climate scenarios because of a lack of information in some of the climate projection datasets. For example, the rainfall projections for the statistical downscaling group of warming scenarios (Elison Timm and others, 2015) do not describe projected changes in daily rainfall frequency, magnitude, or duration. In a separate analysis for CMIP3 A1B and A2 emission scenarios, Elison Timm and others (2011) projected small changes in the number of late-21st century heavy rainfall days using a statistical downscaling approach. However, their projections have large uncertainties resulting from disparities related to the mean state of ENSO and PNA during the 21st century simulated by the GCMs for A1B and A2 emission scenarios. Because of the lack of information on projected changes in daily rainfall for the SD RCP8.5 2041–71 and SD RCP8.5 2071–99 scenarios, separate datasets of rainfall fragments for these particular scenarios were not developed. However, the climate projection datasets for the HRCM1 and HRCM2 scenarios do include gridded daily rainfall that can be used to develop separate sets of daily rainfall fragments. Projected changes in daily rainfall frequency have a relatively small but non-negligible effect on recharge estimates. An analysis of the effects of potential changes in daily rainfall frequency on water-budget results for the selected HRCM1 and HRCM2 scenarios is included in the "Effects of Selected Climate Inputs on Estimated Recharge" section of this report.

Fog Interception

In Hawai‘i, clouds can form when a moist air mass cools and condenses as it is forced upslope by trade winds or by thermal circulation systems such as sea breezes. As clouds flow near the land surface, some of the cloud moisture might accumulate on vegetation through a process called cloud-water interception or fog interception. At places where fog interception is frequent, intercepted cloud moisture that reaches the ground can be a substantial part of the water budget (Ekern, 1964; Juvik and Ekern, 1978; Juvik and Nullet, 1995; Health, 1998; Giambelluca and others, 2011; Takahashi and others, 2011). The cloud zone includes areas with frequent clouds between altitudes of about 2,000 and 8,200 ft in Hawai‘i (DeLay and Giambelluca, 2010). However, clouds can also form in areas above the generalized cloud zone (Juvik and Ekern, 1978).

For Kaua‘i, O‘ahu, Moloka‘i, and Lāna‘i, all areas at or above 2,000 ft were assumed to be within the cloud zone and all areas below 2,000 ft were assumed to be below the cloud zone

(fig. 4*A–D*), consistent with earlier studies (Hardy, 1996; Izuka and others, 2018; Oki and others, 2020). For Maui, the base altitude of the cloud zone for windward areas was assumed to be 2,000 ft (fig. 4*E*), the same as that assumed by earlier studies (Johnson and others, 2018; Brewington and others, 2019; Mair and others, 2019). Field measurements by Juvik and Hughes (1997) indicate that the base altitude of the cloud zone for leeward areas of Maui lies between 2,600 ft and 4,400 ft. For this study, the base altitude of the cloud zone for leeward areas of Maui was assumed to be 2,600 ft. For leeward and windward areas of Maui, the top altitude of the cloud zone was assumed to be 7,800 ft, approximately equal to the mean of the average base altitudes of the trade-wind inversion layer estimated by Zhang and others (2012) on the basis of radiosonde observations for two sites—one on Kauaʻi and one on the Island of Hawaiʻi—during wet weather conditions between 1998 and 2011. The average base altitudes determined from radiosonde observations during wet conditions were selected because wet conditions were assumed to be most representative of conditions when fog interception occurs, given that the model estimates fog interception using fog-to-rainfall ratios. The fog-zone delineation used for Maui in this study was also assumed by Johnson and others (2023).

For the Island of Hawaiʻi, Engott (2011) assumed an altitude of 2,500 ft for the base of the cloud zone in his water-budget study. However, DeLay and Giambelluca (2010) described the presence of fog at 2,000 ft for windward areas in the eastern side of the island. Therefore, for this study, the base altitude of the cloud zone for windward areas of the Island of Hawaiʻi was assumed to be 2,000 ft (fig. 4*F*). Field measurements by Juvik and Ekern (1978) indicate that the base altitude of the cloud zone for leeward areas of the Island of Hawaiʻi lies between 2,300 ft and 3,200 ft. For this study, the base altitude of the cloud zone for leeward areas of the Island of Hawaiʻi was assumed to be the same as Maui and set to 2,600 ft (fig. 4*F*).

For each island, fog interception was assumed to be limited to non-native forest, native forest, tree plantation, kiawe, shrubland, coffee, and macadamia land-cover types above the base of the cloud zone. Fog interception was assumed to be negligible for all areas below the cloud zone. In the water budget, fog interception was assumed to occur only on days with rainfall where daily fog interception was computed as a fraction of daily rainfall using fog-to-rainfall ratios. For Oʻahu, Molokaʻi, and the Island of Hawaiʻi, daily fog interception was computed using ratios of mean monthly fog-to-rainfall from earlier studies (table 5; Engott, 2011; Engott and others, 2017; Oki and others, 2020). For Lānaʻi, daily fog interception was computed using ratios of mean monthly fog-to-rainfall derived from Juvik and others (2011; table 5). For Kauaʻi and Maui, mean annual fog-interception rates from earlier studies (table 6; Izuka and others, 2018; Johnson and others, 2018) were used to determine fog-to-rainfall ratios so that daily fog interception in each subarea was computed as a fraction of daily rainfall equivalent to the ratio of mean monthly fog interception and mean monthly rainfall during 1978–2007. The fog-to-rainfall ratios determined for Kauaʻi and Maui can vary spatially within the cloud zone, which can result in a different set of ratios for subareas with the same altitude. These resulting fog-to-rainfall ratios were used to compute recharge for each water-budget scenario.

Irrigation

Irrigation rates were estimated in the model for subareas on each island that were classified as agriculture (coffee, diversified agriculture, pineapple, seed corn, and taro) or other land-cover types (golf course, high-intensity developed, and medium-intensity developed) that were assumed to contain irrigated lawns and landscapes. Irrigation rates were also estimated for subareas with macadamia land cover on the Island of Hawaiʻi consistent with the approach of Engott (2011). However, no irrigation was applied in the model to subareas with macadamia on Maui consistent with the approach of earlier studies (Johnson and others, 2018; Brewington and others, 2019; Mair and others, 2019). Irrigation rates were estimated in the model using two methods: (1) a supply-based approach in which monthly irrigation depths were specified, and (2) a demand-based approach that considers rainfall and potential ET. A supply-based approach was used for taro, whereas a demand-based approach was used for all other irrigated land-cover types. The approach described herein was used to estimate irrigation rates for each of the modeled scenarios.

All subareas of the taro land-cover type were assigned an irrigation rate of 455 in/yr in the model on the basis of water-use studies for various taro pond fields in Hawaiʻi (Miles, 1931, as cited in Penn, 1997; Watson, 1964; De la Pena and Melchor, 1984; Berg and others, 1997), although local differences in irrigation rates likely exist. For this study, taro was further classified based on data from the United States Department of Agriculture National Agriculture Statistics Service (National Agricultural Statistics Service, 2014). Census data from individual taro growers were utilized to summarize the total area of irrigated taro harvested in 2012 for each county in the state (National Agricultural Statistics Service, 2014). For this study, the percentage of subareas with taro cultivated at any given time, hereinafter taro-cultivation rate, was estimated as the quotient of the total area of irrigated taro harvested in 2012 (National Agricultural Statistics Service, 2014) and the total area of taro land cover in 2015 (Melrose and others, 2016). For example, a total of 53 acres of irrigated taro was harvested in 2012 on the Island of Hawaiʻi (National Agricultural Statistics Service, 2014) and a total of 61 acres of taro land cover was delineated on the island in 2015 (Melrose and others, 2016), which results in a taro-cultivation rate of 87 percent. A similar computation for the other islands in this study produced the following taro-cultivation rates: (1) 30 percent for Kauaʻi, (2) 100 percent for Oʻahu, and (3) 93 percent for Molokaʻi, Lānaʻi, and Maui. The remaining taro land cover percentage of each subarea was treated in the model as developed open space land cover. In the model calculations, the monthly irrigation depths for taro were evenly distributed on each day of the month, about 1.25 inches per day (in/d), regardless of the rainfall on the day.

Table 5. Mean monthly fog-to-rainfall ratios used in the water-budget scenarios for Oʻahu, Molokaʻi, Lānaʻi, and the Island of Hawaiʻi.

[<, less than; base of cloud zone shown in figure 4; leeward-windward boundary for each island shown in figure 5; fog-to-rainfall ratios for Oʻahu, Molokaʻi, and Island of Hawaiʻi from Engott and others (2017), Oki and others (2020), and Engott (2011), respectively; fog-to-rainfall ratios for Lānaʻi derived from Juvik and others (2011); fog-to-rainfall ratios set to zero for all areas except where noted below]

Island	Description	Altitude range, in feet	Jan	Feb	Mar	Apr	May	June	July	Aug	Sept	Oct	Nov	Dec
Oʻahu	Leeward Oʻahu within cloud zone	2,000 and above	0.12	0.15	0.08	0.08	0.02	0.12	0.09	0.06	0.05	0.01	0.05	0.11
	Windward Oʻahu within cloud zone	2,000 and above	0.20	0.20	0.20	0.20	0.20	0.20	0.20	0.20	0.20	0.20	0.20	0.20
Molokaʻi	Molokaʻi within cloud zone	2,000 and above	0.16	0.16	0.16	0.16	0.16	0.16	0.16	0.16	0.16	0.16	0.16	0.16
Lānaʻi	Lānaʻi within cloud zone	2,000 and above	0.11	0.11	0.11	0.11	0.11	0.11	0.11	0.11	0.11	0.11	0.11	0.11
Island of Hawaiʻi	Leeward Kohala and Mauna Kea within and above cloud zone	2,600 and above	0.20	0.60	0.45	0.40	0.32	0.50	0.34	0.23	0.23	0.15	0.19	0.45
	Windward Island of Hawaiʻi within and above cloud zone	2,000 to <3,000	0.01	0.01	0.02	0.04	0.05	0.07	0.10	0.13	0.11	0.09	0.05	0.03
		3,000 to <4,000	0.04	0.05	0.07	0.12	0.16	0.22	0.30	0.39	0.31	0.25	0.16	0.10
		4,000 to <5,000	0.14	0.14	0.16	0.20	0.30	0.36	0.51	0.66	0.51	0.40	0.27	0.25
		5,000 to <6,000	0.23	0.25	0.28	0.35	0.53	0.66	0.83	0.91	0.68	0.51	0.37	0.36
		6,000 to <7,000	0.34	0.39	0.49	0.68	0.93	1.31	1.41	1.09	0.77	0.52	0.42	0.40
		7,000 to <8,000	0.41	0.49	0.70	0.87	1.09	1.59	1.68	1.17	0.85	0.60	0.51	0.51
		8,000 to <9,000	0.43	0.54	0.82	0.97	1.10	1.57	1.65	1.11	0.83	0.62	0.58	0.57
		9,000 to <10,000	0.37	0.49	0.73	0.86	0.78	0.88	0.98	0.73	0.58	0.51	0.59	0.48
		10,000 and above	0.40	0.56	0.78	0.79	0.45	0.42	0.51	0.37	0.35	0.46	0.60	0.48
	Leeward Hualālai, Mauna Loa, and Kīlauea within and above cloud zone	2,600 to <3,000	0.03	0.03	0.03	0.03	0.03	0.03	0.03	0.03	0.03	0.03	0.03	0.03
		3,000 to <4,000	0.07	0.07	0.07	0.07	0.07	0.07	0.07	0.07	0.07	0.07	0.07	0.07
		4,000 to <5,000	0.12	0.12	0.12	0.12	0.12	0.12	0.12	0.12	0.12	0.12	0.12	0.12
		5,000 to <6,000	0.17	0.17	0.17	0.17	0.17	0.17	0.17	0.17	0.17	0.17	0.17	0.17
		6,000 to <7,000	0.23	0.23	0.23	0.23	0.23	0.23	0.23	0.23	0.23	0.23	0.23	0.23
		7,000 to <8,000	0.27	0.27	0.27	0.27	0.27	0.27	0.27	0.27	0.27	0.27	0.27	0.27
		8,000 to <9,000	0.27	0.27	0.27	0.27	0.27	0.27	0.27	0.27	0.27	0.27	0.27	0.27
		9,000 and above	0.27	0.27	0.27	0.27	0.27	0.27	0.27	0.27	0.27	0.27	0.27	0.27

Table 6. Mean annual fog-interception rates used to compute fog-to-rainfall ratios for the water-budget scenarios for Kauaʻi and Maui.

[<, less than; base of cloud zone shown in figure 4; leeward-windward boundary shown in figure 5; mean annual fog-interception rates set to zero for all areas except where noted below]

Island	Description	Altitude range, in feet	Mean annual potential fog interception, in inches
Kauaʻi	Leeward Kauaʻi within cloud zone	2,000 and above	14
	Windward Kauaʻi within cloud zone	2,000 and above	30
Maui	Leeward Maui within cloud zone	2,600 to <7,800	14
	Windward Maui within cloud zone	2,000 to <7,800	30
	Leeward Maui above cloud zone	7,800 to <9,000	10
	Windward Maui above cloud zone	7,800 to <9,000	18
	Leeward and Windward Maui above cloud zone	9,000 and above	6

For irrigated land-cover types other than taro, irrigation rates were estimated using a demand-based approach. For the demand-based approach, irrigation demand for a given subarea was estimated on the basis of monthly rainfall, runoff, potential ET, and irrigation-method efficiency.

$$I_m = \left[\left(PE \right)_m + U_m - R_m \right] / g$$

$$\text{for } (PE)_m + U_m > R_m \text{ and} \tag{7}$$

$$I_m = 0$$

$$\text{for } (PE)_m + U_m \le R_m$$

where

I_m is amount of irrigation for month m [L, unit length],
$(PE)_m$ is potential ET for month m [L] (varies by location and land cover),
U_m is amount of runoff for month m [L],
R_m is amount of rainfall for month m [L], and
g is irrigation-method efficiency [dimensionless].

Irrigation-method efficiency is the fraction of applied irrigation water available for plant consumption. Taro was assumed to have an irrigation efficiency of 1. The irrigation efficiencies used in the water budget for all remaining irrigation methods were obtained from University of Hawai‘i (2008). Seed corn, diversified agriculture, and pineapple were assumed to use drip irrigation, which has an irrigation-method efficiency of 0.85. Coffee and macadamia (Island of Hawai‘i only) were assumed to use micro-spray irrigation, which has an irrigation efficiency of 0.80. Golf course, and medium- and high-intensity developed land-cover types were assumed to use sprinkler irrigation, which has an irrigation efficiency of 0.70.

For all irrigated land-cover types other than pineapple and taro, total monthly irrigation estimated from equation 7 was allocated in equal amounts for each day of a given month. Monthly irrigation volume calculated for pineapple using equation 16 in Johnson and others (2018) was uniformly distributed on days 1, 2, 3, 8, 9, 10, 15, 16, 17, 22, 23, 24, and 28 of each month (see Engott and Vana, 2007). Simulation of the pineapple-cultivation cycle was not represented in this study and irrigation was assumed for all months of the simulation period. Seed corn irrigation for this study is based on a cultivation cycle of two crops per year. Only 25 percent of subareas classified as seed corn are cultivated at any one time, leaving the remaining 75 percent of subareas to be classified as bare soil. This approach was developed on the basis of cultivation practices on Moloka‘i and used by Engott and others (2017) for estimating seed corn irrigation.

The model’s irrigation estimates for diversified agriculture and medium- and high-intensity developed land-cover types were adjusted with irrigation-adjustment factors (table 7). These factors ensured that the model’s irrigation estimates were, on average, consistent with reported irrigation and water availability rates. The estimated irrigation rates for coffee, golf course, pineapple, taro, and seed corn were not adjusted.

Septic Effluent

Septic effluent for this study consisted of cesspool seepage and septic-system leachate. In the model calculations, cesspool seepage was considered direct recharge, whereas septic-system leachate was added to the plant-root zone where it was subject to ET (Oki, 2022). Some buildings and premises on each island use onsite sewage disposal systems (OSDS) to dispose of wastewater. Whittier and El-Kadi (2009, 2014) compiled an inventory of OSDS on Kaua‘i, O‘ahu, Moloka‘i, Maui, and the Island of Hawai‘i. Whittier and El-Kadi (2014) omitted Lāna‘i from their inventory of OSDS because the island has few OSDS owing to a centralized wastewater collection and treatment system servicing the resorts and populated area of Lāna‘i City. Therefore, cesspool seepage and septic-system leachate were considered negligible on Lāna‘i for this study.

For each tax map key (TMK) parcel, Whittier and El-Kadi (2009, 2013) specified the number and type (class of OSDS) and the total estimated septic effluent discharge for each. They documented four types of OSDS including (1) any system that receives soil treatment such as bed, trench, and infiltration/chamber type, (2) septic systems discharging to a seepage pit, (3) aerobic units discharging to a seepage pit, and (4) all cesspools. Each TMK parcel might contain one or more model subareas. Hence, all subareas, including those not in the developed land-cover type, within a TMK parcel that contains an OSDS can have septic effluent.

For TMK parcels with cesspools, the total estimated septic effluent discharge was converted into an equivalent depth per unit time (inches/day) by dividing the rate of discharge (volume per unit time) by the parcel area, hereinafter the cesspool seepage rate. For subareas within TMK parcels with cesspools, the cesspool seepage rate was applied as direct recharge (see “Direct Recharge” section). For TMK parcels with other types of OSDS, the effluent discharge was converted into an equivalent depth per unit time by dividing the rate of discharge (volume per unit time) by the parcel area, hereinafter the septic-system leachate rate.

For subareas within TMK parcels with other types of OSDS, the septic-system leachate rate applied only to the portions of the subarea that are pervious, hereinafter pervious fraction. The pervious fraction of each subarea was calculated as the pervious area within a subarea divided by total area of the same subarea. However, the septic-system leachate rate in each subarea was adjusted by dividing the rate by the pervious fraction. The adjustment to the septic-system leachate rate was required to maintain the same volumetric rate of discharge within each subarea. In the model calculations, the adjusted septic-system leachate rate was added to the plant-root zone of the subarea to determine the amount of water available for ET, recharge, and soil-moisture storage (Oki, 2022).

The area of each TMK parcel was determined in a geographic information system using polygon shapefiles of TMK parcels (City and County of Honolulu, 2020; State of Hawai‘i, 2020). Dozens of TMK parcel numbers in the shapefiles of OSDS points did not match any TMK numbers in the shapefiles of TMK parcels. To account for TMK number

Table 7. Land-cover parameters used in the model calculations.

[Crop coefficients for forests were used to compute the sum of transpiration and ground evaporation, canopy evaporation was computed separately; crop coefficients for non-forest land covers were used to compute the sum of all evaporative components; --, not applicable]

Land-cover description	Root depth, in inches	Depletion fraction	Crop coefficient	Canopy capacity, in inches	Trunk-storage capacity, in inches	Fraction of precipitation diverted to stemflow	Fog-catch efficiency	Irrigation method	Irrigation-method efficiency	Irrigation-adjustment factor
Forest land covers										
Native forest, fog	30	0.50	0.30	0.05	0.01	0.04	1	--	--	--
Native forest, no fog	30	0.50	0.30	0.05	0.01	0.04	1	--	--	--
Non-native forest, fog	60	0.50	0.33	0.05	0.01	0.04	1	--	--	--
Non-native forest, no fog	60	0.50	0.44	0.05	0.01	0.04	1	--	--	--
Tree plantation, fog	60	0.50	0.33	0.05	0.01	0.04	1	--	--	--
Tree plantation, no fog	60	0.50	0.44	0.05	0.01	0.04	1	--	--	--
Kiawe, fog	71	0.50	0.23	0.05	0.01	0.04	1	--	--	--
Kiawe, no fog	71	0.50[b], 1.00[c]	0.23	0.05	0.01	0.04	1	--	--	--
Non-forest land covers										
Bare soil[a]	6	0.60	0.25	0	0	0	0	--	--	--
Coffee	48	0.40	0.91	0	0	0	0.50	Micro-spray	0.80	1
Diversified agriculture	10	0.35	1.00	0	0	0	0	Drip	0.85	0.40[f], 0.41[g], 0.55[h]
Macadamia	60	0.50	0.91	0	0	0	1	Micro-spray	0.80[e]	1[e]
Pineapple	18	0.50	0.30	0	0	0	0	Drip	0.85	1
Seed corn	18	0.60	0.29–1.20[d]	0	0	0	0	Drip	0.85	1
Taro	15	1.00	1.25	0	0	0	0	Flood	1	1
Developed, open space	12	0.50	1.18	0	0	0	0	--	--	--
Developed, low intensity	12	0.50	1.18	0	0	0	0	--	--	--
Developed, medium intensity	12	0.50	1.18	0	0	0	0	Sprinkler	0.70	0.37
Developed, high intensity	12	0.50	1.18	0	0	0	0	Sprinkler	0.70	0.37
Golf course	30	0.50	0.85	0	0	0	0	Sprinkler	0.70	1
Grassland	39	0.60	0.95	0	0	0	0	--	--	--
Shrubland, fog	12	0.50	1.00	0	0	0	0.50	--	--	--
Shrubland, no fog	12	0.50	1.00	0	0	0	0	--	--	--
Sparsely vegetated	5	0.50	1.18	0	0	0	0	--	--	--
Reservoir, lined	0	1.00	1.05	0	0	0	0	--	--	--
Reservoir, unlined	0	1.00	1.05	0	0	0	0	--	--	--
Coastal water body	0	1.00	1.05	0	0	0	0	--	--	--
Inland water body	0	1.00	1.05	0	0	0	0	--	--	--
Wetland	39	0.50	1.18	0	0	0	0	--	--	--

[a] Used in mixed land-cover designation for seed corn (see "Irrigation" section for explanation)

[b] Value used at altitude above 49 feet (see appendix 1 for explanation)

[c] Value used at altitude equal to or less than 49 feet (see appendix 1 for explanation)

[d] Varies by month (see" Potential Evapotranspiration" section for explanation)

[e] Island of Hawai'i only; irrigation of macadamia on Maui considered negligible

[f] Value used on Moloka'i

[g] Value used on Lāna'i, Maui, and Island of Hawai'i

[h] Value used on Kaua'i and O'ahu

disparity, the cesspool and septic-leachate for an OSDS point was included in the leachate values assigned to the nearest TMK parcel.

Storm-Drain Systems

Some developed areas in the Hawaiian Islands have storm-drain systems that collect and divert stormwater runoff. In the water-budget calculation, only subareas with developed, medium- and high-intensity land cover on each island were considered to have storm-drain systems (fig. 4). For the subareas with developed, medium- and high-intensity land cover, excess water that flows off impervious surfaces was assumed to be collected by storm-drain systems instead of flowing to adjacent pervious surfaces.

Soil-Infiltration Capacity

Infiltration is the process by which water at the ground surface enters the soil and is affected by precipitation and ET rates, surface topography, soil characteristics, and land cover. The soil-infiltration capacity is the maximum rate of infiltration and typically decreases during the early part of a precipitation event due to increases in soil moisture. Measurements of field-saturated hydraulic conductivity, K_{fs} (Nimmo and others, 2009), an indicator of soil-infiltration capacity, have been conducted at multiple sites across the Hawaiian Islands (Perkins and others, 2018; Kennedy and others, 2019a, b; Fortini and others, 2020). Results of these studies indicate that median values of K_{fs} vary from about 8 in/d at a site dominated by bare soil (Perkins and others, 2018) to about 14,000 in/d at a site dominated by forest species (Kennedy and others, 2019a). These data indicate a wide range of variability in soil-infiltration capacities across the Hawaiian Islands. WATRMod requires a specified value for soil-infiltration capacity for each subarea, which limits infiltration during a day. However, direct runoff in this study is computed separately using runoff-to-rainfall ratios, an approach intended to include runoff generated from precipitation rates in excess of the soil-infiltration capacity (see the "Direct Runoff" section of this report for a full description). Therefore, the soil-infiltration capacity in the water budget was set to 9,999 in/d, a value greater than the daily maximum precipitation rate, to ensure that direct runoff computed by the model was derived solely from the application of runoff-to-rainfall ratios.

Direct Runoff

Direct runoff from rainfall, hereinafter runoff, is the fraction of rainfall that does not contribute to net moisture gain within the plant-root zone. Runoff consists of overland flow and subsurface storm flow that rapidly returns infiltrated water to the stream (Oki, 2003). In the water-budget simulations, runoff was estimated as a fraction of rainfall using runoff-to-rainfall ratios, hereinafter runoff ratios. This approach was used in previous water-budget studies of Hawai‘i and other Pacific islands (Izuka and others, 2005; Engott and Vana, 2007; Engott, 2011; Johnson, 2012; Engott and others, 2017; Izuka and others, 2018; Johnson and others, 2018; Brewington and others, 2019; Mair and others, 2019; Oki and others, 2020) and provides reasonable estimates of regional average runoff with a limited level of complexity. For this study, the approaches by Izuka and others (2018), Mair and others (2019), and Oki and others (2020) were modified for use in the water-budget simulations. Initially, the method was developed for Kaua‘i, O‘ahu, and Maui (Engott and others, 2017; Izuka and others, 2018; Johnson and others, 2018), and modified for Maui (Mair and others, 2019) and Moloka‘i (Oki and others, 2020). This study extends the runoff analysis to Lāna‘i and the Island of Hawai‘i.

The spatial variability of runoff ratios depends on numerous factors including geology, climate, soil type, topography, and land cover. Runoff ratios are expected to be highest where the rainfall amount and intensity are high, permeability of the soils and substrate is low, slopes are steep, and soil moisture is high (Oki, 2003). Temporal variability in runoff ratios reflects factors including antecedent soil moisture and rainfall intensity. In Hawai‘i, runoff ratios generally follow a seasonal pattern, with highest values during the wet-season months and lowest during the dry-season months. For this study, November through April and May through October were considered the wet and dry seasons, respectively. In the water-budget model, daily runoff was calculated by multiplying daily rainfall by seasonal (wet and dry season) runoff ratios assigned to catchment zones.

Separate runoff ratios were computed for the 1978–2007 scenario and future-climate scenarios for each island and the drought scenario for Lāna‘i. Runoff ratios within a catchment zone for the 1978–2007 scenario are either observed seasonal ratios, mean seasonal ratios, estimated ratios using a regional-regression analysis that considered 30 different basin characteristics including soil, climate, land-cover, and morphometric characteristics (Izuka and others, 2018), or a combination of these ratios (Oki and others, 2020). For the drought and future-climate scenarios, runoff ratios were adjusted for interannual variability to account for differences in rainfall compared to 1978–2007 mean rainfall (Mair and others, 2019).

Catchment Zones and Drainage Basins of Streamgages

Catchment zones of a stream network for Kaua‘i, O‘ahu, Moloka‘i, and Maui were delineated by Rea and Skinner (2012) using a 10-m digital elevation model (DEM) with a flow-accumulation threshold of 20,000 cells. Thus, for a 10-m DEM, the minimum catchment size is at least 2 square kilometers. For Lāna‘i, catchment zones were developed using a 10-m DEM (U.S. Geological Survey, 2013f, g) by generally following the methods of Rea and Skinner (2012). Additionally, boundaries of the catchment zones were modified to be consistent with the boundaries of the catchment areas and associated surface-water flowlines of the National Hydrography Dataset Plus High Resolution (NHDPlus HR) for Lāna‘i (U.S. Geological Survey, 2019). Catchment zones for the Island of Hawai‘i

were developed using a 10-m DEM (U.S. Geological Survey, 2013a, b, c, d, e) by generally following the methods of Rea and Skinner (2012). Flowlines from the NHDPlus HR for the Island of Hawai‘i (U.S. Geological Survey, 2020) were modified to follow the clearly incised drainage channels visible from aerial imagery. The topography of the DEM was modified to conform to the flowlines. Catchment zones were delineated using a flow-accumulation threshold of a minimum of 20,000 cells (Rea and Skinner, 2012). The number of catchments on the Island of Hawai‘i differs from those published by Rea and Skinner (2012) because they applied a flow-accumulation threshold of 40,000 cells for the Island of Hawai‘i. The higher threshold resulted in a smaller number of catchments.

Some catchment zones are closed and form topographic sinks that cause runoff to converge to a point(s) within the catchment zone and not contribute to downstream flow. The sinks are areas where, at the lowest point along the preferred drainage path, water would have to fill to a depth of 10 m before overflowing and proceeding downstream. The closed catchment zones were identified with a 10-m fill limit, using the most recent DEMs and the associated flow-direction raster. Catchment delineations downstream of the closed catchment zones exclude the upstream closed catchment zones because they do not provide any flow to the downstream catchment zone (see for example, small catchments within the drainage basins 16701800 and 16764000 in figures 5*K* and 5*L*). The percentage of land surface area within closed catchment zones on each island is below 1 percent, except for the Island of Hawai‘i, where about 4 percent of the island’s land-surface area is within closed catchment zones. Runoff from closed catchment zones was also computed using runoff ratios and removed from the water budget. Estimated runoff ratios for closed catchment zones are generally low; therefore, we assume runoff in these zones has a negligible effect on the overall water budget. Though removing runoff from closed catchment zones ignores the potential for ponding-induced soil infiltration, it is conservative from a recharge-estimation perspective.

Each catchment zone was assigned two attributes: (1) gaged or ungaged, and (2) windward or leeward (fig. 5). For this study, a gaged catchment zone is defined by the drainage-basin boundary of a streamgage station. Selected gaged catchment zones on Kaua‘i, O‘ahu, Moloka‘i, and Maui were delineated by Rosa and Oki (2010). No streamgages exist on Lāna‘i, leaving the island without any gaged catchment zones. The drainage basins for selected streamgages on the Island of Hawai‘i were delineated using the same conditioned DEM that was used for the catchment-zone delineation. Gaged drainage basins were further distinguished as either consisting of a single catchment zone or multiple catchment zones. For this study, a total of 41 streamgage stations have single catchment zone drainage basins and 33 streamgage stations have multiple catchment zone drainage basins. The boundary between windward and leeward regions was based on topographic divides (Yamanaga, 1972; Cheng and others, 2021). Because Lāna‘i is in the rain shadow of Maui, the entire island was considered a leeward region.

Observed Seasonal Runoff Ratios

Runoff and rainfall data for the drainage basins of 74 streamgages (13 on Kaua‘i, 17 on O‘ahu, 11 on Moloka‘i, 25 on Maui, and 8 on the Island of Hawai‘i; fig. 5; table 8) were used to estimate seasonal runoff ratios for the water-budget model. Streamgages selected for the runoff analysis had (1) at least eight complete years of daily mean discharge records between 1920 and 2012, (2) a drainage-basin area greater than 0.2 mi^2, and (3) unregulated streamflow or regulated streamflow with complete and reliable records of daily mean diverted flow available to reconstruct total streamflow at the gage (Izuka and others, 2018). Adjusted monthly rainfall maps were used to estimate rainfall within each drainage basin (Frazier and others, 2016).

Because streamflow measured at most streamgages consists of direct runoff and base flow, the base-flow component was estimated and subtracted from streamflow to estimate the direct runoff component at each streamgage. Streamflow at streamgages 16013000 on Kaua‘i, 16500100 and 16660000 on Maui, 16411400 and 16414000 on Moloka‘i, and 16759200, 16759800, 16764000, 16765000, 16767000, and 16770500 on the Island of Hawai‘i was considered ephemeral and assumed to have no base flow component. Base flow at the other gaged basins was estimated using a computerized base-flow separation method (Wahl and Wahl, 1995). This method has been used in numerous studies in Hawai‘i (for example, Izuka and others, 2005; Engott and Vana, 2007; Engott, 2011) and provides a reasonable estimate of base flow for perennial streams in Hawai‘i. The base-flow separation method defines local streamflow minimums within consecutive, non-overlapping *n*-day periods and requires two parameters: (1) the turning-point test factor, *f*, and (2) the number of days in a test window, *n*. In this study, the *f* value used for all streamgages was 0.9, which is appropriate in most applications (Wahl and Wahl, 1995). The *n* values were determined for each streamgage using the method described by Wahl and Wahl (1995) and ranged from 3 to 5 days (table 8). Daily base flow was subtracted from daily streamflow to determine daily runoff. Streamflow measured at streamgages 16415000 and 16419500 on Moloka‘i is intermittent and likely reflects the effects of diversions, and therefore base flow might be underestimated at these streamgages.

Observed seasonal runoff ratios were calculated for the 74 gaged drainage basins (table 8) for any particular season and year. Each observed seasonal runoff ratio for the gaged drainage basins was calculated as the quotient of cumulative direct runoff and cumulative rainfall during each season with a complete record of daily runoff estimates and corresponding gridded monthly rainfall from Frazier and others (2016). For example, the observed runoff ratio of a gaged basin during the dry season of 2001 was calculated as the quotient of cumulative runoff and cumulative rainfall during May–October 2001. Runoff at streamgages along the Hāmākua region of the Island of Hawai‘i (Waimanu to Onomea aquifer system; fig. 2*F*) is consistently greater than rainfall and might reflect an underestimation of rainfall in these areas. Therefore, gaged drainage basins along the Hāmākua region and observed runoff ratios from those streamgages are not included in this analysis.

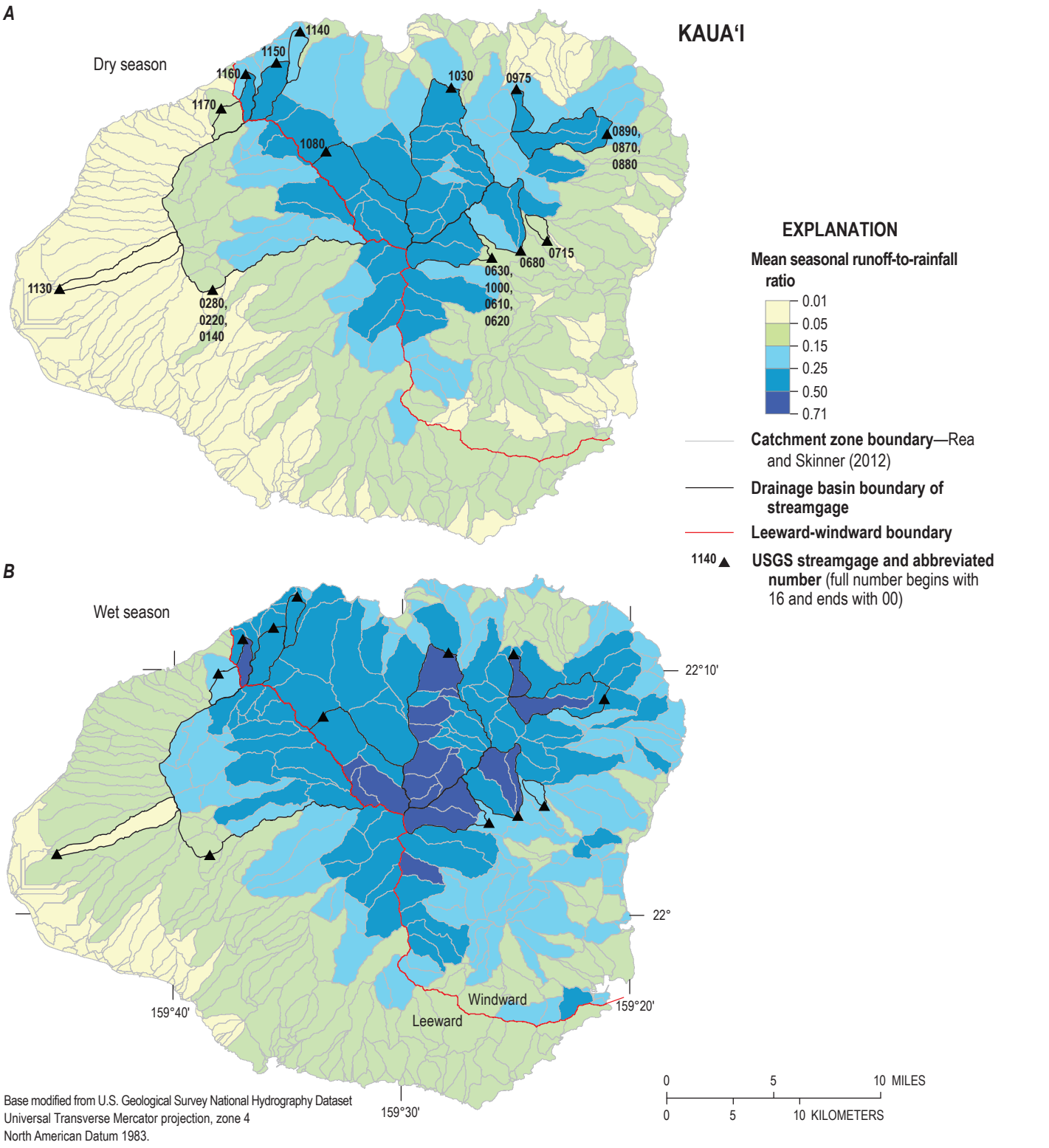


Figure 5. Maps showing the mean runoff-to-rainfall ratios for (*A*) dry-season months (May through October) on Kaua'i, (*B*) wet-season months (November through April) on Kaua'i, (*C*) dry-season months on O'ahu, (*D*) wet-season months on O'ahu, (*E*) dry-season months on Moloka'i, (*F*) wet-season months on Moloka'i, (*G*) dry-season months on Lāna'i, (*H*) wet-season months on Lāna'i, (*I*) dry-season months on Maui, (*J*) wet-season months on Maui, (*K*) dry-season months on the Island of Hawai'i, and (*L*) wet-season months on the Island of Hawai'i from drainage basins of selected streamgages and catchment zones (modified from Rea and Skinner, 2012).

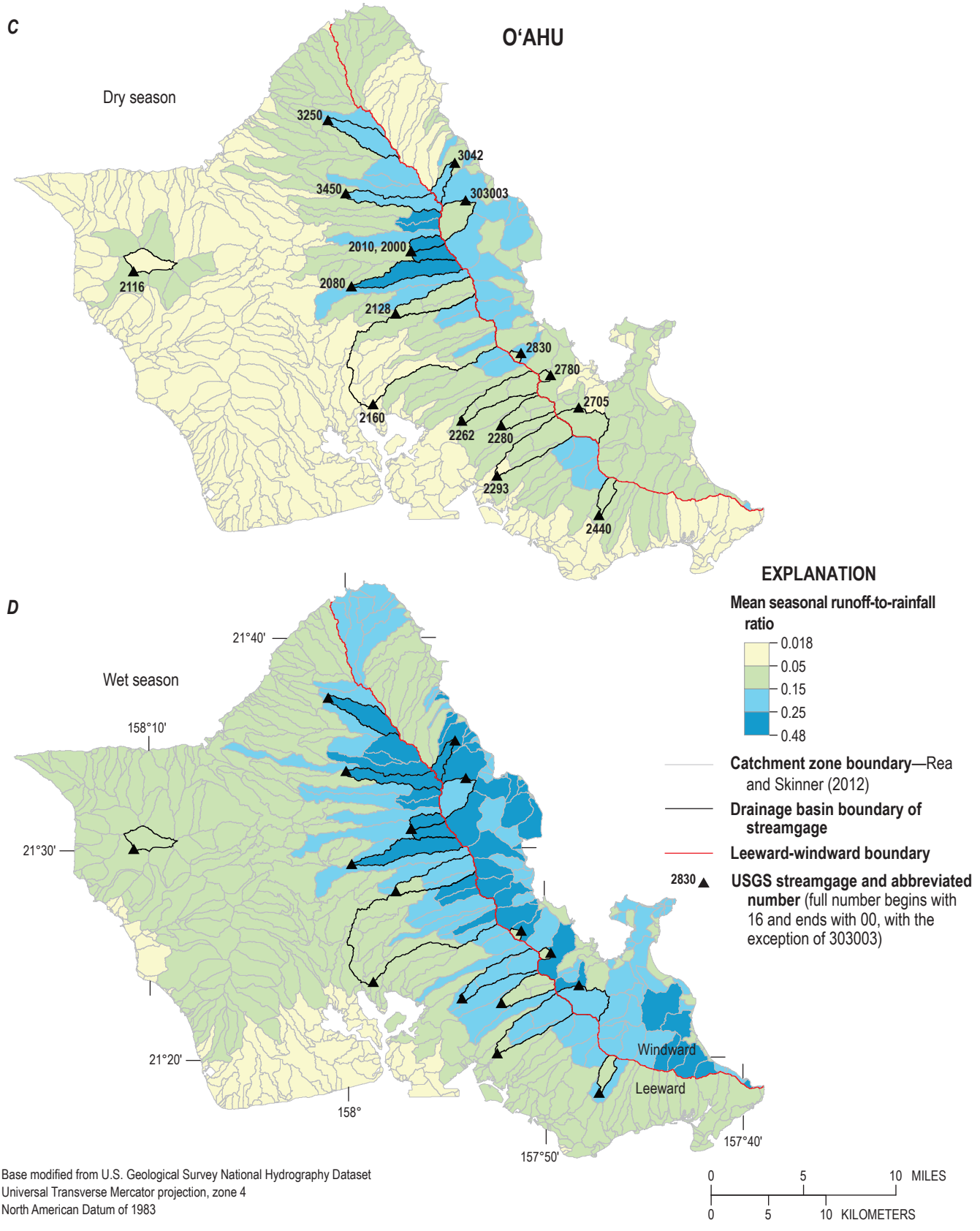
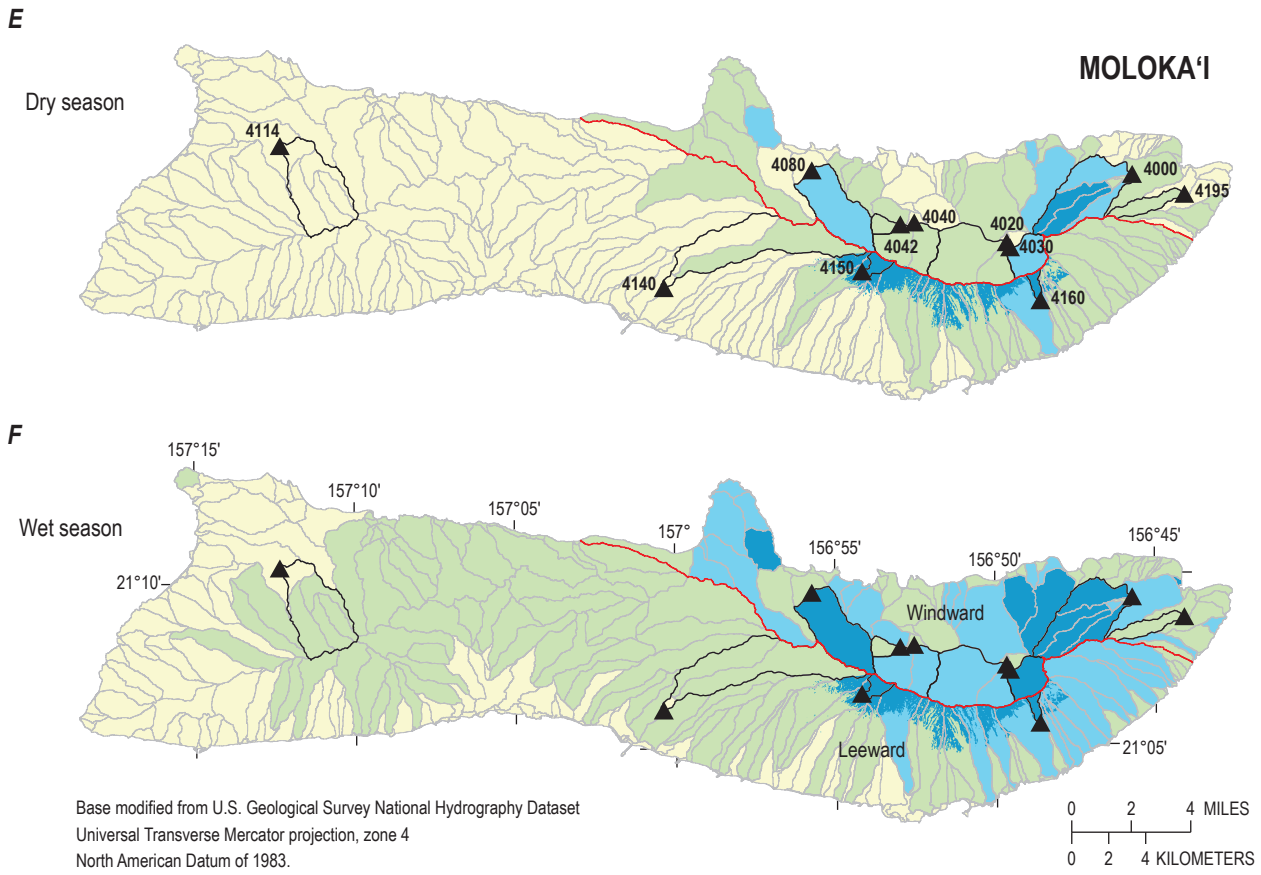


Figure 5.—Continued



EXPLANATION

Mean seasonal runoff-to-rainfall ratio

- 0.004
- 0.05
- 0.15
- 0.25
- 0.40

Catchment zone boundary—Rea and Skinner (2012)

Drainage basin boundary of streamgage

Leeward-windward boundary

4140 ▲ **USGS streamgage and abbreviated number** (full number begins with 16 and ends with 00)

Figure 5.—Continued

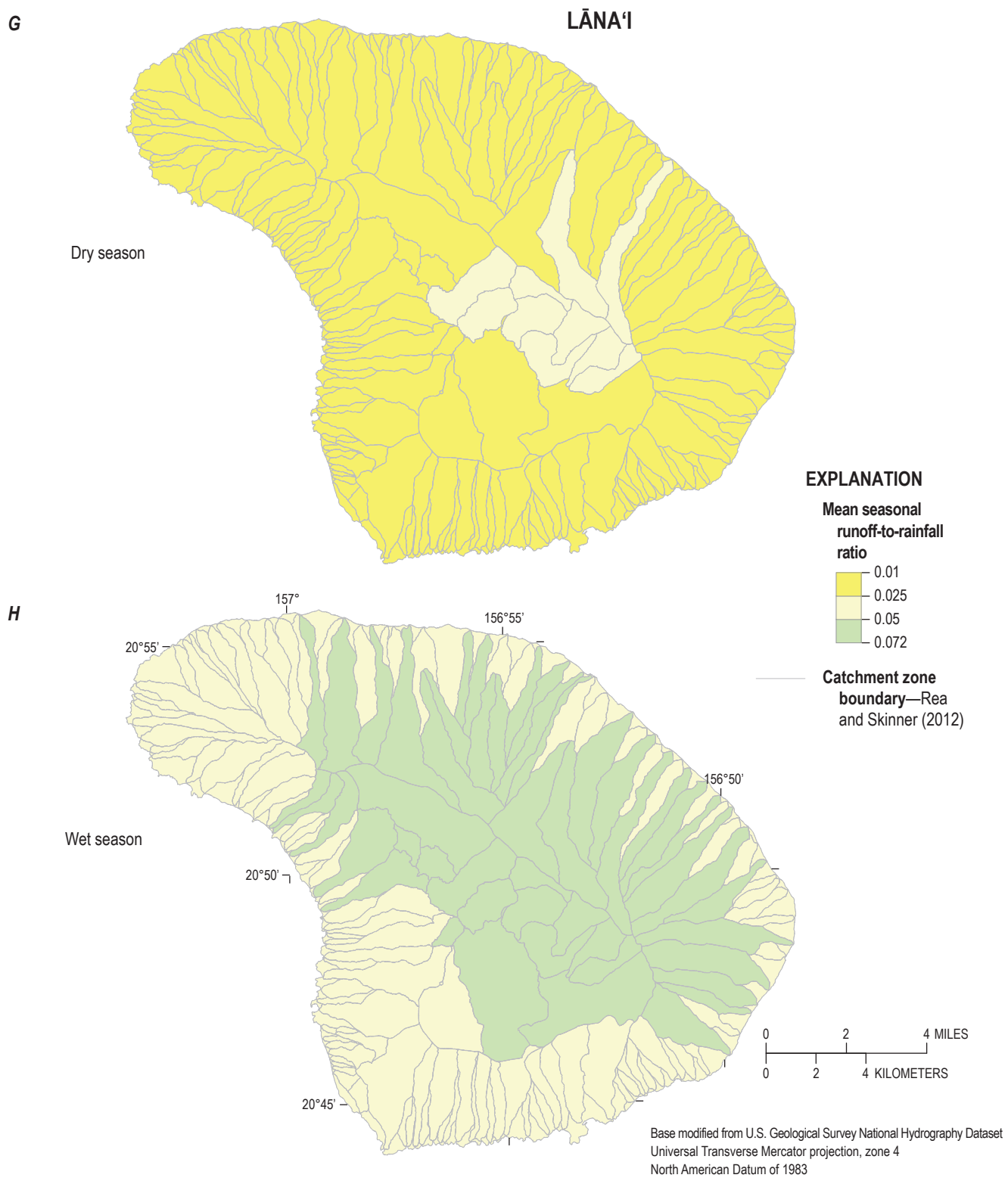


Figure 5.—Continued

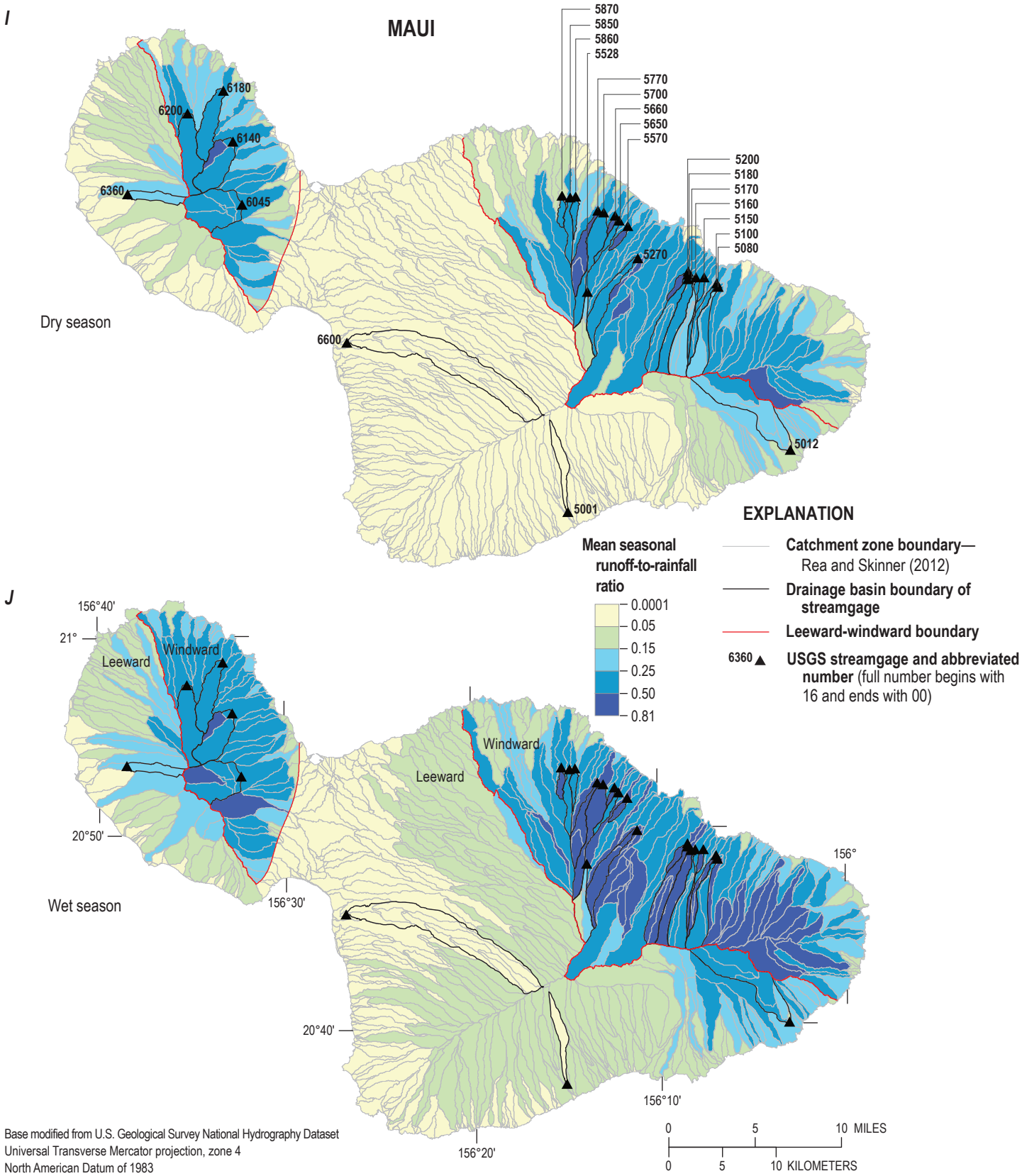
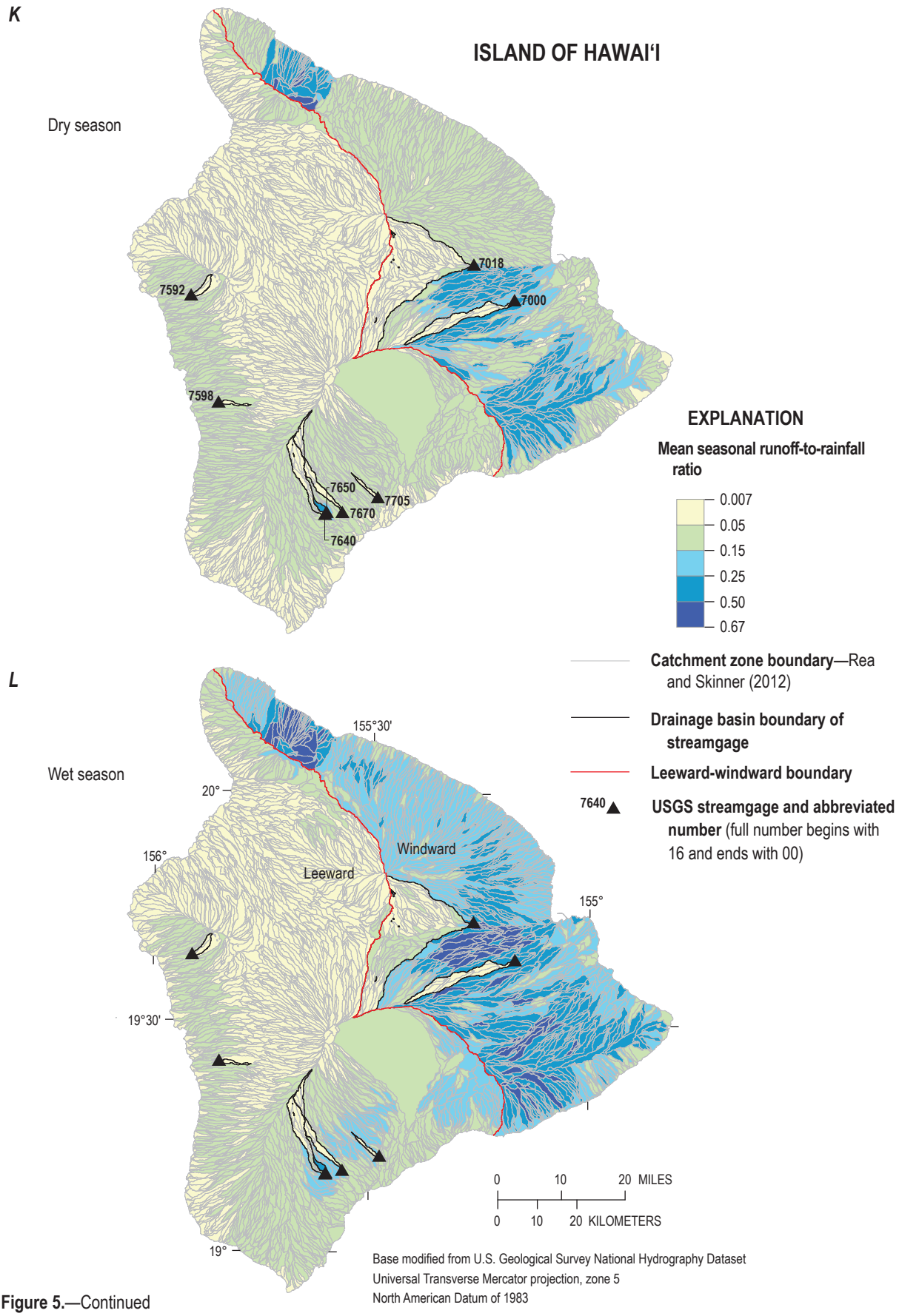


Figure 5.—Continued



Base modified from U.S. Geological Survey National Hydrography Dataset
Universal Transverse Mercator projection, zone 5
North American Datum of 1983

Figure 5.—Continued

Table 8. Mean seasonal runoff-to-rainfall ratios and characteristics for drainage basins of streamgages on Kauaʻi, Oʻahu, Maui, Molokaʻi, and the Island of Hawaiʻi.

[Ratios for Maui modified from Izuka and others, 2018; Moloka'i modified from Oki and others, 2020; See figure 5 for locations of drainage basins; *n*, number of days used for determining minimum flows in the base flow-separation program (Wahl and Wahl, 1995); E, the base flow-separation program (Wahl and Wahl, 1995) was not used to estimate direct runoff because the streamflow measured at station was ephemeral and was assumed to equal direct runoff; wet season is November through April; dry season is May through October]

Abbreviated streamgage number[a]	Stream, river, or gulch	Island	Aspect	Area of drainage basin, in square miles	*n*	Period of record selected to calculate mean seasonal runoff-to-rainfall ratios			Mean seasonal runoff-to-rainfall ratios	
						Start	End	Number of years	Wet season	Dry season
0280	Waimea River	Kauaʻi	Leeward	43.68	5	1927	1947	21	0.29	0.22
0630	North fork Wailua River	Kauaʻi	Windward	5.34	4	1957	1984	28	0.63	0.48
0680	East bank of north fork Wailua River	Kauaʻi	Windward	6.24	4	1978	2007	30	0.50	0.27
0715	Left branch ʻŌpaekaʻa Stream	Kauaʻi	Windward	0.75	4	1978	2007	30	0.22	0.09
0890	Anahola Stream	Kauaʻi	Windward	5.45	4	1956	1985	30	0.48	0.35
0975	Halaulani Stream	Kauaʻi	Windward	1.20	4	1978	2007	30	0.62	0.41
1030	Hanalei River	Kauaʻi	Windward	18.51	4	1993	2007	15	0.57	0.34
1080	Wainiha River	Kauaʻi	Windward	10.44	5	1978	2007	30	0.54	0.39
1140	Limahuli Stream	Kauaʻi	Windward	1.37	4	1995	2005	11	0.35	0.21
1150	Hanakāpīʻai Stream	Kauaʻi	Windward	2.75	5	1941	1949	9	0.41	0.28
1160	Hanakoa Stream	Kauaʻi	Windward	1.09	5	1939	1947	9	0.53	0.31
1170	Kalalau Stream	Kauaʻi	Leeward	1.60	5	1932	1948	17	0.24	0.09
1300	Nahomalu Valley	Kauaʻi	Leeward	3.79	E	1964	1971	8	0.04	0.01
2000	North bank and fork Kaukonahua Stream	Oʻahu	Leeward	1.38	4	1978	2007	30	0.48	0.49
2010	North fork Kaukonahua Stream	Oʻahu	Leeward	1.15	4	1940	1951	12	0.34	0.36
2080	South fork Kaukonahua Stream	Oʻahu	Leeward	4.10	5	1978	2007	29	0.32	0.28
2116	Mākaha Stream	Oʻahu	Leeward	2.12	5	1978	2007	30	0.13	0.04
2128	Kīpapa Stream	Oʻahu	Leeward	4.25	4	1978	2004	27	0.19	0.13
2160	Waiawa Stream	Oʻahu	Leeward	25.19	4	1978	2004	27	0.21	0.12
2262	North Hālawa Stream	Oʻahu	Leeward	4.02	5	1983	2007	24	0.19	0.10
2280	Moanalua Stream	Oʻahu	Leeward	2.71	4	1949	1978	30	0.15	0.07
2293	Kalihi Stream	Oʻahu	Leeward	5.21	4	1978	2003	26	0.21	0.11
2440	Pūkele Stream	Oʻahu	Leeward	1.15	5	1953	1982	30	0.14	0.08
2705	Kamoʻoaliʻi Stream	Oʻahu	Windward	3.20	3	1967	1976	9	0.18	0.06
2780	ʻIolekaʻa Stream	Oʻahu	Windward	0.32	3	1944	1968	25	0.09	0.07
2830	Kahaluʻu Stream	Oʻahu	Windward	0.28	3	1941	1970	30	0.11	0.08
303003	Punaluʻu Stream	Oʻahu	Windward	2.78	4	1978	2006	29	0.21	0.13
3042	Kaluanui Stream	Oʻahu	Windward	1.09	4	1978	2007	30	0.34	0.25
3250	Kamananui Stream	Oʻahu	Leeward	3.15	4	1971	2000	30	0.29	0.15
3450	ʻŌpaeʻula Stream	Oʻahu	Leeward	3.00	4	1978	2007	30	0.37	0.24
4000	Hālawa Stream	Molokaʻi	Windward	4.68	3	1978	2007	30	0.39[b]	0.34[b]
4020	Pūlena Stream	Molokaʻi	Windward	4.39	3	1938	1952	15	0.32[b]	0.22[b]
4030	Waiakeakua Stream	Molokaʻi	Windward	1.41	3	1938	1956	19	0.27[b]	0.20[b]

Table 8.—Continued

Abbreviated streamgage number[a]	Stream, river, or gulch	Island	Aspect	Area of drainage basin, in square miles	*n*	Period of record selected to calculate mean seasonal runoff-to-rainfall ratios			Mean seasonal runoff-to-rainfall ratios	
						Start	End	Number of years	Wet season	Dry season
4040	Pelekunu Stream	Molokaʻi	Windward	2.64	3	1938	1947	10	0.23[b]	0.15[b]
4042	Pilipililau Stream	Molokaʻi	Windward	0.47	3	1969	1996	28	0.14[b]	0.06[b]
4080	Waikolu Stream	Molokaʻi	Windward	3.78	3	1949	1958	10	0.27[b]	0.18[b]
4114	Kakaʻako Gulch	Molokaʻi	Leeward	5.04	E	1964	1971	8	0.05[b]	0.03[b]
4140	Kaunakakai Gulch	Molokaʻi	Leeward	6.62	E	1951	1964	14	0.12[b]	0.04[b]
4150	East fork Kawela Gulch	Molokaʻi	Leeward	0.44	3	1947	1970	23	0.40[b]	0.34[b]
4160	Punaʻula Gulch	Molokaʻi	Leeward	0.28	3	1948	1971	24	0.32[b]	0.26[b]
4195	Pāpio Gulch	Molokaʻi	Windward	0.99	3	1978	2005	28	0.13[b]	0.07[b]
5001	Kepuni Gulch	Maui	Leeward	1.93	E	1963	1972	10	0.04	0.01
5012	ʻOheʻo Gulch	Maui	Leeward	8.53	4	1990	1997	8	0.38	0.25
5080	Hanawī Stream	Maui	Windward	3.29	4	1978	2007	30	0.42	0.24
5100	Kapāʻula Gulch	Maui	Windward	1.35	4	1933	1962	30	0.61	0.38
5150	Waiohue Gulch	Maui	Windward	0.74	4	1933	1962	30	0.52	0.32
5160	Kopiliula Stream	Maui	Windward	3.56	4	1938	1957	20	0.43	0.24
5170	East Wailuaiki Stream	Maui	Windward	3.14	4	1928	1957	30	0.58	0.35
5180	West Wailuaiki Stream	Maui	Windward	3.59	4	1978	2007	30	0.64	0.39
5200	East Wailuanui Stream	Maui	Windward	0.53	4	1928	1957	30	0.75	0.55
5270	Honomanū Stream	Maui	Windward	2.94	4	1939	1963	25	0.55	0.45
5528	Waikamoi Strream	Maui	Windward	2.50	4	1957	1967	11	0.17	0.12
5570	Alo Stream	Maui	Windward	0.47	4	1928	1957	30	0.75	0.65
5650	Kaʻaiea Gulch	Maui	Windward	0.68	4	1933	1962	30	0.50	0.44
5660	ʻOʻopuola Stream	Maui	Windward	0.25	4	1931	1957	27	0.64	0.54
5700	Naʻiliʻilihaele Stream	Maui	Windward	3.69	5	1945	1974	30	0.60	0.50
5770	Kailua Stream	Maui	Windward	2.41	4	1928	1957	30	0.75	0.53
5850	Hoʻolawanui Stream	Maui	Windward	1.34	5	1942	1971	30	0.50	0.32
5860	Hoʻolawaliʻiliʻi Stream	Maui	Windward	0.62	4	1931	1957	27	0.43	0.33
5870	Honopou Stream	Maui	Windward	0.59	4	1978	2007	30	0.57	0.38
6045	Wailuku River	Maui	Windward	6.13	4	1983	2007	24	0.48	0.40
6140	Waiheʻe River	Maui	Windward	4.27	3	1984	2007	24	0.49	0.45
6180	Kahakuloa Stream	Maui	Windward	3.40	4	1978	2007	30	0.43	0.33
6200	Honokōhau Stream	Maui	Windward	4.18	4	1978	2007	28	0.35	0.31
6360	Kanahā Stream	Maui	Leeward	1.57	3	1920	1931	11	0.17	0.16
6600	Kūlanihākoʻi Gulch	Maui	Leeward	15.03	E	1963	1970	8	0.01	0.00
7000	Waiākea Stream	Hawaiʻi	Windward	19.13	5	1966	1995	30	0.03	0.02
7018	Wailuku River	Hawaiʻi	Windward	128.15	3	1966	1982	16	0.06	0.02
7592	Right branch Waiʻaha Stream	Hawaiʻi	Leeward	3.22	E	1960	1982	22	0.03	0.03

Table 8.—Continued

Abbreviated streamgage number[a]	Stream, river, or gulch	Island	Aspect	Area of drainage basin, in square miles	n	Period of record selected to calculate mean seasonal runoff-to-rainfall ratios			Mean seasonal runoff-to-rainfall ratios	
						Start	End	Number of years	Wet season	Dry season
7598	Kiʻilae Stream	Hawaiʻi	Leeward	1.47	E	1958	1982	24	0.03	0.04
7640	Tributary Hīlea Gulch	Hawaiʻi	Leeward	1.70	E	1966	1991	25	0.44	0.49
7650	Upstream tributary Hīlea Gulch	Hawaiʻi	Leeward	14.82	E	1966	1982	16	0.05	0.04
7670	Nīnole Gulch	Hawaiʻi	Leeward	11.92	E	1966	1982	16	0.02	0.04
7705	Pāʻauʻau Gulch	Hawaiʻi	Leeward	1.85	E	1978	2007	7	0.03	0.02

[a]Full number begins with 16 and ends with 00, with the exception of 303003.

[b]Average of the observed mean seasonal runoff-to-rainfall ratio and the estimated mean seasonal runoff-to-rainfall ratio derived from the regional-regression equations (table 9).

Observed Mean Seasonal Runoff Ratios

For each of the 74 selected gaged drainage basins on Kauaʻi, Oʻahu, Molokaʻi, Maui, and the Island of Hawaiʻi, observed mean seasonal runoff ratios were calculated for a period with rainfall and runoff conditions generally representative of those during 1978–2007 (see tables 1–5 in Izuka and others, 2018). Criteria for selecting representative periods of record were determined by examining temporal variations in seasonal runoff ratios of streamgages that were operational during 1978–2007. For these streamgages, the 1978–2007 mean seasonal runoff values were compared with those calculated for smaller, subset periods during 1920–2007. The results of these comparisons show that the difference between the 1978–2007 and smaller subset mean seasonal runoff values decreases with increasing subset record period length. Accordingly, the period of record selected to calculate observed mean seasonal runoff values consisted of either (1) the entire period during 1978–2007 for streamgages with at least 7 (Island of Hawaiʻi), 24 (Kauaʻi, Oʻahu, Maui), or 28 (Molokaʻi) complete years of record during 1978–2007, or (2) the longest contiguous period during 1920–2007 that had less than 5-percent difference in mean annual rainfall, relative to rainfall for 1978–2007, for all remaining streamgages on perennial streams with base flow, or (3) the entire period of record for the streamgages on ephemeral or intermittent streams (Izuka and others, 2018; Oki and others, 2020). Each observed mean seasonal runoff ratio for a gaged drainage basin was calculated as the quotient of cumulative runoff and cumulative rainfall during the appropriate season of the selected period. For example, the observed mean dry-season runoff ratio for streamgage 2280 was calculated as the quotient of cumulative runoff and cumulative rainfall during May–October between 1949 and 1978.

Adjusted Seasonal Runoff Ratios

The runoff analysis was initially developed for the period 1978–2007 (Izuka and others, 2018). For periods other than 1978–2007 (the Lānaʻi drought scenario and all future-climate scenarios), regression analyses were conducted to develop the following equation for adjusting observed and estimated mean seasonal runoff ratios to account for interannual rainfall variability to the mean seasonal rainfall during 1978–2007 (Mair and others, 2019):

$$rr_{t,j} = \overline{rr}_t \times \left((s-c) \times \left(1 - \exp\left(-\frac{rf_{t,j} / \overline{rf}_t}{a} \right) \right) + c \right) \qquad (8)$$

$$\text{for } rr_{t,j} \le 0$$

$$rr_{t,j} = 0,$$

where

$rr_{t,j}$ is the adjusted seasonal runoff ratio for season t (wet or dry) and year j of interest [dimensionless],

$\overline{rr}_t$ is the observed mean seasonal runoff ratio for season t (wet or dry) during period

representative of average climate conditions for gaged basins, or estimated mean seasonal runoff ratio for season t (wet or dry) from regional-regression equations for ungaged basins [dimensionless],

$rf_{t,j}$ is the cumulative rainfall for season t (wet or dry) and year j of interest [inches],

$\overline{rf}_t$ is the mean rainfall for season t (wet or dry) during period representative of average climate conditions for gaged basins and during 1978–2007 for ungaged basins [inches],

t is the subscript designating season (wet or dry) of interest, where the wet season is November through April and the dry season is May through October,

a is the range coefficient (varied by region and season), with bounds of [0, +∞],

s is the sill coefficient (varied by region and season), with bounds of [0, +∞], and

c is the intercept coefficient (varied by region and season), with bounds of [0, +∞].

The form of the empirical equation 8 was selected because it could be used to satisfy the following set of conditions:

$$\begin{aligned} &\text{for } rf_{t,j}/\overline{rf}_t = 1, \\ &rr_{t,j} = \overline{rr}_t, \\ &\text{for } rf_{t,j}/\overline{rf}_t \gg 1, \\ &rr_{t,j} \cong s \times \overline{rr}_t, \end{aligned} \tag{9}$$

Four sets of range, sill, and intercept coefficients were developed for equation 8, one for each region (windward and leeward) and season (wet and dry; table 9). Coefficient (a, s, and c) values were derived from regression analyses of observed seasonal runoff ratios, seasonal rainfall, observed mean seasonal runoff ratios, and mean seasonal rainfall values for the drainage basins of 55 selected streamgages on Kaua‘i, O‘ahu, and Maui (Izuka and others, 2018). For each of the 55 streamgages, observed seasonal runoff and rainfall values were computed for 1920–2007. Seasonal rainfall values, computed using monthly rainfall maps from Frazier and others (2016), and observed seasonal runoff values were normalized by mean seasonal rainfall and observed mean seasonal runoff ratios, respectively. For each region-season pairing, the normalized rainfall ratio and normalized runoff ratios were binned into a minimum of 23 equally spaced intervals with a minimum of five data pairs per bin. Values for a, s, and c were estimated as the least-squares fit through the median of each bin (fig. 6).

In summary, the adjusted seasonal runoff ratios varied temporally and were (1) greater than observed mean seasonal runoff ratios for seasons when rainfall was greater than the seasonal 1978–2007 average, (2) equal to observed mean seasonal runoff ratios for seasons when rainfall was equal to the seasonal 1978–2007 average, and (3) less than observed mean seasonal runoff ratios for seasons when rainfall was less than the seasonal 1978–2007 average.

Table 9. Coefficients for equation 8 that were derived from 55 drainage basins on Kaua‘i, O‘ahu, and Maui and used to adjust seasonal runoff-to-rainfall ratios to changes in rainfall for catchment zones on Kaua‘i, O‘ahu, Moloka‘i, Lāna‘i, Maui, and the Island of Hawai‘i.

[Data for 55 drainage basins from Izuka and others (2018), changes in rainfall from Mair and others (2019). See figure 5 for locations of leeward and windward regions; wet season is November through April; dry season is May through October; see “Adjusted Seasonal Runoff Ratios” section for equation 8]

Region	Season	Coefficient for equation 8		
		a (range)	*s* (sill)	*c* (intercept)
Windward	Wet	0.470	1.192	–0.416
Windward	Dry	0.695	1.444	–0.427
Leeward	Wet	0.560	1.400	–0.990
Leeward	Dry	0.601	1.673	–1.881

Runoff Ratios Assigned to Ungaged Catchment Zones

Ungaged catchment zones are geographically outside of the drainage basins of the 74 selected streamgages (table 5). Runoff for ungaged catchment zones was calculated using estimated mean seasonal runoff ratios derived from four regional-regression equations (table 10). These equations are the same as those used for Kaua‘i, O‘ahu, and Maui by Izuka and others (2018), Engott and others (2017), and Johnson and others (2018), in which the development method is described. Separate windward and leeward regression equations were derived for the wet and dry seasons. The regional-regression analysis tested the relation between 30 basin characteristics—including those related to climate, soil, vegetation, and morphometric characteristics—and observed mean seasonal runoff ratios and were developed with methods consistent with discussion given in Helsel and Hirsch (1992).

For leeward catchment zones, wet-season runoff ratios were estimated on the basis of mean wet-season rainfall during 1978–2007 (Giambelluca and others, 2013) and dry-season runoff ratios were estimated on the basis of mean dry-season rainfall during 1978–2007. Soils are classified by the Natural Resource Conservation Service into four Hydrologic Soil Groups (A, B, C, and D) based on the soil’s runoff potential. For windward catchment zones, wet-season runoff ratios were estimated on the basis of mean annual reference ET (Giambelluca and others, 2014) and the percentage of the catchment zone’s area containing hydrologic soil groups C and D, which have a relatively slow rate of infiltration when thoroughly wet (Natural Resources Conservation Service, 2006a, b, c, d; 2008; 2019). Dry-season runoff ratios were

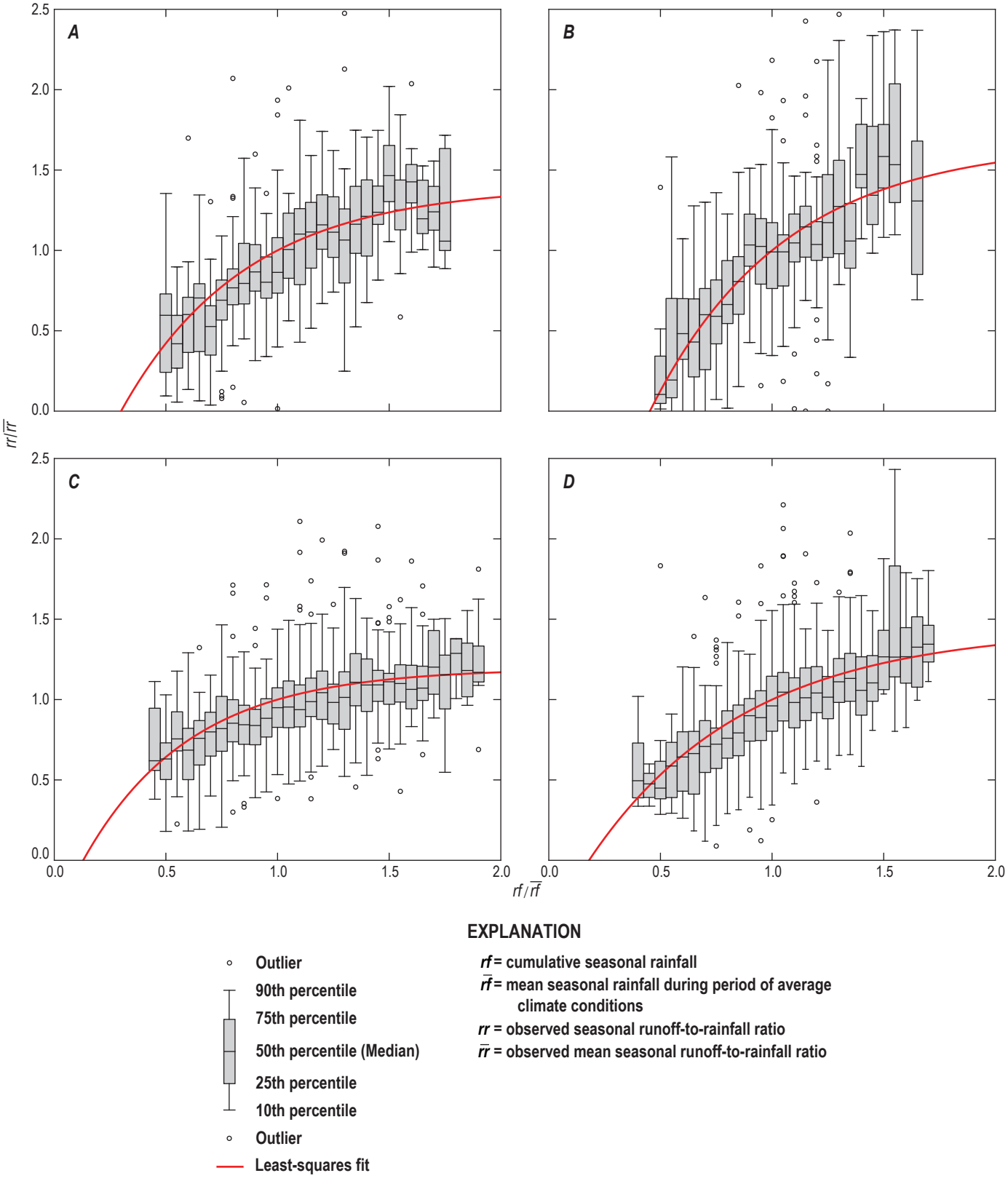


Figure 6. Least-squares fit curves of normalized rainfall and normalized runoff-to-rainfall ratios derived from 55 drainage basins on Kaua‘i, O‘ahu, and Maui (Izuka and others, 2018). (*A*) Leeward region and wet season, (*B*) leeward region and dry season, (*C*) windward region and wet season, and (*D*) windward region and dry season.

Table 10. Regional-regression equations derived from 55 drainage basins on Kauaʻi, Oʻahu, and Maui and used to calculate mean seasonal runoff-to-rainfall ratios for ungaged catchment zones on Kauaʻi, Oʻahu, Molokaʻi, Lānaʻi, Maui, and the Island of Hawaiʻi.

[Data for 55 drainage basins from Izuka and others (2018). See figure 5 for locations of leeward and windward regions; wet season is November through April; dry season is May through October; mean wet- and dry-season rainfall, in inches, are from Giambelluca and others (2013); mean annual and dry-season ETo, reference evapotranspiration, in inches, are from Giambelluca and others (2014); % HSG C is the percentage of area in a catchment zone that contains hydrologic soil group C; % HSG D is the percentage of area in a catchment zone that contains hydrologic soil group D; areas of hydrologic soil groups were derived from Natural Resources Conservation Service (2006a, b, c, d, 2008, 2019)]

Region	Season	Regional-regression equations used to compute mean seasonal runoff-to-rainfall ratios
Leeward	Wet	0.00320 [mean wet-season rainfall]
Leeward	Dry	0.00293 [mean dry-season rainfall]
Windward	Wet	[% HSG C]$^{0.185}$ [% HSG D]$^{0.247}$ [mean annual ETo]$^{(-0.590)}$
Windward	Dry	[% HSG C]$^{0.186}$ [% HSG D]$^{0.346}$ [mean dry-season ETo]$^{(-0.947)}$

estimated on the basis of mean dry-season reference ET and the percentage of the zone's area containing hydrologic soil groups C and D. The estimated mean seasonal runoff ratios computed using the regional-regression equations (table 10) were used to estimate runoff in ungaged catchments on Kauaʻi, Oʻahu, Molokaʻi, Lānaʻi, Maui, and the Island of Hawaiʻi during 1978–2007. For the Lānaʻi drought scenario and all future-climate scenarios, the adjusted seasonal runoff ratios computed with equation 8 were used to estimate runoff in ungaged catchment zones.

Runoff Ratios Assigned to Gaged Basins with a Single Catchment Zone

A total of 41 streamgages on Kauaʻi, Oʻahu, Molokaʻi, Maui, and the Island of Hawaiʻi have drainage basins with a single catchment zone (fig. 5). Runoff during 1978–2007 for each single-catchment-zone basin on Kauaʻi, Oʻahu, Maui, and the Island of Hawaiʻi was calculated using (1) observed seasonal runoff ratios when available, and (2) observed mean seasonal runoff ratios when observed seasonal runoff ratios were not available (Izuka and others, 2018). Rainfall is uncertain in northeastern Molokaʻi, owing to limited gage coverage and rainfall data, therefore runoff for each single-catchment-zone basin on Molokaʻi during 1978–2007 was estimated using an average of the observed mean seasonal runoff ratio and estimated mean seasonal runoff ratio (Oki and others, 2020). For the future-climate scenarios, runoff for each single-catchment-zone basin on Kauaʻi, Oʻahu, Molokaʻi, Maui, and the Island of Hawaiʻi was calculated using adjusted seasonal runoff ratios computed with equation 8.

Runoff Ratios Assigned to Gaged Basins with Multiple Catchment Zones

A total of 33 streamgages have drainage basins containing multiple catchment zones (fig. 5). Runoff for each of the catchment zones within these gaged basins was calculated using the same seasonal runoff ratios described for gaged basins with a single catchment and further spatially disaggregating runoff ratios of a gaged basin for each catchment zone within a gaged basin determined from equation 10 (Engott and others, 2017; Izuka and others, 2018; Johnson and others, 2018; Oki and others, 2020).

$$\widetilde{rr_{a,t}} = \widetilde{RR}_a \times \frac{rr_{b,t} \times rf_{b,t}}{\sum_{i=1}^{j} \widetilde{RR}_i \times rf_{i,t}} \qquad (10)$$

where

$\widetilde{rr_{a,t}}$ is the disaggregated runoff ratio for catchment zone a during season t [dimensionless],

$\widetilde{RR}_a$ is the mean runoff ratio for catchment zone a estimated using the regional-regression equations (table 10) [dimensionless],

j is the number of catchment zones in the gaged basin b [dimensionless],

$\widetilde{RR}_i$ is the mean runoff ratio for catchment zone i estimated using the regional-regression equations (table 10) [dimensionless],

$rf_{i,t}$ is the rainfall for catchment zone i during season t [L^3],

$rr_{b,t}$ is the runoff ratio for gaged basin b during season t [dimensionless], and

$rf_{b,t}$ is the rainfall for gaged basin b during season t [L^3].

Equation 10 uses the regional-regression models and seasonal rainfall to spatially disaggregate observed and mean seasonal runoff ratios of a gaged basin for each catchment zone within that gaged basin. This approach creates seasonal runoff ratios that are spatially variable across the catchment zones within gaged drainage basins containing multiple catchment zones.

Modifications to Runoff Ratios in Kawela and ʻUalapuʻe Aquifer Systems, Molokaʻi

At the higher altitudes of the Kawela and ʻUalapuʻe aquifer systems on Molokaʻi, an area consisting largely of native cloud forest (figs. 2*C,* 4*C*), the runoff ratios derived for ungaged basins using the regional-regression equations are lower than the ratios calculated

for the drainage basins of streamgages 16415000 and 16416000 in that area (figs. 5*E, F*). Assuming that the local streamgage data are more reliable indicators of runoff conditions for the higher altitude area than the regional-regression equations, native-forest land cover above the cloud zone was assigned the runoff ratios from (1) streamgage 16415000 in the Kawela aquifer system and (2) streamgage 16416000 in the ʻUalapuʻe aquifer system.

Evapotranspiration

Evapotranspiration (ET) is the sum of all water that is evaporated or transpired from a vegetated surface and plant-root zone. Evapotranspiration can be divided into three main evaporative processes: (1) canopy evaporation, or evaporation of intercepted rain and fog from the surface of vegetation; (2) ground evaporation, or evaporation of water from the soil surface and overlying litter and mulch layers; and (3) transpiration, the process by which soil moisture or groundwater taken up by vegetation is eventually evaporated through plant pores (Viessman and Lewis, 2003). All three ET processes are represented in the total-ET estimates of the water-budget model.

Canopy evaporation in forested areas can substantially reduce the rainfall that reaches the ground beneath a forest canopy (Gaskill, 2004; DeLay, 2005; McJannet and others, 2007; Giambelluca and others, 2011; Safeeq and Fares, 2014). Because of the height of trees and their canopy structure, turbulent diffusion is much more efficient at removing intercepted water from forest canopies than from shorter vegetation. Moreover, canopy evaporation in forests tends to operate on much shorter time scales (hours) than transpiration (weeks or longer; Savenije, 2004). Thus, accurate estimates of ET from forests generally require transpiration and canopy evaporation to be evaluated separately (Shuttleworth, 1993).

For this study, total ET from subareas of the forest land-cover type is computed by separately estimating forest-canopy evaporation and combined ground evaporation and transpiration from the plant-root zone. Evaporation from the forest canopy and ET from the plant-root zone are added together to yield total ET. For subareas of non-forest land-cover types, ET is computed using an approach in which canopy evaporation, ground evaporation, and transpiration are not separately estimated. The concept of potential ET, combined with empirical models when soil moisture is limited, is used to estimate ground evaporation and transpiration in forests and total ET for all other land-cover types. A similar approach to estimating total ET was used in other studies for Kauaʻi, Oʻahu, Molokaʻi, and Maui (Engott and others, 2017; Izuka and others, 2018; Johnson and others, 2018; Mair and others, 2019; Brewington and others, 2019; Oki and others, 2020).

Forest-Canopy Evaporation and Net Precipitation

As rain falls on a vegetated surface, a fraction of the droplets will accumulate on the leaves, trunks, or stems of the vegetation. Additional moisture from fog interception might supplement the amount of water that accumulates on vegetation. Canopy evaporation is the part of precipitation that accumulates on and then evaporates from the vegetation (Gerrits and Savenije, 2010). Net precipitation is the part of precipitation that reaches the forest floor.

WATRMod computes daily forest-canopy evaporation and net precipitation using one of three methods (Oki, 2022). In this study, the method derived from the Gash model (Gash and others, 1995) was used to calculate forest-canopy evaporation for the following reasons. First, the Gash model accounts for gaps in the forest canopy, allowing a sparse canopy to be differentiated from a dense canopy. Second, canopy evaporation during a period of precipitation is dependent on the amount of precipitation during that period. Third, the Gash model has the capacity to account for spatial differences in climate, including climate differences between windward and leeward forests. One disadvantage of the Gash model, however, is that it is theoretical. Therefore, one of the parameters of the Gash model used to calculate forest-canopy evaporation in the water budget was calibrated to the wet-canopy evaporation maps of Giambelluca and others (2014) as described below. Use of the Gash model, instead of the mean wet-canopy evaporation maps, in the water budget allows forest-canopy evaporation to be calculated on a daily basis in response to daily variations in precipitation.

The estimation of forest-canopy evaporation in the water budget requires values for rainfall and the following five parameters: (1) canopy cover, (2) canopy capacity, (3) trunk-storage capacity, (4) fraction of precipitation diverted to stemflow (precipitation that flows to the ground along tree trunks or plant stems), and (5) the ratio of the mean evaporation rate to mean precipitation rate during saturated conditions, V. The values assigned to the first four parameters in the water-budget calculations were derived from published data for areas in the Hawaiian Islands. The values assigned to V were derived using a method developed in a previous study (Johnson and others, 2018) that uses maps of mean wet-canopy evaporation for Kauaʻi, Oʻahu, and Maui (Giambelluca and others, 2014).

Canopy cover of forest land-cover types varies spatially across each island. The canopy cover of each forest land-cover type subarea in the water budget was estimated from a map of mean annual vegetation cover fraction (Giambelluca and others, 2014) and was assumed to be the same for each time period simulated. The map quantifies the vegetation cover fraction at a spatial resolution of about 14 acres. A canopy cover of 0 implies an absence of canopy cover, whereas a value of 1 implies a dense canopy with no gaps.

Canopy capacity, trunk-storage capacity, and the fraction of precipitation diverted to stemflow (stemflow rate) were set to 0.05 in, 0.01 in, and 0.04, respectively, for each subarea with a forest land-cover classification (table 7). These values were estimated by Johnson and others (2018) on the basis of published data for the Hawaiian Islands (Gaskill, 2004; DeLay, 2005; Takahashi and others, 2011; Safeeq and Fares, 2014). Forest sites of Takahashi and others (2011) and Safeeq and Fares (2014) dominated by *Psidium cattleianum* (strawberry guava) had relatively high stemflow rates ranging from 0.29 to 0.34. In order to have conservative stemflow estimates in the water budget, the high stemflow rates from these sites were excluded from the calculated mean stemflow rate of 0.04 used in the water budget.

The spatial distribution of V was estimated using the method described in Johnson and others (2018) and used in other studies (Izuka and others, 2018; Brewington and others, 2019; Mair and others, 2019; Oki and others, 2020). The method is as follows:

$$\begin{aligned} V &= 0.01 \\ &\text{for } w < 0.009, \\ V &= (2.677 \times (w)) - 0.014 \\ &\text{for } 0.009 \le w \le 0.192, \text{ and} \\ V &= 0.50 \\ &\text{for } w > 0.192 \end{aligned} \tag{11}$$

where

V is the ratio of the mean evaporation rate to mean precipitation rate during saturated conditions [dimensionless], and

w is the mean annual wind speed divided by mean annual rainfall [(meters/second)/inch].

The variable w is the quotient of (1) mean annual wind speed, in meters per second, derived from a map of mean annual wind speed at a height of about 100 ft above the land surface (AWS Truewind LLC, 2004), and (2) 1978–2007 mean annual rainfall, in inches (Giambelluca and others, 2013). Values of V ranged from 0.01 to 0.50, with wetter areas typically having lower values than drier areas.

The spatial distribution of V estimated with equation 11 was used for the 1978–2007 scenario and all future-climate scenarios because of a lack of information to estimate potential changes in V for future-climate scenarios. For example, projected changes in wind speed are needed to estimate potential changes in V; however, this information is lacking for the SD RCP8.5 2041–71 and SD RCP8.5 2071–99 scenarios. The climate projection datasets for the HRCM1 and HRCM2 scenarios include wind speed information that can be used to estimate potential changes in the spatial distribution of V. An analysis of the effects of potential changes in the spatial distribution of V on water-budget results for selected HRCM1 and HRCM2 scenarios is included in the "Effects of Selected Climate Inputs on Estimated Recharge" section of this report. Estimated changes in the spatial distribution of V have a small but non-negligible effect on recharge estimates.

Potential Evapotranspiration

The potential-ET rate is the maximum rate that water can be removed from the plant-root zone by ET if soil moisture is non-limiting (Giambelluca, 1983). The actual plant-root zone rate of ET, referred to as the actual-ET rate, is computed in the water budget as a function of potential ET, soil-moisture content, and threshold soil-moisture content (see equation 14 in Oki [2022]). The actual-ET rate becomes less than the potential-ET rate with the onset of soil-moisture stress. As the soil dries, capillary and adsorptive forces bind the remaining water to the soil matrix, reducing water available to roots. Soil-moisture stress occurs when the reduced availability of water to the root system induces a response in the plant to slow down transpiration and prevent desiccation. The threshold-moisture content at which a plant begins to react to soil drying varies with the type of plant species.

The potential-ET rate is needed to compute the actual-ET rate in WATRMod. WATRMod computes the potential-ET rate for each subarea as the product of its crop coefficient and reference-ET rate. Crop coefficients are used to represent the integrated effects of land-cover and vegetation characteristics on the potential-ET rate. The reference-ET rate is the ET rate of a hypothetical grass surface of uniform height that is completely shading the ground and has optimum soil-water conditions for given climatic conditions (Allen and others, 1998). The mean annual reference-ET rate ranges from about 26 to 114 inches across the Hawaiian Islands (Giambelluca and others, 2014; map of grass reference surface potential ET available at http://evapotranspiration.geography.hawaii.edu/interactivemap.html). In the water-budget calculation, the monthly reference-ET rate was not varied from year to year, and was assumed to equal the mean monthly reference-ET rate. Reference ET was assumed to be the same each day of a given month.

Crop coefficients were assigned to each subarea on the basis of land-cover type (table 7). The crop coefficients used for this study's scenarios were the same as those used by Johnson and others (2023), except for kiawe as described in the next paragraph. For non-forest land-cover types, crop coefficients integrated the effects of transpiration, ground evaporation, and canopy evaporation. For forest land-cover types, crop coefficients integrated the effects of transpiration and ground evaporation only because canopy evaporation was accounted for separately.

The crop coefficient and other model input parameters for kiawe were derived for this study using the results from recent studies of transpiration and groundwater use by one species of kiawe (Dudley and others, 2014, 2020; Miyazawa and others, 2016; see appendix 1). The crop coefficients for corn vary monthly and are based on information in the Hawai‘i Agricultural Water Use and Development Plan (University of Hawai‘i, 2008). The crop coefficients used for corn were 0.85, 0.50, 0.29, 0.40, 0.80, 1.20, 0.85, 0.50, 0.29, 0.40, 0.80, and 1.20 for the months of January through December, in respective monthly order. The values for seed corn were the same as those used in recent water-budget analyses for Kaua‘i, O‘ahu, Maui, and Moloka‘i (Engott and others, 2017; Izuka and others, 2018; Mair and others, 2019; Oki and others, 2020).

Monthly reference ET for each model subarea was estimated for the 1978–2007 scenario using maps of mean monthly grass reference surface ET (Giambelluca and others, 2014). The monthly reference ET for each model subarea for the 1978–2007 scenario was also used for each of the future-climate scenarios because of a lack of information needed to estimate potential changes in reference ET. For example, the climate projection datasets for the SD RCP8.5 2041–71 and SD RCP8.5 2071–99 scenarios describe the spatial distribution of potential changes to seasonal rainfall and mean annual surface temperature (Elison Timm and others,

2015; Elison Timm, 2017). However, additional consistently and independently derived meteorological data, such as hourly or daily air temperature, humidity, wind speed, and net radiation are required to estimate changes in reference-ET rates using the Food and Agriculture Organization of the United Nations Penman-Monteith (FAO-PM) equation of Allen and others (2006). Other approaches for estimating reference-ET rates using limited meteorological data are available but many of these methods do not perform adequately across a wide range of locations and climates (Allen and others, 1998). For example, the Blaney-Criddle method estimates reference ET using only inputs of mean temperature. However, the Blaney-Criddle method underestimates reference-ET rates by as much as 60 percent in windy, dry, and sunny areas and overestimates reference-ET rates by as much as 40 percent in calm, humid, and cloudy areas (Brouwer and Heibloem, 1986). Therefore, Allen and others (1998) recommend that missing climate data be estimated, whenever feasible, to enable the use of the FAO-PM equation for estimating reference-ET rates. Estimating missing climate data for the SD RCP8.5 2041–71 and SD RCP8.5 2071–99 scenarios to estimate reference-ET rates using the FAO-PM equation is beyond the scope of this study. The climate projection datasets for the HRCM1 and HRCM2 scenarios do include gridded meteorological information needed to estimate potential changes in reference-ET rates using the FAO-PM equation of Allen and others (2006). An analysis of the effects of potential changes in reference-ET rates on WATRMod results for selected HRCM1 and HRCM2 scenarios is included in the "Effects of Selected Climate Inputs on Estimated Recharge" section of this report. Estimated changes in reference-ET rates have a moderate to large effect on recharge estimates.

Total Evapotranspiration

In this study, total ET in each subarea represents the sum of transpiration, ground evaporation, and canopy evaporation. However, the approach used to compute the amount attributable to each ET process varies with land-cover type. For subareas of the non-forest land-cover type, WATRMod estimates of plant-root zone ET were used to represent all three ET processes and quantify total ET. For subareas of the forest land-cover type, WATRMod estimates of plant-root zone ET were used to represent transpiration and ground evaporation, whereas estimates of ET losses from the Gash model were used to represent canopy evaporation. Total ET for subareas of the forest land-cover type, including tree plantation, was computed as the sum of plant-root zone ET and canopy evaporation.

Plant-root zone ET for each subarea was computed by WATRMod on the basis of potential ET, moisture storage in the plant-root zone, and threshold-moisture storage, as described in Oki (2022). When moisture storage was greater than or equal to the threshold-moisture storage, the plant-root-zone-ET rate was assumed to equal the potential-ET rate. When moisture storage was less than the threshold-moisture storage, the plant-root-zone-ET rate was less than the potential-ET rate and declined linearly with soil-moisture content. Threshold-moisture storage was computed on the basis of the depletion fraction and moisture-storage capacity of the plant-root zone (see equation 25 in Oki [2022]). The depletion fraction, which ranges from 0 to 1, is defined as the fraction of moisture-storage capacity that can be depleted from the plant-root zone before moisture stress causes a reduction in the actual-ET rate (Johnson and others, 2018). Values for depletion fraction assigned to each land-cover type (table 7) were the same as those used by Mair and others (2019), except for bare soil, kiawe, and taro. The values for bare soil, kiawe, and taro were the same as those used by Johnson and others (2023).

Moisture-Storage Capacity of the Plant-Root Zone

The moisture-storage capacity of the plant-root zone for each subarea was computed by WATRMod as the product of its assigned root depth and the available water capacity of its soil unit (see equation 12 in Johnson and others [2018]). The root depth of each subarea was set to the root depth assigned to its land-cover type (table 7). The available water capacities of soils were assumed to be time-invariant and were estimated from the Natural Resources Conservation Service (2019) soil map and corresponding tables of available water capacities. For each soil unit, the water capacity tables listed the minimum and maximum available water capacities for various ranges of depth. A depth-weighted mean of available water capacity was computed for each soil type in WATRMod. All depths of the "rock outcrop" soil unit, however, had zero available water capacity according to Natural Resources Conservation Service (2019). For this study, zero available water capacity was considered too low for this soil unit because many areas with this soil unit were mapped as grassland or shrubland. Therefore, in the WATRMod calculations, a value of 0.01 was substituted for all subareas with the "rock outcrop" soil unit.

Direct Recharge

Direct recharge in WATRMod represents water seepage that bypassed the plant-root zone and was therefore not subject to the direct runoff or ET process. Direct recharge in this study was estimated for subareas with cesspool seepage and for subareas with reservoir and water body land-cover types (fig. 4). Cesspool seepage estimation methods are explained in the "Septic Effluent" section. Specified rates of constant recharge were assigned as daily values to subareas of reservoir and water body land-cover types in the water-budget calculations (table 11).

Subareas with coastal water body (water bodies along the perimeter of the island in the tidal zone) and reservoir, lined land-cover types were assigned a constant recharge rate of 0 in/yr. For this study, subareas with inland water body land cover were assigned a constant recharge rate of 12 in/yr on the basis of previous water-budget studies (Engott and Vana, 2007; Izuka and others, 2018; Johnson and others, 2018; Mair and others, 2019). Unlined reservoirs on Kauaʻi were assigned a constant recharge rate of 56 in/yr, the average of annual seepage rages reported for two unlined reservoirs by Sustainable Resources Group Intn'l, Inc. (2009). For Molokaʻi, unlined reservoirs were assigned a constant recharge rate of 528 in/yr on the basis of previous studies for Maui due to the proximity of Molokaʻi to Maui and similarities in

Table 11. Constant direct recharge rates used in the water-budget calculations for reservoir and water-body land covers.

[Constant recharge was allocated as a daily value in the water-budget model simulations]

Land Cover	Constant recharge rates, in inches per year					
	Kauaʻi	Oʻahu	Molokaʻi	Lānaʻi	Maui	Island of Hawaiʻi
Reservoir, unlined	56[a]	90[b]	528[c]	528[c]	528[c], 1,268[d]	528[c]
Reservoir, lined	0	0	0[e], 75[f]	0	0	0
Inland water body	12[g]	12[g]	12[g]	12[g]	12[g]	12[g]
Coastal water body	0	0	0	0	0	0

[a]Value computed as the average of annual seepage rates reported for two unlined reservoirs on Kauaʻi by Sustainable Resources Group Intn'l, Inc. (2009).

[b]Value computed as the estimated percolation to groundwater from Wahiawa Reservoir divided by the total water surface area from Dugan and others (1975).

[c]Value used for unlined reservoirs in all aquifer systems on Maui except Pāʻia, Kahului, Waikapū, and ʻĪao, and used by Izuka and others (2018), Johnson and others (2018), and Mair and others (2019).

[d]Value used for unlined reservoirs in the Pāʻia, Kahului, Waikapū, and ʻĪao aquifer systems and used by Izuka and others (2018), Johnson and others (2018), and Mair and others (2019).

[e]Value used for all lined reservoirs on Molokaʻi except Kualapuʻu reservoir.

[f]Value used for Kualupuʻu reservoir and used by Oki and others (2020).

[g]Value used for water body land cover on Kauaʻi, Molokaʻi, and Maui in previous water budget studies by Engott and Vana (2007), Izuka and others (2018), and Oki and others (2020).

volcanic age (Johnson and others, 2018; Mair and others, 2019). The Kualapuʻu reservoir on Molokaʻi is a lined reservoir that was assigned a constant recharge rate of 75 in/yr due to leakage (Oki and others, 2020). For Oʻahu, unlined reservoirs were assigned a constant recharge rate of 90 in/yr, which was computed from the estimated percolation to groundwater from Wahiawa reservoir as reported by Dugan and others (1975). For Maui, unlined reservoirs within the Pāʻia, Kahului, Waikapū, and ʻĪao aquifer systems were assigned a constant recharge rate of 1,268 in/yr (Izuka and others, 2018), whereas unlined reservoirs in the other 21 aquifer systems were assigned a value of 528 in/yr on the basis of previous studies (Johnson and others, 2018; Mair and others, 2019). Unlined reservoirs on the Island of Hawaiʻi were assigned a constant recharge rate of 528 in/yr on the basis of previous studies conducted on nearby islands (Izuka and others, 2018; Johnson and others, 2018; Mair and others, 2019).

Initial Conditions

In addition to water-budget inputs, other parameter inputs were required. The initial moisture storage for the pervious fraction of subareas was set at 50 percent of the soil moisture-storage capacity. The initial moisture storage for the impervious fraction of subareas was set at 0.125 in, equivalent to 50 percent of the rainfall-retention capacity of impervious surfaces. These values were also used for other recent water budgets for Kauaʻi, Oʻahu, Molokaʻi, and Maui (Izuka and others, 2018; Oki and others, 2020), and in each of the climate scenarios. The effects of these inputs on regional-scale mean annual recharge were generally minor because they either pertained to only a small area or were only applicable during a small amount of time.

Iteration for WATRMod

To mitigate any arbitrary effects of the initial soil-moisture conditions and random selection of monthly rainfall fragment sets (see "Initial Conditions" and "Rainfall" sections), for each island WATRMod was run for 10 simulations for each scenario and the results were averaged. To evaluate differences in recharge estimates between simulations, the average absolute percentage change in cumulative mean recharge for each island was computed for each successive simulation of the 1978–2007 scenario (fig. 7). A series of cumulative mean recharge values for each island was computed by averaging the values from each successive and all preceding simulations. The series of average absolute percentage change values for each island was computed by comparing the cumulative mean recharge from each successive simulation with the value from the preceding simulation. The average percentage change for each island generally decreased with each successive simulation and was less than 0.002 percent for all islands after nine simulations. This small value, 0.002 percent, was also used in other recent water-budget studies (Izuka and others, 2018; Mair and others, 2019) as the threshold for determining the minimum number of simulations needed.

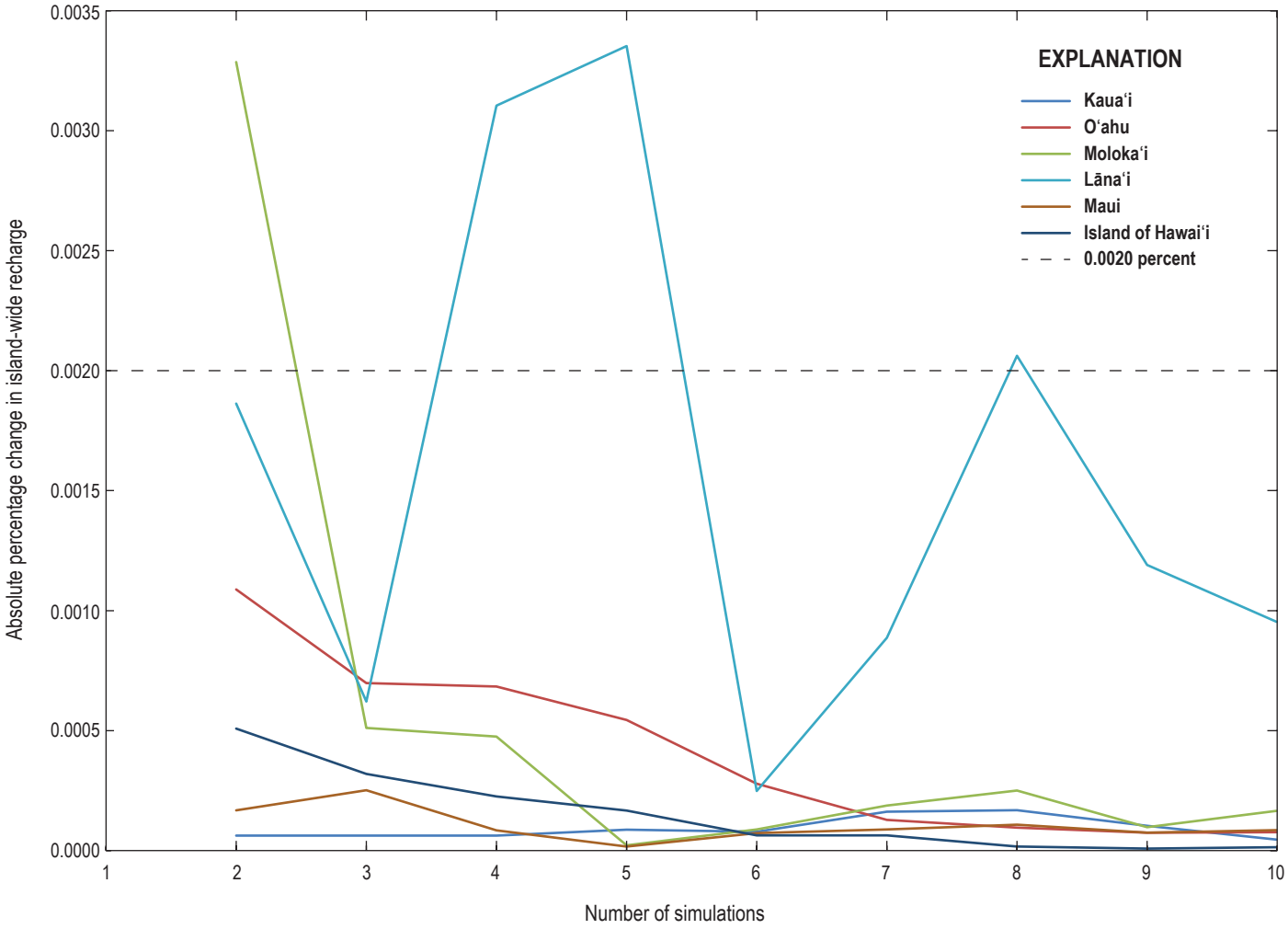


Figure 7. Graph of the absolute percentage change in island-wide recharge with each successive WATRMod simulation for Kaua‘i, O‘ahu, Moloka‘i, Lāna‘i, Maui, and the Island of Hawai‘i. The dashed line at 0.0020 percent change represents the threshold for determining the minimum number of simulations needed to mitigate the possible effects of arbitrary initial-moisture conditions and random selection of monthly rainfall fragments.

Water-Budget and Groundwater-Recharge Estimates

WATRMod was used to compute groundwater recharge and other water-budget components for a set of future-climate scenarios for Kaua‘i, O‘ahu, Moloka‘i, Lāna‘i, Maui, and the Island of Hawai‘i. A drought scenario was added for Lāna‘i to assess the effect of episodic dry periods on groundwater recharge and other water-budget components. Water-budget estimates for each future-climate and the drought scenario were compared with estimates from the 1978–2007 scenario to quantify potential climate-change effects on groundwater recharge for each island. All scenarios used 2020 land-cover conditions in the water-budget simulations.

Kaua‘i

Four water-budget scenarios were selected for Kaua‘i: (1) recent conditions represented by the 1978–2007 scenario, (2) a mid-century climate scenario using projected rainfall conditions for the SD RCP8.5 2041–71 scenario, (3) a dry-climate scenario using projected rainfall conditions for the SD RCP8.5 2071–99 scenario, and (4) a wet-climate scenario using projected rainfall conditions for the HRCM2 RCP4.5 2080–99 scenario (fig. 8; table 4). The water-budget components estimated for

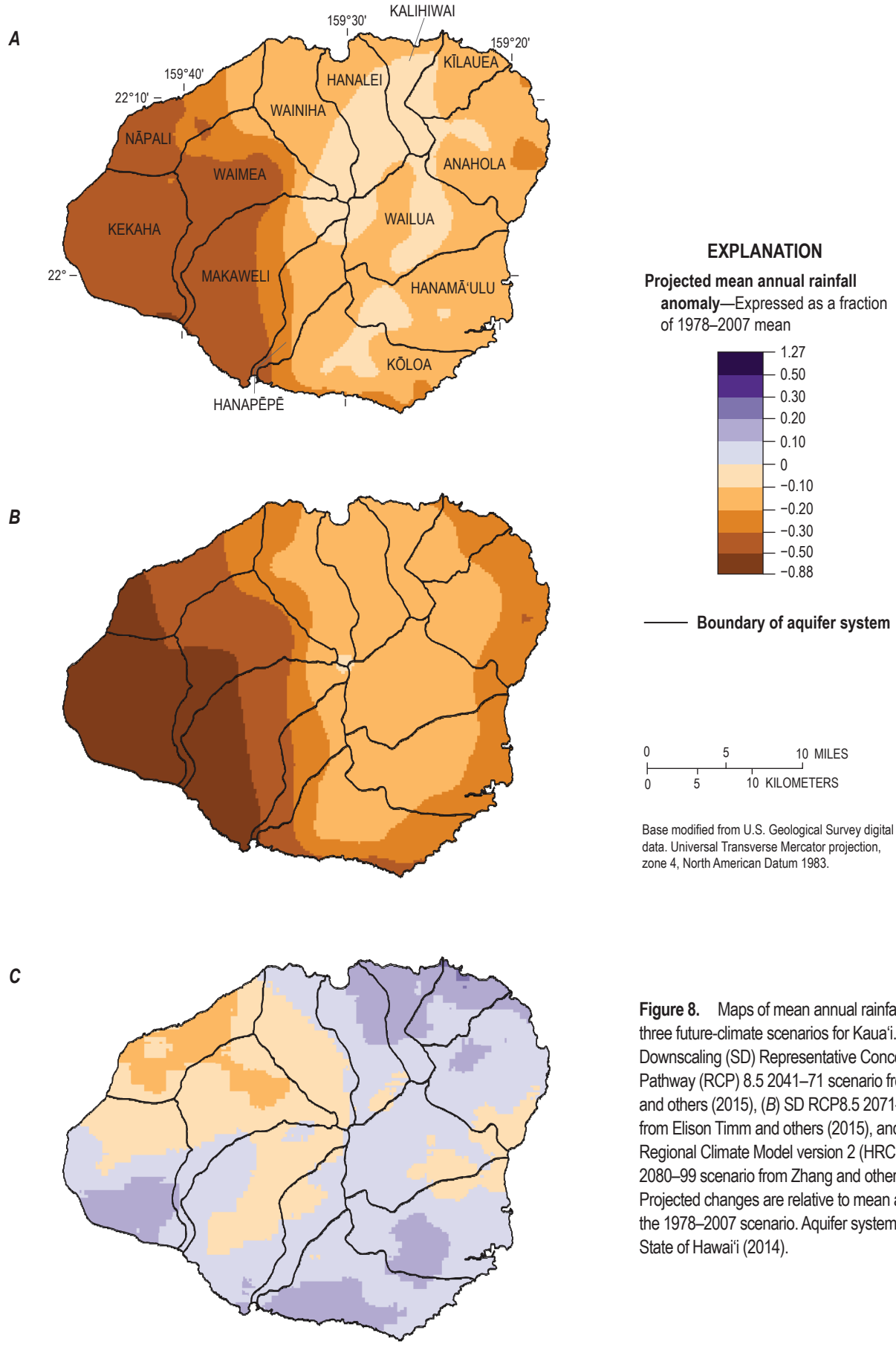


Figure 8. Maps of mean annual rainfall anomalies for three future-climate scenarios for Kauaʻi. (*A*) Statistical Downscaling (SD) Representative Concentration Pathway (RCP) 8.5 2041–71 scenario from Elison Timm and others (2015), (*B*) SD RCP8.5 2071–99 scenario from Elison Timm and others (2015), and (*C*) Hawai'i Regional Climate Model version 2 (HRCM2) RCP4.5 2080–99 scenario from Zhang and others (2017). Projected changes are relative to mean annual rainfall for the 1978–2007 scenario. Aquifer system boundaries from State of Hawai'i (2014).

the 1978–2007 scenario and three future-climate scenarios for Kauaʻi are shown in table 2-1 (appendix 2) and summarized by subarea in Kāne and others (2024).

1978–2007 Scenario

Mean annual recharge for Kauaʻi is approximately 807 Mgal/d for the 1978–2007 scenario (appendix 2; table 2-1). Mean annual recharge is about 30.6 inches, expressed as a depth of water uniformly distributed over the island. For Kauaʻi's 13 aquifer systems (fig. 2*A*), mean annual recharge ranges from about 15 to 138 Mgal/d for the 1978–2007 scenario (appendix 2; table 2-1). In terms of recharge expressed as an average depth of water, the aquifer systems with the greatest recharge are Wainiha, Hanapēpē, and Hanalei, located in the central and northern parts of Kauaʻi. The aquifer systems with the least recharge are Kīlauea, Anahola, and Kekaha, located in the leeward and northeastern parts of the island. The spatial pattern of mean annual recharge on Kauaʻi for the 1978–2007 scenario (fig. 9*A*) resembles the pattern of mean annual rainfall during the same period (fig. 3).

The island-wide mean annual recharge for Kauaʻi is less than the recharge calculated by Izuka and others (2018) by roughly 69 Mgal/d (7.8 percent) for the same time period (1978–2007) mainly due to increases in canopy evaporation and decreases in constant recharge (appendix 2; table 2-1). Island-wide canopy evaporation increased by about 36 Mgal/d when compared to the canopy evaporation estimated by Izuka and others (2018) and indicates a change in the daily rainfall pattern to more frequent, low-magnitude daily rainfall. The change in daily rainfall pattern is likely related to a difference in the rainfall fragment sets used to disaggregate monthly rainfall during 1978–2007 into daily rainfall for the water-budget simulations. Izuka and others (2018) utilized historical rain-gage data from a nearby gage to create rainfall fragment sets for every month in which complete daily rainfall measurements were available. The record of historical rain gage data might extend back as far as the early 20th century at some gage locations. However, in this study, the gridded daily rainfall dataset (250-m resolution) for the period 1990–2014 by Longman and others (2019) was used to disaggregate monthly rainfall for each subarea. Hence, the rainfall fragment sets used in this study represent daily rainfall patterns across a uniform time period (1990–2014) and higher spatial resolution (250 m), and indicate a change to more frequent, low-magnitude daily rainfall when compared to Izuka and others (2018). However, the effect of these different rainfall fragment datasets on recharge was not evaluated as part of this study. The indicated changes in daily rainfall frequency are consistent with identified changes in rainfall extremes across Hawaiʻi (Chu and others, 2010; Elison Timm and others, 2011). Chu and others (2010) analyzed changes in rainfall intensity, frequency, and magnitude from the 1950s to 2007 using daily observational records at selected rain gage locations on Kauaʻi, Oʻahu, Maui, and the Island of Hawaiʻi. Chu and others (2010) reported a shift to more frequent, low-magnitude rainfall events and to less frequent moderate and heavy rainfall events at many rain gage locations on all four islands including four rain gages on Kauaʻi. Elison Timm and others (2011) analyzed changes in the frequency of heavy rainfall events from 1958 to 2005 using daily observational records from 12 rain gage locations across Kauaʻi, Oʻahu, Lānaʻi, Maui, and the Island of Hawaiʻi. Elison Timm and others (2011) reported a reduction in heavy rainfall events at 9 rain gage locations.

The water-budget simulations by Izuka and others (2018) used 2010 land-cover conditions, whereas this study utilized 2020 land-cover conditions that included detailed information on lined and unlined reservoirs. For example, in this study, subareas with reservoir land cover were distinguished as lined or unlined, and subareas with lined reservoirs were assumed to have no seepage (except in certain cases). Furthermore, subareas with unlined reservoirs were assigned a constant recharge rate of 56 in/yr on the basis of seepage rate estimates at two unlined reservoirs on Kauaʻi, whereas all subareas with reservoir land cover in Izuka and others (2018) were assigned a constant recharge rate of 528 in/yr on the basis of information from Maui. As a result of these differences in constant recharge rates assigned to reservoir land cover, island-wide estimates of total direct recharge decreased by about 42 Mgal/d when compared to Izuka and others (2018).

SD RCP8.5 2041–71 Scenario

The projected mean annual rainfall anomalies for SD RCP8.5 2041–71 scenario indicates drying across all Kauaʻi when compared to the 1978–2007 scenario (fig. 8*A*). The greatest drying occurs across western Kauaʻi including parts of the Nāpali, Waimea, Kekaha, and Makaweli aquifer systems where mean annual rainfall is projected to decrease by greater than 30 percent for a SD RCP8.5 2041–71 scenario. Island-wide mean annual rainfall is projected to decrease by 15 percent (table 2).

Mean annual recharge is estimated to decrease across most of Kauaʻi for the SD RCP8.5 2041–71 scenario (fig. 9*B*). The island-wide mean annual recharge for the SD RCP8.5 2041–71 scenario is less than recharge for the 1978–2007 scenario by 167 Mgal/d (about 21 percent) (appendix 2; table 2-1). The greatest estimated decrease occurs in the upland areas of Kōloa, Makaweli, Waimea, and Wainiha aquifer systems, where decreases in recharge range from 15 to 30 inches (fig. 9*B*). Small regions in the upland areas of Makaweli, Waimea, Wainiha and Wailua aquifer systems show slight increases in recharge (0.1 to 15 inches). In these wet areas, WATRMod estimates proportionally greater decrease in runoff than projected rainfall, and estimates of total ET remaining nearly the same. Hence, the amount of water available for recharge increases relative to the 1978–2007 scenario, which results in slight increases in recharge estimates.

Decreases in aquifer-system recharge are estimated for all Kauaʻi's aquifer systems (fig. 10*A*). In terms of absolute difference in mean annual recharge, the greatest decreases in aquifer-system recharge are estimated for the Kekaha,

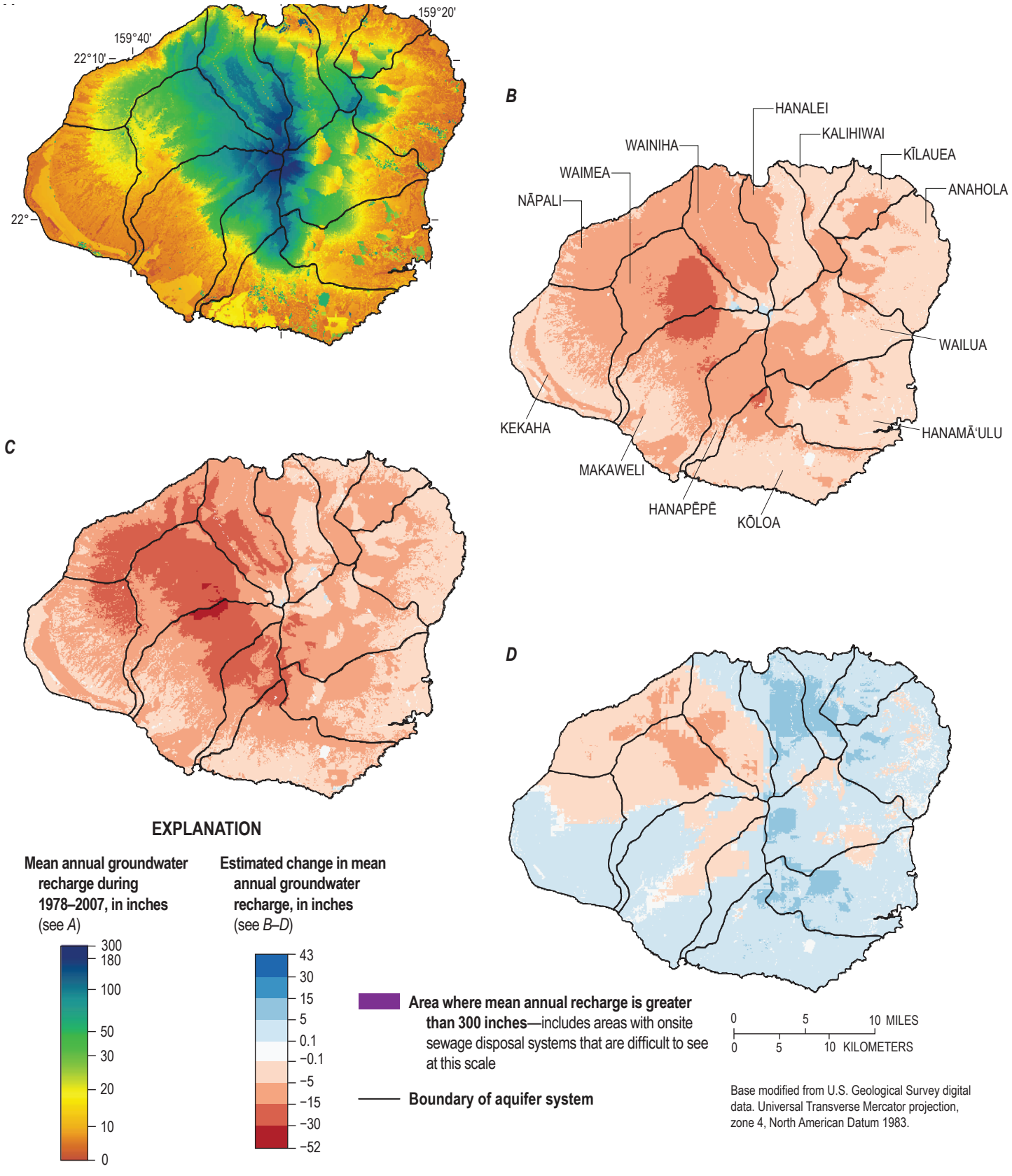


Figure 9. Maps showing the distribution of mean annual groundwater recharge for the 1978–2007 scenario and 2020 land-cover conditions, and estimated change in mean annual groundwater recharge for three future-climate scenarios for Kauaʻi. (*A*) Mean annual recharge for the 1978–2007 scenario. Estimated mean annual recharge for (*B*) Statistical Downscaling (SD) Representative Concentration Pathway (RCP) 8.5 2041–71 scenario from Elison Timm and others (2015), (*C*) SD RCP8.5 2071–99 scenario from Elison Timm and others (2015), and (*D*) Hawaiʻi Regional Climate Model version 2 (HRCM2) RCP4.5 2080–99 scenario from Zhang and others (2017). Estimated changes are relative to mean annual recharge for the 1978–2007 scenario. Aquifer system boundaries from State of Hawaiʻi (2014).

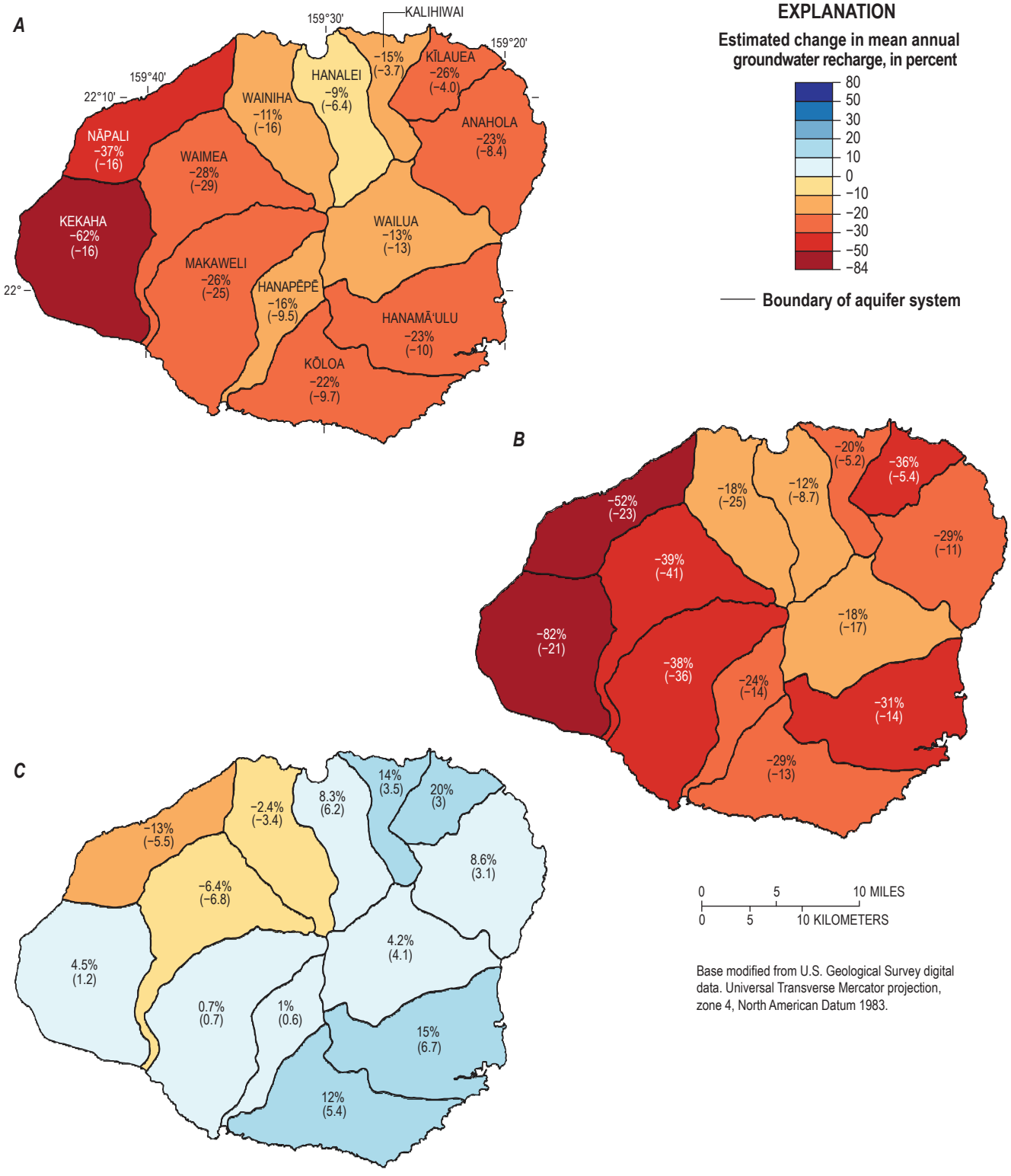


Figure 10. Maps showing estimated change in mean annual groundwater recharge by aquifer system for three future-climate scenarios for Kauaʻi. (*A*) Statistical Downscaling (SD) Representative Concentration Pathway (RCP) 8.5 2041–71 scenario from Elison Timm and others (2015), (*B*) SD RCP8.5 2071–99 scenario from Elison Timm and others (2015), and (*C*) Hawaiʻi Regional Climate Model version 2 (HRCM2) RCP4.5 2080–99 scenario from Zhang and others (2017). Estimated changes are relative to mean annual recharge for the 1978–2007 scenario. Aquifer system boundaries from State of Hawaiʻi (2014).

Makaweli, Nāpali, Waimea, and Wainiha aquifer systems, where decreases range from 16 to 29 Mgal/d less than the corresponding recharge for the 1978–2007 scenario. In terms of percentage change from the 1978–2007 scenario, the greatest decreases in aquifer-system recharge are estimated for the Kekaha, Nāpali, Waimea, Makaweli, and Kīlauea aquifer systems, where decreases range from 26 to 62 percent.

SD RCP8.5 2071–99 Scenario

The estimated mean annual rainfall anomalies for the SD RCP8.5 2071–99 scenario indicates drying across all Kauaʻi compared to the 1978–2007 scenario (fig. 8*B*). The greatest drying occurs across western Kauaʻi including parts of the Nāpali, Waimea, Kekaha, and Makaweli aquifer systems where mean annual rainfall is projected to decrease by greater than 50 percent for a SD RCP8.5 2071–99 scenario.

Mean annual recharge is estimated to decrease across most of Kauaʻi for the SD RCP8.5 2071–99 scenario (fig. 9*C*). The island-wide mean annual recharge for the SD RCP8.5 2071–99 scenario is less than recharge for the 1978–2007 scenario by 234 Mgal/d (29 percent; appendix 2, table 2-1). The greatest estimated decrease occurs in the upland areas of the Makaweli and Waimea aquifer systems, where decreases in recharge exceed 30 inches (fig. 9*C*). Small increases in recharge are estimated in the Wainiha, Hanalei, and Wailua aquifer systems. In these wet areas, WATRMod estimates proportionally greater decrease in runoff than projected rainfall and total ET to remain nearly the same. Hence, the amount of water available for recharge increases relative to the 1978–2007 scenario, resulting in slight recharge increases estimated by WATRMod.

Decreases in aquifer-system recharge are estimated for all Kauaʻi's aquifer systems (fig. 10*B*). In terms of absolute difference in mean annual recharge, the greatest decreases in aquifer-system recharge are estimated for the Kekaha, Makaweli, Nāpali, Waimea, and Wainiha aquifer systems, where decreases range from 21 to 41 Mgal/d less than the corresponding recharge for the 1978–2007 scenario. In terms of percentage change from the 1978–2007 scenario, the greatest decreases in aquifer-system recharge are estimated for the Kekaha, Makaweli, Nāpali, and Waimea aquifer systems, where decreases range from 38 to 82 percent (fig. 10*B*).

HRCM2 RCP4.5 2080–99 Scenario

The projected mean annual rainfall anomalies for HRCM2 RCP4.5 2080–99 scenario show a pattern of drying and wetting across Kauaʻi (fig. 8*C*). The greatest drying occurs in the Nāpali and Waimea aquifer systems where mean annual rainfall is projected to decrease by greater than 10 percent for a HRCM2 RCP4.5 2080–99 scenario. The greatest wetting occurs along coastal areas of the Kalihiwai and Kīlauea aquifer systems where mean annual rainfall is projected to increase by greater than 20 percent. Other areas of the Anahola, Hanalei, Hanamāʻulu, Kalihiwai, Kekaha, Kīlauea, Kōloa, Makaweli, and Waimea aquifer systems show projected increases in mean annual rainfall of over 10 percent. This scenario indicates a slight wetting trend for the entire island (table 2, fig. 8*C*).

The estimated changes in mean annual recharge include a wide range of increases and decreases across large areas of Kauaʻi for the HRCM2 RCP4.5 2080–99 scenario (fig. 9*D*). Island-wide mean annual recharge is greater than recharge for the 1978–2007 scenario by roughly 19 Mgal/d (2.3 percent; appendix 2, table 2-1). Mean annual recharge is estimated to increase in the northeastern, southeastern, and southwestern parts of the island, where increases of 5 to 15 inches are estimated. Mean annual recharge is estimated to decrease across much of northwestern Kauaʻi, and parts of eastern and southwestern Kauaʻi. The greatest estimated decrease occurs in the upland areas of northwestern Kauaʻi where decreases of 5 to 15 inches are estimated.

Increases in aquifer-system recharge are estimated for 10 of Kauaʻi's 13 aquifer systems (fig. 10*C*). In terms of absolute change, the greatest increases in aquifer-system recharge occur in the Hanalei, Hanamāʻulu, and Kōloa aquifer systems, where increases range from 5.4 to 6.7 Mgal/d. In terms of percentage change from the 1978–2007 scenario, the greatest increases in aquifer-system recharge occur in the Hanamāʻulu, Kalihiwai, Kīlauea, and Kōloa aquifer systems, where increases range from 12 to 20 percent. Decreases in aquifer-system recharge occur in the Nāpali, Waimea, and Wainiha aquifer systems, where decreases range from 3.4 to 6.8 Mgal/d (or about 2 to 13 percent; fig. 10*C*).

Oʻahu

Four water-budget scenarios were selected for Oʻahu: (1) recent conditions represented by the 1978–2007 scenario, (2) a mid-century climate scenario using projected rainfall conditions for the SD RCP8.5 2041–71 scenario, (3) a dry-climate scenario using projected rainfall conditions for the SD RCP8.5 2071–99 scenario, and (4) a wet-climate scenario using projected rainfall conditions for the HRCM2 RCP8.5 2080–99 scenario (fig. 11, table 4). The water-budget components estimated for the 1978–2007 scenario and three future-climate scenarios for Oʻahu are shown in table 2-2 (appendix 2) and summarized by subarea in Kāne and others (2024).

1978–2007 Scenario

Mean annual recharge for Oʻahu is approximately 597 Mgal/d for the 1978–2007 scenario (appendix 2, table 2-2). Mean annual recharge is about 21 inches, expressed as a depth of water uniformly distributed over the island. For Oʻahu's 23 aquifer systems (fig. 2*B*), mean annual recharge ranges from about 1.0 to 119 Mgal/d for the 1978–2007 scenario. In terms of recharge expressed as an average depth of water, the aquifer systems with the greatest recharge are Kahana, Koʻolauloa, and Wahiawa, whereas the aquifer systems with the least recharge are Lualualei, Makaīwa, and ʻEwa-Kunia. The spatial pattern of mean annual recharge on Oʻahu for the 1978–2007 scenario (fig. 12*A*) resembles the pattern of mean annual rainfall during the same period (fig. 3).

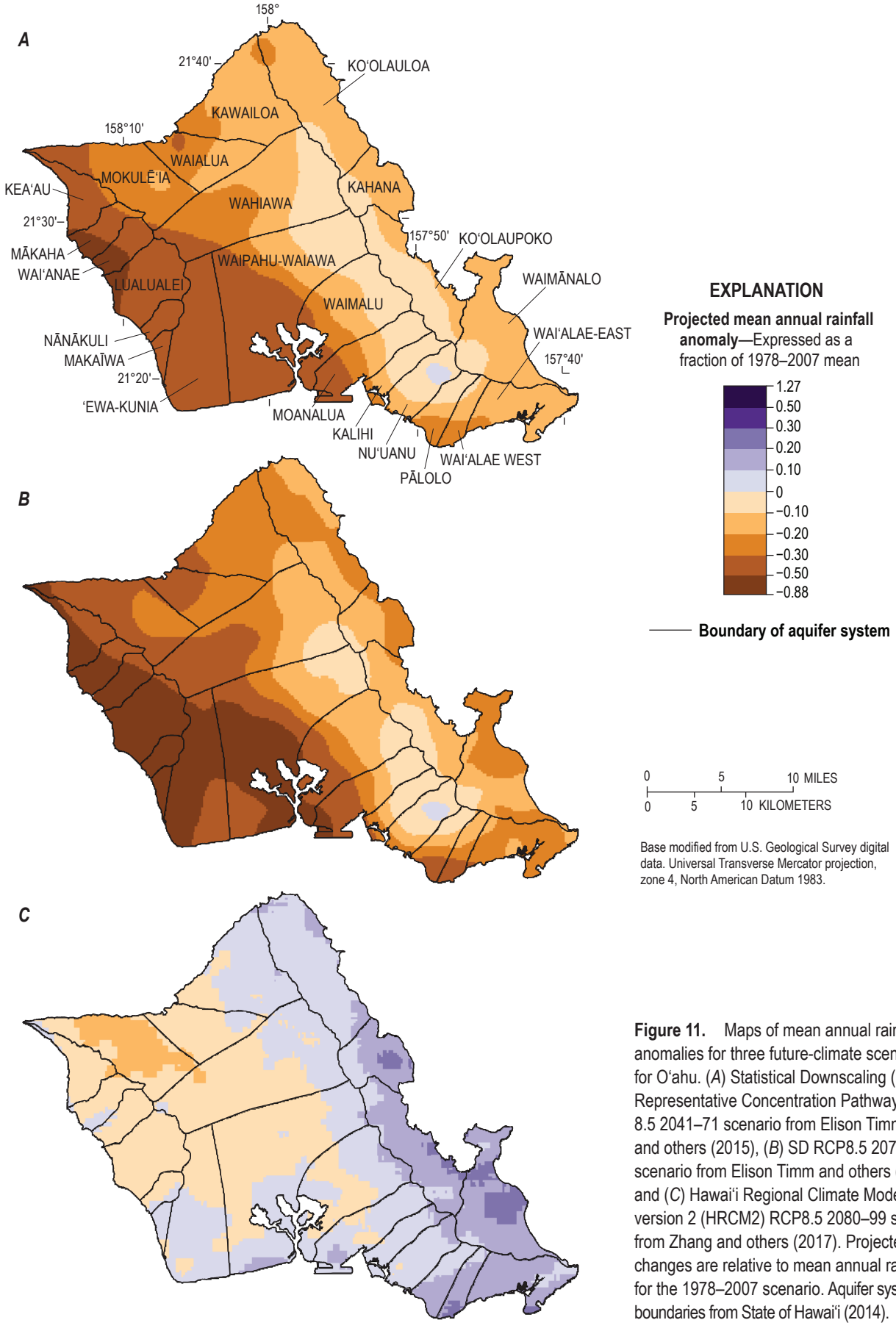


Figure 11. Maps of mean annual rainfall anomalies for three future-climate scenarios for O'ahu. (*A*) Statistical Downscaling (SD) Representative Concentration Pathway (RCP) 8.5 2041–71 scenario from Elison Timm and others (2015), (*B*) SD RCP8.5 2071–99 scenario from Elison Timm and others (2015), and (*C*) Hawai'i Regional Climate Model version 2 (HRCM2) RCP8.5 2080–99 scenario from Zhang and others (2017). Projected changes are relative to mean annual rainfall for the 1978–2007 scenario. Aquifer system boundaries from State of Hawai'i (2014).

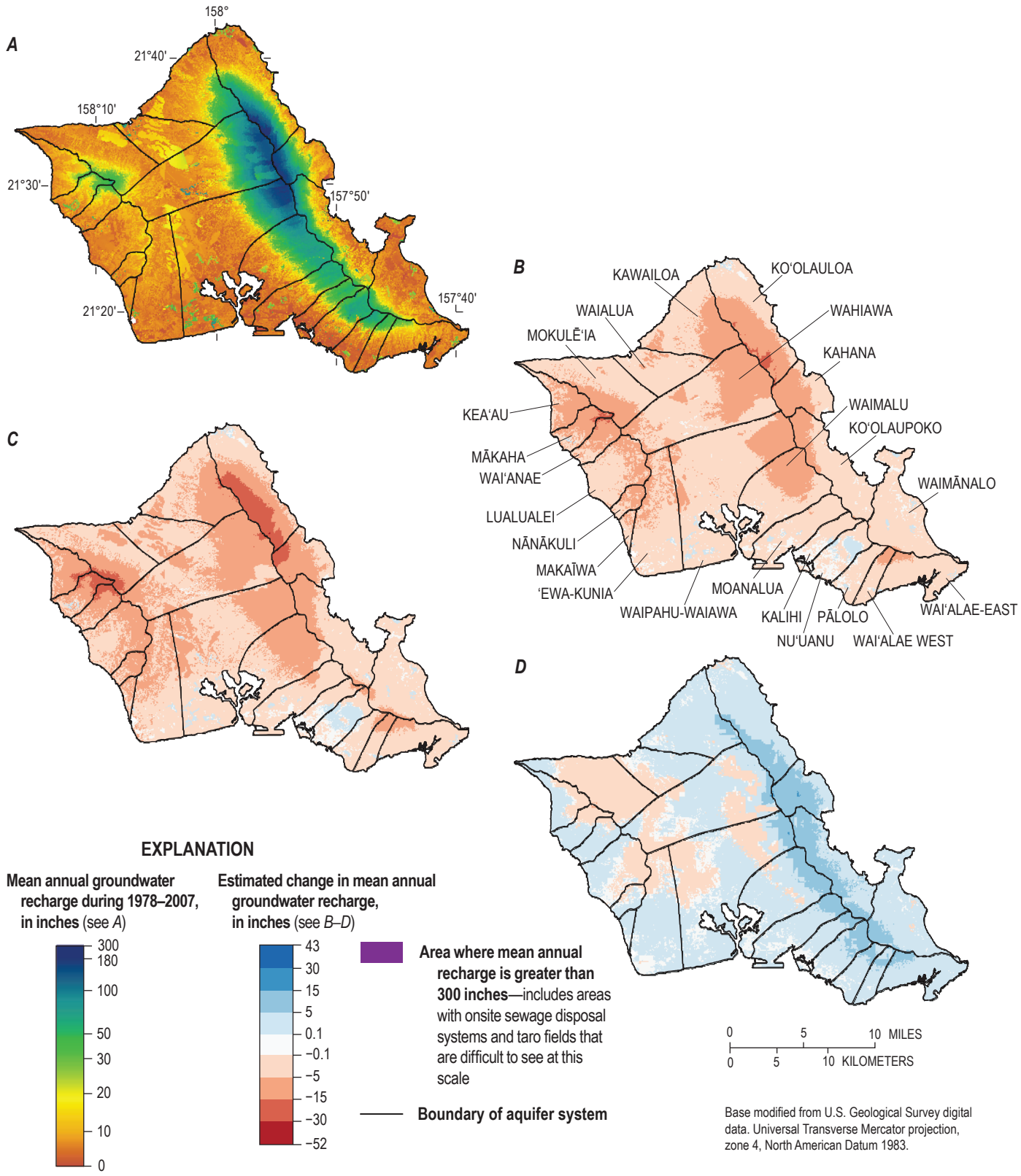


Figure 12. Maps showing the distribution of mean annual groundwater recharge for the 1978–2007 scenario and 2020 land-cover conditions, and estimated change in mean annual groundwater recharge for three future-climate scenarios for O'ahu. (*A*) Mean annual recharge for the 1978–2007 scenario. Estimated change in mean annual recharge for (*B*) Statistical Downscaling (SD) Representative Concentration Pathway (RCP) 8.5 2041–71 scenario from Elison Timm and others (2015), (*C*) SD RCP8.5 2071–99 scenario from Elison Timm and others (2015), and (*D*) Hawai'i Regional Climate Model version 2 (HRCM2) RCP8.5 2080–99 scenario from Zhang and others (2017). Estimated changes are relative to mean annual recharge for the 1978–2007 scenario. Aquifer system boundaries from State of Hawai'i (2014).

The island-wide mean annual recharge for O‘ahu is less than the recharge calculated by Engott and others (2017) by roughly 63 Mgal/d (9.6 percent) mainly due to increases in canopy evaporation and decreases in constant recharge and irrigation (appendix 2, table 2-2). Island-wide canopy evaporation increased by about 45 Mgal/d when compared to the canopy evaporation estimated by Engott and others (2017) and indicates a change in the daily rainfall pattern to more frequent, low-magnitude daily rainfall. The change in daily rainfall pattern is likely related to a difference in the rainfall fragment sets used to disaggregate monthly rainfall during 1978–2007 into daily rainfall for the water-budget simulations. Engott and others (2017) utilized historical rain gage data from a nearby gage to create rainfall fragment sets for every month in which complete daily rainfall measurements were available. The record of historical rain gage data might extend back as far as the early 20th century at some gage locations. However, in this study, the gridded daily rainfall dataset (250-m resolution) for the period 1990–2014 by Longman and others (2019) was used to disaggregate monthly rainfall for each subarea. Hence, the rainfall fragment sets used in this study represent daily rainfall patterns across a uniform time period (1990–2014) and higher spatial resolution (250 m), and indicate a change to more frequent, low-magnitude daily rainfall when compared to the daily rainfall pattern of the rainfall fragment sets used by Engott and others (2017). However, the effect of these different rainfall fragment sets on recharge estimates was not evaluated as part of this study. The indicated changes in daily rainfall frequency are consistent with identified changes in rainfall extremes across O‘ahu and other parts of Hawai‘i, where a shift to more frequent, low-magnitude rainfall events and to less frequent moderate and heavy rainfall events has been observed at numerous locations from the 1950s to the 2000s (Chu and others, 2010; Elison Timm and others, 2011).

The water-budget simulations by Engott and others (2017) used 2010 land-cover conditions, whereas this study utilized 2020 land-cover conditions that include an additional distinction for subareas with reservoir land cover. In this study, reservoirs were distinguished as either a lined or unlined reservoir, and subareas with lined reservoir land cover were assumed to have no seepage. Furthermore, subareas with inland water body and unlined reservoir land cover types were assigned constant recharge rates of 12 and 90 in/yr, respectively, on the basis of seepage rate estimates at the Wahiawa reservoir on O‘ahu and other published studies (Dugan and others, 1975; Engott and Vana, 2007; Izuka and others, 2018; Johnson and others, 2018; Mair and others, 2019). Engott and others (2017) assigned constant recharge rates of 64 and 528 in/yr to subareas with water body and reservoir land cover, respectively. As a result of these differences in constant recharge rates assigned to water body and reservoir land cover, island-wide estimates of total direct recharge decreased by about 11 Mgal/d when compared to Engott and others (2017).

Island-wide estimates of irrigation decreased by about 12 Mgal/d when compared to the irrigation estimated by Engott and others (2017). The decrease in irrigation rates might be related to changes in the amount of irrigated land cover when compared to 2010 land-cover conditions. Irrigation rates were estimated in WATRMod for subareas that were classified as agriculture (pineapple, diversified agriculture, and taro) or other land-cover types (golf course; developed, high intensity; and developed, medium intensity) that were assumed to contain irrigated lawns and landscapes. The fraction of O‘ahu’s land area classified as agriculture increased from 0.04 for 2010 land-cover conditions to 0.06 for 2020 land-cover conditions, which indicates an increase in irrigated area for agricultural land cover when compared to Engott and others (2017). However, the fraction of land area with developed and golf course land cover decreased from 0.29 for 2010 land-cover conditions to 0.27 for 2020 land-cover conditions, which indicates a decrease in irrigated area for developed land cover when compared to Engott and others (2017). Also, use of rainfall fragment sets that are characterized by a daily rainfall pattern of more frequent, low-magnitude daily rainfall, as described previously, might decrease daily irrigation needs calculated by WATRMod and decrease island-wide estimates of irrigation when compared to Engott and others (2017). The effects of changes in land cover and rainfall fragment sets on irrigation estimates were not evaluated as part of this study.

SD RCP8.5 2041–71 Scenario

Mean annual rainfall anomalies for the SD RCP8.5 2041–71 scenario indicate drying across most of O‘ahu, except for a small upland area in the Nu‘uanu aquifer system (fig. 11*A*). The greatest drying occurs along the leeward coast of the island in the Mākaha, Wai‘anae, and Lualualei aquifer systems where mean annual rainfall is projected to decrease by greater than 50 percent.

For this mid-century climate scenario, island-wide recharge for O‘ahu is about 486 Mgal/d, an estimated decrease of 111 Mgal/d (19 percent) from the 1978–2007 scenario. The greatest estimated decrease occurs in the upper regions of the Ko‘olauloa, Mākaha, Mokulē‘ia, and Wai‘anae aquifer systems (fig. 12*B*). Estimates of recharge increase slightly (0.1 to 5 inches) in isolated areas around the island due to projected increases in rainfall (Nu‘uanu aquifer system only) and increases in irrigation for golf courses.

Decreases in aquifer-system recharge are estimated for all O‘ahu’s aquifer systems (fig. 13*A*). In terms of absolute difference in mean annual recharge, the greatest decreases in aquifer-system recharge are estimated for the Waipahu-Waiawa, Ko‘olauloa, and Wahiawa aquifer systems, where decreases range from 12 to 16 Mgal/d less relative to recharge for the 1978–2007 scenario. In terms of percentage change from the 1978–2007 scenario, the greatest decreases in aquifer-system recharge are estimated for Makaīwa, Lualualei, and Nānākuli aquifer systems, where decreases range from 64 to 66 percent.

SD RCP8.5 2071–99 Scenario

Mean annual rainfall anomalies for the SD RCP8.5 2071–99 scenario indicate drying across most of O‘ahu, except for a small upland area in the Nu‘uanu aquifer system (fig. 11*B*). The greatest drying occurs along the leeward coast and extends from the western part of Mokulē‘ia aquifer system to the southwestern

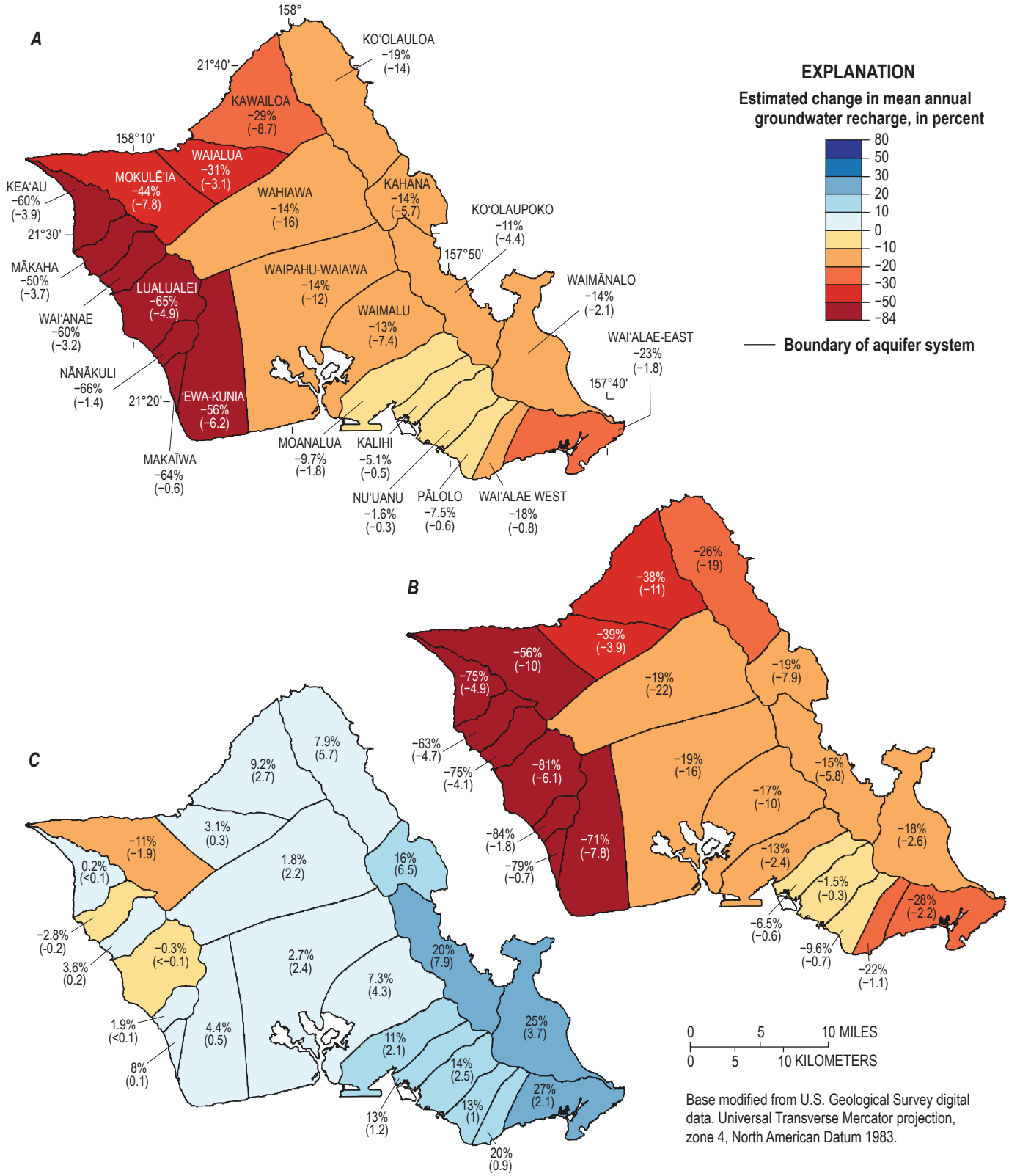


Figure 13. Maps showing estimated change in mean annual groundwater recharge by aquifer system for three future-climate scenarios for Oʻahu. (*A*) Statistical Downscaling (SD) Representative Concentration Pathway (RCP) 8.5 2041–71 scenario from Elison Timm and others (2015), (*B*) SD RCP8.5 2071–99 scenario from Elison Timm and others (2015), and (*C*) Hawaiʻi Regional Climate Model version 2 (HRCM2) RCP8.5 2080–99 scenario from Zhang and others (2017). Estimated changes are relative to mean annual recharge for the 1978–2007 scenario. Values in parentheses represent change in millions of gallons per day. Aquifer system boundaries from State of Hawaiʻi (2014).

part of Moanalua aquifer system, where mean annual rainfall is projected to decrease by greater than 50 percent.

For this dry-climate scenario, recharge for O‘ahu is about 451 Mgal/d, which is an estimated decrease of 146 Mgal/d (24 percent) from the 1978–2007 scenario. The greatest estimated decreases in recharge occur in the upland areas of Kahana, Ko‘olauloa, Kea‘au, Mākaha, Mokulē‘ia, Wahiawa, and Wai‘anae aquifer systems (fig. 12*C*). Estimates of recharge increase slightly (0.1 to 5 inches) in isolated areas around the island due to projected increases in rainfall (Nu‘uanu aquifer system only) and increases in irrigation for golf courses.

Decreases in aquifer-system recharge are estimated for all O‘ahu’s aquifer systems (fig. 13*B*). In terms of absolute difference in mean annual recharge, the greatest decreases in aquifer-system recharge are estimated for the Waipahu-Waiawa, Ko‘olauloa, and Wahiawa aquifer systems, where decreases range from 16 to 22 Mgal/d relative to recharge for the 1978–2007 scenario. In terms of percentage change from the 1978–2007 scenario, the greatest decreases in aquifer-system recharge are estimated for the leeward coastline in the Kea‘au, Mākaha, Wai‘anae, Lualualei, Nānākuli, Makaīwa, and ‘Ewa-Kunia aquifer systems, where decreases range from 63 to 84 percent (fig. 13*B*).

HRCM2 RCP8.5 2080–99 Scenario

The estimated mean annual rainfall anomalies for HRCM2 RCP8.5 2080–99 scenario show a pattern of drying and wetting across O‘ahu (fig. 11*C*). The greatest drying occurs across parts of northwestern O‘ahu where mean annual rainfall is projected to decrease by greater than 10 percent for a HRCM2 RCP8.5 2080–99 scenario. The greatest wetting occurs in small, isolated areas of eastern O‘ahu where mean annual rainfall increases by greater than 10 percent. Island-wide mean annual rainfall is projected to increase by 4 percent (table 2).

The estimated changes in mean annual recharge include a wide range of increases and decreases across large areas of O‘ahu for the HRCM2 RCP8.5 2080–99 scenario (fig. 12*D*). Island-wide mean annual recharge is greater than recharge for the 1978–2007 scenario by roughly 44 Mgal/d (7.4 percent; appendix 2, table 2-2). Mean annual recharge is estimated to increase across much of O‘ahu with the greatest estimated increase in recharge occurring in the upland regions of eastern O‘ahu, where increases range from 5 to 15 inches. A decrease in recharge is estimated for parts of central and western O‘ahu where recharge decreases by as much as 5 inches.

Increases in aquifer-system recharge are estimated for all but three of O‘ahu’s aquifer systems (fig. 13*C*). In terms of absolute change, the greatest increases in aquifer-system recharge occur in the Ko‘olauloa, Kahana, and Ko‘olaupoko aquifer systems, where increases range from 5.7 to 7.9 Mgal/d. In terms of percentage change from the 1978–2007 scenario, the greatest increases of aquifer-system recharge occur in the Ko‘olaupoko, Waimānalo, and Wai‘alae-East aquifer systems aquifer systems, where increases range from 20 to 27 percent. The greatest decrease in aquifer-system recharge occurs in the Mokulē‘ia aquifer system, where the decrease is 1.9 Mgal/d (11 percent).

Moloka‘i

Four water-budget scenarios were selected for Moloka‘i: (1) recent conditions represented by the 1978–2007 scenario, (2) a mid-century climate scenario using projected rainfall conditions for the SD RCP8.5 2041–71 scenario, (3) a dry-climate scenario using projected rainfall conditions for the SD RCP8.5 2071–99 scenario, and (4) a wet-climate scenario using projected rainfall conditions for the HRCM2 RCP8.5 2080–99 scenario (fig. 14, table 4). The water-budget components estimated for the 1978–2007 reference climate scenario and three future-climate scenarios for Moloka‘i are shown in table 2-3 (appendix 2) and summarized by subarea in Kāne and others (2024).

1978–2007 Scenario

Mean annual recharge for Moloka‘i is approximately 228 Mgal/d for the 1978–2007 scenario (appendix 2, table 2-3). Mean annual recharge is about 18.4 inches (fig. 15*A*), expressed as a depth of water uniformly distributed over the island. For Moloka‘i’s 16 aquifer systems (fig. 2*C*), mean annual recharge ranges from about 1.3 to 43 Mgal/d for the 1978–2007 scenario (appendix 2, table 2-3). In terms of recharge expressed as an average depth of water, the aquifer systems with the greatest recharge are Waikolu, Pelekunu, and Wailau aquifer systems located in the windward area of eastern Moloka‘i. The aquifers with the least recharge are the Ho‘olehua, Punakou, and Kaluako‘i aquifer systems in the drier, western part of the island. The spatial pattern of mean annual recharge on Moloka‘i for the 1978–2007 scenario (fig. 15*A*) resembles the pattern of mean annual rainfall during the same period (fig. 3).

The island-wide mean annual groundwater recharge for Moloka‘i is greater than island-wide recharge calculated by Oki and others (2020) by about 1.4 Mgal/d. The slight increase in island-wide recharge is mainly due to increases in irrigation, total direct recharge, and water available for recharge owing to decreased runoff when compared to Oki and others (2020; appendix 2, table 2-2). However, island-wide total ET (canopy evaporation plus actual ET) increased when compared to Oki and others (2020), which decreased water available for recharge in the WATRMod calculations and partly offset the increases in recharge due to greater irrigation and total direct recharge, and lower runoff.

SD RCP8.5 2041–71 Scenario

The projected mean annual rainfall anomalies for SD RCP8.5 2041–71 scenario indicates drying across the entire island when compared to the 1978–2007 scenario (fig. 14*A*). The greatest drying occurs across western Moloka‘i including parts of the Kaluako‘i and Punakou aquifer systems where mean annual rainfall is projected to decrease by greater than 30 percent.

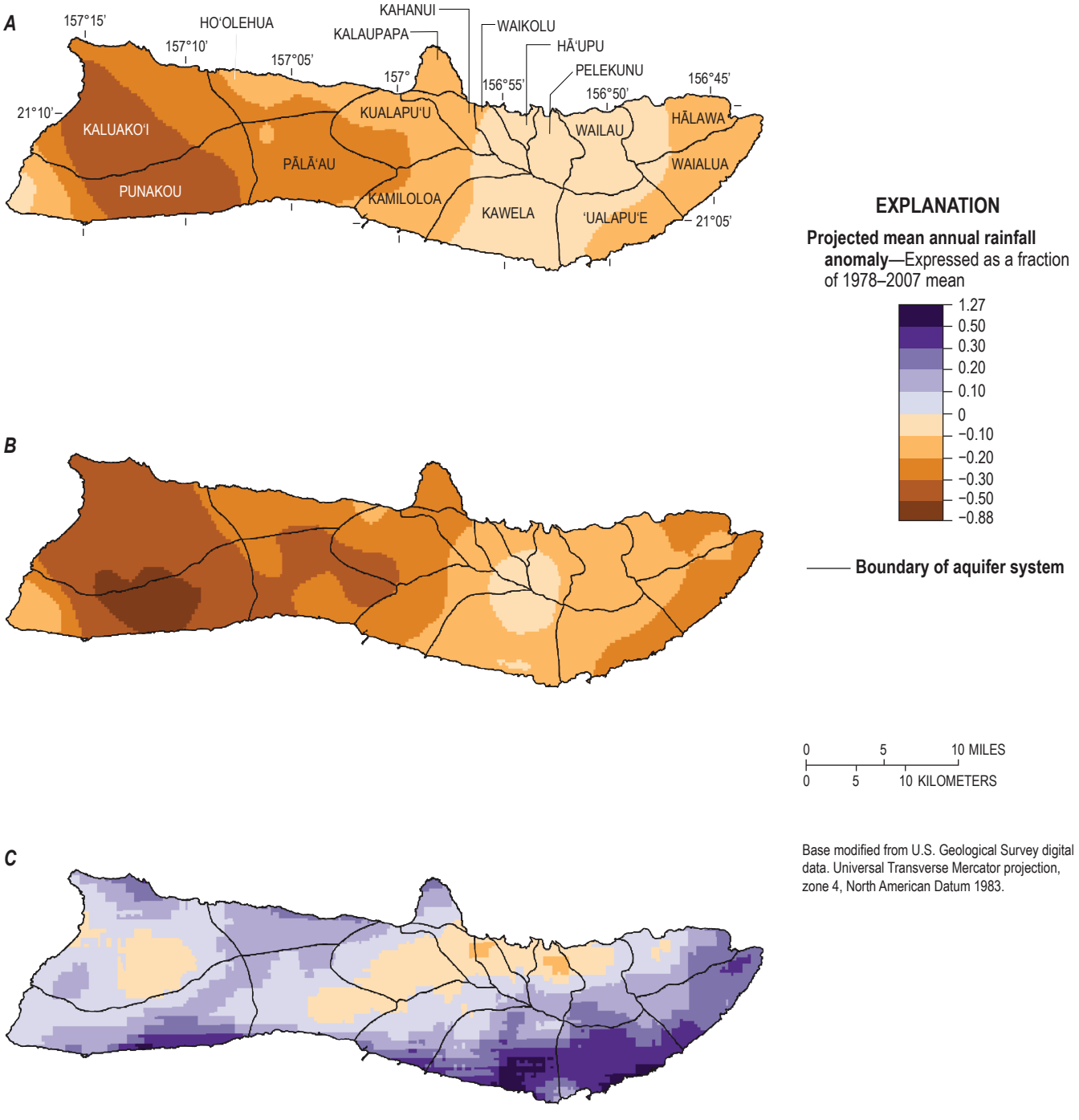


Figure 14. Maps of mean annual rainfall anomalies for three future-climate scenarios for Moloka'i. (*A*) Statistical Downscaling (SD) Representative Concentration Pathway (RCP) 2041–71 scenario from Elison Timm and others (2015), (*B*) SD RCP8.5 2071–99 scenario from Elison Timm and others (2015), and (*C*) Hawai'i Regional Climate Model version 2 (HRCM2) RCP8.5 2080–99 scenario from Zhang and others (2017). Projected changes are relative to mean annual rainfall for the 1978–2007 scenario. Aquifer system boundaries from State of Hawai'i (2014).

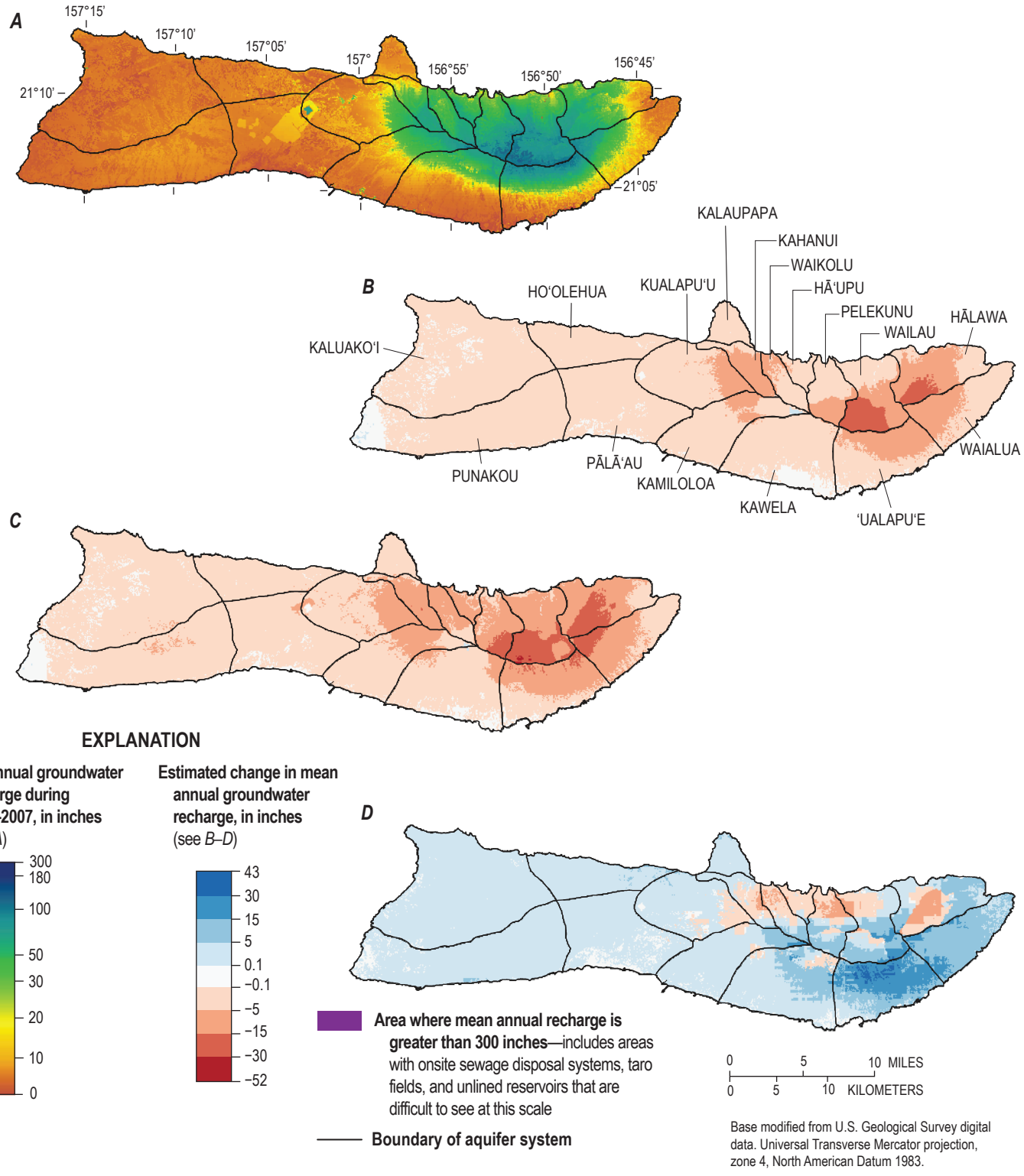


Figure 15. Maps showing the distribution of mean annual groundwater recharge for the 1978–2007 scenario and 2020 land-cover conditions, and estimated change in mean annual groundwater recharge for three future-climate scenarios for Moloka'i. (*A*) Mean annual recharge for the 1978–2007 scenario, (*B*) Statistical Downscaling (SD) Representative Concentration Pathway (RCP) 8.5 2041–71 scenario from Elison Timm and others (2015), (*C*) SD RCP8.5 2071–99 scenario from Elison Timm and others (2015), and (*D*) Hawai'i Regional Climate Model version 2 (HRCM2) RCP8.5 2080–99 scenario from Zhang and others (2017). Estimated changes are relative to mean annual recharge for the 1978–2007 scenario. Aquifer system boundaries from State of Hawai'i (2014).

For this mid-century climate scenario, recharge for Molokaʻi is about 190 Mgal/d, which is an estimated decrease of 38 Mgal/d (17 percent) from the 1978–2007 scenario. The greatest estimated decrease occurs across many upland areas of eastern Molokaʻi. Small regions in the upland area of the Kamiloloa aquifer system and the coastal area of the Punakou aquifer system show slight increases in recharge (0.1 to 5 inches; fig. 15*B*). In both areas, WATRMod estimates proportionally greater decreases in runoff than projected decreases in rainfall and total ET. Hence, the amount of water available for recharge increases relative to the 1978–2007 scenario, which results in slight increases in recharge estimated by WATRMod.

Decreases in aquifer-system recharge are estimated for all of Molokaʻi's aquifer systems (fig. 16*A*). In terms of absolute difference in mean annual recharge, the greatest decreases in aquifer-system recharge are estimated for the ʻUalapuʻe, Hālawa, and Wailau aquifer systems, where decreases range from 3.6 to 6.2 Mgal/d relative to recharge for the 1978–2007 scenario. In terms of percentage change from the 1978–2007 scenario, the greatest decreases in aquifer-system recharge are estimated for Hoʻolehua, Punakou, and Kaluakoʻi aquifer systems, where decreases range from 37 to 51 percent.

SD RCP8.5 2071–99 Scenario

The projected mean annual rainfall anomalies for SD RCP8.5 2071–99 scenario indicate drying across the entire island when compared to the 1978–2007 scenario (fig. 14*B*). The greatest drying occurs across western Molokaʻi including parts of the Kaluakoʻi and Punakou aquifer systems where mean annual rainfall is projected to decrease by greater than 50 percent.

For this dry-climate scenario, recharge for Molokaʻi is about 172 Mgal/d. This is an estimated decrease of 56 Mgal/d (about 24 percent) from the 1978–2007 scenario. The greatest estimated decrease occurs in the upland areas of eastern Molokaʻi (fig. 15*C*). Small regions in the upland areas of the Kamiloloa and Kawela aquifer systems, and the coastal area of the Punakou aquifer system show slight increases in recharge (0.1 to 5 inches). In these areas, WATRMod estimates proportionally greater decreases in runoff than projected decreases in rainfall and total ET. Hence, the amount of water available for recharge increases relative to the 1978–2007 scenario, resulting in slight increases in recharge estimated by WATRMod.

Decreases in aquifer-system recharge are estimated for all Molokaʻi's aquifer systems (fig. 16*B*). In terms of absolute difference in mean annual recharge, the greatest decreases in aquifer-system recharge are estimated for the ʻUalapuʻe, Hālawa, and Wailau aquifer systems, where decreases range from 5.9 to 9.8 Mgal/d relative to recharge for the 1978–2007 scenario. In terms of percentage change from the 1978–2007 scenario, the greatest decreases in aquifer-system recharge are estimated for the leeward region in the Pālāʻau, Hoʻolehua, Punakou, and Kaluakoʻi aquifer systems, where decreases range from 42 to 64 percent.

HRCM2 RCP8.5 2080–99 Scenario

The projected mean annual rainfall anomalies for the HRCM2 RCP8.5 2080–99 scenario show a pattern of drying and wetting across Molokaʻi (fig. 14*C*). The greatest wetting occurs in parts of southeastern Molokaʻi where mean annual rainfall is projected to increase by greater than 50 percent. Parts of central and northeastern Molokaʻi show decreases in mean annual rainfall of up to 10 percent or more. Island-wide mean annual rainfall is projected to increase by 12 percent (table 2).

The estimated changes in mean annual recharge include a wide range of increases and decreases across large areas of Molokaʻi for the HRCM2 RCP8.5 2080–99 scenario (fig. 15*D*). Island-wide mean annual recharge is greater than recharge for the 1978–2007 scenario by roughly 40 Mgal/d (17 percent; appendix 2, table 2-3). Mean annual recharge is estimated to increase across much of western and southeastern Molokaʻi with the greatest increases occurring in the upland areas of eastern Molokaʻi where recharge is estimated to increase by greater than 30 inches.

Increases in aquifer-system recharge are estimated for all but two of Molokaʻi's aquifer systems (fig. 16*C*). In terms of absolute change, the greatest increases in aquifer-system recharge occur in the ʻUalapuʻe and Waialua aquifer systems, where increases range from 7.2 to 13 Mgal/d. In terms of percentage change from the 1978–2007 scenario, the greatest increases in aquifer-system recharge from the 1978–2007 scenario occur in the Punakou and Hoʻolehua aquifer systems, where increases range from 54 to 62 percent. Aquifer-system recharge decreases in the Hāʻupu and Kahanui aquifer systems by 0.2 Mgal/d or less (4.4 percent or less).

Lānaʻi

Five water-budget scenarios were selected for Lānaʻi: (1) recent conditions represented by the 1978–2007 scenario, (2) a mid-century climate scenario using projected rainfall conditions for the SD RCP8.5 2041–71 scenario, (3) a dry-climate scenario using projected rainfall conditions for the SD RCP8.5 2071–99 scenario, (4) a wet-climate scenario using projected rainfall conditions for the HRCM2 RCP4.5 2080–99 scenario, and (5) a drought scenario represented by dry conditions during 2008–12 (fig. 17, table 4). The water-budget components estimated for all five scenarios for Lānaʻi are shown in table 2-4 (appendix 2) and summarized by subarea in Kāne and others (2024).

1978–2007 Scenario

Mean annual recharge for Lānaʻi is approximately 27 Mgal/d for the 1978–2007 scenario (appendix 2, table 2-4). Expressed as a depth of water uniformly distributed over the island, mean annual recharge is about 4.0 inches (fig. 18). For Lānaʻi's 9 aquifer systems (fig. 2*D*), mean annual recharge ranges from about 1.9 to about 4.4 Mgal/d for the 1978–2007 scenario (appendix 2, table 2-4). In terms of recharge expressed as an average depth of water, the aquifer systems with the greatest recharge are Mānele,

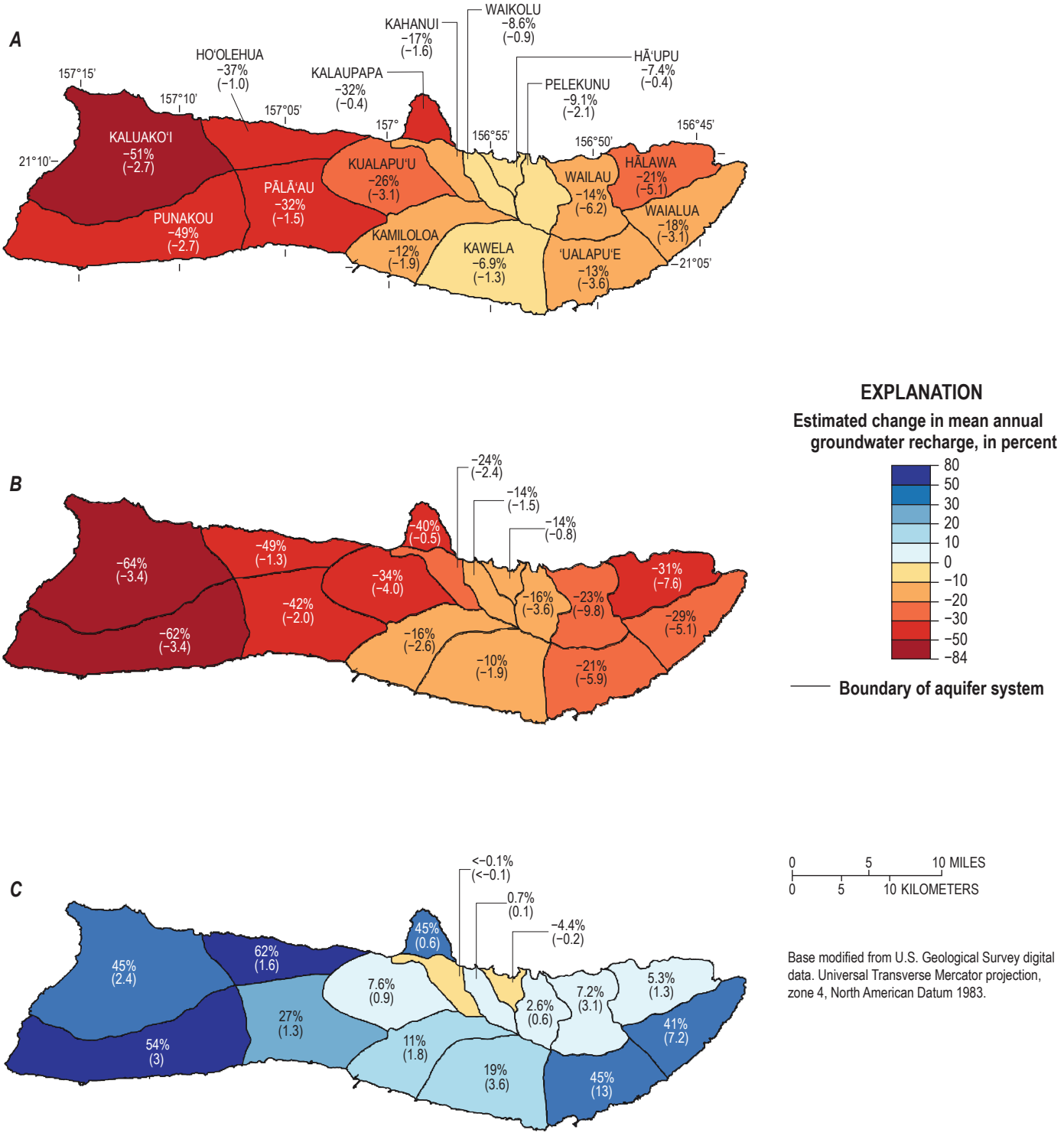


Figure 16. Maps showing estimated change in mean annual groundwater recharge by aquifer system for three future-climate scenarios for Moloka'i. (*A*) Statistical Downscaling (SD) Representative Concentration Pathway (RCP) 8.5 2041–71 scenario from Elison Timm and others (2015), (*B*) SD RCP8.5 2071–99 scenario from Elison Timm and others (2015), and (*C*) Hawai'i Regional Climate Model version 2 (HRCM2) RCP8.5 2080–99 scenario from Zhang and others (2017). Estimated changes are relative to mean annual recharge for the 1978–2007 scenario. Values in parentheses represent change in millions of gallons per day. Aquifer system boundaries from State of Hawai'i (2014).

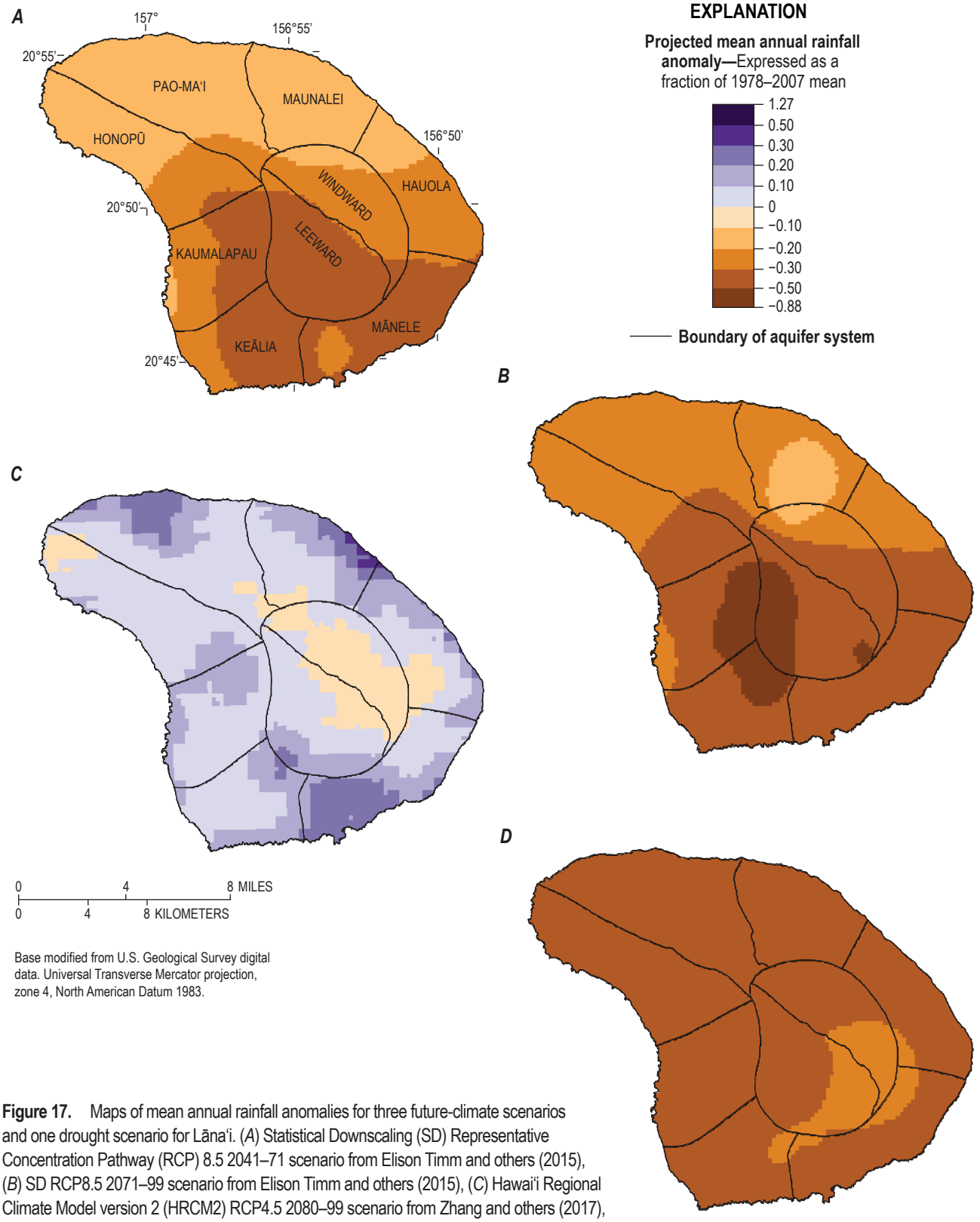


Figure 17. Maps of mean annual rainfall anomalies for three future-climate scenarios and one drought scenario for Lānaʻi. (*A*) Statistical Downscaling (SD) Representative Concentration Pathway (RCP) 8.5 2041–71 scenario from Elison Timm and others (2015), (*B*) SD RCP8.5 2071–99 scenario from Elison Timm and others (2015), (*C*) Hawaiʻi Regional Climate Model version 2 (HRCM2) RCP4.5 2080–99 scenario from Zhang and others (2017), and (*D*) 2008–12 drought scenario. Projected changes are relative to mean annual rainfall for the 1978–2007 scenario. Aquifer system boundaries from State of Hawaiʻi (2014).

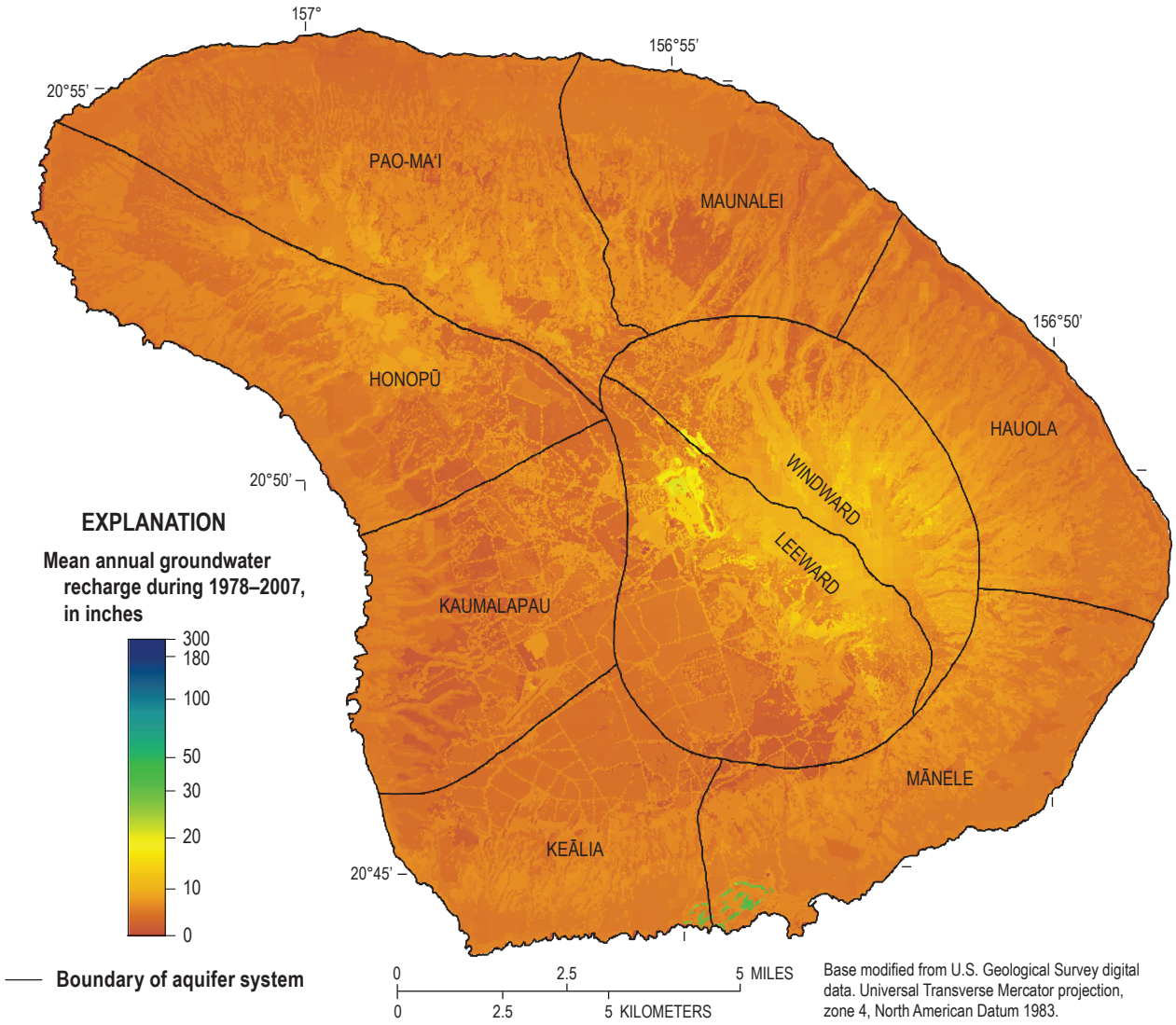


Figure 18. Map showing the distribution of mean annual groundwater recharge for the 1978–2007 scenario and 2020 land-cover conditions for Lānaʻi. Aquifer system boundaries from State of Hawaiʻi (2014).

Leeward, and Windward aquifer systems which are located in the southeastern and central regions of Lānaʻi; the aquifer systems with the least recharge are Kaumalapau, Keālia, and Maunalei which are located in the northern and southwestern regions of the island. The spatial pattern of mean annual recharge on Lānaʻi for the 1978–2007 scenario (fig. 18) resembles the pattern of mean annual rainfall during the same period (fig. 3). Areas with golf courses, an irrigated land cover, in central and southern Lānaʻi are recognizable because of their higher recharge rates.

The island-wide mean annual recharge for Lānaʻi is less than the recharge calculated by Hardy (1996) by roughly 35 Mgal/d (56 percent) mainly owing to decreases in total precipitation, increases in total ET, and differences in the modeling approach. Island-wide total precipitation decreased by 12 Mgal/d and total ET increased by 33 Mgal/d when compared to the values estimated by Hardy (1996) owing, in part, to a difference in input datasets. For example, Hardy (1996) used maps of mean monthly rainfall that were developed by Giambelluca and others (1986) using rainfall measurements collected during 1916–1983. Also, Hardy (1996) used the pan evaporation dataset developed by Ekern and Chang (1985) to define the potential-ET rate. Differences in other input datasets including soils, land cover, and runoff also likely contributed to differences in recharge estimates for Lānaʻi. Finally, the study by Hardy (1996) used a modeling approach derived from earlier studies (Thornthwaite and Mather, 1955; Shade, 1995). Determining the effect of the differences in input datasets and modeling approach on recharge estimates for Lānaʻi is beyond the scope of this study.

SD RCP8.5 2041–71 Scenario

The projected mean annual rainfall anomalies for SD RCP8.5 2041–71 scenario indicate drying across the entire island relative to the 1978–2007 scenario (fig. 17*A*). The greatest drying occurs across southeastern Lāna‘i where mean annual rainfall is projected to decrease by greater than 30 percent.

For this mid-century scenario, recharge for Lāna‘i is about 16 Mgal/d (appendix 2, table 2-4), an estimated decrease of about 11 Mgal/d (41 percent) from the 1978–2007 scenario. The greatest estimated decrease in recharge occurs in the Leeward and Windward aquifer systems where recharge decreases as much as 15 inches (fig. 19*A*).

Decreases in aquifer-system recharge are estimated for all Lāna‘i’s aquifer systems (fig. 20*A*). In terms of absolute difference in mean annual recharge, the greatest decreases in aquifer-system recharge are estimated for the Mānele, Pao-ma‘i, Windward, and Leeward aquifer systems, where decreases range from 1.2 to 2.3 Mgal/d relative to recharge for the 1978–2007 scenario. In terms of percentage change from the 1978–2007 scenario, the greatest decreases in aquifer-system recharge are estimated for Keālia, Kaumalapau, and Leeward aquifer systems, where decreases range from 47 to 54 percent.

SD RCP8.5 2071–99 Scenario

The projected mean annual rainfall anomalies for SD RCP8.5 2071–99 scenario indicate drying across Lāna‘i when compared to the 1978–2007 scenario (fig. 17*B*). The greatest drying occurs across south-central Lāna‘i where mean annual rainfall is projected to decrease by greater than 50 percent. Island-wide mean annual rainfall is projected to decline by 35 percent (table 2).

For this dry-climate scenario, recharge for Lāna‘i is about 12 Mgal/d, an estimated decrease of 15 Mgal/d (about 55 percent) from the 1978–2007 scenario. The greatest estimated decrease occurs mostly within the Leeward and Windward aquifer systems where recharge decreases by as much as 15 inches (fig. 19*B*).

Decreases in aquifer-system recharge are estimated for all Lāna‘i’s aquifer systems. In terms of absolute difference in mean annual recharge, the greatest decreases in aquifer-system recharge are estimated for the Pao-ma‘i, Windward, and Leeward aquifer systems, where decreases range from 1.8 to 3 Mgal/d relative to recharge for the 1978–2007 scenario (fig. 20*B*). In terms of percentage change from the 1978–2007 scenario, the greatest decreases in aquifer-system recharge are estimated for the Kaumalapau and Leeward aquifer systems, where decreases range from 65 to 70 percent.

HRCM2 RCP4.5 2080–99 Scenario

The projected mean annual rainfall anomalies for the HRCM2 RCP4.5 2080–99 scenario show wetting across most of Lāna‘i (fig. 17*C*). The greatest wetting occurs in coastal region of the Maunalei and Hauola aquifer systems where mean annual rainfall is projected to increase by greater than 30 percent. Limited drying occurs across central and western Lāna‘i where mean annual rainfall is projected to decrease as much as 10 percent. Island-wide mean annual rainfall is projected to increase by 7.2 percent (table 2).

The estimated changes in mean annual recharge show increases across most of Lāna‘i except for upland areas of Lāna‘ihale (figs. 1, 19*C*). Island-wide mean annual recharge is about 38 Mgal/d, which is greater than recharge for the 1978–2007 scenario by roughly 12 Mgal/d (about 43 percent; appendix 2, table 2-4). Mean annual recharge is estimated to increase by as much as 5 inches or more across much of the lowland area of the island. However, recharge is estimated to decrease by as much as 5 inches across the upland areas of Lāna‘ihale. Island-wide mean annual recharge increases by 43 percent despite an increase in island-wide mean annual rainfall of only 7.2 percent. This disparity is caused, in part, by large variations in projected changes to mean monthly rainfall across Lāna‘i that effectively increase the amount of water available for recharge. For example, mean monthly rainfall is projected to increase by 35 to 130 percent during the months of November, January, and March on Lāna‘i but decrease by as much as 55 percent during most of the remaining months of the year (Zhang and Wang, 2017; fig. 21). These changes indicate that annual rainfall across Lāna‘i is projected to become more concentrated in selected months for the HRCM2 RCP4.5 2080–99 scenario and correspond to increases of 59 to 367 percent in mean monthly recharge during November, January, and March, and decreases in mean monthly recharge by as much as 84 percent during most of the remaining months of the year. The estimated increases in mean monthly recharge in November, January, and March indicate that a greater proportion of additional water from increased rainfall is converted to recharge. This increased recharge more than compensates for the decreases in recharge during most of the remaining months, resulting in an enhanced positive effect on island-wide mean annual recharge for Lāna‘i. The amount of water available for recharge is also controlled by the amount of rainfall diverted to runoff. The amount of rainfall diverted to runoff for the future-climate scenarios is computed by WATRMod using adjusted seasonal runoff ratios. The adjusted seasonal runoff ratios were estimated using wet- and dry-season rainfall, which does not reflect the month-to-month variability in projected rainfall for each future-climate scenario. Hence, where month-to-month variations in projected changes to rainfall are large, adjusted seasonal runoff ratios might underestimate actual runoff during months with projected increases in rainfall and overestimate runoff during months with decreased rainfall. Determining the cause of the large variations in projected changes to mean monthly rainfall is beyond the scope of this study. The effect of using projected changes in seasonal rainfall for the HRCM2 RCP4.5 2080–99 scenario for Lāna‘i, instead of projected changes in monthly rainfall, on recharge estimates is described in the “Effects of Selected Climate Inputs on Estimated Recharge” section of the report.

Increases in aquifer-system recharge are estimated for all Lāna‘i’s aquifer systems (fig. 20*C*). In terms of absolute change, the greatest increases in aquifer-system recharge occur in the Pao-ma‘i, Leeward, and Mānele aquifer systems, where recharge is estimated to increase by 1.7 to 1.9 Mgal/d relative to recharge for the 1978–2007 scenario. In terms of percentage change from the 1978–2007 scenario, the greatest increases in aquifer-system recharge are estimated for the Kaumalapau, Keālia, and Mānele aquifer systems, where increases range from 65 to 80 percent.

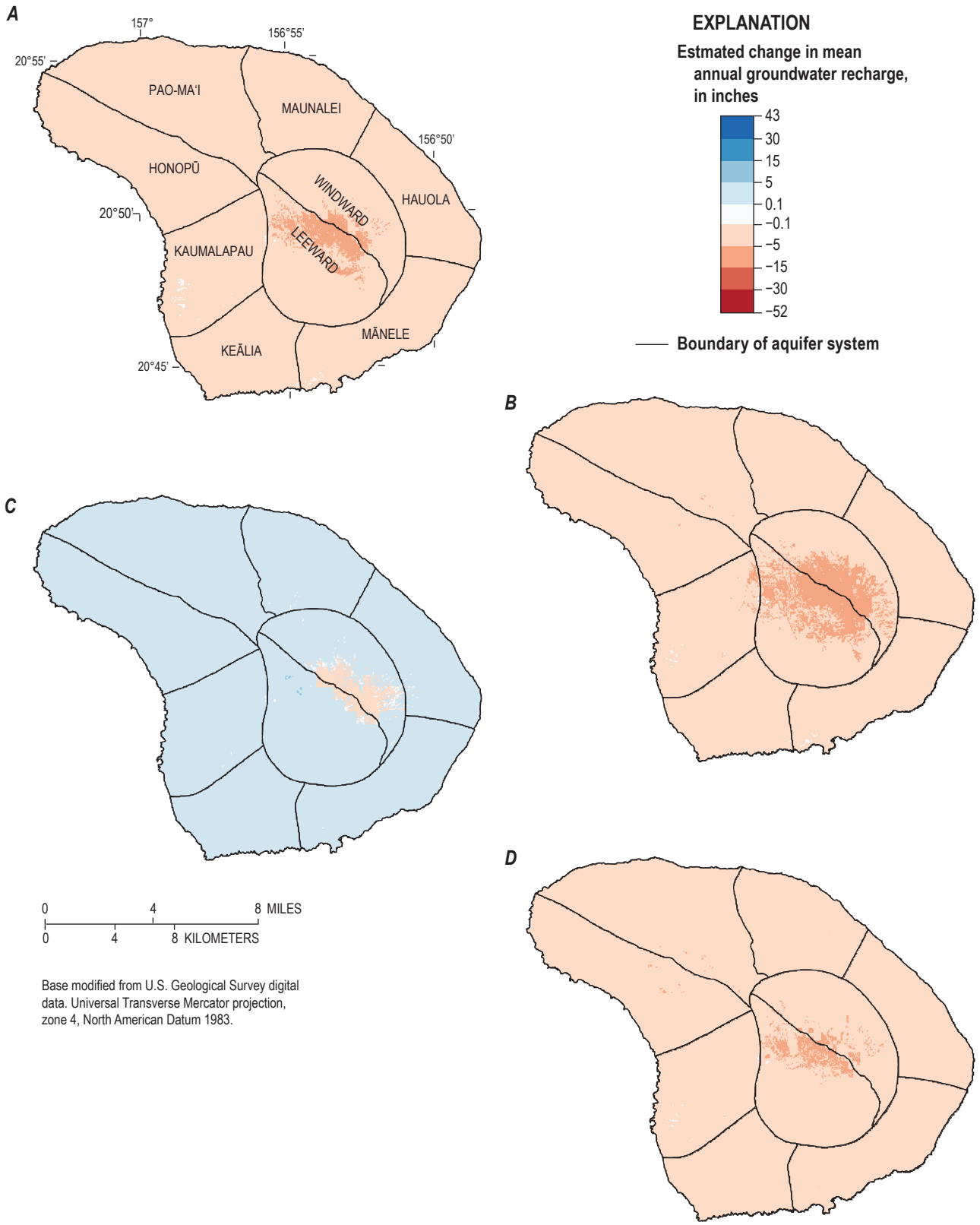


Figure 19. Maps showing estimated change in mean annual groundwater recharge for three future-climate scenarios and one drought scenario for Lānaʻi. (*A*) Statistical Downscaling (SD) Representative Concentration Pathway (RCP) 8.5 2041–71 scenario from Elison Timm and others (2015), (*B*) SD RCP8.5 2071–99 scenario from Elison Timm and others (2015), (*C*) Hawaiʻi Regional Climate Model version 2 (HRCM2) RCP4.5 2080–99 scenario from Zhang and others (2017), and (*D*) 2008–12 drought scenario. Estimated changes are relative to mean annual recharge for the 1978–2007 scenario. Aquifer system boundaries from State of Hawaiʻi (2014).

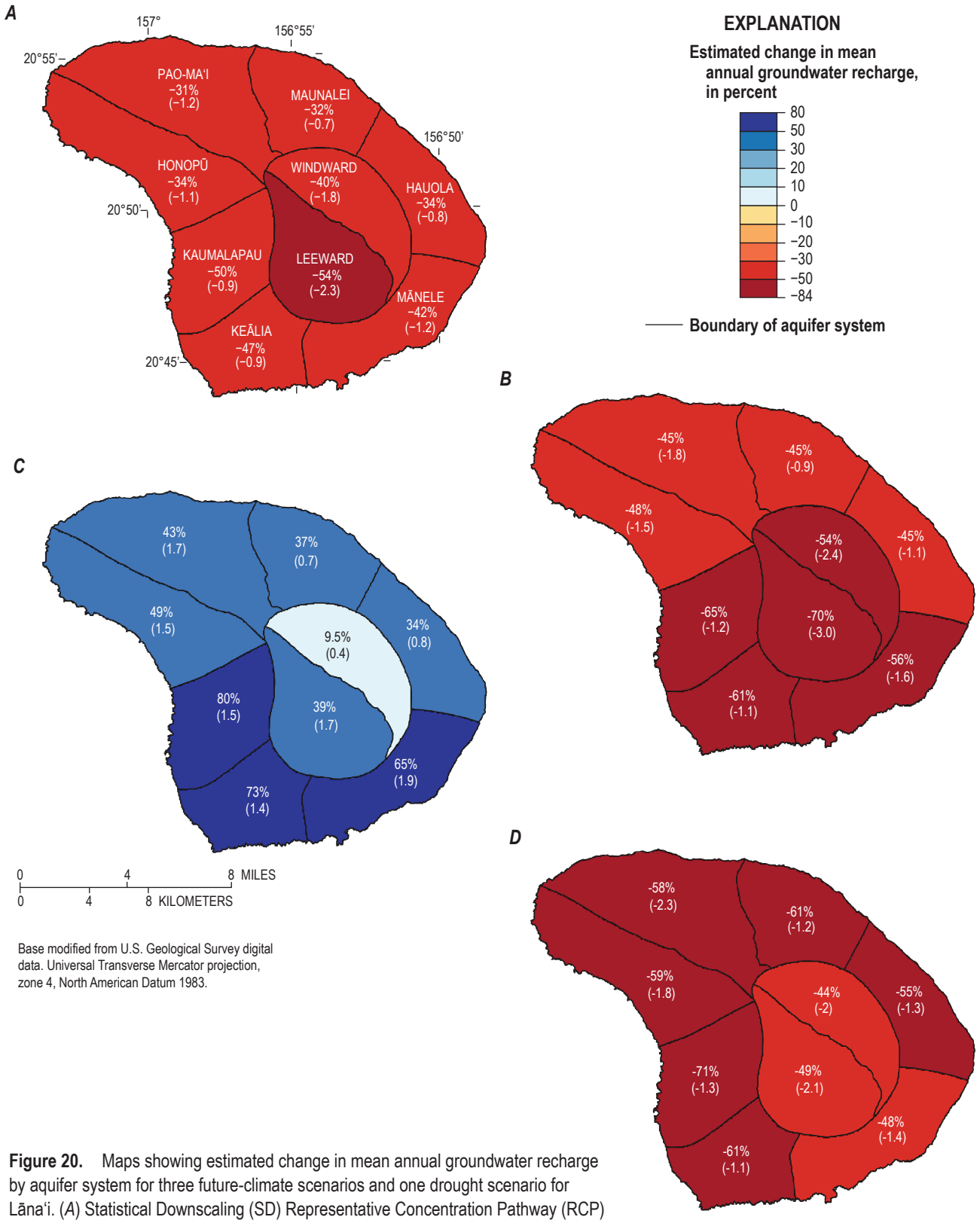


Figure 20. Maps showing estimated change in mean annual groundwater recharge by aquifer system for three future-climate scenarios and one drought scenario for Lāna'i. (*A*) Statistical Downscaling (SD) Representative Concentration Pathway (RCP) 8.5 2041–71 scenario from Elison Timm and others (2015), (*B*) SD RCP8.5 2071–99 scenario from Elison Timm and others (2015), (*C*) Hawai'i Regional Climate Model version 2 (HRCM2) RCP4.5 2080–99 scenario from Zhang and others (2017), and (*D*) 2008–12 drought scenario. Estimated changes are relative to mean annual recharge for the 1978–2007 scenario. Values in parentheses represent change in millions of gallons per day. Aquifer system boundaries from State of Hawai'i (2014).

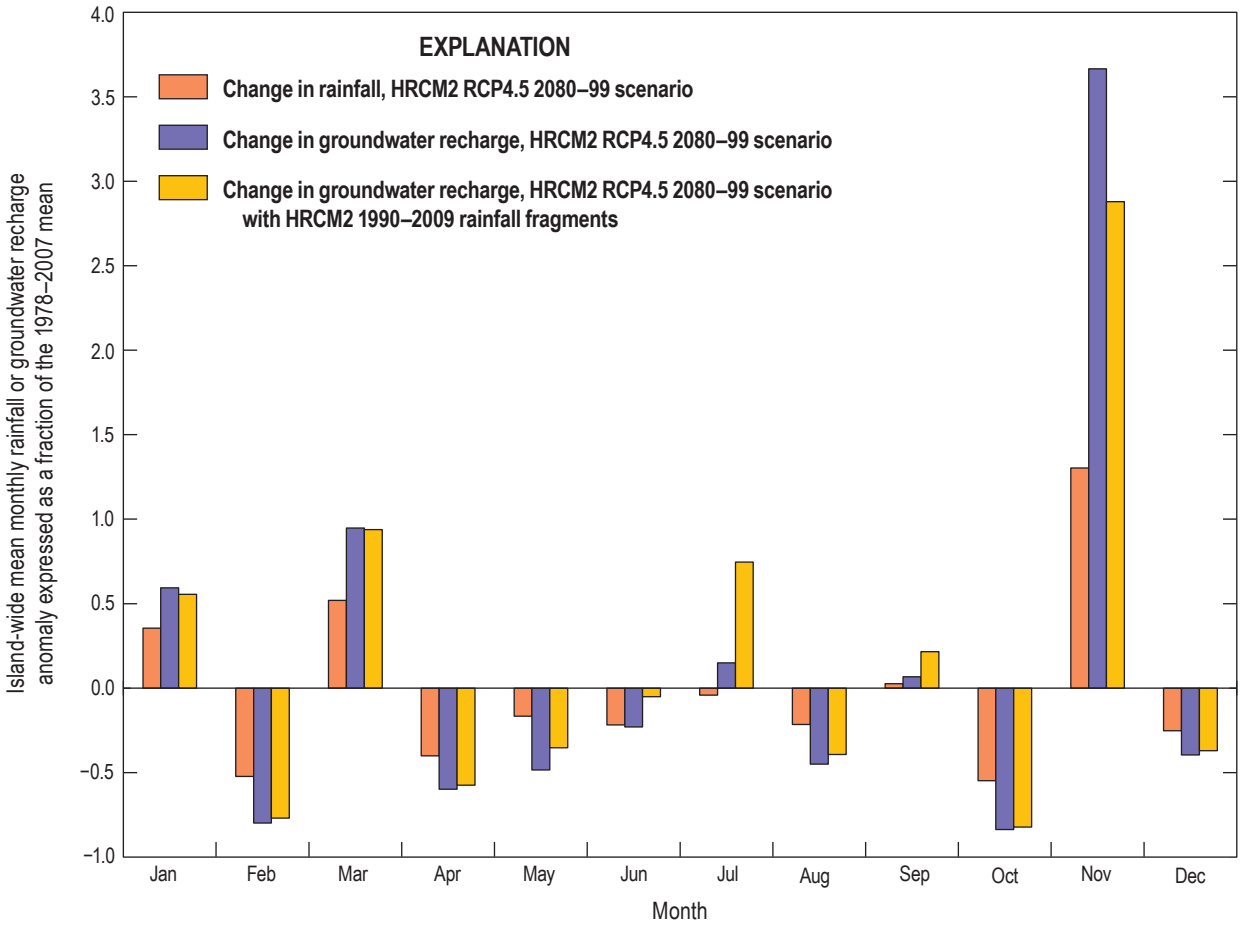


Figure 21. Graph of projected changes in mean monthly rainfall for the Hawai'i Regional Climate Model version 2 (HRCM2) Representative Concentration Pathway (RCP) 4.5 2080–99 scenario, and estimated changes in mean monthly groundwater recharge for the HRCM2 RCP4.5 2080–99 scenario and mean monthly groundwater recharge for the HRCM2 RCP4.5 2080–99 scenario with HRCM2 1990–2009 rainfall fragments for Lāna'i. Projected changes in rainfall and estimated changes in groundwater recharge for the HRCM2 RCP4.5 2080–99 scenario are relative to mean monthly values for the 1978–2007 scenario. Estimated changes in groundwater recharge for the HRCM2 RCP4.5 2080–99 scenario with HRCM2 1990–2009 rainfall fragments are relative to mean monthly values for the 1978–2007 scenario with HRCM2 1990–2009 rainfall fragments.

2008–12 Drought Scenario

To assess the effect of episodic dry conditions on recharge, rainfall during 2008–12 was selected to represent drought conditions on Lāna'i (see the "Selected Drought Scenario for Lāna'i" section). During the 2008–12 drought scenario, island-wide mean annual rainfall computed by WATRMod is less than rainfall for the 1978–2007 scenario by about 56 Mgal/d (38 percent; appendix 2, table 2-4). Mean annual rainfall across Lāna'i during the 2008–12 drought scenario ranges from 20 to greater than 30 percent less than rainfall during the 1978–2007 scenario (fig. 17*D*)

For this drought scenario, recharge for Lāna'i is about 12 Mgal/d (appendix 2, table 2-4). This is an estimated decrease of approximately 15 Mgal/d (about 54 percent) from the 1978–2007 scenario. The greatest decrease in recharge occurs across parts of the Honopū, Pao-ma'i, Windward, and Leeward aquifer systems where recharge decreases by greater than 5 inches (fig. 19*D*).

Decreases in aquifer-system recharge are estimated for all Lāna'i's aquifer systems (fig. 20*D*). In terms of absolute difference in mean annual recharge, the greatest decreases in aquifer-system recharge are estimated for the Pao-ma'i, Windward, and Leeward aquifer systems, where decreases range from 2 to 2.3 Mgal/d relative to recharge for the 1978–2007 scenario. In terms of percentage change from the 1978–2007 scenario, the greatest decrease in aquifer-system recharge is estimated for the Kaumalapau aquifer system, where recharge decreases by 71 percent.

Maui

Four water-budget scenarios were selected for Maui: (1) recent conditions represented by the 1978–2007 scenario, (2) a mid-century climate scenario using projected rainfall conditions for the SD RCP8.5 2041–71 scenario, (3) a dry-climate scenario using projected rainfall conditions for the SD RCP8.5 2071–99 scenario, and (4) a wet-climate scenario using projected rainfall conditions for the HRCM1 A1B 2080–99 scenario (fig. 22, table 4). The water-budget components estimated for the

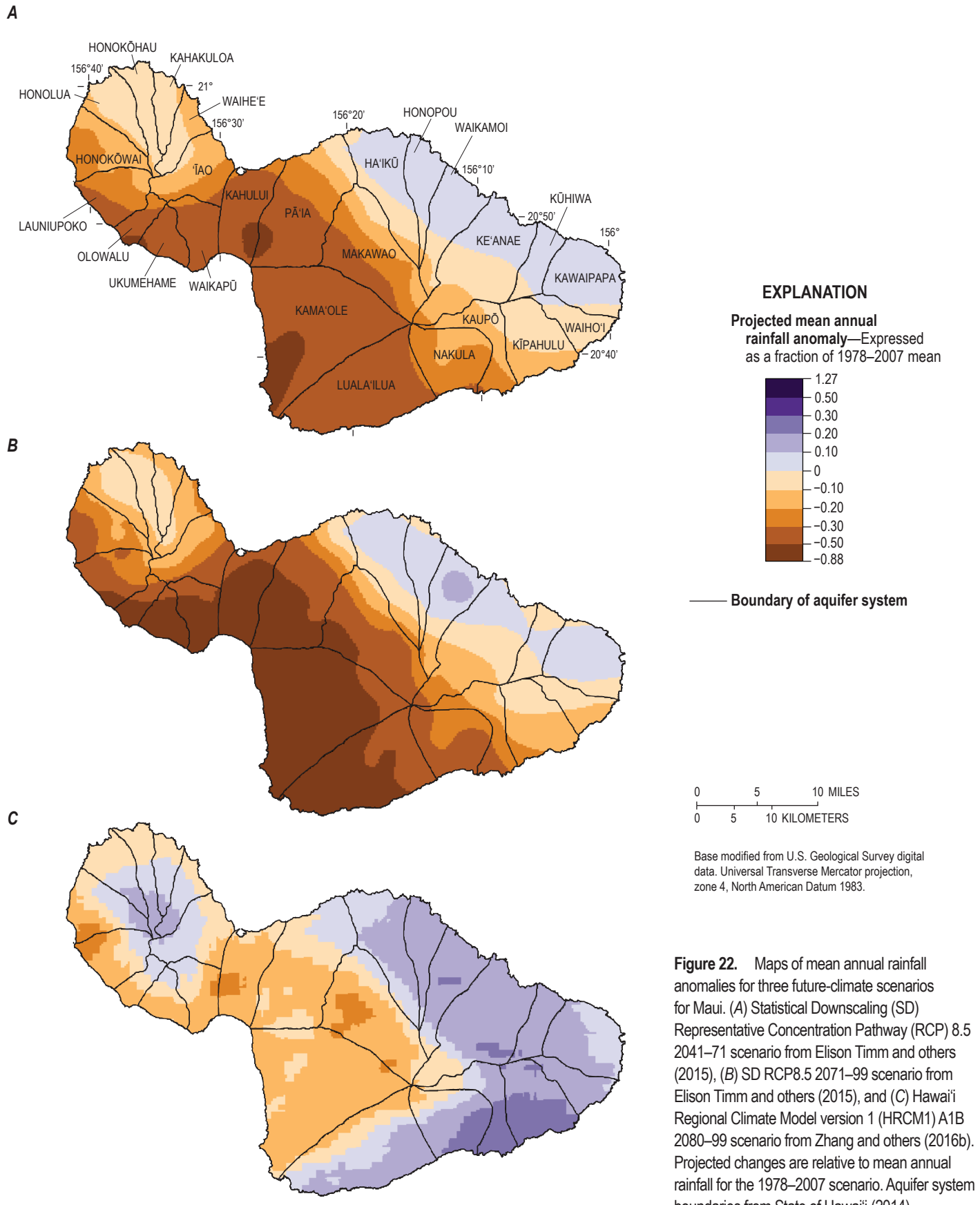


Figure 22. Maps of mean annual rainfall anomalies for three future-climate scenarios for Maui. (*A*) Statistical Downscaling (SD) Representative Concentration Pathway (RCP) 8.5 2041–71 scenario from Elison Timm and others (2015), (*B*) SD RCP8.5 2071–99 scenario from Elison Timm and others (2015), and (*C*) Hawai'i Regional Climate Model version 1 (HRCM1) A1B 2080–99 scenario from Zhang and others (2016b). Projected changes are relative to mean annual rainfall for the 1978–2007 scenario. Aquifer system boundaries from State of Hawai'i (2014).

1978–2007 scenario and three future-climate scenarios for Maui are shown in table 2-5 (appendix 2) and summarized by subarea in Kāne and others (2024).

1978–2007 Scenario

Mean annual recharge for Maui is about 1,192 Mgal/d for the 1978–2007 scenario (fig. 23*A*, appendix 2, table 2-5). Mean annual recharge is about 34.4 inches, expressed as a depth of water uniformly distributed over the island. For Maui's 25 aquifer systems (fig. 2*E*), mean annual recharge ranges from about 6 to about 217 Mgal/d for the 1978–2007 scenario (appendix 2, table 2-5). In terms of recharge expressed as an average depth of water, the aquifer systems with the greatest recharge are Keʻanae, Kīpahulu, and Kūhiwa, which are located in eastern Maui; the aquifer systems with the least recharge are Kahului, Lualaʻilua, and Kamaʻole, which are located in central Maui and along the southern slope of Haleakalā. The spatial pattern of mean annual recharge on Maui for the 1978–2007 scenario resembles the pattern of mean annual rainfall during the same period (fig. 3).

Island-wide mean annual recharge is less than recharge estimated by Mair and others (2019) by about 39 Mgal/d (3.2 percent) mainly owing to decreases in total precipitation, and increases in canopy evaporation and actual ET that are partially offset by increases in total direct recharge (appendix 2, table 2-5). Island-wide total precipitation decreased by about 11 Mgal/d when compared to Mair and others (2019) mostly owing to reduced fog interception computed by the model. In this study, the base of the cloud zone in leeward areas of Maui was set to an altitude of 2,600 ft, whereas Mair and others (2019) used an altitude of 2,000 ft. Furthermore, an altitudinal inversion boundary for reduced fog-interception rates for both leeward and windward areas was lowered from 8,200 ft used by Mair and others (2019) to 7,800 ft for this study (table 6). These changes in the cloud-zone delineation for Maui effectively reduced the island-wide fog interception estimates when compared to Mair and others (2019).

Island-wide canopy evaporation and actual ET increased by about 24 and 20 Mgal/d, respectively, when compared to the canopy evaporation and actual ET estimated by Mair and others (2019). The increase in canopy evaporation indicates a change in the daily rainfall pattern to more frequent, low-magnitude daily rainfall, which might also increase water available for actual ET and reduce the frequency and magnitude of recharge from less frequent, high-magnitude daily rainfall. The changes in the daily rainfall pattern are likely related to a difference in the rainfall fragment sets used to disaggregate monthly rainfall during 1978–2007 into daily rainfall for the water-budget simulations. Mair and others (2019) utilized historical rain gage data from a nearby gage to create rainfall fragment sets for every month in which complete daily rainfall measurements were available. The record of historical rain gage data may extend as far back as the early 20th century at some gage locations. However, in this study, the gridded daily rainfall dataset (250-m resolution) for the period 1990–2014 by Longman and others (2019) was used to disaggregate monthly rainfall for each subarea. Hence, the rainfall fragment sets used in this study represent daily rainfall patterns across a uniform time period (1990–2014) and higher spatial resolution (250 m), and indicate a change to more frequent, low-magnitude daily rainfall when compared to the daily rainfall pattern of the rainfall fragment sets used by Mair and others (2019). The effect of these different rainfall fragment sets on recharge was not evaluated as part of this study. The indicated changes in daily rainfall frequency are consistent with identified changes in rainfall extremes across Maui and other parts of Hawaiʻi, where a shift to more frequent, low-magnitude rainfall events and to less frequent moderate and heavy rainfall events has been observed across the Hawaiian Islands from the 1950s to the 2000s (Chu and others, 2010; Elison Timm and others, 2011).

The water-budget simulations by Mair and others (2019) used 2017 land-cover conditions, whereas this study utilized 2020 land-cover conditions that include (1) many reservoirs that were converted to grassland in the 2017 land-cover map due to the cessation of sugarcane cultivation in December 2016, and (2) a distinction between lined and unlined reservoirs. In this study, reservoirs in the map of 2020 land cover were identified using the latest available land-cover information and aerial imagery, which indicated the ongoing use of many sugarcane-era reservoirs. Furthermore, reservoirs in the map of 2020 land cover were distinguished as either lined or unlined. Subareas with lined reservoir land cover were assumed to have no seepage and assigned a constant recharge rate of zero (table 11). These changes to land cover for 2020 land-cover conditions effectively increased the island-wide estimate of total direct recharge by about 16 Mgal/d when compared to Mair and others (2019).

SD RCP8.5 2041–71 Scenario

The projected mean annual rainfall anomalies for SD RCP8.5 2041–71 scenario indicate drying across much of central and leeward Maui along with wetting across parts of the windward slopes of Haleakalā (fig. 22*A*). The greatest drying occurs across leeward Maui and includes aquifer systems extending from Launiupoko to Lualaʻilua where mean annual rainfall is projected to decrease by more than 30 percent. Windward areas of Haleakalā show increases in projected rainfall of as much as 10 percent. Island-wide mean annual rainfall is projected to decrease by 8.3 percent (table 2).

For this mid-century scenario, recharge for Maui is about 1,078 Mgal/d, an estimated decrease of 115 Mgal/d (9.6 percent) from the 1978–2007 scenario. The greatest estimated decrease occurs in the upland areas of the Launiupoko, Ukumehame, and Waikapū aquifer systems where recharge is estimated to decrease by greater than 30 inches (fig. 23*B*). Mean annual recharge is estimated to increase along the windward region of Haleakalā with the greatest increases occurring in the Honopou, Waikamoi, and Keʻanae aquifer systems.

Decreases in aquifer-system recharge are estimated for all but three of Maui's aquifer systems (fig. 24*A*). In terms of absolute difference in mean annual recharge, the greatest decreases in aquifer-system recharge are estimated for the Kamaʻole, Makawao, and Lualaʻilua aquifer systems, where decreases range from 11 to 17 Mgal/d relative to recharge for the 1978–2007 scenario. In

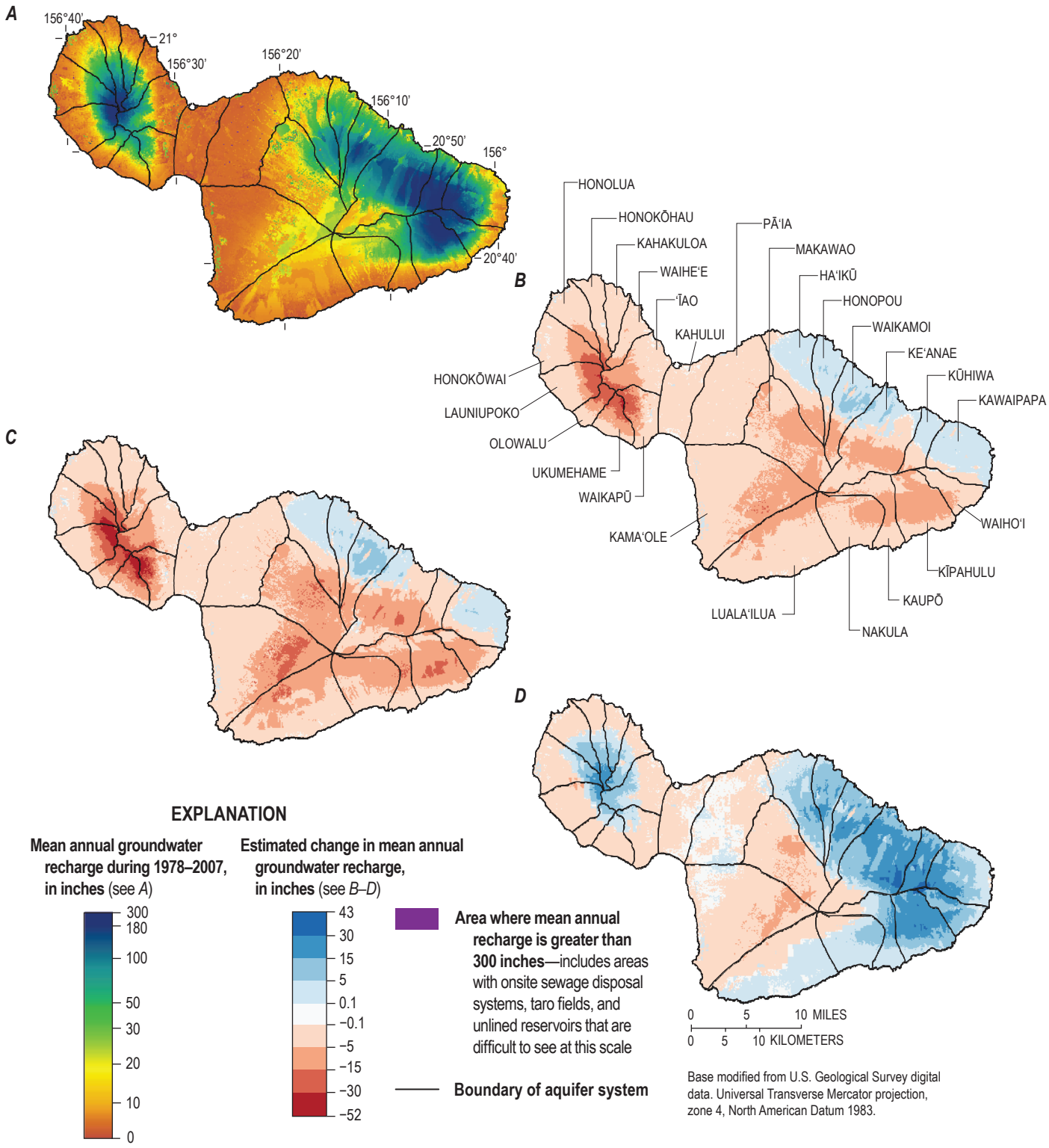


Figure 23. Maps showing the distribution of mean annual groundwater recharge for the 1978–2007 scenario and 2020 land-cover conditions, and estimated change in mean annual groundwater recharge for three future-climate scenarios for Maui. (*A*) Mean annual recharge for the 1978–2007 scenario. Estimated change in mean annual recharge for (*B*) Statistical Downscaling (SD) Representative Concentration Pathway (RCP) 8.5 2041–71 scenario from Elison Timm and others (2015), (*C*) SD RCP8.5 2071–99 scenario from Elison Timm and others (2015), and (*D*) Hawai'i Regional Climate Model version 1 (HRCM1) A1B 2080–99 scenario from Zhang and others (2016b). Estimated changes are relative to mean annual recharge for the 1978–2007 scenario. Aquifer system boundaries from State of Hawai'i (2014).

terms of percentage change from the 1978–2007 scenario, the greatest decreases in aquifer-system recharge are estimated for Ukumehame, Kama‘ole, and Lula‘ilua aquifer systems where decreases range from 51 to 56 percent. Aquifer-system recharge increased in the Honopou, Waikamoi, and Kawaipapa aquifer systems by 1.1 to 1.4 Mgal/d (1.3 to 3.4 percent).

SD RCP8.5 2071–99 Scenario

The projected mean annual rainfall anomalies for the SD RCP8.5 2071–99 scenario indicate drying across most of Maui, except for the windward slopes of Haleakalā (fig. 22*B*). The greatest drying occurs in southern and central Maui where for most of the aquifer systems rainfall is projected to decrease by greater than 50 percent. The greatest wetting occurs across parts of the Waikamoi aquifer system where mean annual rainfall is projected to increase by greater than 10 percent. Island-wide mean annual rainfall is projected to decrease by 13 percent (table 2).

For this dry-climate scenario, mean annual recharge is estimated to decrease across most of Maui for the SD RCP8.5 2071–99 scenario (fig. 23*C*). The island-wide mean annual recharge for the SD RCP8.5 2071–99 scenario is less than recharge for the 1978–2007 scenario by 162 Mgal/d (14 percent; appendix 2, table 2-5). The greatest estimated decrease occurs in the upland areas of the Honokōwai, Launiupoko, Olowalu, Ukumehame, and Waikapū aquifer systems, where mean annual recharge decreases by more 30 inches. Increases in recharge are estimated for the windward slopes of Haleakalā and areas of land-cover types with estimated increases in irrigation, such as coffee and golf courses (fig. 4*E*), where increases of greater than 5 inches are estimated. The greatest estimated increase in recharge occurs across parts of the windward slopes of Haleakalā where increases of between 5 and 15 inches are estimated.

Decreases in aquifer-system recharge are estimated for all but three of Maui’s 25 aquifer systems (fig. 24*B*). In terms of absolute changes in mean annual recharge, the greatest decreases in aquifer-system recharge are estimated for the Kama‘ole, Lula‘ilua, and Makawao aquifer systems, where decreases range from 15 to 22 Mgal/d relative to recharge for the 1978–2007 scenario. In terms of percentage difference from the 1978–2007 scenario, the greatest decreases in aquifer-system recharge are estimated for the Ukumehame, Waikapū, Kama‘ole, and Lula‘ilua aquifer systems, where decreases range from 54 to 72 percent. Increases in aquifer-system recharge ranging from 0.7 to 1.8 Mgal/d (or about 0.6 to 2.6 percent) are estimated for the Honopou, Kawaipapa, and Waikamoi aquifer systems.

HRCM1 A1B 2080–99 Scenario

The projected mean annual rainfall anomalies for HRCM1 A1B 2080–99 scenario show a pattern of limited drying and wetting across Maui (fig. 22*C*). Mean annual rainfall is projected to increase in the upland areas of western Maui, and northeastern and southeastern slopes of Haleakalā. The greatest wetting occurs in an area of southeastern Maui where mean annual rainfall is projected to increase by greater than 20 percent. Drying is projected for the lowland areas of western Maui, central Maui, and western slopes of Haleakalā, where mean annual rainfall decreases by as much as 20 percent or more in some locations. Island-wide mean annual rainfall is projected to increase by 9.8 percent (table 2).

The estimated changes in mean annual recharge include a range of increases and decreases across large areas of Maui for the HRCM1 A1B 2080–99 scenario (fig. 23*D*). Island-wide mean annual recharge is greater than recharge for the 1978–2007 scenario by about 126 Mgal/d (11 percent; appendix 2, table 2-5). Mean annual recharge is estimated to increase in the upland areas of western Maui, parts of central Maui, and across the northeastern and southeastern slopes of Haleakalā. The greatest estimated increase in mean annual recharge occurs in the eastern slopes of Haleakalā, where increases range from 30 to 43 inches. A decrease in recharge occurs in the coastal areas of western Maui, parts of central Maui, and the western slopes of Haleakalā, where estimated recharge rates decrease by 5 to 15 inches.

Increases in aquifer-system recharge are estimated for 18 of Maui’s 25 aquifer systems (fig. 24*C*). In terms of absolute change, the greatest increases in aquifer-system recharge occur in the Kawaipapa, Ke‘anae, and Kīpahulu aquifer systems, where increases range from 15 to 34 Mgal/d. In terms of percentage change from the 1978–2007 scenario, the greatest increases in aquifer-system recharge occur in the aquifer systems along the northeastern and southeastern slopes of Haleakalā, where increases range from 13 to 21 percent. The greatest decreases in aquifer-system recharge occur in the Makawao and Kama‘ole aquifer systems, where decreases are each 7 Mgal/d (16 and 23 percent, respectively).

Island of Hawai‘i

Four water-budget scenarios were selected for the Island of Hawai‘i: (1) recent conditions represented by the 1978–2007 scenario, (2) a mid-century climate scenario using projected rainfall conditions for the SD RCP8.5 2041–71 scenario, (3) a dry-climate scenario using projected rainfall conditions for the SD RCP8.5 2071–99 scenario, and (4) a wet-climate scenario using projected rainfall conditions for the HRCM2 RCP4.5 2080–99 scenario (fig. 25, table 4). The water-budget components estimated for the 1978–2007 scenario and three future-climate scenarios for the Island of Hawai‘i are shown in table 2-6 (appendix 2) and summarized by subarea in Kāne and others (2024).

1978–2007 Scenario

Mean annual recharge for the Island of Hawai‘i is about 6,885 Mgal/d for the 1978–2007 scenario (appendix 2, table 2-6). Mean annual recharge is about 35.9 inches, expressed as a depth of water uniformly distributed over the island. For the Island of Hawai‘i’s 24 aquifer systems (fig. 2*F*), mean annual recharge ranges from about 26 to about 895 Mgal/d for the 1978–2007 scenario (appendix 2, table 2-6). In terms of recharge expressed

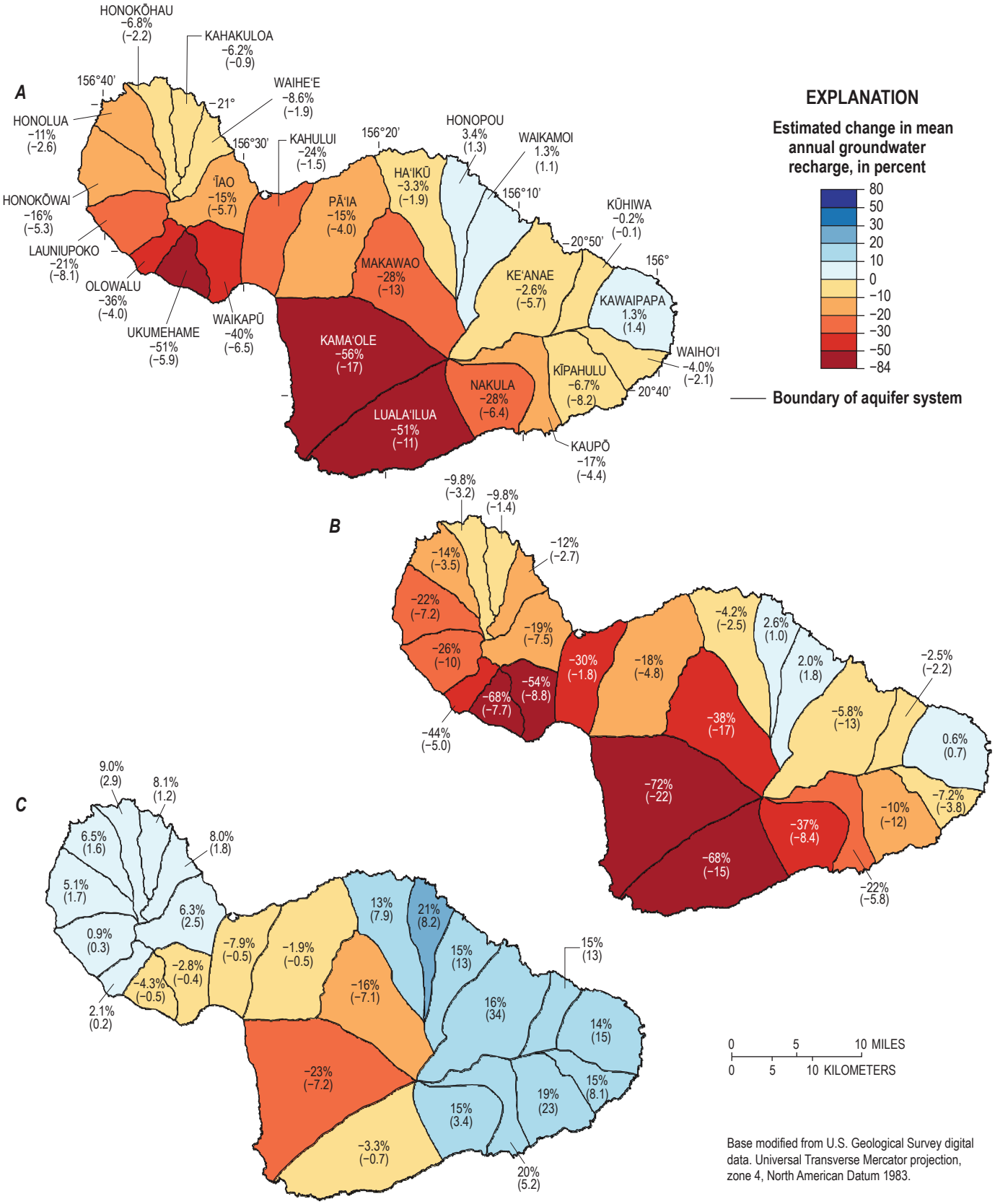


Figure 24. Maps showing estimated change in mean annual groundwater recharge by aquifer system for three future-climate scenarios for Maui. (*A*) Statistical Downscaling (SD) Representative Concentration Pathway (RCP) 8.5 2041–71 scenario from Elison Timm and others (2015), (*B*) SD RCP8.5 2071–99 scenario from Elison Timm and others (2015), and (*C*) Hawai'i Regional Climate Model version 1 (HRCM1) A1B 2080–99 scenario from Zhang and others (2016b). Estimated changes are relative to mean annual recharge for the 1978–2007 scenario. Values in parentheses represent change in millions of gallons per day. Aquifer system boundaries from State of Hawai'i (2014).

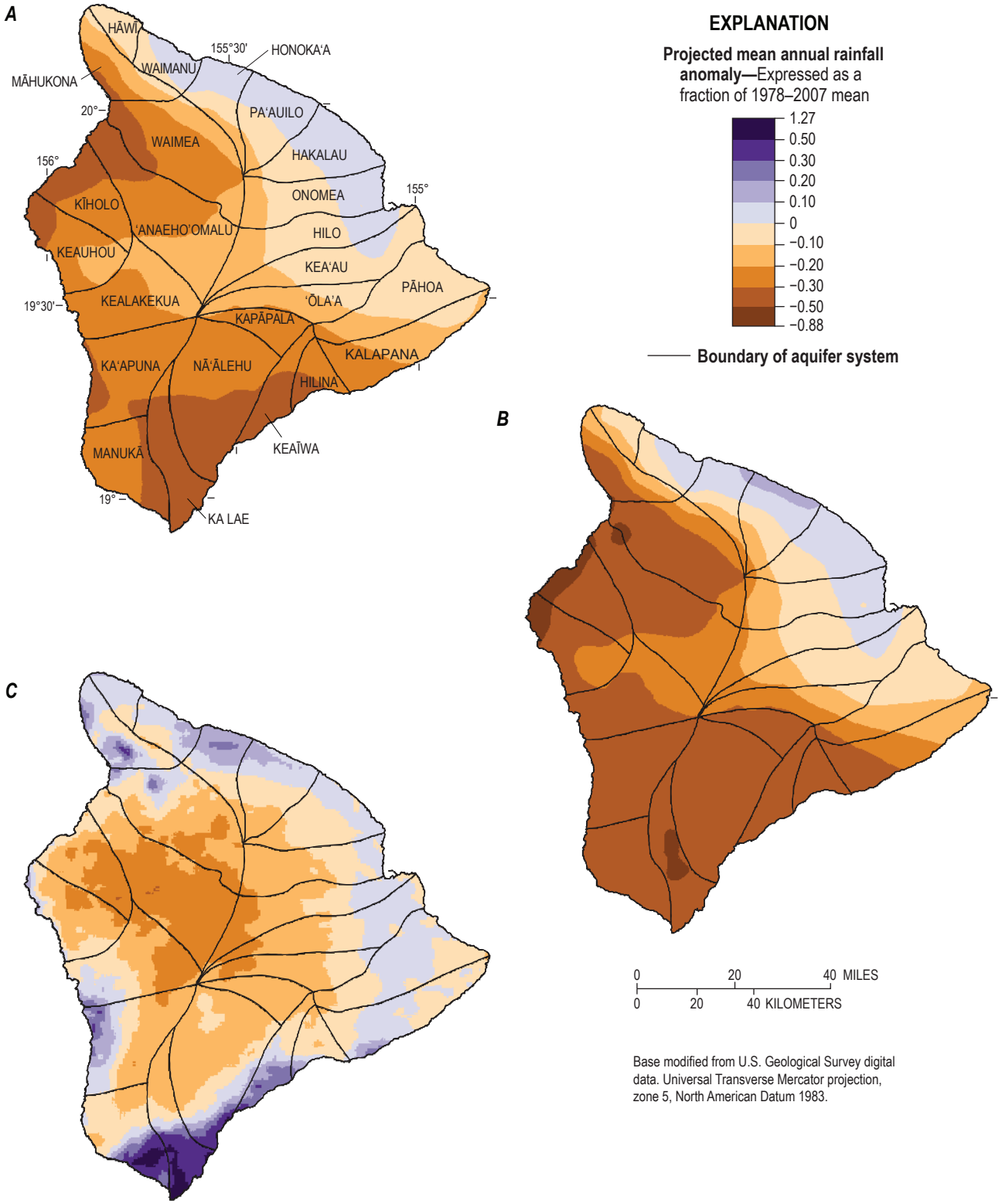


Figure 25. Maps of mean annual rainfall anomalies for three future-climate scenarios for the Island of Hawai'i. (*A*) Statistical Downscaling (SD) Representative Concentration Pathway (RCP) 8.5 2041–71 scenario from Elison Timm and others (2015), (*B*) SD RCP8.5 2071–99 scenario from Elison Timm and others (2015), and (*C*) Hawai'i Regional Climate Model version 2 (HRCM2) RCP4.5 2080–99 scenario from Zhang and others (2016b). Projected changes are relative to mean annual rainfall for the 1978–2007 scenario. Aquifer system boundaries from State of Hawai'i (2014).

as an average depth of water, the aquifer systems with the greatest recharge are Kea‘au, Hakalau, and Onomea, which are all on the windward side; the aquifer systems with the least recharge are Waimea, Kīholo, and ‘Anaeho‘omalu which are on the leeward part of the island. The spatial pattern of mean annual recharge on the Island of Hawai‘i for the 1978–2007 scenario (fig. 26) resembles the pattern of mean annual rainfall during the same period (fig. 3).

Island-wide mean annual recharge is greater than recharge estimated by Engott and others (2011) by about 290 Mgal/d (4.4 percent). In this study, updates to several model input datasets developed by Engott (2011) were used in the WATRMod simulations including datasets related to the following: (1) monthly and annual rainfall, (2) land cover, (3) daily rainfall frequency, (3) soil physical properties, (4) runoff, (5) reservoir seepage, and (6) cloud-zone delineation. For example, Engott (2011) used a rainfall dataset representative of conditions during 1916–1983 and assumed an altitude of 2,500 ft for the base of the cloud zone except in leeward areas of Kohala where there was no assumed fog interception. In this study, rainfall datasets used are representative of base conditions during 1978–2007. An altitude of 2,000 ft was used for the base of the cloud zone in areas with a windward aspect; an altitude of 2,600 ft was used in areas with a leeward aspect including leeward Kohala. These altitude differences caused total precipitation (rainfall plus fog) to increase by 548 Mgal/d compared to the total precipitation estimated by Engott (2011). Determining the effect of the differences in model inputs and assumptions on recharge estimates for the Island of Hawai‘i is beyond the scope of this study.

SD RCP8.5 2041–71 Scenario

The projected mean annual rainfall anomalies for the SD RCP8.5 2041–71 scenario indicate drying across much of the Island of Hawai‘i except for areas along the northeastern coast compared to the 1978–2007 scenario (fig. 25*A*). On the Island of Hawai‘i, the greatest drying occurs across parts of western and southern regions where mean annual rainfall is projected to decrease by greater than 30 percent. Mean annual rainfall along the northeastern coast of the Island of Hawai‘i is projected to increase by as much as 10 percent. Island-wide mean annual rainfall is projected to decrease by 11 percent (table 2).

For this mid-century scenario, mean annual recharge for the Island of Hawai‘i is about 5,945 Mgal/d (fig. 27*A*), an estimated decrease of 940 Mgal/d (14 percent) relative to the 1978–2007 scenario. Mean annual recharge is estimated to decrease across much of the Island of Hawai‘i except for areas along the northeastern coast. The greatest estimated decrease occurs in the mid-altitude region of Nā‘ālehu aquifer system where recharge is estimated to decrease by more than 30 inches. Recharge is estimated to increase along the northeastern coast of the Island of Hawai‘i with the greatest increases occurring in the Honoka‘a, Pa‘auilo, and Hakalau aquifer systems.

Decreases in aquifer-system recharge are estimated for all but 3 of the Island of Hawai‘i’s aquifer systems (fig. 28*A*). In terms of absolute difference in mean annual recharge, the greatest decreases in aquifer-system recharge are estimated for the Pāhoa, Kalapana, and Nā‘ālehu aquifer systems, where decreases range from 72 to 239 Mgal/d relative to recharge for the 1978–2007 scenario. In terms of percentage change from the 1978–2007 scenario, the greatest decreases in aquifer-system recharge are estimated for the Ka‘apuna, Ka Lae, Kīholo, and Manukā aquifer systems, where decreases range from 45 to 46 percent. Increases in aquifer-system recharge in the Hakalau, Honoka‘a, and Pa‘auilo aquifer systems range from 3.1 to 13 Mgal/d (0.5 to 5.8 percent).

SD RCP8.5 2071–99 Scenario

The projected mean annual rainfall anomalies for the SD RCP8.5 2071–99 scenario indicates drying across much of the Island of Hawai‘i, except for the northeastern coast, relative to the 1978–2007 scenario (fig. 25*B*). The greatest drying occurs across large parts of western and southern sections of the island where mean annual rainfall is projected to decrease by greater than 30 percent including areas with decreases of greater than 50 percent. Mean annual rainfall is projected to increase along the northeastern coast of the Island of Hawai‘i by as much as 10 percent or more. Island-wide mean annual rainfall is projected to decrease by 16 percent (table 2).

For this dry-climate scenario, mean annual recharge is estimated to decrease across most of the Island of Hawai‘i for the SD RCP8.5 2071–99 scenario (fig. 27*B*). The island-wide mean annual recharge for the SD RCP8.5 2071–99 scenario is less than the recharge for the 1978–2007 scenario by 1,349 Mgal/d (20 percent; appendix 2, table 2-6). The greatest estimated decrease occurs in upland areas of the Pāhoa, Kalapana, ‘Ōla‘a, and Nā‘ālehu aquifer systems, where recharge decreases by greater than 30 inches. Recharge is estimated to increase along the northeastern coast of the Island of Hawai‘i by as much as 5 inches or more.

Decreases in aquifer-system recharge are estimated for all but three of the Island of Hawai‘i’s aquifer systems (fig. 28*B*). In terms of absolute difference in mean annual recharge, the greatest decreases in aquifer-system recharge are estimated for the Pāhoa, Kalapana, and Nā‘ālehu aquifer systems, where decreases range from 102 to 339 Mgal/d relative to recharge for the 1978–2007 scenario. In terms of percentage change from the 1978–2007 scenario, the greatest decreases in aquifer-system recharge are estimated for Ka‘apuna, Ka Lae, Kīholo, and Manukā aquifer systems, where decreases ranged from 60 to 65 percent. Increases in recharge in the Hakalau, Honoka‘a, and Pa‘auilo aquifer systems ranged from 1.6 to 16 Mgal/d (0.2 to 7 percent).

HRCM2 RCP4.5 2080–99 Scenario

The projected mean annual rainfall anomalies for HRCM2 RCP4.5 2080–99 scenario indicate a mixture of drying and wetting across the Island of Hawai‘i relative to the 1978–2007 scenario (fig. 25*C*). The greatest drying occurs in the central region of the island where mean annual rainfall is projected to decrease by greater than 30 percent. Mean annual rainfall in parts of northwestern, western, and southern sections of the island is

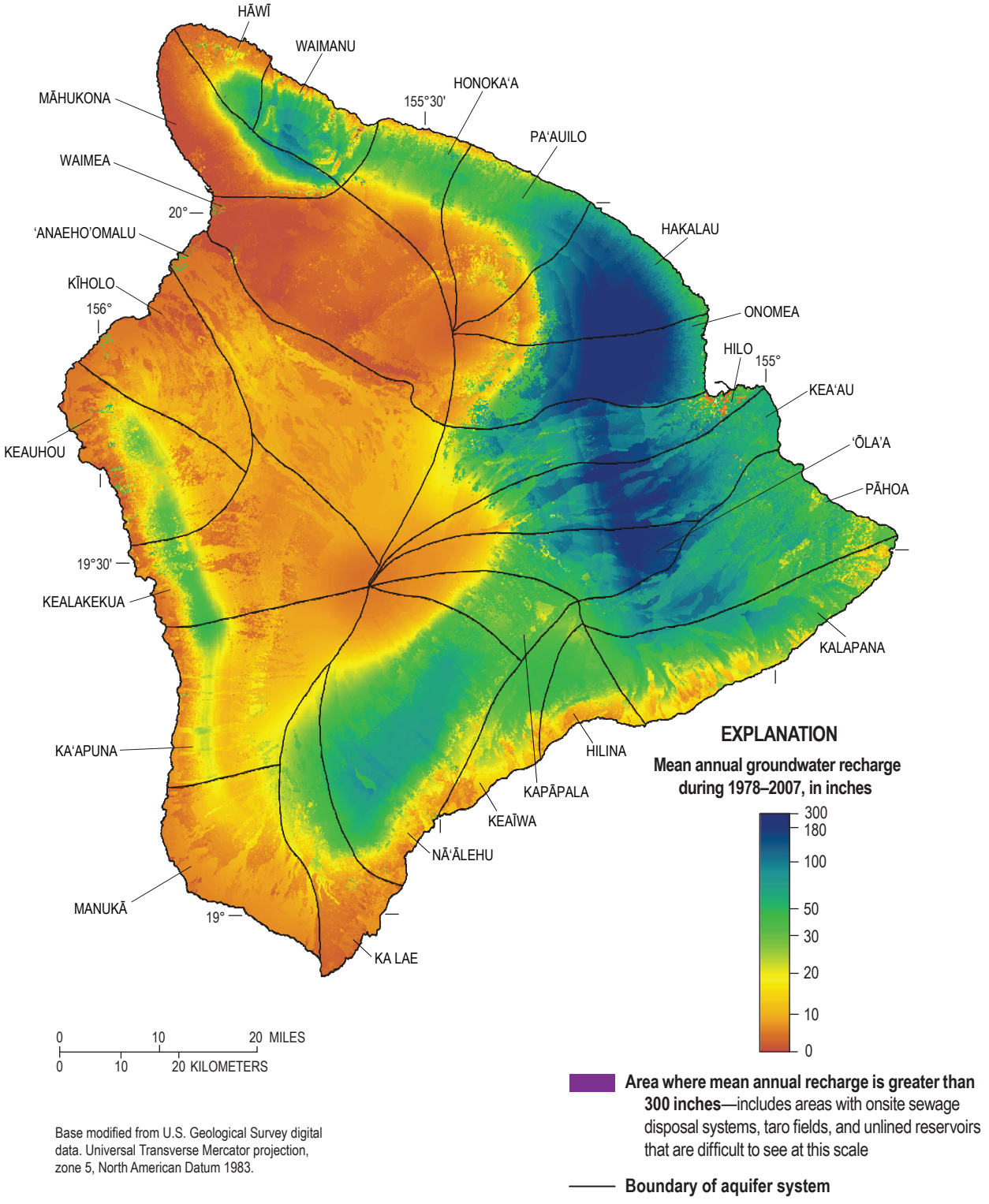


Figure 26. Map showing the distribution of mean annual groundwater recharge for the 1978–2007 scenario and 2020 land-cover conditions for the Island of Hawai'i. Aquifer system boundaries from State of Hawai'i (2014).

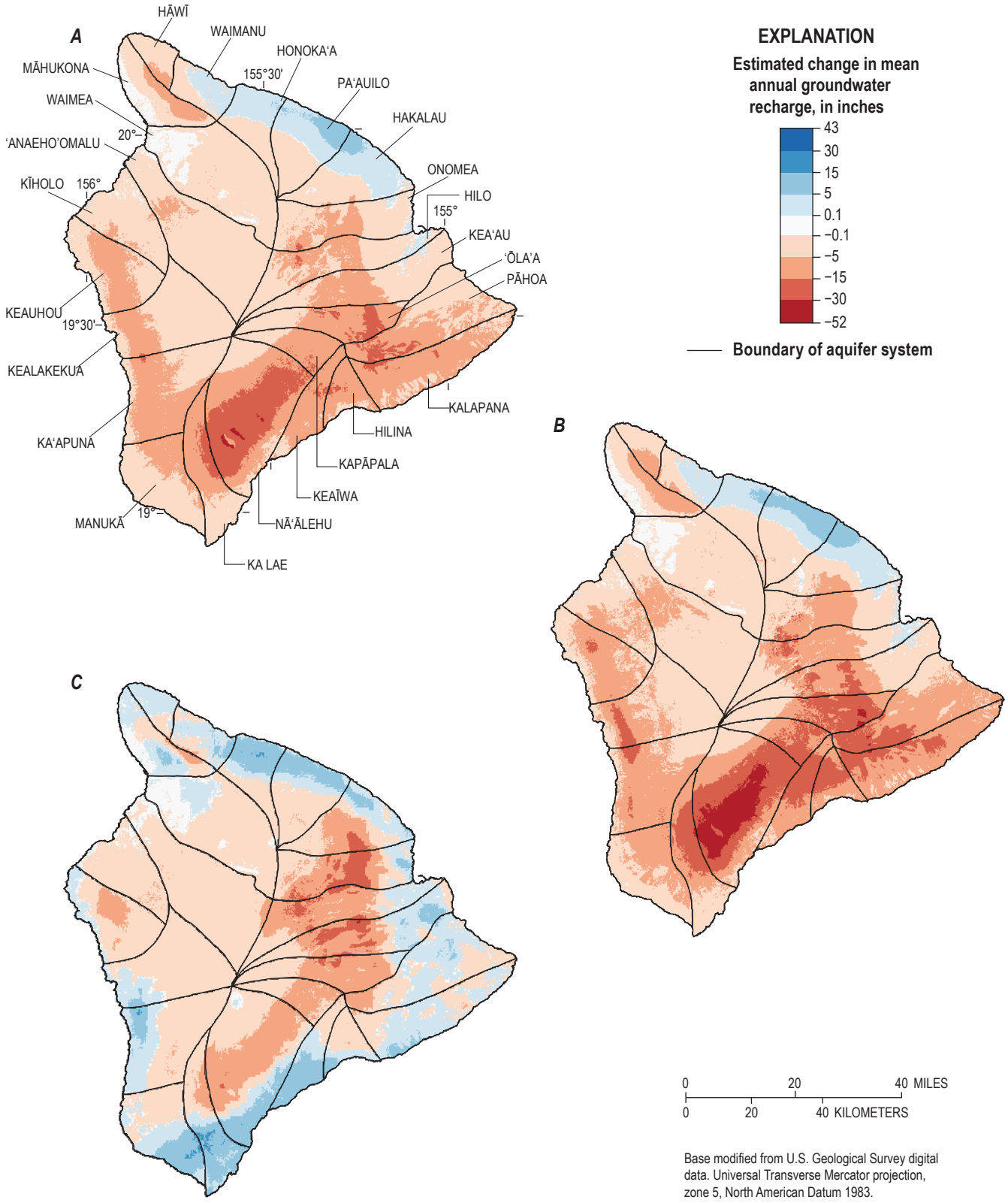


Figure 27. Maps showing estimated change in mean annual groundwater recharge for three future-climate scenarios for the Island of Hawai'i. (*A*) Statistical Downscaling (SD) Representative Concentration Pathway (RCP) 8.5 2041–71 scenario from Elison Timm and others (2015), (*B*) SD RCP8.5 2071–99 scenario from Elison Timm and others (2015), and (*C*) Hawai'i Regional Climate Model version 2 (HRCM2) RCP4.5 2080–99 scenario from Zhang and others (2017). Estimated changes are relative to mean annual recharge for the 1978–2007 scenario. Aquifer system boundaries from State of Hawai'i (2014).

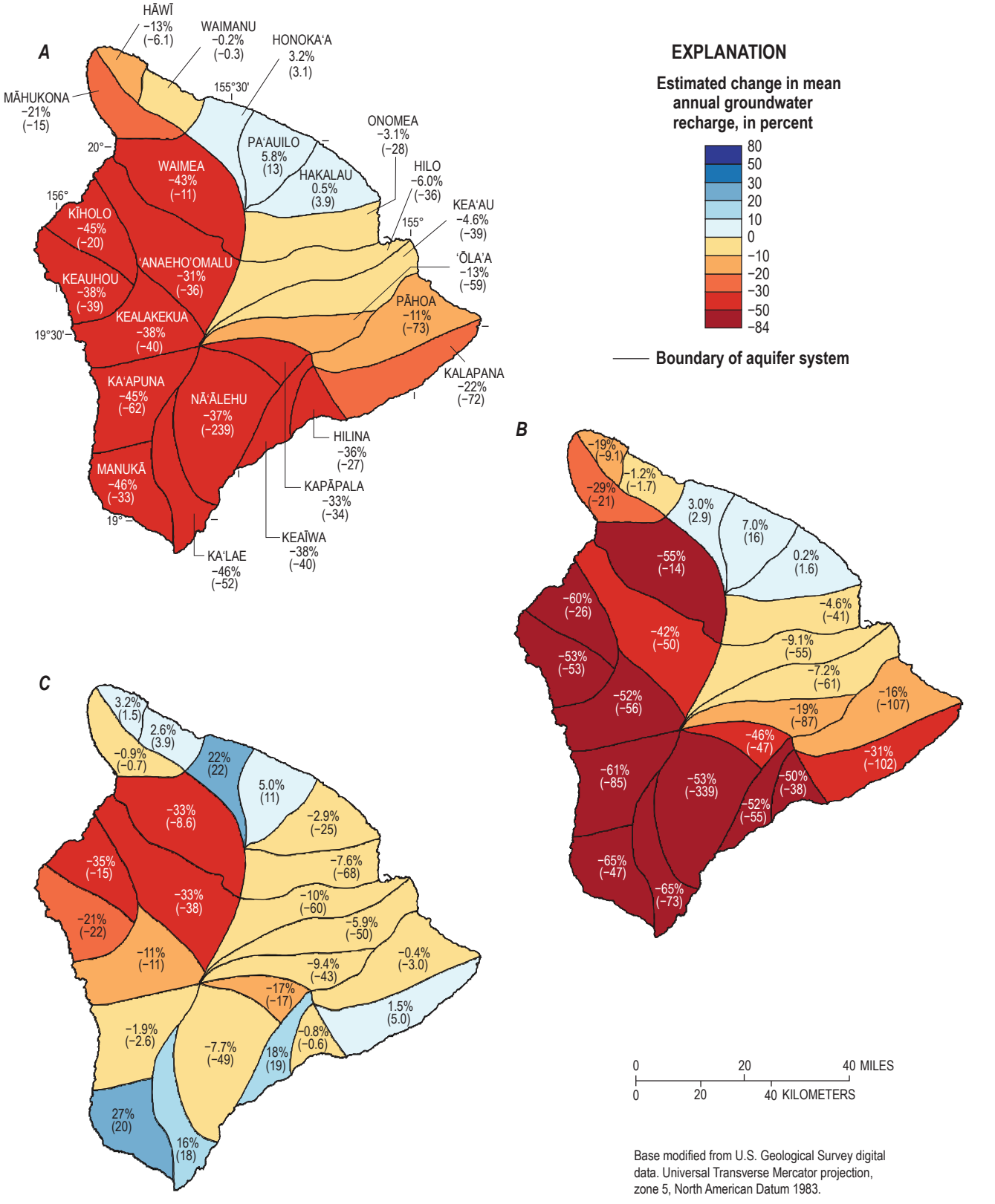


Figure 28. Maps showing estimated change in mean annual groundwater recharge by aquifer system for three future-climate scenarios for the Island of Hawai'i. (*A*) Statistical Downscaling (SD) Representative Concentration Pathway (RCP) 8.5 2041–71 scenario from Elison Timm and others (2015), (*B*) SD RCP8.5 2071–99 scenario from Elison Timm and others (2015), and (*C*) Hawai'i Regional Climate Model version 2 (HRCM2) RCP4.5 2080–99 scenario from Zhang and others (2017). Estimated changes are relative to mean annual recharge for the 1978–2007 scenario. Values in parentheses represent change in millions of gallons per day. Aquifer system boundaries from State of Hawai'i (2014).

projected to increase by greater than 50 percent. Island-wide mean annual rainfall is projected to decrease by 3.6 percent (table 2).

The estimated changes in mean annual recharge include a wide range of increases and decreases across large areas of the Island of Hawai‘i for the HRCM2 RCP4.5 2080–99 scenario (fig. 27*C*). Island-wide mean annual recharge is less than the recharge for the 1978–2007 scenario by roughly 314 Mgal/d (4.6 percent; appendix 2, table 2-6). Mean annual recharge is estimated to increase the most in the northeastern and southern parts of the island. The greatest estimated increase in mean annual recharge occurs in isolated areas of the Nā‘ālehu, Honoka‘a, Ka Lae, Manukā, and Ka‘apuna aquifer systems (fig. 2*F*), where recharge increases by greater than 15 inches (fig. 27*C*). Mean annual recharge is estimated to decrease across much of the Island of Hawai‘i with the greatest decreases occurring in the Hilo and Kea‘au aquifer systems where recharge decreases by greater than 30 inches.

Decreases in aquifer-system recharge are estimated for all but eight of the Island of Hawai‘i’s aquifer systems (fig. 28*C*). In terms of absolute change, the greatest decreases in aquifer-system recharge occur in the Hilo, Kea‘au, Onomea, and Nā‘ālehu aquifer systems, where decreases range from 49 to 68 Mgal/d. In terms of percentage change from the 1978–2007 scenario, the greatest decreases in aquifer-system recharge occur in the Waimea, ‘Anaeho‘omalu, and Kīholo aquifer systems, where decreases range from 33 to 35 percent. The greatest increases in aquifer-system recharge occur in the Honoka‘a, Keaīwa, Ka Lae, and Manukā aquifer systems, where increases range from 18 to 22 Mgal/d (16 to 27 percent).

Implications of Estimated Changes in Groundwater Recharge

The water-budget simulations for the mid-century SD RCP8.5 2041–71 scenario and dry-climate SD RCP8.5 2071–99 scenario estimate decreases in mean annual groundwater recharge across much of Kaua‘i, O‘ahu, Moloka‘i, Lāna‘i, Maui, and the Island of Hawai‘i. Island-wide recharge for these six islands is estimated to decrease by 9.6 to 41 percent for the mid-century SD RCP8.5 2041–71 scenario and by 14 to 55 percent for the dry-climate SD RCP8.5 2071–99 scenario (fig. 29). The greatest decreases in island-wide recharge occur on Lāna‘i, where island-wide recharge is estimated to decrease by 41 percent for the SD RCP8.5 2041–71 scenario and 55 percent for the SD RCP8.5 2071–99 scenario. In terms of aquifer-system recharge, decreases are estimated for all aquifer systems on each island for the SD RCP8.5 2041–71 and SD RCP8.5 2071–99 scenarios, except for three aquifer systems on Maui (Honopou, Kawaipapa, and Waikamoi) and three aquifer systems on the Island of Hawai‘i (Hakalau, Honoka‘a, and Pa‘auilo), where nominal increases in aquifer-system recharge are estimated for both scenarios. The greatest decreases in aquifer-system recharge occur in the drier, leeward parts of each island, where aquifer-system recharge is estimated to decrease by 50 percent or more on Kaua‘i (Kekaha), O‘ahu (Kea‘au, Wai‘anae, Lualualei, Nānākuli, Makaīwa, and ‘Ewa-Kunia), Moloka‘i (Kaluako‘i), Lāna‘i (Leeward and Kaumalapau), and Maui (Ukumehame, Kama‘ole, and Lualaʻilua). The estimated changes in mean annual recharge for the SD RCP8.5 2041–71 and SD RCP8.5 2071–99 scenarios are consistent with the projected changes in mean annual rainfall.

Increases and decreases in mean annual groundwater recharge are estimated for each island for the wet-climate scenarios, including the HRCM1 A1B 2080–99 scenario for Maui, HRCM2 RCP4.5 2080–99 scenario for Kaua‘i, Lāna‘i, and the Island of Hawai‘i, and HRCM2 RCP8.5 2080–99 scenario for O‘ahu and Moloka‘i. Island-wide recharge is estimated to increase on Kaua‘i, O‘ahu, Moloka‘i, Lāna‘i, and Maui by 2.3 to 43 percent, but decrease on the Island of Hawai‘i by 4.6 percent for the wet-climate scenarios (fig. 29). Increases and decreases in aquifer-system recharge are also estimated for the aquifer systems on each island, except for Lāna‘i where recharge is estimated to increase in all aquifer systems. Aquifer-system recharge is estimated to increase for the wet-climate scenario by as much as 80 percent in 79 aquifer systems—Kaua‘i (10 aquifer systems, fig. 10*C*), O‘ahu (20 aquifer systems, fig. 13*C*), Moloka‘i (14 aquifer systems, fig. 16*C*), Lāna‘i (9 aquifer systems, fig. 20*C*), Maui (18 aquifer systems, fig. 24*C*), and the Island of Hawai‘i (8 aquifer systems, fig. 28*C*). Aquifer-system recharge is estimated to decrease by as much as 35 percent in 31 aquifer systems across the islands—Kaua‘i (3 aquifer systems, fig. 10*C*), O‘ahu (3 aquifer systems, fig. 13*C*), Moloka‘i (2 aquifer systems, fig. 16*C*), Maui (7 aquifer systems, fig. 24*C*), and the Island of Hawai‘i (16 aquifer systems, fig. 28*C*).

In terms of a consistent trend among all three future-climate scenarios, 30 of the 31 aquifer systems with estimated decreases in aquifer-system recharge on all six islands for the wet-climate scenarios also show estimated decreases in recharge for the mid-century and dry-climate scenarios. In contrast, only 5 of the 79 aquifer systems with estimated increases in recharge for the wet-climate scenarios also show estimated increases in recharge for the mid-century and dry-climate scenarios. About one-third of the aquifer systems, 35 out of a total of 110, on all six islands show similar changes (in terms of direction of change) in estimated aquifer-system recharge for the mid-century, dry-climate, and wet-climate scenarios. The remaining 75 aquifer systems on all six islands show contrasting changes in estimated aquifer-system recharge. The contrast in change direction and change magnitude among the three future-climate scenarios provides a range of uncertainty among available climate projections and their estimated effects on groundwater recharge, although the range might not capture the true future conditions.

The contrast in estimated changes to aquifer-system recharge is also evident when distinguishing the future-climate scenarios by the approach used to downscale rainfall projections for the Hawaiian Islands. The mid-century and dry-climate scenarios both use rainfall projections derived from a statistical downscaling approach, whereas the wet-climate

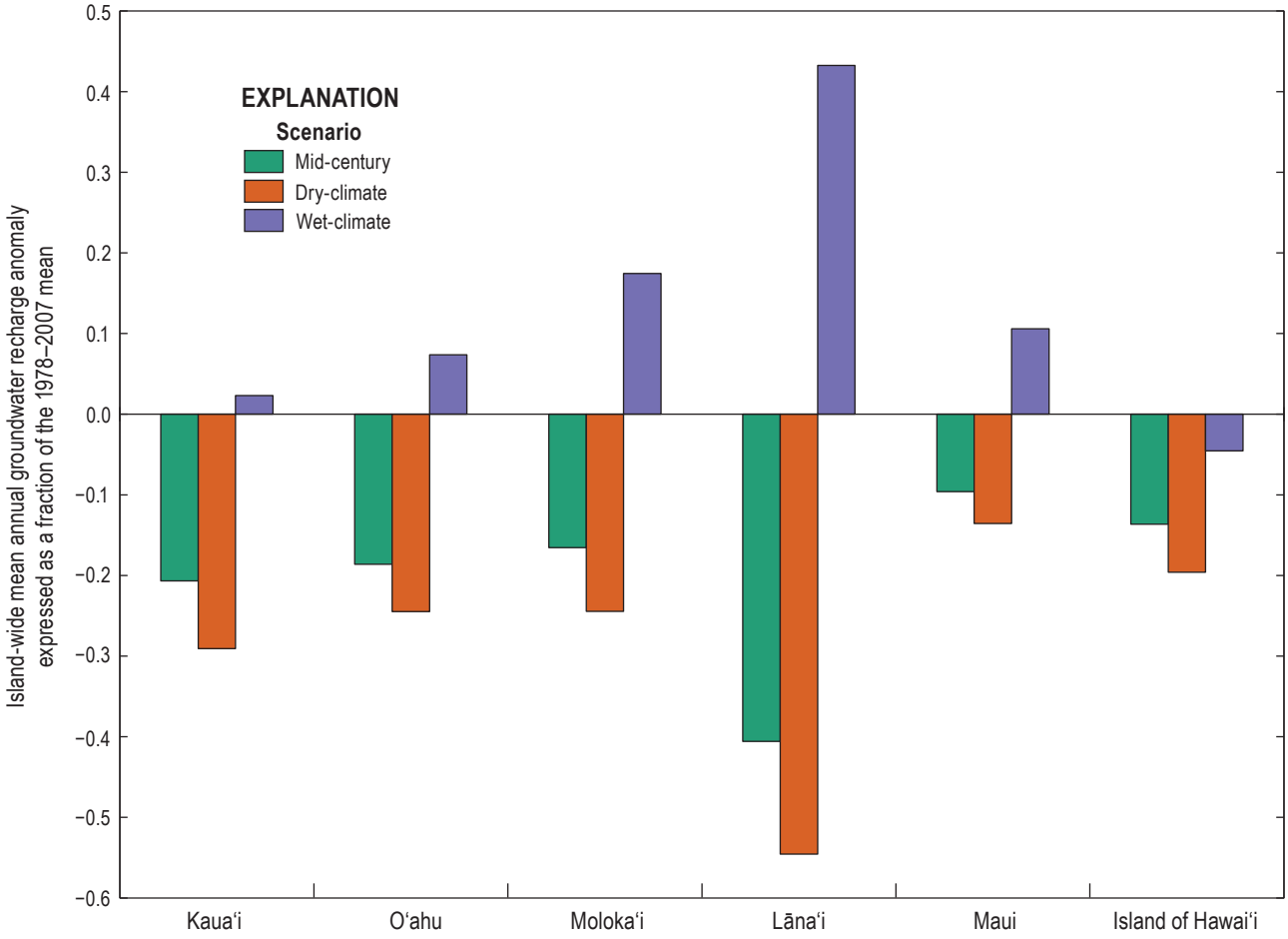


Figure 29. Graph of island-wide mean annual groundwater recharge anomalies for the mid-century, dry-climate, and wet-climate scenarios for Kauaʻi, Oʻahu, Molokaʻi, Lānaʻi, Maui, and the Island of Hawaiʻi.

scenarios use rainfall projections derived from a dynamical downscaling approach (see the "Downscaled Climate Projections for the Hawaiian Islands" section of this report). In terms of a contrasting trend between the two downscaling approaches, 75 out of 110 aquifer systems show contrasting changes (in terms of direction or sense) in estimated aquifer-system recharge for the projections derived from statistical and dynamical downscaling approaches. The 75 aquifer systems with contrasting changes in estimated recharge are spread across all six islands—Kauaʻi (10 aquifer systems), Oʻahu (20 aquifer systems), Molokaʻi (14 aquifer systems), Lānaʻi (9 aquifer systems), Maui (15 aquifer systems), and the Island of Hawaiʻi (7 aquifer systems). In 74 of the 75 aquifer systems with contrasting changes (in terms of direction or sense), the scenarios derived from statistical downscaling show decreases in recharge and the scenarios derived from dynamical downscaling show increases in recharge. The Hakalau aquifer system on the Island of Hawaiʻi is the only aquifer system where scenarios derived from statistical downscaling show an increase in recharge and the scenario derived from dynamical downscaling shows a decrease in recharge. The contrasts in recharge changes (in both direction and magnitude) extend to multiple aquifer systems on each island and reflect uncertainty in projected mid- and end-of-century rainfall derived from statistical and dynamical downscaling. Determining the cause of the differences in rainfall projections derived from statistical and dynamical downscaling is beyond the scope of this study.

The results of this study indicate estimated changes in the amount and distribution of groundwater recharge on Kauaʻi, Oʻahu, Molokaʻi, Lānaʻi, Maui, and the Island of Hawaiʻi that could affect groundwater availability. Furthermore, these changes in recharge could also affect groundwater levels and salinity in production wells that supply potable drinking water. Changes in recharge could also affect aquatic ecosystems and traditional cultural practices on these islands that rely on groundwater discharge to streams and nearshore marine environments. The scope of this study does not include an assessment of groundwater availability related to the estimated changes in groundwater recharge for the future-climate scenarios included in this study.

Effects of Selected Climate Inputs on Estimated Recharge

In this study, a set of water-budget model inputs was developed for each island on the basis of projected climate change for three future-climate scenarios and one drought scenario (Lānaʻi only). The model inputs were then used in WATRMod for each island to estimate groundwater recharge and other water-budget components. To analyze the effect that future-scenario model inputs have on estimated recharge, the water budget for each island was rerun successively, changing only one model input per run for a future-climate condition while keeping all other model inputs the same as the reference climate condition. Two separate sets of future-scenario model input parameters were tested to assess the effect of future-scenario model inputs on estimated recharge. A set of five input parameters characterizing rainfall, runoff, canopy evaporation, and reference ET were tested for each future-climate scenario to compare island-wide recharge estimates for each island to reference climate conditions. Inputs of future monthly rainfall, developed using a set of seasonal change factors derived from projected changes in rainfall, were tested for each future-climate scenario for Lānaʻi and Oʻahu.

Input Parameters Characterizing Rainfall, Runoff, Canopy Evaporation, and Reference ET

The five input parameters tested for each selected future-climate scenario were (1) monthly rainfall and seasonal runoff-to-rainfall ratios, (2) mean monthly reference ET adjusted for projected warming, (3) mean monthly reference ET adjusted for projected warming and increases in atmospheric CO_2 concentrations, (4) the ratio of mean evaporation rate to mean precipitation rate during saturated conditions, *V*, and (5) daily rainfall fragments. The combined effect of these input parameters was also tested using the following sets of parameters: (1) a combination of monthly rainfall and seasonal runoff-to-rainfall ratios, mean monthly reference ET adjusted for projected warming, the ratio of mean evaporation rate to mean precipitation rate during saturated conditions, *V*, and daily rainfall fragments, and (2) a combination of monthly rainfall and seasonal runoff-to-rainfall ratios, mean monthly reference ET adjusted for projected warming and increases in atmospheric CO_2 concentrations, the ratio of mean evaporation rate to mean precipitation rate during saturated conditions, *V*, and daily rainfall fragments.

The monthly rainfall and seasonal runoff-to-rainfall ratio datasets were treated as one input parameter because the estimation of runoff for the future-climate and drought scenarios is calculated from rainfall in this study (see "Direct Runoff" section). The model input datasets of mean monthly reference ET, the ratio of mean evaporation rate to mean precipitation rate during saturated conditions, *V*, and daily rainfall fragments were developed using the climate datasets from the HRCM1 1990–2009 and HRCM2 1990–2009 simulations, and the HRCM1 A1B 2080–99, HRCM2 RCP4.5 2080–99, and HRCM2 RCP8.5 2080–99 scenarios (see appendix 3 for a description of the development of model inputs for the future-climate scenarios). The recharge estimates for these analyses were compared with island-wide recharge estimates of a reference climate condition for each island (table 12).

For this analysis, the reference climate condition for all islands was the 1978–2007 scenario and daily rainfall fragments were from the HRCM2 1990–2009 simulation, with exception of Maui which used daily rainfall fragments from the HRCM1 1990–2009 simulation. The HRCM1 A1B 2080–99 scenario was used as the future-climate scenario for Maui, the HRCM2 RCP4.5 2080–99 scenario was used as the future-climate scenario for Kauaʻi, Lānaʻi, and the Island of Hawaiʻi, and the HRCM2 RCP8.5 2080–99 scenario was used as the future-climate scenario for Oʻahu and Molokaʻi. The future-climate scenario selected for each island represents the wettest scenario, in terms of projected changes in island-wide rainfall, among the available set of projections (table 2).

HRCM1 A1B 2080–99 Scenario for Maui

Monthly rainfall and seasonal runoff-to-rainfall ratios have the largest influence on recharge estimates among the five parameter inputs for Maui (table 12). The dependence of recharge on rainfall is evident in the similarities between projected increases in rainfall and estimated increases in recharge. For example, projected rainfall and estimated recharge increase by 10 and 11 percent, respectively (tables 2 and 12). Recharge estimates are also affected by mean monthly reference ET adjusted for projected warming and increases in atmospheric CO_2 concentrations. For example, adjusting future mean monthly reference ET for projected warming of 2.1 to 3 °C decreases recharge by 0.6 percent (tables 3 and 12). However, adjusting future mean monthly reference ET for projected warming and an increase of 315 parts per million (ppm) in mean atmospheric CO_2 concentrations increases recharge by 1.4 percent. Increases in atmospheric CO_2 concentrations might act to suppress ET by decreasing plant leaf stomatal conductance, which can decrease actual ET and increase soil moisture available for recharge in WATRMod calculations. These results indicate that decreases in island-wide recharge owing to projected warming are more than offset by enhanced recharge owing to projected increases in mean atmospheric CO_2 concentrations. Hence, projected increases in CO_2 concentrations result in a net positive effect on island-wide recharge.

If monthly rainfall and seasonal runoff-to-rainfall ratios are the only changes included in the future-scenario simulation, then estimated changes in recharge for Maui come close to matching the values estimated when projected changes for three model input parameters (monthly rainfall and seasonal runoff-to-rainfall ratios; *V*; daily rainfall fragments) plus mean monthly reference ET adjusted for projected warming are included. For example, the estimated changes in recharge for Maui differ by only 1.7 Mgal/d from the estimated changes considering

Table 12. Results of testing for selected water-budget model input parameters.

[HRCM1, Hawai‘i Regional Climate Model version 1 from Zhang and others (2016a, b); HRCM2, Hawai‘i Regional Climate Model version 2 from Zhang and Wang (2017); RCP4.5 and RCP8.5, Representative Concentration Pathway with total radiative forcing of 4.5 and 8.5 Watts per square meter, respectively, by the year 2100 from Intergovernmental Panel on Climate Change (2014); A1B, A1B emission scenario from Nakićenović and others (2000) and the Intergovernmental Panel on Climate Change (2014); reference climate condition for Maui is 1978–2007 scenario and daily rainfall fragments from HRCM1 1990–2009 simulation from Zhang and others (2016a); reference climate condition for all other islands is 1978–2007 scenario and daily rainfall fragments from HRCM2 1990–2009 simulation from Zhang and Wang (2017); future climate condition for Maui is HRCM1 A1B 2080–99 scenario from Zhang and others (2016b); future climate condition for Kaua‘i, Lāna‘i, and Island of Hawai‘i is HRCM2 RCP4.5 2080–99 scenario from Zhang and Wang (2017); future climate condition for O‘ahu and Moloka‘i is HRCM2 RCP8.5 2080–99 scenario from Zhang and Wang (2017); ET, evapotranspiration; CO_2, carbon dioxide; ppm, parts per million; <, less than]

Parameter	Change in island-wide recharge for future climate condition relative to reference climate condition, in percent; (change in million gallons per day)					
	Kaua‘i	O‘ahu	Moloka‘i	Lāna‘i	Maui	Island of Hawai‘i
	HRCM2 RCP4.5 2080–99	HRCM2 RCP8.5 2080–99	HRCM2 RCP8.5 2080–99	HRCM2 RCP4.5 2080–99	HRCM1 A1B 2080–99	HRCM2 RCP4.5 2080–99
Monthly rainfall, seasonal runoff-to-rainfall ratios	2.3 (20.8)	6.8 (45.7)	16.6 (44)	33.8 (12.2)	10.7 (128.2)	–4.4 (–325.7)
Mean monthly reference ET, adjusted for projected warming	–1.4 (–12.7)	–2.9 (–19.8)	–2.4 (–6.4)	–1.7 (–0.6)	–0.6 (–7.6)	–1.5 (–111.9)
Mean monthly reference ET, adjusted for projected warming and increased atmospheric CO_2 concentrations[a]	–0.4 (–3.2)	0.3 (2.3)	0.4 (0.9)	–0.4 (–0.2)	1.4 (16.3)	–0.7 (–50)
Ratio of the mean evaporation rate to mean precipitation rate during saturated conditions, *V*	0.5 (4)	1.1 (7.4)	0.9 (2.4)	–1.2 (–0.4)	0.9 (11.2)	<–0.1 (–3.1)
Daily rainfall fragments	–0.4 (–3.4)	–1.8 (–12.4)	–1.2 (–3.1)	–1.0 (–0.4)	–0.3 (–4.0)	0.4 (31.9)
Monthly rainfall; seasonal runoff-to-rainfall ratios; daily rainfall fragments; mean monthly reference ET, adjusted for projected warming; ratio of the mean evaporation rate to mean precipitation rate during saturated conditions, *V*	1.1 (9.8)	3.3 (22.3)	14.4 (38)	30.6 (11.1)	10.8 (129.9)	–5.4 (–402.7)
Monthly rainfall; seasonal runoff-to-rainfall ratios; daily rainfall fragments; mean monthly reference ET, adjusted for projected warming and increased atmospheric CO_2 concentrations[a]; ratio of the mean evaporation rate to mean precipitation rate during saturated conditions, *V*	2.2 (19.7)	6.6 (44.7)	17.3 (45.8)	31.9 (11.5)	12.9 (154.7)	–4.6 (–344.8)

[a]An increase of 164 ppm in atmospheric CO_2 above HRCM2 1990–2009 simulation conditions was used for the HRCM2 RCP4.5 2080-99 scenario; an increase of 472 ppm in atmospheric CO_2 concentrations above HRCM2 1990–2009 simulation conditions was used for the HRCM2 RCP8.5 2080-99 scenario; an increase of 315 ppm in atmospheric CO_2 concentrations above HRCM1 1990–2009 simulation conditions was used for the HRCM1 A1B 2080-99 scenario.

all combined effects and no change in mean atmospheric CO_2 concentrations (table 12). However, the estimated increase in recharge is 17 percent (26 Mgal/d) less than the estimated increase when considering all combined effects plus mean monthly reference ET adjusted for projected warming and increased mean atmospheric CO_2 concentrations.

Including future estimates of *V* increased recharge by 0.9 percent (table 12). A decrease in *V* effectively reduces canopy evaporation and increases recharge in WATRMod calculations, whereas an increase in *V* enhances canopy evaporation and decreases recharge. Increases in *V* are estimated for dry areas around western Maui and along the western slope of Haleakalā (fig. 3-1*E*, appendix 3). Decreases in *V* are estimated across much of the wet, forested areas of western Maui and eastern part of Haleakalā. Increases in estimated recharge in wet, forested areas with reduced *V* more than compensate for the decreased recharge rates in areas with increased *V,* resulting in a net positive effect on island-wide recharge.

Daily rainfall fragments control the magnitude and timing of daily rainfall within a given month in the WATRMod simulations for each scenario. Therefore, including future daily rainfall fragments can induce changes in the patterns of daily rainfall and increase or decrease recharge estimates in a given month. Future patterns of daily rainfall that concentrate the amount of rainfall during a given month over a fewer number of days (higher fragment values, fewer non-zero fragments) can result in decreased canopy evaporation and increased recharge when compared to the reference climate condition. Conversely, future patterns that distribute the amount of rainfall during a given month over a greater number of days (lower fragment values, more non-zero fragments) can increase canopy evaporation and decrease recharge. For example, the inclusion of future daily rainfall fragments decreases island-wide recharge for Maui by 0.3 percent (table 12).

HRCM2 RCP4.5 2080–99 Scenario for Kaua‘i, Lāna‘i, and the Island of Hawai‘i

Monthly rainfall and seasonal runoff-to-rainfall ratios have the largest influence on recharge estimates among the five parameter inputs for Kaua‘i, Lāna‘i, and the Island of Hawai‘i (table 12). The dependence of recharge on rainfall is evident in the similarities between projected increases in rainfall and estimated increases in recharge for Kaua‘i, and between projected decreases in rainfall and estimated decreases in recharge for the Island of Hawai‘i. For example, projected rainfall and estimated recharge increase by 1.9 and 2.3 percent, respectively, on Kaua‘i and decrease by 3.6 and 4.4 percent, respectively, on the Island of Hawai‘i (tables 2 and 12). For Lāna‘i, projected rainfall increases by 7.2 percent whereas estimated recharge increases by 34 percent. The disproportional increase in estimated mean annual recharge for Lāna‘i is due, in part, to non-uniform projected changes in mean monthly rainfall for selected months during the wet season. For example, mean monthly rainfall is projected to increase by 35 to 130 percent during the months of November, January, and March on Lāna‘i but decrease by as much as 55 percent during most of the remaining months of the year (fig. 21). These changes in mean monthly rainfall correspond to increases of 56 to 288 percent in mean monthly recharge during November, January, and March, and decreases of up to 82 percent in mean monthly recharge during most of the remaining months of the year (fig. 21). The estimated increases in mean monthly recharge in November, January, and March more than compensate for the decrease in recharge during most of the remaining months, resulting in a net positive effect on island-wide mean annual recharge for Lāna‘i.

Recharge estimates for Kaua‘i, Lāna‘i, and the Island of Hawai‘i are also affected by adjustments in mean monthly reference ET for projected warming and decrease by 1.4 to 1.7 percent. Decreases in island-wide recharge owing to projected warming for Kaua‘i, Lāna‘i, and the Island of Hawai‘i are partly offset by increases in recharge owing to increased atmospheric CO_2 concentrations. For example, adjusting future mean monthly reference ET for projected warming and an increase of 164 ppm in atmospheric CO_2 concentrations decreases recharge by 0.4 to 0.7 percent.

Including future estimates of *V* increases island-wide recharge by 0.5 percent on Kaua‘i but decreases recharge by 1.2 percent or less on Lāna‘i and the Island of Hawai‘i (table 12). Increases in *V* are estimated for the northwestern part of Kaua‘i, whereas decreases in *V* are estimated across much of eastern and southern Kaua‘i (fig. 3-1A, appendix 3). The increases in estimated recharge in forested areas with reduced *V* more than compensate for the decreased recharge rates in areas with increased *V* resulting in a net positive effect on island-wide recharge for Kaua‘i. For Lāna‘i, estimated increases in *V* around Lāna‘ihale result in a negative effect on island-wide recharge (fig. 3-1*D*, appendix 3). Increases in *V* are estimated for much of the central parts of the Island of Hawai‘i including the upland slopes of Mauna Kea and Mauna Loa, whereas decreases in *V* are estimated for coastal areas northeast of Mauna Kea and south and west of Mauna Loa (fig. 3-1*F*, appendix 3). Increases in estimated recharge in areas with decreased *V* are offset by decreases in recharge in forested areas with increased *V*, resulting in a slight net negative effect on island-wide recharge for the Island of Hawai‘i.

The inclusion of future daily rainfall fragments decreases island-wide recharge by 0.4 to 1 percent on Kaua‘i and Lāna‘i but increases recharge by 0.4 percent on the Island of Hawai‘i (table 12). These changes indicate that projected changes in daily rainfall frequency for the HRCM2 RCP4.5 2080–99 scenario have a small but non-negligible effect on island-wide recharge estimates.

HRCM2 RCP8.5 2080–99 Scenario for O‘ahu and Moloka‘i

Monthly rainfall and seasonal runoff-to-rainfall ratios have the largest influence on recharge estimates among the five parameter inputs for O‘ahu and Moloka‘i. For example, projected rainfall and estimated recharge increase by 4 and 7 percent, respectively, on O‘ahu and increase by 12 and 17 percent, respectively, on Moloka‘i (tables 2 and 12). These results indicate some disproportional increases in recharge are associated with the projected increases in rainfall for O‘ahu and Moloka‘i.

The recharge estimates for O‘ahu and Moloka‘i are also affected by mean monthly reference ET adjusted for projected warming and decrease by 2.4 to 2.9 percent. The adjustments in mean monthly reference ET have a larger effect on recharge estimates when compared to the HRCM1 A1B 2080–99 scenario for Maui and HRCM2 RCP4.5 2080–99 scenario for Kaua‘i, Lāna‘i, and the Island of Hawai‘i. The larger effect of these adjustments are expected with the HRCM2 RCP8.5 2080–99 scenario owing to greater projected rates of warming (table 3). Decreases in island-wide recharge owing to projected warming for O‘ahu and Moloka‘i are completely offset by increases in recharge owing to increased mean atmospheric CO_2 concentrations. For example, adjusting future mean monthly reference ET for projected warming and an increase of 472 ppm in atmospheric CO_2 concentrations increases recharge by 0.3 and 0.4 percent for O‘ahu and Moloka‘i, respectively (tables 3 and 12).

Including future estimates of *V* increases island-wide recharge by 1.1 and 0.9 percent on O‘ahu and Moloka‘i, respectively (table 12). Increases in *V* are estimated for central and leeward parts of O‘ahu and Moloka‘i, whereas decreases in *V* are estimated across much of the windward areas (figs. 3-1B and 3-1C, appendix 3). The increases in estimated recharge in forested areas with reduced *V* more than compensate for the decreased recharge rates in areas with increased *V*, resulting in a net positive effect on island-wide recharge for O‘ahu and Moloka‘i.

The inclusion of future daily rainfall fragments decreases island-wide recharge by 1.8 and 1.2 percent on O‘ahu and Moloka‘i, respectively (table 12). The inclusion of future daily rainfall fragments has a larger effect on island-wide recharge estimates when compared to the HRCM1 A1B 2080–99 and HRCM2 RCP4.5 2080–99 scenarios for Kaua‘i, Lāna‘i, and Maui, which indicates changes in daily rainfall frequency for the HRCM2 RCP8.5 2080–99 scenario that further enhance canopy evaporation and reduce net precipitation.

Future Monthly Rainfall

In this study, the change-factor method was used to develop inputs of future monthly rainfall by adjusting datasets of monthly rainfall during the reference period with relative change factors. For the statistical downscaling set of projections (SD RCP8.5 2041–71 and SD RCP8.5 2071–99 scenarios), change factors were calculated for each island using wet- and dry-season rainfall anomalies (Elison Timm and others, 2015). For the HRCM2 set of projections, change factors were calculated using mean monthly rainfall anomalies computed from the differences in monthly rainfall means between the HRCM2 1990–2009 simulation and the HRCM2 RCP4.5 2080–99 or HRCM2 RCP8.5 2080–99 scenarios (Zhang and Wang, 2017). To analyze the effect of using projected changes in mean seasonal rainfall to estimate monthly rainfall change for the HRCM2 set of projections, relative change factors were calculated using mean wet- and dry-season rainfall anomalies computed from the differences in seasonal rainfall means between the HRCM2 1990–2009 simulation and the HRCM2 RCP4.5 2080–99 or HRCM2 RCP8.5 2080–99 scenarios. The relative change factors were then used to develop model inputs of future monthly rainfall. WATRMod was rerun using inputs of seasonally adjusted monthly rainfall, keeping all other model inputs the same including the seasonal runoff-to-rainfall ratios. O‘ahu was selected for these analyses because of the large demand for groundwater. Lāna‘i was also selected because of the pronounced variability in projected changes to mean monthly rainfall and mean monthly recharge for the HRCM2 RCP4.5 2080–99 scenario (fig. 21). The recharge estimates for these analyses were compared with island-wide and aquifer-system recharge estimates of a reference climate condition for Lāna‘i (table 13) and O‘ahu (table 14).

Table 13. Results of testing for effect of monthly rainfall change factors for aquifer systems and all of Lāna‘i for the HRCM2 RCP4.5 2080–99 scenario.

[See figure 2 for locations of aquifer systems; HRCM2, Hawai‘i Regional Climate Model version 2 from Zhang and Wang (2017); RCP4.5, Representative Concentration Pathway with total radiative forcing of 4.5 Watts per square meter by the year 2100 from Intergovernmental Panel on Climate Change (2014); ETo, reference evapotranspiration for grass surface; *V*, ratio of the mean evaporation rate to mean precipitation rate during saturated conditions; reference climate condition is 1978–2007 rainfall from Frazier and others (2016), daily rainfall fragments from Longman and others (2019), ETo from Giambelluca and others (2014), *V* ratios computed using equation 11 for a 1978–2007 scenario, and 2020 land cover]

Aquifer system	Change in recharge for the HRCM2 RCP4.5 2080–99 scenario relative to reference climate condition, in percent (million gallons per day)	
	Monthly rainfall input computed using monthly rainfall change factors	Monthly rainfall input computed using seasonally adjusted monthly rainfall change factors
Island of Lāna‘i	43.3 (11.6)	30.6 (8.2)
Leeward	38.9 (1.7)	27.2 (1.2)
Windward	9.7 (0.4)	4.7 (0.2)
Honopū	48.9 (1.5)	32.2 (1)
Kaumalapau	80.1 (1.5)	59.1 (1.1)
Keālia	73.1 (1.4)	50.5 (0.9)
Mānele	65.2 (1.9)	50.2 (1.4)
Hauola	34.2 (0.8)	23.2 (0.6)
Maunalei	36.6 (0.7)	30.2 (0.6)
Pao-ma‘i	42.9 (1.7)	29.8 (1.2)

Table 14. Results of testing for effect of monthly rainfall change factors for aquifer systems and all of O‘ahu for the HRCM2 RCP8.5 2080–99 scenario.

[See figure 2 for locations of aquifer systems; Water-budget area descriptions: A, entire aquifer-system area; B, entire aquifer-system area excluding areas seaward of the 0-ft altitude contour of the top of the volcanic aquifer (see fig. 2); HRCM2, Hawai‘i Regional Climate Model version 2 from Zhang and Wang (2017); RCP8.5, Representative Concentration Pathway with total radiative forcing of 8.5 Watts per square meter by the year 2100 from Intergovernmental Panel on Climate Change (2014); ETo, reference evapotranspiration for grass surface; *V*, ratio of the mean evaporation rate to mean precipitation rate during saturated conditions; reference climate condition is 1978–2007 rainfall from Frazier and others (2016), daily rainfall fragments from Longman and others (2019), ETo from Giambelluca and others (2014), *V* ratios computed using equation 11 for a 1978–2007 scenario, and 2020 land cover; <, less than]

Aquifer system	Water-budget area description	Change in recharge for the HRCM2 RCP8.5 2080–99 scenario relative to reference climate condition, in percent (million gallons per day)	
		Monthly rainfall input computed using monthly rainfall change factors	Monthly rainfall input computed using seasonally adjusted monthly rainfall change factors
Island of O‘ahu	A	7.4 (44.1)	5.3 (31.7)
	B	7.1 (39.7)	5.3 (29.7)
Wahiawa	A	1.8 (2.2)	0.8 (1)
Kalihi	A	12.6 (1.2)	10.8 (1)
	B	12.6 (1.1)	11.2 (1)
Moanalua	A	11.3 (2.1)	9.3 (1.8)
	B	11 (1.9)	9.7 (1.6)
Nu‘uanu	A	13.8 (2.5)	11.7 (2.1)
	B	13.4 (2.3)	11.8 (2)
Pālolo	A	13 (1)	10 (0.8)
	B	12.2 (0.8)	10.1 (0.7)
Wai‘alae-East	A	26.5 (2.1)	19.2 (1.5)
	B	26.3 (1.7)	19.8 (1.3)
Wai‘alae-West	A	19.7 (0.9)	15.1 (0.7)
	B	19 (0.8)	15.5 (0.7)
Kawailoa	A	9.2 (2.8)	4.9 (1.5)
	B	9.2 (2.7)	5.1 (1.5)
Mokulē‘ia	A	–10.7 (–1.9)	–15 (–2.7)
	B	–11.1 (–1.7)	–15 (–2.2)
Waialua	A	3.2 (0.3)	–4 (–0.4)
	B	3.8 (0.3)	–3.9 (–0.3)
‘Ewa-Kunia	A	4.4 (0.5)	–0.1 (<–0.1)
	B	3.7 (0.3)	–1 (–0.1)
Makaīwa	A	7.6 (0.1)	4.3 (<0.1)
	B	8.6 (0.1)	5.7 (<0.1)
Waimalu	A	7.4 (4.3)	6.5 (3.8)
	B	7.3 (4.2)	6.5 (3.8)
Waipahu-Waiawa	A	2.7 (2.4)	1.7 (1.5)
	B	2.3 (1.8)	1.4 (1.1)
Kea‘au	A	0.2 (<0.1)	–6.5 (–0.4)
	B	–0.6 (<–0.1)	–7 (–0.4)
Lualualei	A	–0.3 (<–0.1)	–6.1 (–0.5)
	B	–1.7 (–0.1)	–7.7 (–0.4)
Mākaha	A	–2.8 (–0.2)	–6.1 (–0.5)
	B	–3.9 (–0.3)	–6.9 (–0.5)
Nānākuli	A	1.9 (<0.1)	–1.4 (<–0.1)
	B	1.6 (<0.1)	–1.6 (<–0.1)
Wai‘anae	A	3.5 (0.2)	0.2 (<0.1)
	B	3.1 (0.1)	–0.2 (<–0.1)
Kahana	A	15.6 (6.5)	14.8 (6.2)
	B	15.2 (6.2)	14.5 (5.9)
Ko‘olauloa	A	7.9 (5.7)	7 (5)
	B	7.4 (5)	6.7 (4.5)
Ko‘olaupoko	A	20.3 (7.9)	17.3 (6.7)
	B	20.3 (7.4)	17.5 (6.4)
Waimānalo	A	25.3 (3.7)	18.6 (2.7)
	B	27 (2.9)	20.8 (2.2)

HRCM2 RCP4.5 2080–99 Scenario for Lānaʻi

The future monthly rainfall dataset developed using seasonally adjusted monthly rainfall change factors indicated less variability compared to future monthly rainfall developed using monthly rainfall change factors. Including future monthly rainfall developed using seasonally adjusted monthly rainfall change factors increased island-wide recharge by 31 percent relative to the reference climate condition, whereas including future monthly rainfall developed using monthly rainfall change factors increased island-wide recharge by 43 percent (table 13). Including future monthly rainfall computed using seasonally adjusted monthly rainfall change factors increased aquifer-system recharge by 5 to 59 percent relative to the reference climate condition, whereas including future monthly rainfall computed using monthly rainfall change factors increased aquifer-system recharge by 10 to 80 percent (table 13). These differences indicate a reduction of water available for recharge in the WATRMod calculations when including future monthly rainfall developed using projected changes in mean seasonal rainfall for Lānaʻi and that the magnitude of estimated increases in recharge are diminished.

HRCM2 RCP8.5 2080–99 Scenario for Oʻahu

The future monthly rainfall dataset developed using seasonally adjusted monthly rainfall change factors indicated less variability when compared to future monthly rainfall developed using monthly rainfall change factors. Including future monthly rainfall developed using seasonally adjusted monthly rainfall change factors increased island-wide recharge by 5.3 percent relative to the reference climate condition, whereas including future monthly rainfall developed using monthly rainfall change factors increased island-wide recharge by 7.4 percent (table 14). In terms of aquifer-system recharge, including future monthly rainfall computed using seasonally adjusted monthly rainfall change factors have diminished estimated recharge increases in 16 aquifer systems relative to the estimated recharge increases derived from monthly rainfall change factors (table 14). Results for four aquifer systems (Waialua, ʻEwa-Kunia, Keaʻau, and Nānākuli) indicate a switch from estimated increases to estimated decreases in aquifer-system recharge if future monthly rainfall is computed using seasonally adjusted monthly rainfall change factors instead of monthly rainfall change factors. The remaining three aquifer systems (Mokulēʻia, Lualualei, and Mākaha) show larger decreases in recharge. These differences indicate a reduction of water available for recharge in the WATRMod calculations when including future monthly rainfall developed using projected changes in mean seasonal rainfall for Oʻahu.

Study Limitations

This study relies on the availability of various types of physical data for simulating hydrological processes and physical conditions that affect groundwater recharge for a set of recent and future conditions. Lack of data or sparse, uneven distribution of data in space and time and poor understanding of some hydrologically relevant processes limit the precision and accuracy of study results. This section reviews the limitations imposed by selected model input datasets, the present set of climate projections, and the transformation of these climate datasets into model inputs for simulating the estimated effects on groundwater recharge and other water-budget components. Information that could advance future efforts to reduce uncertainty associated with recharge estimates and to assess the estimated effects of future-climate conditions on groundwater availability is also discussed. Descriptions of WATRMod limitations are provided in Oki (2022).

Dissimilar simulation periods between the sets of climate projections—The climate projection datasets of the future scenarios reflect departures from climate conditions during different reference periods (1978–2007 and 1990–2009). The rainfall projections of the SD RCP8.5 2041–99 and SD RCP8.5 2071–99 scenarios represent 2041–71 and 2071–99 time periods, respectively, relative to a 1978–2007 reference period. However, the climate projection datasets for the HRCM1 A1B 2080–99, HRCM2 RCP4.5 2080–99, and HRCM2 RCP8.5 2080–99 scenarios characterize future climate for a 2080–99 time period relative to a 1990–2009 reference period. Although the reference periods overlap during 1990–2007, the 1990–2009 reference period represents a drier baseline condition compared to the 1978–2007 reference period. The differences in baseline conditions might be a contributing factor to the identified changes for each future-climate scenario. However, assessing the extent to which these differences in baseline conditions affect the drying and wetting changes identified in this analysis for each future-climate scenario is beyond the scope of this study. These differences in future time periods and baseline conditions preclude direct comparisons of the two future-climate scenarios if adjustments are not made to account for the different baseline conditions. In this study, the approach by Mair and others (2019) was used to modify rainfall projections for the HRCM1 A1B 2080–99, HRCM2 RCP4.5 2080–99, and HRCM2 RCP8.5 2080–99 scenarios to reflect the same 1978–2007 reference period used by the SD RCP8.5 2041–71 and SD RCP8.5 2071–99 scenarios. This approach enabled a direct comparison of both future scenario WATRMod simulations to a common reference period. However, the approach used in this analysis might not completely compensate for the potential inherent biases of the two downscaling approaches and their respective reference climate condition. An evaluation of the potential biases of each downscaling approach with respect to their reference climate condition is beyond the scope of this study. Recharge estimates for multiple future-climate scenarios could be improved with the availability of climate projections that use common reference and projection time periods.

Rainfall—Rainfall is an essential dataset for estimating inflow to the water budget. Although Hawaiʻi has a considerable number of rain gages, the placement of gages and their periods of operation have commonly been

dictated by the water-supply needs of humans—particularly for agriculture (Giambelluca and others, 1986). One of the primary sources of rainfall data used in this study, the Rainfall Atlas of Hawai‘i by Giambelluca and others (2013), discusses limitations to rainfall regionalization related to the number and distribution of rain gages and the limited periods of record. These limitations translate to limitations in the recharge results of this study. The online version of the rainfall atlas provides maps of rain-gage locations and uncertainty in the extrapolated rainfall estimates. These maps indicate that rain gages are scarce in areas that are less populated or unfavorable for agriculture, such as in the high-altitude interiors of islands or areas that have rugged terrain. Rainfall estimates for these areas have a greater degree of uncertainty and tend to be the wettest areas of the islands and the areas where most recharge occurs. For example, the effect of rain gage scarcity on rainfall estimates is evident in some drainage basins along the windward region of the Island of Hawai‘i where direct runoff estimates derived from streamgage station data exceed concurrent rainfall represented in the monthly rainfall maps by Frazier and others (2016). The streamgage data from these wet drainage basins indicate that the monthly rainfall maps by Frazier and others (2016) might be underestimating rainfall across the area. Additionally, sparse rain gage data can also have an effect on the interpolation of daily rainfall for the rainfall fragments. Spatial variability in rainfall, particularly at short time scales, is large, which might not be captured well using interpolation methods. Recharge estimates can be improved with rainfall data from these sparsely gaged areas.

The climate projection datasets used in this study characterize a wide range of projected changes in rainfall, resulting in opposing drying and wetting changes across parts of each island. Reduced uncertainty in projected rainfall changes would reduce uncertainty in quantifying potential influences on recharge and groundwater availability. Also, the projected effects of potential changes to daily rainfall frequency on recharge and other water-budget components could not be simulated for the SD RCP8.5 2041–71 and SD RCP8.5 2071–99 scenarios because consistently developed estimates of daily rainfall frequency were not available for these scenarios. Recharge estimates for future conditions can be improved with information on projected changes in daily rainfall frequency for each future-climate scenario.

Fog interception—Fog-interception data are particularly sparse. Fog interception is believed to provide a substantial part of the water available for groundwater recharge over large areas of the Hawaiian Islands, but has been quantified by only a few studies in a few places. As a result, fog-interception distribution was generalized in the recharge calculation. For example, fog-to-rainfall ratios were used in WATRMod to compute fog interception in forested areas within the cloud zone as a fraction of daily rainfall. The use of fog-to-rainfall ratios in the WATRMod calculations indicates that fog interception is represented only on days with rainfall. Therefore, the fog-to-rainfall ratio approach precludes representing fog interception in the absence of rainfall. Recharge estimates can be improved with a better quantification of the spatial and temporal variability of fog-interception rates and fog-zone altitudes, and the capacity of different types of vegetation to intercept fog.

Projected effects of potential changes to fog-interception rates and cloud-zone altitudes, and their effect on recharge and other water-budget components could not be simulated for the selected future-climate scenarios. Fog interception might provide an important part of the water available for recharge on each island. Therefore, recharge estimates for multiple future-climate scenarios can be improved with information that quantifies the potential changes in fog-interception rates and fog-zone altitudes for future conditions.

Runoff—Runoff in gaged areas is estimated in WATRMod using seasonal runoff-to-rainfall ratios that are derived from monitoring data from a limited number of streamgage stations on each of the islands considered in this study. In ungaged areas, runoff is estimated using runoff-to-rainfall ratios derived from statistical analyses of available streamgage and rain-gage data. For the future-climate scenarios, seasonal runoff-to-rainfall ratios derived from statistical analyses are also used to estimate runoff in all gaged and ungaged areas. Streamgage data indicate that rainfall might be underestimated in some parts of Hawai‘i owing to sparse rain gages and that runoff might be underestimated in these affected areas. Therefore, runoff estimates can be improved with streamgage data from ungaged areas and from rainfall data in sparsely gaged areas, which also would improve recharge estimates for recent- and future-climate conditions.

Canopy evaporation, net precipitation, and transpiration—The current understanding of the regional effects of invasive non-native and dominant native forest species on water resources is incomplete. Limited data restricts WATRMod simulations of canopy evaporation, net precipitation, and transpiration processes in this study to broad categories of forest land cover, such as non-native forest, native forest, and tree plantation land cover. For example, only one set of model input parameters for root depth, depletion fraction, and crop coefficient are used for simulating these processes for non-native forest below the fog zone (table 7). Recharge estimates could be improved with development of these model input parameters for specific types of forest species or forest environments. In this study, an approach was developed to estimate model input parameters needed to simulate transpiration and groundwater use by kiawe land cover on the basis of recent studies on the Island of Hawai‘i (see appendix 1 for a description of the model input parameter estimation for kiawe). Transpiration and groundwater use by kiawe land cover on each island were then simulated with WATRMod using model input parameters developed for kiawe land cover. Recharge estimates could be improved with additional sets of model input parameters developed for other dominant native and non-native forest species.

The estimated effects of potential changes to canopy-evaporation and reference-ET rates on recharge and other water-budget components could not be simulated for the SD RCP8.5 2041–71 and SD RCP8.5 2071–99 scenarios because consistently developed projections for canopy evaporation and reference ET were not available for these scenarios. Improved spatial estimates of *V*, the ratio of the mean evaporation rate to the mean precipitation rate during saturated canopy conditions, would improve estimates of the spatial distribution of canopy evaporation and estimated effects of future-climate conditions on canopy evaporation.

Seepage rates from unlined reservoir and inland water body land covers—Seepage rates from subareas with unlined reservoir and inland water body land covers were estimated from limited available data. Seepage rates were assigned to model subareas with unlined reservoir land cover on the basis of available information and generally decrease with age of the soils and substrate associated with each island. For example, published seepage rates for unlined reservoirs range from relatively low values on Kauaʻi, the oldest island, to relatively high values on Maui, one of the youngest islands. However, these studies were conducted at a limited number of locations on Kauaʻi, Molokaʻi, and Maui, and might not capture the full range of reservoir seepage rates for unlined reservoirs throughout the Hawaiian Islands. For example, the Island of Hawaiʻi lacked published data on reservoir seepage rates. An estimated seepage rate was assigned based on data available for other islands. The seepage rate assigned to inland water body land cover in this study is based on estimates from published studies (Engott and Vana, 2007; Izuka and others, 2018; Johnson and others, 2018; Mair and others, 2019). The seepage rates used for unlined reservoir and inland water body land covers were considered constant values, regardless of possible substantial fluctuations in surface-water availability from episodic dry or wet periods. Recharge estimates could be improved with better quantification of seepage rates from unlined reservoir and inland water body land covers on each island and for a range of climate conditions.

Taro cultivation and irrigation—For this study, only a subset of the wetland taro areas delineated in the land-cover map for each island was considered to be irrigated and cultivated at any one time. The subset of wetland taro areas was determined from the number of acres of irrigated taro in each county that were harvested during a single year as reported in a National Agricultural Statistics Service census of crop production. The information used in the census is provided voluntarily by commercial taro farmers and does not include non-commercial taro farming. Therefore, the actual percentage of wetland taro areas being irrigated and cultivated on each island might differ from the census information. Also, the irrigation rate for wetland taro was estimated as the mean value from a limited number of water-use studies of wetland taro in the State of Hawaiʻi. Recharge estimates could be improved with better quantification of wetland taro cultivation areas and of irrigation rates needed to support wetland taro across the range of current and potential cultivation areas.

Land Cover—The effects of future land-cover conditions on recharge and other water-budget components were not considered in this study. Maps of future land-cover conditions on each island would be helpful for evaluating the effects of (1) planned development, (2) climate-induced changes in the distribution of the dominant types of forest species, and (3) watershed management and restoration efforts. The approach used for this study, however, did allow for isolating the effects of climate change on recharge by excluding the complicating factor of changing land cover.

Impervious surfaces—For subareas without storm-drain systems, the calculations in WATRMod require that excess water from the impervious fraction be distributed to a pervious part of each subarea. This requirement indicates that a subarea cannot be completely impervious and is a limitation of the approach. Therefore, all subareas were reassigned a value of 0.99 for impervious fraction whenever the value computed from the map of impervious surfaces exceeded 0.99.

Summary

The main objectives of this study were to (1) produce spatially distributed estimates of mean groundwater recharge and other water-budget components for four climate scenarios for Kauaʻi, Oʻahu, Molokaʻi, Lānaʻi, Maui, and the Island of Hawaiʻi, and (2) quantify the regional hydrologic changes between a reference-climate condition and the future-climate scenarios. The four climate scenarios include (1) a reference-climate condition represented by recent climate conditions during 1978–2007, (2) a mid-century scenario using projected rainfall conditions for a SD RCP8.5 2041–71 scenario, (3) a dry-climate scenario using projected rainfall conditions for a SD RCP8.5 2071–99 scenario, and (4) a wet-climate scenario using projected rainfall conditions for a HRCM1 A1B 2080–99 scenario for Maui, a HRCM2 RCP4.5 2080–99 scenario for Kauaʻi, Lānaʻi, and the Island of Hawaiʻi, and a HRCM2 RCP8.5 2080–99 scenario for Oʻahu and Molokaʻi. A fifth climate scenario was added for Lānaʻi to assess the effect of drought conditions during 2008–12 on groundwater recharge. The State of Hawaiʻi has divided the islands included in this study into 110 groundwater management areas called aquifer systems (State of Hawaiʻi, 2014). Recharge was computed island-wide and for the 110 aquifer systems with WATRMod, a water-budget model, that used a daily computation interval. Hydrologic processes and physical conditions that affect recharge on Kauaʻi, Oʻahu, Molokaʻi, Lānaʻi, Maui, and the Island of Hawaiʻi for recent and future-climate conditions were simulated in WATRMod using the most recent datasets available—including maps of 2020 land cover, 1978–2007 monthly rainfall, 1990–2014 daily rainfall, and mean monthly reference ET (Giambelluca and others, 2013; Longman and others, 2019). Published high-resolution downscaled climate projections were used to modify inputs of monthly rainfall and seasonal runoff for simulating the effects of future-climate conditions (Elison Timm and others, 2015; Zhang and others, 2016a, b; Zhang and Wang, 2017).

For Kauaʻi, island-wide mean annual recharge is approximately 807 millions of gallons per day (Mgal/d) for the 1978–2007 scenario, with mean annual recharge for Kauaʻi's 13 aquifer systems, ranging from about 15 to 138 Mgal/d. For the SD RCP8.5 2041–71 scenario, recharge for Kauaʻi is about 640 Mgal/d, an estimated decrease of 167 Mgal/d (21 percent) from the 1978–2007 scenario. For the SD RCP8.5 2071–99 scenario, recharge for Kauaʻi is about 572 Mgal/d, an estimated decrease of 234 Mgal/d (29 percent) from the 1978–2007 scenario. Recharge for the HRCM2 RCP4.5 2080–99 scenario is estimated to increase by about 19 Mgal/d (2.3 percent) relative to the 1978–2007 scenario.

For Oʻahu, the mean annual recharge is approximately 597 Mgal/d for the 1978–2007 scenario, with mean annual recharge for Oʻahu's 23 aquifer systems ranging from about 1 to 119 Mgal/d. For the SD RCP8.5 2041–71 scenario, recharge for Oʻahu is about 486 Mgal/d, an estimated decrease of 111 Mgal/d (19 percent) from the 1978–2007 scenario. For the SD RCP8.5 2071–99 scenario, recharge for Oʻahu is about 451 Mgal/d, an estimated decrease of 146 Mgal/d (24 percent) from the 1978–2007 scenario. Recharge for the HRCM2 RCP4.5 2080–99 scenario is estimated to increase by about 44 Mgal/d (7.4 percent) relative to the 1978–2007 scenario.

For Molokaʻi, mean annual recharge is approximately 228 Mgal/d for the 1978–2007 scenario. For Molokaʻi's 16 aquifer systems, mean annual recharge ranges from about 1.3 to 43 Mgal/d for the 1978–2007 scenario. For the SD RCP8.5 2041–71 scenario, recharge for Molokaʻi is about 190 Mgal/d, an estimated decrease of 38 Mgal/d (17 percent) from the 1978–2007 scenario. For the SD RCP8.5 2071–99 scenario, recharge for Molokaʻi is about 172 Mgal/d, an estimated decrease of 56 Mgal/d (24 percent) from the 1978–2007 scenario. Recharge for the HRCM2 RCP8.5 2080–99 scenario is estimated to increase by 40 Mgal/d (17 percent) relative to the 1978–2007 scenario.

Mean annual recharge for Lānaʻi is approximately 27 Mgal/d for the 1978–2007 scenario. For Lānaʻi's nine aquifer systems, mean annual recharge ranges from about 1.9 to 4.5 Mgal/d for the 1978–2007 scenario. For the SD RCP8.5 2041–71 scenario, recharge for Lānaʻi is about 16 Mgal/d, an estimated decrease of approximately 11 Mgal/d (41 percent) from the 1978–2007 scenario. For the SD RCP8.5 2071–99 scenario, recharge for Lānaʻi's is about 12 Mgal/d, an estimated decrease of 15 Mgal/d (55 percent) from the 1978–2007 scenario. For the HRCM2 RCP4.5 2080–99 scenario island-wide mean annual recharge is greater than recharge for the 1978–2007 scenario by roughly 12 Mgal/d (43 percent). For the 2008–12 drought scenario, recharge for Lānaʻi is about 12 Mgal/d, an estimated decrease of about 15 Mgal/d (54 percent) from the 1978–2007 scenario.

For Maui, mean annual recharge is about 1,192 Mgal/d for the 1978–2007 scenario. For Maui's 25 aquifer systems, mean annual recharge ranges from about 6 to 217 Mgal/d for the 1978–2007 scenario. For the SD RCP8.5 2041–71 scenario, recharge for Maui is about 1,078 Mgal/d, an estimated decrease of 115 Mgal/d (9.6 percent) from the 1978–2007 scenario. For the SD RCP8.5 2071–99 scenario, recharge for Maui is about 1,031 Mgal/d, an estimated decrease of 162 Mgal/d (14 percent). For the HRCM1 A1B 2080–99 scenario, mean annual recharge is greater than recharge for the 1978–2007 scenario by about 126 Mgal/d (11 percent).

For the Island of Hawaiʻi, mean annual recharge is about 6,885 Mgal/d for the 1978–2007 scenario. For the island's 24 aquifer systems, mean annual recharge ranges from about 26 to 895 Mgal/d for the 1978–2007 scenario. For the SD RCP8.5 2041–71 scenario, recharge for the Island of Hawaiʻi is about 5,945 Mgal/d, an estimated decrease of 940 Mgal/d (14 percent) from the 1978–2007 scenario. For the SD RCP8.5 2071–99 scenario, mean annual recharge is less than recharge for the 1978–2007 scenario by 1,349 Mgal/d (20 percent). For the HRCM2 RCP8.5 2080–99 scenario, mean annual recharge is less than the recharge for the 1978–2007 scenario by roughly 314 Mgal/d (4.6 percent).

Agreement in drying and wetting between two future-climate scenarios, the mid-century SD RCP8.5 2041–71 and dry-climate SD RCP8.5 2071–99 scenarios, is evident in the estimated changes to island-wide recharge and aquifer-system recharge on all six islands. Island-wide recharge is estimated to decrease on each island for both mid-century and dry-climate scenarios. Estimated decreases in aquifer-system recharge are indicated for both mid-century and dry-climate scenarios in 104 of 110 aquifer systems from all six islands, whereas estimated increases are indicated for the remaining six aquifer systems from Maui and the Island of Hawaiʻi. However, among the mid-century, dry-climate, and wet-climate HRCM1/HRCM2 scenarios, agreement in estimated changes to island-wide recharge and aquifer-system recharge is limited to the Island of Hawaiʻi and agreement in estimated changes in 35 of 110 aquifer systems. Estimated decreases in aquifer-system recharge are indicated for 30 of 110 aquifer systems for all three future-climate scenarios, whereas estimated increases are indicated for five aquifer systems from Maui and the Island of Hawaiʻi. Despite their agreement in change direction, the magnitude of the changes in island-wide recharge and aquifer-system recharge vary for each future-climate scenario with the greatest decreases estimated for the dry-climate SD RCP8.5 2071–99 scenario and the greatest increases estimated for the wet-climate HRCM1/HRCM2 scenarios. Greater consistency in estimated changes to rainfall between future-climate scenarios would reduce the uncertainty associated with quantifying the effects on recharge and groundwater availability.

Among the five inputs analyzed for their effect on recharge estimates, projected changes in monthly rainfall and seasonal runoff-to-rainfall ratios (considered as a single factor) have the greatest effect on recharge estimates for the HRCM1/HRCM2 scenarios. The HRCM2 RCP4.5 2080–99 scenario projects increases in island-wide mean annual rainfall for Lānaʻi but non-uniform changes in mean monthly rainfall. The non-uniform changes in mean monthly rainfall correspond to substantial increases in mean monthly recharge during selected months and decreases in mean monthly recharge during most of the remaining months. The estimated increases in mean monthly recharge during months with increased rainfall more than compensate for the decreases in recharge during most of the remaining months resulting in a net positive effect on island-wide recharge for Lānaʻi. Estimated changes in aquifer-system recharge and island-wide recharge are also affected by estimated changes in reference ET owing to projected warming. Estimated increases in reference ET due to projected warming results in decreases in island-wide recharge on all islands. However, in WATRMod calculations, increases in atmospheric CO_2 concentrations act to suppress ET,

which decreases actual ET and increases soil moisture available for recharge. The decreases in reference ET owing to projected increases in atmospheric CO_2 concentrations partially or completely offset increases in reference ET due to projected warming for each future-climate scenario. Therefore, decreases in island-wide recharge owing to projected warming are partially or completely offset by enhanced recharge owing to projected increases in mean atmospheric CO_2 concentrations for each future-climate scenario. Estimated changes in aquifer-system recharge and island-wide recharge are affected less by estimated changes in forest-canopy evaporation and projected changes in daily rainfall frequency.

Testing of the effect of future monthly rainfall inputs on recharge indicates that seasonally adjusted monthly rainfall change factors, relative to monthly rainfall change factors, diminish the estimated increases in recharge across all of Lānaʻi for the HRCM2 RCP4.5 2080–99 scenario and most of Oʻahu for the HRCM2 RCP8.5 2080–99 scenario. The application of seasonally adjusted monthly rainfall change factors has a greater relative effect on estimated changes in recharge estimates for Lānaʻi compared to Oʻahu. The application of seasonally adjusted monthly rainfall change factors to develop future monthly rainfall reduces the amount of water available for recharge in the WATRMod calculations when compared to simulations using future monthly rainfall developed from monthly rainfall change factors.

References Cited

Allen, R.G., Pereira, L.S., Raes, D., and Smith, M., 1998, Crop evapotranspiration—Guidelines for computing crop water requirements: Food and Agriculture Organization of the United Nations, FAO Irrigation and Drainage Paper 56, 300 p.

Allen, R.G., Pruitt, W.O., Wright, J.L., Howell, T.A., Ventura, F., Snyder, R., Itenfisu, D., Steduto, P., Berengena, J., Yrisarry, J.B., Smith, M., Pereira, L.S., Raes, D., Perrier, A., Alves, I., Walter, I., and Elliott, R., 2006, A recommendation on standardized surface resistance for hourly calculation of reference ETo by the FAO56 Penman-Monteith method: Agricultural Water Management, v. 81, nos. 1–2, p. 1–22.

AWS Truewind LLC, 2004, Wind energy resource maps of Hawaii: Prepared for Hawaiʻi Department of Business, Economic Development, and Tourism Strategic Industries Division.

Bassiouni, M., and Oki, D.S., 2013, Trends and shifts in streamflow in Hawaiʻi, 1913–2008: Hydrological Processes, v. 27, no. 10, p. 1484–1500, https://doi.org/10.1002/hyp.9298.

Berg, N., McGurk, B., and Calhoun, R., 1997, Hydrology and land use effects on the Hanalei National Wildlife Refuge, Kauai, Hawaii, Final Report: USDA Forest Service, Pacific Southwest Research Station, Interagency Agreement 14-48-0001-94588, 62 p.

Blumenstock, D.I., and Price, S., 1967, Climates of the states—Hawaii, *in* Climatology of the United States: U.S. Department of Commerce, Washington, D.C., p. 27.

Brewington, L., Keener, V., and Mair, A., 2019, Simulating land cover change impacts on groundwater recharge under selected climate projections, Maui, Hawaiʻi: Remote Sensing, v. 11, no. 24, https://doi.org/10.3390/rs11243048.

Brouwer, C., and Heibloem, M., 1986, Irrigation water management—Irrigation water needs, Training Manual no. 3: Food and Agriculture Organization of the United Nations.

Cheng, C.L., Izuka, S.K., Kennedy, J.J., Frazier, A.G., and Giambelluca, T.W., 2021, Water-resource management monitoring needs, State of Hawaiʻi: U.S. Geological Survey Scientific Investigations Report 2020–5115, 114 p., https://doi.org/10.3133/sir20205115.

Chu, P.-S., and Chen, H., 2005, Interannual and interdecadal rainfall variations in the Hawaiian Islands: Journal of Climate, v. 18, no. 2, p. 4796–4813, https://doi.org/10.1175/JCLI3578.1.

Chu, P.-S., Chen, Y.R., and Schroeder, T.A., 2010, Changes in precipitation extremes in the Hawaiian Islands in a warming climate: Journal of Climate, v. 23, no. 18, p. 4881–4900, https://doi.org/10.1175/2010JCLI3484.1.

City and County of Honolulu, 2020, Tax parcels for the City and County of Honolulu: City and County of Honolulu GIS, Honolulu GIS Office (HOLIS), accessed February 13, 2020, at https://honolulu-cchnl.opendata.arcgis.com/datasets/cchnl::parcels-tax/about.

Clilverd, H.M., Tsang, Y.-P., Infante, D.M., Lynch, A.J., and Strauch, A.M., 2019, Long-term streamflow trends in Hawaiʻi and implications for native stream fauna: Hydrological Processes, v. 33, no. 5, p. 699–719, https://doi.org/10.1002/hyp.13356.

De la Pena, R., and Melchor, F., 1984, Water use and efficiency in lowland taro production, *in* Sixth Symposium of the International Society for Tropical Root Crops, February 21–26, 1983: Lima, Peru, p. 97–101.

DeLay, J.K., 2005, Canopy water balance of an elfin cloud forest at Alakahi, Hawaiʻi: Honolulu, Hawaiʻi, University of Hawaiʻi at Mānoa, USA, Masters Thesis, 78 p.

DeLay, J.K., and Giambelluca, T.W., 2010, History of cloud water interception research in Hawaiʻi, *in* Bruijnzeel, L.A., Scatena, F.N., and Hamilton, L.S., eds., Tropical montane cloud forests—Science for conservation and management: Cambridge University Press, Cambridge, UK, p. 332–341.

Diaz, H.F., Giambelluca, T.W., and Eischeid, J.K., 2011, Changes in the vertical profiles of mean temperature and humidity in the Hawaiian Islands: Global and Planetary Change, v. 77, nos. 1–2, p. 21–25, https://doi.org/10.1016/j.gloplacha.2011.02.007.

Dudley, B.D., Hughes, R.F., and Ostertag, R., 2014, Groundwater availability mediates the ecosystem effects of an invasion of *Prosopis pallida*: Ecological Applications, v. 24, no. 8, p. 1954–1971, https://doi.org/10.1890/13-1262.1.

Dudley, B.D., Hughes, R.F., Asner, G.P., Baldwin, J.A., Miyazawa, Y., Dulai, H., Waters, C., Bishop, J., Vaughn, N.R., Yeh, J., Kettwich, S., MacKenzie, R.A., Ostertag, R., and Giambelluca, T., 2020, Hydrological effects of tree invasion on a dry coastal Hawaiian ecosystem: Forest Ecology and Management, v. 458, article 117653, https://doi.org/10.1016/j.foreco.2019.117653.

Dugan, G.L., Lau, L.S., and Yamauchi, H., 1975, Eutrophication and fish toxicity potentials in a multiple-use reservoir: University of Hawai‘i Water Resources Research Center Technical Report no. 89, 175 p., accessed February 24, 2024, at http://hdl.handle.net/10125/1888.

Ekern, P.C., 1964, Direct interception of cloud water on Lanaihale, Hawaii: Soil Science Society of America Journal, v. 28, no. 3, p. 419–421, https://doi.org/10.2136/sssaj1964.03615995002800030035x.

Ekern, P.C., and Chang, J.H., 1985, Pan evaporation—State of Hawai‘i, 1894–1983: Hawai‘i Department of Land and Natural Resources, Division of Water and Land Development, Report R74, 172 p.

Elison Timm, O., 2017, Future warming rates over the Hawaiian Islands based on elevation-dependent scaling factors: International Journal of Climatology, v. 37, no. S1, p. 1093–1104, https://doi.org/10.1002/joc.5065.

Elison Timm, O., Diaz, H.F., Giambelluca, T.W., and Takahashi, M., 2011, Projection of changes in the frequency of heavy rain events over Hawaii based on leading Pacific climate modes: Journal of Geophysical Research, Atmospheres, v. 116, no. D4, 12 p., https://doi.org/10.1029/2010JD014923.

Elison Timm, O., Giambelluca, T.W., and Diaz, H.F., 2015, Statistical downscaling of rainfall changes in Hawai‘i based on the CMIP5 global model projections: Journal of Geophysical Research, Atmospheres, v. 120, no. 1, p. 92–112, https://doi.org/10.1002/2014JD022059.

Engott, J.A., 2011, A water-budget model and assessment of groundwater recharge for the Island of Hawai‘i: U.S. Geological Survey Scientific Investigations Report 2011–5078, 53 p., https://doi.org/10.3133/sir20115078.

Engott, J.A., Johnson, A.G., Bassiouni, M., Izuka, S.K., and Rotzoll, K., 2017, Spatially distributed groundwater recharge for 2010 land cover estimated using a water-budget model for the Island of O‘ahu, Hawai‘i (ver 2.0, December 2017): U.S. Geological Survey Scientific Investigations Report 2015–5010, 49 p., https://doi.org/10.3133/sir20155010.

Engott, J.A., and Vana, T.T., 2007, Effects of agricultural land-use changes and rainfall on ground-water recharge in Central and West Maui, Hawai‘i, 1926–2004: U.S. Geological Survey Scientific Investigations Report 2007–5103, 56 p., https://doi.org/10.3133/sir20075103.

Fortini, L.B., Leopold, C.R., Perkins, K., Chadwick, O.A., Yelenik, S.G., Jacobi, J.D., Bishaw, K., II, Gregg, M., and Rosa, S., 2020, Hawaiian Islands datasets quantifying the effects of invasive animals and plants on native forests across the archipelago 2019 (ver. 2.0 April 2020): U.S. Geological Survey data release, https://doi.org/10.5066/P9J35LMQ.

Frazier, A.G., Elison Timm, O., Giambelluca, T.W., and Diaz, H.F., 2018, The influence of ENSO, PDO and PNA on secular rainfall variations in Hawai‘i: Climate Dynamics, v. 51, no. 5, p. 2127–2140. https://doi.org/10.1007/s00382-017-4003-4.

Frazier, A.G., and Giambelluca, T.W., 2017, Spatial trend analysis of Hawaiian rainfall from 1920 to 2012: International Journal of Climatology, v. 37, no. 5, p. 2522–2531, https://doi.org/10.1002/joc.4862.

Frazier, A.G., Johnson, M.-V.V., Berio Fortini, L., Giardina, C.P., Grecni, Z.N., Kane, H.H., Keener, V.W., King, R., MacKenzie, R.A., Nobrega-Olivera, M., Oleson, K.L.L., Shuler, C.K., Singeo, A.K., Storlazzi, C.D., Wallsgrove, R.J., and Woodworth-Jefcoats, P.A., 2023, Chapter 30—Hawai‘i and US-Affiliated Pacific Islands, *in* Crimmins, A.R., Avery, C.W., Easterling, D.R., Kunkel, K.E., Stewart, B.C., and Maycock, T.K., eds., Fifth National Climate Assessment: U.S. Global Change Research Program, Washington, DC, USA, https://doi.org/10.7930/NCA5.2023.CH30.

Frazier, A.G., Giambelluca, T.W., Diaz, H.F., and Needham, H.L., 2016, Comparison of geostatistical approaches to spatially interpolate month-year rainfall for the Hawaiian Islands: International Journal of Climatology, v. 36, no. 3, p. 1459–1470, https://doi.org/10.1002/joc.4437.

Gash, J.H.C., Lloyd, C.R., and Lachaud, G., 1995, Estimating sparse forest rainfall interception with an analytical model: Journal of Hydrology, v. 170, no. 1–4, p. 79–86, https://doi.org/10.1016/0022-1694(95)02697-N.

Gaskill, T.G.R., 2004, Hydrology of forest ecosystems in the Honouliuli Preserve—Implications for groundwater recharge and watershed restoration: University of Hawai‘i at Mānoa PhD Dissertation, 177 p., accessed April 21, 2021 at http://scholarspace.manoa.hawaii.edu/handle/10125/12116.

Gerrits, A.M.G., and Savenije, H.H.G., 2010, Interception, *in* Wilderer, P. ed., Treatise on water science—The science of hydrology: Elsevier Science, v. 2, p. 2102.

Ghazal, K.A., Leta, O.T., El-Kadi, A.I., and Dulai, H., 2019, Assessment of wetland restoration and climate change impacts on water balance components of the Heeia coastal wetland in Hawaii: Hydrology, v. 6, no. 2, 16 p., https://doi.org/10.3390/hydrology6020037.

Giambelluca, T.W., 1983, Water balance of the Pearl Harbor-Honolulu basin, Hawaii, 1946–1975: University of Hawai‘i Water Resources Research Center Technical Report no. 151, 151 p., accessed February 24, 2024, at http://hdl.handle.net/10125/1992.

Giambelluca, T.W., Chen, Q., Frazier, A.G., Price, J.P., Chen, Y.-L., Chu, P.-S., Eischeid, J.K., and Delparte, D.M., 2013, Online rainfall atlas of Hawai‘i: Bulletin of the American Meteorological Society, v. 94, no. 3, p. 313–316, https://doi.org/10.1175/BAMS-D-11-00228.1.

Giambelluca, T.W., DeLay, J.K., Nullet, M.A., Scholl, M.A., and Gingerich, S.B., 2011, Canopy water balance of windward and leeward Hawaiian cloud forests on Haleakalā, Maui, Hawai‘i: Hydrological Processes, v. 25, no. 3, p. 438–447, https://doi.org/10.1002/hyp.7738.

Giambelluca, T.W., Diaz, H.F., and Luke, M.S.A., 2008, Secular temperature changes in Hawai‘i: Geophysical Research Letters, v. 35, no. 12, article L12702, https://doi.org/10.1029/2008GL034377.

Giambelluca, T.W., Nullet, M.A., and Schroeder, T.A., 1986, Rainfall atlas of Hawai‘i: Hawai‘i Department of Land and Natural Resources, Division of Water and Land Development Report R76, 267 p.

Giambelluca, T., and Schroeder, T.A., 1998, Climate, *in* Juvik, S.P., and Juvik, J.O., eds., Atlas of Hawai‘i: University of Hawai‘i Press Climate, 49–59 p.

Giambelluca, T.W., Shuai, X., Barnes, M.L., Alliss, R.J., Longman, R.J., Miura, T., Chen, Q., Frazier, A.G., Mudd, R.G., Cuo, L., and Businger, A.D., 2014, Evapotranspiration of Hawai‘i: University of Hawai‘i at Mānoa, accessed November 12, 2021, at http://evapotranspiration.geography.hawaii.edu/.

Gingerich, S.B., 2008, Ground-water availability in the Wailuku area, Maui, Hawai‘i: U.S. Geological Survey Scientific Investigations Report 2008–5236, 95 p., https://doi.org/10.3133/sir20085236.

Gingerich, S.B., and Engott, J.A., 2012, Groundwater availability in the Lahaina District, west Maui, Hawai‘i: U.S. Geological Survey Scientific Investigations Report 2012–5010, 90 p., https://doi.org/10.3133/sir20125010.

Gingerich, S.B., and Oki, D.S., 2000, Ground water in Hawaii: U.S. Geological Survey Fact Sheet 126-00, 6 p., https://doi.org/10.3133/fs12600.

Graham, L.P., Hagemann, S., Jaun, S., and Beniston, M., 2007, On interpreting hydrological change from regional climate models: Climatic Change, v. 81, no. 1 supplement, p. 97–122, https://doi.org/10.1007/s10584-006-9217-0.

Hardy, W.R., 1996, A numerical ground-water model for the island of Lana’i, Hawaii: Commission on Water Resource Management, Department of Land and Natural Resources, State of Hawai‘i.

Hay, L.E., Wilby, R.L., and Leavesley, G.H., 2000, A comparison of delta change and downscaled GCM scenarios for three mountainous basins in the United States: Journal of the American Water Resources Association, v. 36, no. 2, p. 387–397, https://doi.org/10.1111/j.1752-1688.2000.tb04276.x.

Health, J.A., and Huebert, B.J., 1998, Cloudwater deposition as a source of fixed nitrogen in a Hawaiian montane forest: Biogeochemistry, no. 44, no. 2, p. 119–134, https://doi.org/10.1023/A:1006091429143.

Helsel, D.R., and Hirsch, R.M., 1992, Statistical methods in water resources: Elsevier, Amsterdam, The Netherlands, 522 p.

Hunt, C.D., 2004, Ground-water quality and its relation to land use on Oahu, Hawaii, 2000–01: U.S. Geological Survey Water Resources Investigations Report 03–4305, 76 p., accessed at https://doi.org/10.3133/wri20034305.

Intergovernmental Panel on Climate Change, 2007, Climate change 2007—The physical science basis: Contribution of Working Group I to the Fourth Assessment Report of the Intergovernmental Panel on Climate Change (Solomon, S., Qin, D., Manning, M., Chen, Z., Marquis, M., Averyt, K.B., Tignor, M., and Miller, H.L., eds.), Cambridge University Press, Cambridge, United Kingdom and New York, NY, USA, 996 p.

Intergovernmental Panel on Climate Change, 2014, Climate change 2014—Synthesis report: Contribution of Working Groups I, II and III to the Fifth Assessment Report of the Intergovernmental Panel on Climate Change (Core Writing Team, Pachauri, R.K., and Meyer, L.A., eds.), IPCC, Geneva, Switzerland, 151 p.

Intergovernmental Panel on Climate Change, 2018, Summary for policymakers, *in* Masson-Delmotte, V., Zhai, P., Pörtner, H.-O., Roberts, D., Skea, J., Shukla, P.R., Pirani, A., Moufouma-Okia, W., Péan, C., Pidcock, R., Connors, S., Matthews, J.B.R., Chen, Y., Zhou, X., Gomis, M.I., Lonnoy, E., Maycock, T.K., Tignor, M., and Waterfield, T. eds., Global warming of 1.5°C—An IPCC Special Report on the impacts of global warming of 1.5°C above pre-industrial levels and related global greenhouse gas emission pathways, in the context of strengthening the global response to the threat of climate change, sustainable development, and efforts to eradicate poverty: World Meteorological Organization, Geneva, Switzerland, 32 p.

Intergovernmental Panel on Climate Change, 2021, Summary for policymakers, *in* Masson-Delmotte, V., Zhai, P., Pirani, A., Connors, S.L., Péan, C., Berger, S., Caud, N., Chen, Y., Goldfarb, L., Gomis, M.I., Huang, M., Leitzell, K., Lonnoy, E., Matthews, J.B.R., Maycock, T.K., Waterfield, T., Yelekçi, O., Yu, R., and Zhou, B. eds., Climate change 2021—The physical science basis: Contribution of Working Group I to the Sixth Assessment Report of the Intergovernmental Panel on Climate Change, Cambridge University Press, United Kingdom and New York, NY, USA.

Izuka, S.K., Engott, J.A., Rotzoll, K., Bassiouni, M., Johnson, A.G., Miller, L.D., and Mair, A., 2018, Volcanic aquifers of Hawai‘i—Hydrogeology, water budgets, and conceptual models (ver. 2.0, March 2018): U.S. Geological Survey Scientific Investigations Report 2015–5164, 158 p., https://doi.org/10.3133/sir20155164.

Izuka, S.K., Oki, D.S., and Chen, C., 2005, Effects of irrigation and rainfall reduction on ground-water recharge in the Lihue Basin, Kauai, Hawaii: U.S. Geological Survey Science Investigations Report 2005–5146, 48 p., https://doi.org/10.3133/sir20055146.

Izuka, S.K., Rotzoll, K., and Nishikawa, T., 2021, Volcanic aquifers of Hawai‘i—Construction and calibration of numerical models for assessing groundwater availability on Kaua‘i, O‘ahu, and Maui: U.S. Geological Survey Scientific Investigations Report 2020–5126, 63 p., https://doi.org/10.3133/sir20205126.

Johnson, A.G., 2012, A water-budget model and estimates of groundwater recharge for Guam: U.S. Geological Survey Scientific Investigations Report 2012–5028, 53 p., http://doi.org/10.3133/sir20125028.

Johnson, A.G., Engott, J.A., Bassiouni, M., and Rotzoll, K., 2018, Spatially distributed groundwater recharge estimated using a water-budget model for the Island of Maui, Hawai‘i, 1978–2007 (ver. 2.0, February 2018): U.S. Geological Survey Scientific Investigations Report 2014–5168, 53 p., https://doi.org/10.3133/sir20145168.

Johnson, A.G., Mair, A., and Oki, D.S., 2023, Identifying the relative importance of water-budget information needed to quantify how land-cover change affects recharge, Hawaiian Islands: U.S. Geological Survey Scientific Investigations Report 2023–5022, 28 p., https://doi.org/10.3133/sir20235022.

Juvik, J.O., DeLay, J.K., Kinney, K.M., and Hansen, E.W., 2011, A 50th anniversary reassessment of the seminal “Lāna‘i fog drip study” in Hawai‘i: Hydrological Processes, v. 25, no. 3, p. 402–410, https://doi.org/10.1002/hyp.7803.

Juvik, J.O., and Ekern, P.C., 1978, A climatology of mountain fog on Mauna Loa, Hawaii Island: University of Hawai‘i Water Resources Research Center, Technical Report 118, 63 p., accessed February 29, 2024, at http://hdl.handle.net/10125/2006.

Juvik, J.O., and Hughes, K., 1997, Climatology and water resources at Kahikinui, Maui: Pacific Environment Planning project report, 24 p.

Juvik, J.O., and Nullet, D., 1995, Relationships between rainfall, cloud-water interception, and canopy throughfall in a Hawaiian montane forest, *in* Hamilton, L.S., Juvik, J.O., and Scatena, F.N. eds., Tropical Montane Cloud Forests: Springer US, New York, NY, p. 165–182.

Kāne, H.L., Mair, A., Mifflin, J., 2024, Mean annual water-budget components for Kaua‘i, O‘ahu, Moloka‘i, Lāna‘i, Maui, and the Island of Hawai‘i, for a set of recent and future-climate conditions, and 2020 land cover: U.S. Geological Survey data release, https://doi.org/10.5066/P972KMSL.

Kennedy, J.J., Mair, A., and Perkins, K.S., 2019a, Summary of soil field-saturated hydraulic conductivity, hydrophobicity, preferential-flow, and particle-size measurements collected at four study sites on the island of Maui, Hawaii, September 2017–August 2018: U.S. Geological Survey data release, https://doi.org/10.5066/P9OP3BHM.

Kennedy, J.J., Mair, A., Perkins, K.S., Nullet, M.A., Tseng, H., and Miyazawa, Y., 2019b, Summary of soil field-saturated hydraulic conductivity, hydrophobicity, and preferential-flow measurements and soil laboratory-testing results collected at three sites on the islands of Maui and Hawaii, Hawaii, July 2016–January 2018: U.S. Geological Survey data release, https://doi.org/10.5066/P9J5SESD.

Lauer, A., Zhang, C., Elison-Timm, O., Wang, Y., and Hamilton, K., 2013, Downscaling of climate change in the Hawaii region using CMIP5 results—On the choice of the forcing fields: Journal of Climate, v. 26, no. 24, p. 10006–10030, https://doi.org/10.1175/JCLI-D-13-00126.1.

Leta, O.T., El-Kadi, A.I., and Dulai, H., 2017, Implications of climate change on water budgets and reservoir water harvesting of Nuuanu area watersheds, Oahu, Hawaii: Journal of Water Resources Planning and Management, v. 143, no. 11, article 05017013, 19 p., https://doi.org/10.1061/(ASCE)WR.1943-5452.0000839.

Leta, O.T., El-Kadi, A.I., and Dulai, H., 2018, Impact of climate change on daily streamflow and its extreme values in Pacific Island watersheds: Sustainability, v. 10, no. 6, 22 p., https://doi.org/10.3390/su10062057.

Leta, O.T., El-Kadi, A.I., Dulai, H., and Ghazal, K.A., 2016, Assessment of climate change impacts on water balance components of Heeia watershed in Hawaii: Journal of Hydrology, Regional Studies, v. 8, no. Supplement C, p. 182–197, https://doi.org/10.1016/j.ejrh.2016.09.006.

Longman, R.J., Frazier, A.G., Newman, A.J., Giambelluca, T.W., Schanzenbach, D., Kagawa-Viviani, A., Needham, H., Arnold, J.R., and Clark, M.P., 2019, High-resolution gridded daily rainfall and temperature for the Hawaiian islands (1990–2014): Journal of Hydrometeorology, v. 20, no. 3, p. 489–508, https://doi.org/10.1175/JHM-D-18-0112.1.

Macdonald, G.A., Abbott, A.T., and Peterson, F.L., 1983, Volcanoes in the sea—the geology of Hawaii, 2nd edition: Honolulu, Hawai‘i, University of Hawai‘i Press, 517 p.

Mair, A., Johnson, A.G., Rotzoll, K., and Oki, D.S., 2019, Estimated groundwater recharge from a water-budget model incorporating selected climate projections, Island of Maui, Hawai‘i: U.S. Geological Survey Scientific Investigations Report 2019–5064, 46 p., https://doi.org/10.3133/sir20195064.

McJannet, D., Wallace, J., and Reddell, P., 2007, Precipitation in Australian tropical rainforests—I. Measurement of stemflow, throughfall and cloud interception: Hydrological Processes, v. 21, no. 13, p. 1692–1702, https://doi.org/10.1002/hyp.6347.

McKenzie, M.M., Giambelluca, T.W., and Diaz, H.F., 2019, Temperature trends in Hawai‘i—A century of change, 1917–2016: International Journal of Climatology, v. 39, no. 10, p. 3987–4001, https://doi.org/10.1002/joc.6053.

McLean, J., Cleveland, S.B., Rotzoll, K., Izuka, S.K., Leigh, J., Jacobs, G.A., and Theriot, R., 2022, The Hawai‘i groundwater recharge tool: Concurrency and Computation, Practice and Experience, v. 35, no. 18, https://doi.org/10.1002/cpe.6843.

Melrose, J., Perroy, R., and Cares, S., 2016, Statewide agricultural land use baseline 2015: State of Hawai‘i Department of Agriculture, prepared by University of Hawai‘i at Hilo, Spatial Data Analysis and Visualization Research Lab, 100 p., accessed February 29, 2024, at https://hdoa.hawaii.gov/salub/.

Miles, K., 1931, Report on study of water requirements of taro in Hanapepe Valley, cooperative study by the Territory of Hawai‘i and McBryde Sugar Co.: ‘Ele‘ele, Hawai‘i, 52 p. [As cited in Penn, 1997.]

Miyazawa, Y., Dudley, B.D., Hughes, R.F., Vandemark, J., Cordell, S., Nullet, M.A., Ostertag, R., and Giambelluca, T.W., 2016, Non-native tree in a dry coastal area in Hawai‘i has high transpiration but restricts water use despite phreatophytic trait: Ecohydrology, v. 9, no. 7, p. 1166–1176, https://doi.org/10.1002/eco.1715.

Nakićenović, N., Alcamo, J., Davis, G., de Vries, B., Fenhann, J., Gaffin, S., Gregory, K., Grübler, A., Jung, T.Y., Kram, T., Lebre La Rovere, E., Michaelis, L., Mori, S., Morita, T., Pepper, W., Pitcher, H., Price, L., Riahi, K., Roehrl, A., Rogner, H.-H., Sankovski, A., Schlesinger, M., Shukla, P., Smith, S., Swart, R., van Rooijen, S., Victor, N., and Dadi, Z., 2000, Special report on emissions scenarios—A special report of Working Group III of the Intergovernmental Panel on Climate Change: Cambridge, United Kingdom, Cambridge University Press, 599 p.

National Agricultural Statistics Service, 2014, 2012 Census of agriculture, Hawaii state and county data: U.S. Department of Agriculture, v. 1, Geographic Area Series, Part 11, 262 p., accessed February 1, 2021, at https://agcensus.library.cornell.edu/wp-content/uploads/2012-Hawaii-hiv1.pdf.

National Agricultural Statistics Service, 2021, Pacific region—Hawaii seed crops: U.S. Department of Agriculture, 2 p., accessed October 1, 2021, at https://www.nass.usda.gov/Statistics_by_State/Hawaii/Publications/Sugarcane_and_Specialty_Crops/Seed/2021/2021%20Hawaii%20Seed%20Crops.pdf.

Natural Resources Conservation Service, 2006a, Soil survey geographic (SSURGO) database for the island of Kaua‘i: U.S. Department of Agriculture, accessed September 4, 2012, at https://websoilsurvey.nrcs.usda.gov/app/.

Natural Resources Conservation Service, 2006b, Soil survey geographic (SSURGO) database for the island of Maui: U.S. Department of Agriculture, accessed October 3, 2011, at https://websoilsurvey.nrcs.usda.gov/app/.

Natural Resources Conservation Service, 2006c, Soil survey geographic (SSURGO) database for the island of Moloka‘i: U.S. Department of Agriculture, accessed December 20, 2011, at https://websoilsurvey.nrcs.usda.gov/app/.

Natural Resources Conservation Service, 2006d, Soil survey geographic (SSURGO) database for the island of O‘ahu: U.S. Department of Agriculture, accessed June 8, 2011, at https://websoilsurvey.nrcs.usda.gov/app/.

Natural Resources Conservation Service, 2008, Soil survey geographic (SSURGO) database for the island of Hawai‘i: U.S. Department of Agriculture, accessed November 1, 2008, at https://SoilDataMart.nrcs.usda.gov/.

Natural Resources Conservation Service, 2019, Soil survey geographic (SSURGO) database for the islands of Kaua‘i, O‘ahu, Moloka‘i, Lāna‘i, Maui, and Hawai‘i: U.S. Department of Agriculture, accessed December 5, 2019, at https://websoilsurvey.nrcs.usda.gov/app/.

Navarro-Racines, C., Tarapues, J., Thornton, P., Jarvis, A., and Ramirez-Villegas, J., 2020, High-resolution and bias-corrected CMIP5 projections for climate change impact assessments: Scientific Data, v. 7, no. 7, https://doi.org/10.1038/s41597-019-0343-8.

Nimmo, J.R., Schmidt, K.M., Perkins, K.S., and Stock, J.D., 2009, Rapid measurement of field-saturated hydraulic conductivity for areal characterization: Vadose Zone Journal, v. 8, no. 1, p. 142–149, https://doi.org/10.2136/vzj2007.0159.

Office for Coastal Management, 2020a, C-CAP land cover, Big Island, Hawaii from 2010-06-15 to 2010-08-15: NOAA National Centers for Environmental Information, accessed May 7, 2012, at https://www.fisheries.noaa.gov/inport/item/48346.

Office for Coastal Management, 2020b, C-CAP land cover, Kauai, Hawaii from 2010-06-15 to 2010-08-15: NOAA National Centers for Environmental Information, accessed May 7, 2021, at https://www.fisheries.noaa.gov/inport/item/48298.

Office for Coastal Management, 2020c, C-CAP land cover, Lanai, Hawaii from 2010-06-15 to 2010-08-15: NOAA National Centers for Environmental Information, accessed May 7, 2021, at https://www.fisheries.noaa.gov/inport/item/48348.

Office for Coastal Management, 2020d, C-CAP land cover, Maui, Hawaii from 2010-06-15 to 2010-08-15: NOAA National Centers for Environmental Information, accessed May 7, 2021, at https://www.fisheries.noaa.gov/inport/item/48351.

Office for Coastal Management, 2020e, C-CAP land cover, Molokai, Hawaii from 2010-06-15 to 2010-08-15: NOAA National Centers for Environmental Information, accessed May 7, 2021, at https://www.fisheries.noaa.gov/inport/item/48354.

Office for Coastal Management, 2020f, C-CAP land cover, Oahu, Hawaii from 2010-06-15 to 2010-08-15: NOAA National Centers for Environmental Information, accessed May 7, 2021, at https://www.fisheries.noaa.gov/inport/item/48356.

Oki, D.S., 2002, Reassessment of ground-water recharge and simulated ground-water availability for the Hawi area of north Kohala, Hawaii: U.S. Geological Survey Water-Resources Investigations Report 02–4006, 62 p., https://doi.org/10.3133/wri20024006.

Oki, D.S., 2003, Surface water in Hawaii: U.S. Geological Survey Fact Sheet 045–03, 6 p., https://doi.org/10.3133/fs04503.

Oki, D.S., 2004, Trends in streamflow characteristics at long-term gaging stations, Hawaii: U.S. Geological Survey Scientific Investigations Report 2004–5080, 116 p., https://doi.org/10.3133/sir20045080.

Oki, D.S., 2022, Water-budget accounting for tropical regions model (WATRMod) documentation: U.S. Geological Survey Open File Report 2022–1013, 77 p., https://doi.org/10.3133/ofr20221013.

Oki, D.S., Engott, J.A., and Rotzoll, K., 2020, Numerical simulation of groundwater availability in central Molokaʻi, Hawaiʻi: U.S. Geological Survey Scientific Investigations Report 2019–5150, 95 p., http://doi.org/10.3133/sir20195150.

Penn, D.C., 1997, Water and energy flows in Hawaii taro pondfields: Honolulu, Hawaiʻi, University of Hawaiʻi at Mānoa PhD Dissertation, 376 p.

Perkins, K.S., Stock, J.D., and Nimmo, J.R., 2018, Vegetation influences on infiltration in Hawaiian soils: Ecohydrology, v. 11, no. 5, article 1973, 6 p., https://doi.org/10.1002/eco.1973.

Perroy, R., and Collier, E., 2021, 2020 Update to the Hawaiʻi statewide agricultural land use baseline: State of Hawaiʻi Department of Agriculture, prepared by University of Hawaiʻi at Hilo, Spatial Data Analysis and Visualization Laboratory, 52 p., accessed February 29, 2024, at https://hdoa.hawaii.gov/salub/.

Povak, N.A., Hessburg, P.F., Giardina, C.P., Reynolds, K.M., Heider, C., Salminen, E., Salter, R.B., and MacKenzie, R.A., 2017, A watershed decision support tool for managing invasive species on Hawaiʻi Island, USA: Forest Ecology and Management, v. 400, p. 300–320, https://doi.org/10.1016/j.foreco.2017.05.046.

Rea, A., and Skinner, K.D., 2012, Geospatial datasets for watershed delineation and characterization used in the Hawaii StreamStats web application: U.S. Geological Survey Data Series 680, 12 p., https://doi.org/10.3133/ds680.

van Roosmalen, L., Christensen, J.H., Butts, M.B., Jensen, K.H., and Refsgaard, J.C., 2010, An intercomparison of regional climate model data for hydrological impact studies in Denmark: Journal of Hydrology, v. 380, nos. 3–4, p. 406–419, https://doi.org/10.1016/j.jhydrol.2009.11.014.

van Roosmalen, L., Christensen, B.S.B., and Sonnenborg, T.O., 2007, Regional differences in climate change impacts on groundwater and stream discharge in Denmark: Vadose Zone Journal, v. 6, no. 3, p. 554–571, https://doi.org/10.2136/vzj2006.0093.

van Roosmalen, L., Sonnenborg, T.O., and Jensen, K.H., 2009, Impact of climate and land use change on the hydrology of a large-scale agricultural catchment: Water Resources Research, v. 45, no. 7, 18 p., https://doi.org/10.1029/2007WR006760.

van Roosmalen, L., Sonnenborg, T.O., Jensen, K.H., and Christensen, J.H., 2011, Comparison of hydrological simulations of climate change using perturbation of observations and distribution-based scaling: Vadose Zone Journal, v. 10, no. 1, p. 136–150, https://doi.org/10.2136/vzj2010.0112.

Rosa, S.N., and Oki, D.S., 2010, Hawaii StreamStats—A web application for defining drainage-basin characteristics and estimating peak-streamflow statistics: U.S. Geological Survey Fact Sheet 2010–3052, 4 p., https://doi.org/10.3133/fs20103052.

Rotzoll, K., and Johnson, A.G., 2023, Mean annual water-budget components for Oahu, Hawaii, for future-climate conditions, CMIP5 RCP8.5 2041–70 scenario rainfall and 2010 land cover: U.S. Geological Survey data release, https://www.sciencebase.gov/catalog/item/64435726d34ee8d4ade9d74e.

Rotzoll, K., Oki, D.S., Johnson, A.G., and Souza, W.R., 2022, Long-term groundwater availability in the Waiheʻe, ʻĪao, and Waikapū aquifer systems, Maui, Hawaiʻi: U.S. Geological Survey Scientific Investigations Report 2021–5113, 80 p., https://doi.org/10.3133/sir20215113.

Safeeq, M., and Fares, A., 2012, Hydrologic response of a Hawaiian watershed to future climate change scenarios: Hydrological Processes, v. 26, no. 18, p. 2745–2764, https://doi.org/10.1002/hyp.8328.

Safeeq, M., and Fares, A., 2014, Interception losses in three non-native Hawaiian forest stands: Hydrological Processes, v. 28, no. 2, p. 237–254, https://doi.org/10.1002/hyp.9557.

Sanderson, M. (Ed.), 1993, Prevailing trade winds—weather and climate in Hawaiʻi: University of Hawaiʻi Press, Honolulu, Hawaiʻi, USA, 126 p.

Savenije, H.H.G., 2004, The importance of interception and why we should delete the term evapotranspiration from our vocabulary: Hydrological Processes, v. 18, no. 8, p. 1507–1511, https://doi.org/10.1002/hyp.5563.

Schroeder, T.A., 1993, Climate controls, *in* Sanderson, M. ed., Prevailing trade winds—weather and climate in Hawai‘i: University of Hawai‘i Press, Honolulu, Hawai‘i, p. 12–36.

Shade, P.J., 1995, Water budget for the Island of Kauai, Hawaii: U.S. Geological Survey Water-Resources Investigations Report 95–4128: 25 p, https://doi.org/10.3133/wri954128.

Sherrod, D.R., Sinton, J.M., Watkins, S.E., and Brunt, K.M., 2007, Geologic map of the State of Hawai‘i: U.S. Geological Survey Open-File Report 2007–1089, 83 p., https://doi.org/10.3133/ofr20071089.

Sherrod, D.R., Sinton, J.M., Watkins, S.E., and Brunt, K.M., 2021, Islands of Moloka‘i, Lāna‘i, and Kaho‘olawe, sheet 3 of Geologic map of the State of Hawai‘i: U.S. Geological Survey Scientific Investigations Map 3143, pamphlet, 72 p., 5 sheets, scales 1:100,000 and 1:250,000, https://doi.org/10.3133/sim3143.

Shuttleworth, W.J., 1993, Evaporation, *in* Maidment, D.R. ed., Handbook of Hydrology: McGraw-Hill, New York, p. 4.1–4.53.

Skamarock, W.C., Klemp, J.B, Dudhia, J., Gill D.O., Barker D.M., Wang W., and Powers J.G., 2008, A description of the Advanced Research WRF Version 3. NCAR Technical Note 475+STR, 113 p.

State of Hawai‘i, 2014, Aquifer sector and aquifer system area boundaries for the Islands of Hawaii, Kauai, Lanai, Maui, Molokai, and Oahu: Department of Land and Natural Resources, accessed August 6, 2020, at http://planning.hawaii.gov/gis/download-gis-data/.

State of Hawai‘i, 2017, Golf courses in the State of Hawai‘i as of October, 2014: Hawaii Statewide GIS Program, Office of Planning, accessed October 1, 2021, at https://planning.hawaii.gov/gis/download-gis-data-expanded/.

State of Hawai‘i, 2020, Tax map key parcels (all islands): Office of Planning and Sustainable Development, Download GIS Data (Expanded), accessed February 25, 2020, at http://planning.hawaii.gov/gis/download-gis-data-expanded/.

State of Hawai‘i, 2021, Dam inventory system: Department of Land and Natural Resources, Engineering Division—Dam Safety, accessed October 1, 2021, at https://dams.hawaii.gov/Default.aspx.

Stearns, H.T., and Macdonald, G.A., 1947, Geology and ground-water resources of the island of Molokai, Hawaii, Bulletin 11: Hawai‘i Division of Hydrography 11, 113 p.

Strauch, A.M., Giardina, C.P., MacKenzie, R.A., Heider, C., Giambelluca, T.W., Salminen, E., and Bruland, G.L., 2017, Modeled effects of climate change and plant invasion on watershed function across a steep tropical rainfall gradient: Ecosystems, v. 20, no. 3, p. 583–600, https://doi.org/10.1007/s10021-016-0038-3.

Sustainable Resources Group Intn’l Inc., 2009, Kilauea irrigation water engineering design and monitoring study: County of Kaua‘i, Office of Economic Development, prepared by Sustainable Resources Group Intn’l, Inc.

Takahashi, M., Giambelluca, T.W., Mudd, R.G., DeLay, J.K., Nullet, M.A., and Asner, G.P., 2011, Rainfall partitioning and cloud water interception in native forest and invaded forest in Hawai‘i Volcanoes National Park: Hydrological Processes, v. 25, no. 3, p. 448–464, https://doi.org/10.1002/hyp.7797.

Taylor, K.E., Stouffer, R.J., and Meehl, G.A., 2012, An overview of CMIP5 and the experiment design: Bulletin of the American Meteorological Society, v. 93, no. 4, p. 485–498, https://doi.org/10.1175/BAMS-D-11-00094.1.

Thornthwaite, C.W., and Mather, J.R., 1955, The water balance: Climatology, v. 8, p. 1–104.

Timm, O., and Diaz, H.F., 2009, Synoptic-statistical approach to regional downscaling of IPCC twenty-first century climate projections—Seasonal rainfall over the Hawaiian Islands: Journal of Climate, v. 22, no. 16, p. 4261–4280, https://doi.org/10.1175/2009JCLI2833.1.

Townscape, Inc., 2019, Hawai‘i Water Plan—Water resource protection plan, 2019 update: State of Hawai‘i, Department of Land and Natural Resources, Commission on Water Resource Management, prepared by Townscape, Inc., 746 p., accessed April 22, 2021, at https://dlnr.hawaii.gov/cwrm/planning/hiwaterplan/wrpp/.

Tribble, G., 2008, Ground water on tropical pacific islands—Understanding a vital resource: U.S. Geological Survey Circular 1312, 35 p., https://doi.org/10.3133/cir1312.

University of Hawai‘i, 2008, Hawaii agricultural water use and development plan: State of Hawai‘i Department of Agriculture, prepared by College of Tropical Agriculture and Human Resources, Department of Natural Resources and Environmental Management, University of Hawai‘i at Mānoa, Honolulu, Hawai‘i, 108 p.

U.S. Census Bureau, 2021, Historical population change data (1910–2020), accessed April 26, 2021, at https://www.census.gov/data/tables/time-series/dec/popchange-data-text.html.

U.S. Geological Survey, 2013a, USGS NED n19w156 1/3 arc-second 2013 1 x 1 degree ArcGrid, accessed June 10, 2020, at https://apps.nationalmap.gov/viewer/.

U.S. Geological Survey, 2013b, USGS NED n20w155 1/3 arc-second 2013 1 x 1 degree ArcGrid, accessed June 10, 2020, at https://apps.nationalmap.gov/viewer/.

U.S. Geological Survey, 2013c, USGS NED n20w156 1/3 arc-second 2013 1 x 1 degree ArcGrid, accessed June 10, 2020, at https://apps.nationalmap.gov/viewer/.

U.S. Geological Survey, 2013d, USGS NED n20w157 1/3 arc-second 2013 1 x 1 degree ArcGrid, accessed June 10, 2020, at https://apps.nationalmap.gov/viewer/.

U.S. Geological Survey, 2013e, USGS NED n21w156 1/3 arc-second 2013 1 x 1 degree ArcGrid, accessed June 10, 2020, at https://apps.nationalmap.gov/viewer/.

U.S. Geological Survey, 2013f, USGS NED n21w157 1/3 arc-second 2013 1 x 1 degree ArcGrid, accessed February 7, 2020, at https://apps.nationalmap.gov/viewer/.

U.S. Geological Survey, 2013g, USGS NED n21w158 1/3 arc-second 2013 1 x 1 degree ArcGrid, accessed February 7, 2020, at https://apps.nationalmap.gov/viewer/.

U.S. Geological Survey, 2016, LANDFIRE, 2014, existing vegetation type layer, LANDFIRE 1.4.0: U.S. Department of the Interior, Geological Survey, and U.S. Department of Agriculture, accessed November 2, 2019, at http://landfire.cr.usgs.gov/viewer/.

U.S. Geological Survey, 2019, USGS National Hydrography Dataset Plus High Resolution (NHDPlus HR) for 4-digit Hydrologic Unit—2004, accessed February 7, 2020, at https://apps.nationalmap.gov/viewer/.

U.S. Geological Survey, 2020, USGS National Hydrography Dataset Plus High Resolution (NHDPlus HR) for 4-digit Hydrologic Unit—2001, accessed June 10, 2020, at https://apps.nationalmap.gov/viewer/.

U.S. Geological Survey Gap Analysis Project, 2016, GAP/LANDFIRE National Terrestrial Ecosystems 2011: U.S. Geological Survey data release, https://doi.org/10.5066/F7ZS2TM0.

Viessman, W., and Lewis, G.L., 2003, Introduction to hydrology, fifth edition: Prentice Hall, Pearson Education, Inc., Upper Saddle River, New Jersey.

Wahl, K.L., and Wahl, T.L., 1995, Determining the flow of Comal Springs at New Braunfels, Texas, *in* Proceedings of Texas Water '95, a component conference of the first international conference on water resource engineering: American Society of Civil Engineers, August 16–17, 1995, San Antonio, Texas, p. 77–86.

Watson, L.J. (Ed.), 1964, Observations made with respect to irrigation and growth of taro at certain patches at Waiahole and Kahaluu—Honolulu, Hawai'i, City & County of Honolulu Board of Water Supply Water Resources Division, *in* Reppun, v., Board of Water Supply.

Whittier, R.B., and El-Kadi, A.I., 2009, Human and environmental risk ranking of onsite sewage disposal systems, final: State of Hawai'i Department of Health, Safe Drinking Water Branch, prepared by University of Hawai'i at Mānoa, Department of Geology and Geophysics, 72 p.

Whittier, R.B., and El-Kadi, A.I., 2014, Human and environmental risk ranking of onsite sewage disposal systems for the Hawaiian Islands of Kauai, Molokai, Maui and Hawaii, final: State of Hawai'i Department of Health, Safe Drinking Water Branch, prepared by University of Hawai'i at Mānoa, Department of Geology and Geophysics, 257 p., accessed October 5, 2021, at http://hdl.handle.net/10125/50771.

Wilby, R.L., Hay, L.E., Gutowski, W.J., Arritt, R.W., Takle, E.S., Pan, Z., Leavesley, G.H., and Clark, M.P., 2000, Hydrological responses to dynamically and statistically downscaled climate model output: Geophysical Research Letters, v. 27, no. 8, p. 1199–1202, https://doi.org/10.1029/1999GL006078.

Yamanaga, G., 1972, Evaluation of the streamflow-data program in Hawaii: U.S Geological Survey Open-File Report 72–453, 28 p., https://doi.org/10.3133/ofr72453.

Zhang, C., 2016, Dynamical Downscaled and Projected Climate for the US Pacific Islands: U.S. Geological Survey data release, https://doi.org/10.21429/ss35-k236.

Zhang, C. 2017, Very fine resolution dynamically downscaled climate data for Hawaii: U.S. Geological Survey data release, https://doi.org/10.21429/1vke-9325.

Zhang, C., and Wang, Y., 2017, Projected future changes of tropical cyclone activity over the western north and south Pacific in a 20-km-mesh regional climate model: Journal of Climate, v. 30, no. 15, p. 5923–5941, https://doi.org/10.1175/JCLI-D-16-0597.1.

Zhang, C., Wang, Y., Hamilton, K., and Lauer, A., 2016a, Dynamical downscaling of the climate for the Hawaiian Islands Part I—Present day: Journal of Climate, v. 29, no. 8, p. 3027–3048, https://doi.org/10.1175/JCLI-D-15-0432.1.

Zhang, C., Wang, Y., Hamilton, K., and Lauer, A., 2016b, Dynamical downscaling of the climate for the Hawaiian Islands Part II—Projection for the late twenty-first century: Journal of Climate, v. 29, no. 23, p. 8333–8354, https://doi.org/10.1175/JCLI-D-16-0038.1.

Zhang, C., Wang, Y., Lauer, A., Hamilton, K., and Xie, F., 2012, Cloud base and top heights in the Hawaiian region determined with satellite and ground-based measurements: Geophysical Research Letters, v. 39, no. 15, 6 p., https://agupubs.onlinelibrary.wiley.com/doi/10.1029/2012GL052355.

Appendixes 1–3

Appendix 1. Estimation of Model Input Parameters for Kiawe

This appendix describes the estimation of selected parameter values assigned to kiawe for calculations by the Water-budget Accounting for Tropical Regions Model (WATRMod; Oki, 2022). The selected parameters are crop coefficient, root depth, depletion fraction, depth to groundwater, groundwater-use adjustment factor, and mixed land-cover designation. Values assigned to these parameters were used by WATRMod to compute evapotranspiration (ET) and groundwater use by kiawe.

Prosopis pallida (hereafter *P. pallida*) and other species from the genus *Prosopis*, including *Prosopis juliflora* (hereafter *P. juliflora*), are phreatophytic forest species collectively referred to as kiawe in Hawai‘i (Gallaher and Merlin, 2010). *P. pallida* was introduced to the Hawaiian Islands in 1828 and is now found on all main islands. *P. juliflora* was first observed on O‘ahu in 1978 and is now also found on Kaua‘i, Moloka‘i, and Lāna‘i. The Kiawe, no fog and Kiawe, fog land-cover designations used in this study (fig. 4) only represent *P. pallida* (U.S. Geological Survey Gap Analysis Project, 2016), the more dominant species of kiawe in Hawai‘i, and cover about 2 percent of the combined total land area of Kaua‘i, O‘ahu, Moloka‘i, Lāna‘i, Maui, and the Island of Hawai‘i (Kāne and others, 2024).

In their native habitat of South America, *P. pallida* and *P. juliflora* can survive in areas with mean annual rainfall of less than 200 millimeters per year (mm/yr; 8 inches per year [in/yr]) by accessing shallow groundwater (Pasiecznik and others, 2001). On the Hawaiian Islands, optimal and marginal potential kiawe habitat constitutes 34 percent of the total land area (Gallaher and Merlin, 2010). *P. pallida* and *P. juliflora* have two distinct root systems that form under normal conditions of unimpeded root development: (1) a lateral root system, and (2) a deep root system. The lateral root system of *P. juliflora* can spread up to 18 meters (m; 59 feet [ft]) or more from the base of the tree and extend to a depth of 1.8 m (71 inches [in]) (Pasiecznik and others, 2001). The deep root system for these two species is comprised of 1–3 main tap roots for anchoring the tree and sourcing groundwater. In their native habitat, the tap roots of *P. pallida* can extend to groundwater up to 25 m (82 ft) below the ground surface.

Recent field studies of transpiration and groundwater use by *P. pallida* in a coastal area of the Kīholo aquifer system, Island of Hawai‘i (fig. 2; hereinafter Kīholo) indicate that *P. pallida* uses shallow groundwater and infiltrating rainfall at ground-surface altitudes of 15 m (49 ft) or less but relies solely on infiltrating rainfall at altitudes above 15 m (49 ft; Dudley and others, 2014; Miyazawa and others, 2016, 2020). Miyazawa and others (2016) estimated an annual stand-level transpiration rate of 350 mm/yr (13.8 in/yr) from August 2010 to July 2011 in a lowland stand of *P. pallida* in Kīholo with a ground-surface altitude of 4.3 m (14 ft; B. Dudley, National Institute of Water and Atmospheric Research Ltd, New Zealand, written commun. 2021), hereinafter site L1. Using stable-isotope analyses of stem water, Dudley and others (2020) estimated that groundwater supplied 95 percent of the water transpired by the stand of *P. pallida* at site L1. Dudley and others (2020) also estimated an annual stand-level transpiration rate of 3.1 mm/yr (0.12 in/yr) at an upland stand of *P. pallida* in Kīholo at an altitude of 79.5 m (261 ft).

Dudley and others (2020) noted that above-ground biomass and transpiration of *P. pallida* in Kīholo are controlled by depth to groundwater, indicating that areas of *P. pallida* with greater biomass and greater access to groundwater tend to have higher transpiration. They used light detection and ranging metrics, and other field measurements to develop a statistical regression for modeling transpiration rates by *P. pallida* across the Kīholo area on the basis of maximum tree-canopy height of *P. pallida* and ground-surface altitude, where ground-surface altitude was considered a proxy for depth to groundwater. They estimated that 98 percent of the total transpiration by *P. pallida* in Kīholo occurs across 25 percent of the total study area where the ground-surface altitude is less than 15 m (49 ft) and *P. pallida* can access shallow groundwater.

Crop Coefficient

The crop coefficient, K_c, is an empirically derived ratio of the potential ET of a given land-cover type under standard conditions, ET_c, to reference ET, ET_o (Allen and others, 1998). Standard conditions refer to crops and vegetation grown under ideal agronomic and soil-water conditions where no limitations are placed on crop/vegetation growth or ET due to water shortage and (or) crop density, or disease, weed, insect and (or) salinity pressures. WATRMod calculates potential ET for a particular subarea as the product of ET_o and the K_c assigned to the subarea's land-cover type. The term "crop coefficient" can be misleading because WATRMod uses crop coefficients to calculate potential ET for all subareas, including subareas with non-agricultural types of land cover, such as kiawe.

In WATRMod, crop coefficients for forest land-cover types integrate the effects of transpiration and ground evaporation, whereas canopy evaporation is accounted for separately. For this study, crop coefficients were assumed to be temporally constant for all land-cover types except seed corn which has crop coefficients that vary monthly (see "Potential Evaporation" section of report). Reference ET, as defined in this study, is the potential ET of a hypothetical grass surface with specific characteristics and optimum soil-water conditions for given climatic conditions and is equivalent to Food and Agriculture Organization of the United Nations (FAO) Penman-Monteith ET (Allen and others, 1998) (see "Potential Evapotranspiration" section of report).

The crop coefficient for kiawe, K_{kiawe}, can be estimated from the following equation (modified from Allen and others, 1998):

$$K_{kiawe} = \frac{ET_{kiawe}}{ET_o} \quad (1\text{-}1)$$

where

K_{kiawe} is the crop coefficient for kiawe land cover [dimensionless],

ET_{kiawe} is the annual kiawe potential ET, in inches per year,

ET_o is the annual grass reference-surface potential ET, in inches per year.

ET_{kiawe} can be estimated as:

$$ET_{kiawe} = E_{trans} + E_{ground} \quad (1\text{-}2)$$

where

E_{trans} is the annual transpiration of kiawe assuming no moisture stress and 100 percent tree cover, in inches per year,

E_{ground} is the annual ground evaporation assuming 100 percent tree cover, in inches per year.

The transpiration of kiawe, E_{trans}, was estimated from the annual stand-level transpiration rate (350 mm/yr; 13.8 in/yr) for *P. pallida* at site L1 from August 2010 to July 2011 (Miyazawa and others, 2016). The stand of *P. pallida* at site L1 was assumed to be under no moisture stress due to access to shallow groundwater. The amount of above-ground biomass at site L1 and adjacent areas in Kīholo was evaluated using maps developed by the U.S. Geological Survey (USGS) Landfire program that characterize the vertically projected percentage cover of existing vegetation and fuel vegetation cover at a grid size of 30 m (98 ft; U.S. Geological Survey, 2020a, b), hereinafter Landfire maps. Fuel vegetation cover refers to tree, shrub, and herbaceous lifeforms that can provide "fuel" for wildfire occurrence and propagation. The Landfire maps show a percentage or range of percentages that characterize the vegetation cover percentage at each grid cell. For example, at site L1, the tree cover percentage is defined as "Tree Cover = 75%" (U.S. Geological Survey, 2020a) and "Tree Cover ≥ 70 and < 80%" (U.S. Geological Survey, 2020b), which indicates that site L1 has 75 percent tree cover. For this analysis, the tree cover percentage represented in the Landfire maps was considered a proxy for above-ground biomass. The tree cover percentage shown in the Landfire maps in other areas of Kīholo varies and indicates the potential for higher (or lower) transpiration rates in areas with greater (or less) above-ground biomass and similar access to groundwater when compared to conditions at site L1. Additionally, a map of above-ground biomass in Kīholo, developed by Dudley and others (2020), bears resemblance with the Landfire maps of tree cover percentage. Collectively, these maps indicate that the transpiration rate estimated by Miyazawa and others (2016) at site L1 might represent the transpiration rate of kiawe with 75 percent tree cover, rather than 100 percent tree cover (eq. 1-2). Therefore, the transpiration rate estimated by Miyazawa and others (2016) might underestimate the transpiration rates in areas of Kīholo with greater tree cover and similar access to groundwater. The transpiration of kiawe, for equation 1-2 was computed as the ratio of the transpiration rate reported by Miyazawa and others (2016) for site L1 and an adjustment factor:

$$E_{trans} = \frac{E_{L1}}{TC} \quad (1\text{-}3)$$

where

E_{L1} is the annual kiawe transpiration rate of 350 mm/yr (13.8 in/yr) at site L1 from August 2010 to July 2011 (Miyazawa and others, 2016),

TC is the adjustment factor of 0.75 that represents the tree cover at site L1 from the Landfire maps expressed as a decimal fraction (U.S. Geological Survey, 2020a, b).

Substituting values for E_{L1} and TC into equation 1-3, E_{trans} can be estimated as:

$$E_{trans} = \frac{13.8 \text{ in/yr}}{0.75} = 18.4 \text{ in/yr}$$

The transpiration of kiawe, E_{trans}, from equation 1-3 was assumed to be equivalent to the mean annual transpiration rate of kiawe at site L1 assuming 100 percent tree cover. Ground evaporation, E_{ground}, was assumed to be temporally constant throughout the simulation period and estimated as the mean annual soil evaporation at site L1, 103.55 mm/yr (4.1 in/yr), from Giambelluca and others (2014). Owing to a lack of information on the relation between ground evaporation and vegetative cover at the Kīholo site, no adjustment was made to E_{ground} to reflect the tree cover at site L1. ET_{kiawe} was then estimated from equation 1-2 as:

$$ET_{kiawe} = 18.4 \text{ in/yr} + 4.1 \text{ in/yr} = 22.5 \text{ in/yr}$$

The estimated rates for kiawe transpiration and soil evaporation at the Kīholo site are similar to values reported for other species from the genus *Prosopis*. For example, in a study of *Prosopis velutina* at a site near the San Pedro River in Arizona, Scott and others (2006) reported mean transpiration and mean soil evaporation rates of 20.4 and 3.7 in/yr, respectively, during growing seasons from 2001 to 2003.

Grass reference-surface potential ET, ET_o, was estimated as the mean annual grass reference-surface potential ET at site L1, 2,540.2 mm/yr (100 in/yr), from Giambelluca and others (2014). The crop coefficient for kiawe, K_{kiawe}, was then estimated from equation 1-1 as:

$$K_{kiawe} = \left[\frac{22.5 \text{ in/yr}}{100 \text{ in/yr}}\right] = 0.23$$

Root Depth, Depletion Fraction, and Depth to Groundwater

Information on the root depth of the two species of kiawe, *P. pallida* and *P. juliflora*, across their established range in the Hawaiian Islands is lacking. For WATRMod input, kiawe land cover was assigned a root depth of 1.8 m (71 in), which is the approximate depth of the lateral root zone system of *P. juliflora* in its native habitat (Pasiecznik and others, 2001). This is the same root depth used for the kiawe/phreatophyte land cover in recent water-budget studies in Hawai‘i (Engott and others, 2017; Izuka and others, 2018) and is generally consistent with anecdotal evidence of the lateral root zone depth of *P. pallida* (fig. 1-1). The deep root systems of *P. pallida* and *P. juliflora* were not considered in determining the root depth for kiawe because their primary function is for anchoring the tree and sourcing groundwater (Pasiecznik and others, 2001).

Dudley and others (2020) assumed groundwater to be at an altitude close to that of sea level throughout Kīholo and therefore used ground-surface altitude (in meters above sea level) as a proxy for depth to groundwater. We assumed ground-surface altitude equaled depth to groundwater for subareas of kiawe land cover on Kaua‘i, O‘ahu, Moloka‘i, Lāna‘i, Maui, and the Island of Hawai‘i. On each island, all subareas of kiawe land cover with a ground-surface altitude of 15 m (49 ft) or less were assumed to have year-round access to groundwater, whereas all subareas above 15 m (49 ft) were assumed to have no access to groundwater. A depletion fraction of 1 was used for subareas of kiawe land cover assumed to have year-round access to shallow groundwater. The depletion fraction, which ranges from 0 to 1, is defined as the fraction of moisture-storage capacity that can be depleted from the plant-root zone before moisture stress causes a reduction in the actual-ET rate (see Total Evapotranspiration section of report). A depletion fraction of 0.5 was used for subareas of kiawe land cover assumed to have no access to groundwater, which is consistent with the value used in this study for native and non-native forest land cover. The altitude of each subarea of kiawe land cover was determined using digital elevation models of the Hawaiian Islands with a grid size of 10 m (33 ft; National Oceanic and Atmospheric Administration, 2007).

Groundwater Use-Adjustment Factor

WATRMod was used to simulate actual ET and groundwater use by kiawe by treating (1) subareas of kiawe assumed to have access to groundwater as an irrigated land cover, and (2) groundwater accessed by the deep tap roots of kiawe as the source of irrigation water. Similar to the irrigation rate adjustment approach for other irrigated land cover types, WATRMod's irrigation rate estimates for kiawe land cover were multiplied by an adjustment factor, hereinafter groundwater use-adjustment factor. The groundwater use-adjustment factor ensured that WATRMod's estimates of ET_{kiawe} were consistent with estimates of ET_{kiawe} determined from the Kīholo studies (Miyazawa and others, 2016; Dudley and others, 2020). Subareas of kiawe assumed to have no access to groundwater were simulated as unirrigated land cover, similar to the approach used for native and non-native forest land cover.

Figure 1-1. Photograph of uprooted *Prosopis pallida* tree in southeastern O‘ahu, State of Hawai‘i. U.S. Geological Survey photograph by Alan Mair, February 2021.

First, WATRMod was used to simulate ET_{kiawe} and other water-budget components at site L1 during 1990–2012. A sequential simulation approach was used to replicate the daily rainfall pattern during 1990–2012. For these simulations, a separate set of sequential rainfall fragments was developed using the daily rainfall dataset described by Longman and others (2019; data available at https://waihona.its.hawaii.edu/index.php/s/SkChz33eca4Y6tl). The 93-year gridded dataset of monthly rainfall maps during 1920–2012 (Frazier and others, 2016) was used to identify the 5-year period with the lowest mean annual island-wide rainfall. Mean annual island-wide rainfall on the Island of Hawai‘i was lowest during 2008–12. Given the low rainfall and high permeability of the young surface rocks in the area, runoff from rainfall was assumed to be negligible at site L1 during 1990–2012. A crop coefficient of 0.23 was used for all subareas with kiawe land cover. Values assigned to all other model input parameters were the same as those described earlier in the report. Next, only WATRMod results for the period from August 2010 to July 2011 were analyzed to be consistent with the period used by Miyazawa and others (2016). A groundwater use-adjustment factor of 0.86 was then estimated using WATRMod and a trial-and-error approach. The trial-and-error approach involved repeatedly adjusting the groundwater use-adjustment factor and rerunning WATRMod until the model estimate ET_{kiawe} at site L1 for the period from August 2010 to July 2011 matched the estimate of ET_{kiawe} from Miyazawa and others (2016) during the same period. The groundwater use-adjustment factor of 0.86 was then used in WATRMod for all subareas with kiawe land cover and ground-surface altitudes of 15 m (49 ft) or less on each island.

Mixed Land-Cover Designation for Subareas of Kiawe

A mixed land-cover designation was used in WATRMod for all kiawe subareas with year-round access to groundwater to account for the spatial variability in above-ground biomass. First, the area of each tree cover percentage class (for example, "Tree Cover ≥ 10 and < 20%," "Tree Cover ≥ 20 and < 30%," "Tree Cover ≥ 30 and < 40%," and so forth) within each subarea with kiawe land cover was tabulated in a Geographic Information System using the Landfire fuel vegetation cover map (table 1-1; U.S. Geological Survey, 2020b). Next, the tree cover percentage was computed for each subarea with kiawe land cover using the following equation:

$$M_i = \frac{X_i}{T_i} \times 100 \quad (1\text{-}4)$$

where

M_i is the tree cover percentage in subarea i [dimensionless],

X_i is the area of tree cover in subarea i, in square meters, and

T_i is the total area of subarea i, in square meters.

The area of tree cover, X_i, in each subarea with kiawe land cover was calculated with the following equation:

$$X_i = \sum_{j=1}^{n} x_{i,j} c_j \quad (1\text{-}5)$$

where

j is the tree cover percentage class from the Landfire fuel vegetation map (table 1-1; U.S. Geological Survey, 2020b),

n is the total number of tree cover percentage classes,

$x_{i,j}$ is the area of tree cover percentage class j within subarea i, in square meters, and

c_j is the midpoint value in tree cover percentage class j (table 1-1) [dimensionless].

The percentage tree cover, M_i, was then used to assign a mixed land-cover designation consisting of Kiawe, no fog and Grassland for subareas with kiawe land cover (table 1-1). Grassland land cover was assigned as the secondary land-cover type because kiawe land cover in the Hawaiian Islands is often described as mixed with grassland species (National Oceanic and Atmospheric Administration, 2011). The mixed land cover designation approach was applied to subareas with kiawe land cover and an altitude of 15 m (49 ft) or less. In this study, a mixed land-cover designation was used in WATRMod for only a subset of irrigated land-cover types including seed corn, taro, and kiawe at an altitude of 15 m (49 ft) or less. Evaluating mixed land-cover designations for all other land-cover types, including kiawe at an altitude of greater than 15 m (49 ft), was beyond the scope of this study.

Table 1-1. Summary of percentage tree cover classes, midpoint values, and mixed land-cover designations for subareas with kiawe land cover and ground-surface altitudes of 15 m (49 feet) or less.

[Ground-surface altitudes for these subareas of 15 meters (49 feet) or less; percentage tree cover from U.S. Geological Survey (2020b); >, greater than]

Percentage tree cover in subarea	Midpoint value of percentage tree cover expressed as a decimal fraction	Percentage of subarea assigned to Kiawe, no fog and Grassland land cover	
		Kiawe, no fog	Grassland
10 and below	0.05	5	95
> 10 to 20	0.15	15	85
> 20 to 30	0.25	25	75
> 30 to 40	0.35	35	65
> 40 to 50	0.45	45	55
> 50 to 60	0.55	55	45
> 60 to 70	0.65	65	35
> 70 to 80	0.75	75	25
> 80 to 90	0.85	85	15
> 90	0.95	95	5

References Cited

Allen, R.G., Pereira, L.S., Raes, D., and Smith, M., 1998, Crop evapotranspiration—Guidelines for computing crop water requirements: Food and Agriculture Organization of the United Nations, Irrigation and drainage paper 56, v. 300, no. 9, p. D05109, accessed on April 21, 2021, at https://www.fao.org/3/x0490e/x0490e00.htm.

Dudley, B.D., Hughes, R.F., and Ostertag, R., 2014, Groundwater availability mediates the ecosystem effects of an invasion of Prosopis pallida: Ecological Applications, v. 24, no. 8, p. 1954–1971, https://doi.org/10.1890/13-1262.1.

Dudley, B.D., Hughes, R.F., Asner, G.P., Baldwin, J.A., Miyazawa, Y., Dulai, H., Waters, C., Bishop, J., Vaughn, N.R., Yeh, J., Kettwich, S., MacKenzie, R.A., Ostertag, R., and Giambelluca, T., 2020, Hydrological effects of tree invasion on a dry coastal Hawaiian ecosystem: Forest Ecology and Management, v. 458, no. 117653, https://doi.org/10.1016/j.foreco.2019.117653.

Engott, J.A., Johnson, A.G., Bassiouni, M., Izuka, S.K., and Rotzoll, K., 2017, Spatially distributed groundwater recharge for 2010 land cover estimated using a water-budget model for the Island of O'ahu, Hawai'i (ver 2.0, December 2017): U.S. Geological Survey Scientific Investigations Report 2015–5010, 49 p., https://doi.org/10.3133/sir20155010.

Frazier, A.G., Giambelluca, T.W., Diaz, H.F., and Needham, H.L., 2016, Comparison of geostatistical approaches to spatially interpolate month-year rainfall for the Hawaiian Islands: International Journal of Climatology, v. 36, no. 3, p. 1459–1470, https://doi.org/10.1002/joc.4437.

Gallaher, T., and Merlin, M., 2010, Biology and impacts of Pacific island invasive species—6. *Prosopis pallida* and *Prosopis juliflora* (Algarroba, Mesquite, Kiawe) (Fabaceae): Pacific Science, v. 64, no. 4, p. 489–526, https://doi.org/10.2984/64.4.489.

Giambelluca, T.W., Shuai, X., Barnes, M.L., Alliss, R.J., Longman, R.J., Miura, T., Chen, Q., Frazier, A.G., Mudd, R.G., Cuo, L., and Businger, A.D., 2014, Evapotranspiration of Hawai‘i: University of Hawai‘i at Mānoa, accessed on November 12, 2021, at http://evapotranspiration.geography.hawaii.edu/.

Izuka, S.K., Engott, J.A., Rotzoll, K., Bassiouni, M., Johnson, A.G., Miller, L.D., and Mair, A., 2018, Volcanic aquifers of Hawai‘i—Hydrogeology, water budgets, and conceptual models (ver. 2.0, March 2018): U.S. Geological Survey Scientific Investigations Report 2015–5164, 158 p., https://doi.org/10.3133/sir20155164.

Kāne, H.L., Mair, A., Mifflin, J., 2024, Mean annual water-budget components for Kaua‘i, O‘ahu, Moloka‘i, Lāna‘i, Maui, and the Island of Hawai‘i, for a set of recent and future-climate conditions, and 2020 land cover: U.S. Geological Survey data release, https://doi.org/10.5066/P972KMSL.

Longman, R.J., Frazier, A.G., Newman, A.J., Giambelluca, T.W., Schanzenbach, D., Kagawa-Viviani, A., Needham, H., Arnold, J.R., and Clark, M.P., 2019, High-resolution gridded daily rainfall and temperature for the Hawaiian Islands (1990–2014): Journal of Hydrometeorology, v. 20, no. 3, p. 489–508, https://doi.org/10.1175/JHM-D-18-0112.1.

Miyazawa, Y., Dudley, B.D., Hughes, R.F., Vandemark, J., Cordell, S., Nullet, M.A., Ostertag, R., and Giambelluca, T.W., 2016, Non-native tree in a dry coastal area in Hawai’i has high transpiration but restricts water use despite phreatophytic trait: Ecohydrology, v. 9, no. 7, p. 1166–1176, https://doi.org/10.1002/eco.1715.

National Oceanic and Atmospheric Administration, 2007, Digital elevation models (DEMs) for the main 8 Hawaiian Islands: National Ocean Service, Center for Coastal Monitoring and Assessment, Biogeography Branch, accessed on February 5, 2021, at https://www.fisheries.noaa.gov/inport/item/38724.

National Oceanic and Atmospheric Administration, 2011, Coastal change analysis program (C-CAP) high-resolution land cover map for the Hawaiian Islands: Office of Coastal Management, accessed on September 28, 2020 at www.coast.noaa.gov/htdata/raster1/landcover/bulkdownload/hires/.

Oki, D.S., 2022, Water-budget accounting for tropical regions model (WATRMod) documentation: U.S. Geological Survey Open File Report 2022–1013, 77 p., https://doi.org/10.3133/ofr20221013.

Pasiecznik, N.M., Felker, P., Harris, P.J.C., Harsh, L.N., Cruz, G., Tewari, J.C., Cadoret, K., and Maldonado, L.J., 2001, The *Prosopis juliflora–Prosopis pallida* complex—A monograph: HDRA, Coventry, UK, 172 p.

Scott, R.L., Williams, D.G., Goodrich, D.C., Cable, W.L., Levick, L.R., McGuire, R., Gazal, R.M., Yepez, E.A., Ellsworth, P., and Huxman, T.E., 2006, Determining the riparian ground-water use within the San Pedro Riparian National Conservation Area and the Sierra Vista Subwatershed, Arizona, chap D, *in* Leenhouts, J.M., Stromberg, J.C., and Scott, R.L. eds., Hydrologic requirements of and consumptive ground-water use by riparian vegetation along the San Pedro River, Arizona: U.S. Geological Survey Scientific Investigations Report 2005–5163, p. 107–142, https://doi.org/10.3133/sir20055163.

U.S. Geological Survey, 2020a, LANDFIRE Remap 2016 Existing Vegetation Cover (EVC) HI: Earth Resources Observation and Science Center (EROS), accessed January 27, 2021, at http://landfire.cr.usgs.gov/viewer/.

U.S. Geological Survey, 2020b, LANDFIRE Remap Fuel Vegetation Cover (FVC) HI: Earth Resources Observation and Science Center (EROS), accessed January 22, 2021, at http://landfire.cr.usgs.gov/viewer/.

U.S. Geological Survey (USGS) Gap Analysis Project (GAP), 2016, GAP/LANDFIRE National Terrestrial Ecosystems 2011: U.S. Geological Survey data release, https://doi.org/10.5066/F7ZS2TM0.

Appendix 2. Summary of Mean Annual Water-Budget Components

Table 2-1. Mean annual water-budget components from this study for aquifer systems and all of Kauaʻi.

[See figure 2 for locations of aquifer systems; --, not applicable; Total precipitation, sum of Rain and Fog; Septic, septic-system leachate; Total direct recharge, sum of cesspool seepage and constant recharge from unlined reservoirs and inland water bodies; Runoff, direct runoff; ET, evapotranspiration; Total ET, sum of Canopy evaporation and Actual ET; Groundwater use, kiawe, groundwater use by kiawe land cover; Recharge, groundwater recharge which includes Total direct recharge; ET_o, reference evapotranspiration for grass surface; V, ratio of the mean evaporation rate to mean precipitation rate during saturated conditions; SD, statistical downscaling; HRCM2, Hawai'i Regional Climate Model version 2 from Zhang and Wang (2017); RCP4.5 and RCP8.5, Representative Concentration Pathway with total radiative forcing of 4.5 and 8.5 Watts per square meter, respectively, by the year 2100 from Intergovernmental Panel on Climate Change (2014); 1978–2007 scenario is 1978–2007 rainfall from Frazier and others (2016), daily rainfall fragments from Longman and others (2019), ET_o from Giambelluca and others (2014), V ratios computed using equation 11 for a 1978–2007 scenario, and 2020 land cover; SD RCP8.5 2041–71 scenario is projected 2041–71 rainfall for a RCP8.5 warming scenario from Elison Timm and others (2015), daily rainfall fragments from Longman and others (2019), ET_o from Giambelluca and others (2014), V ratios computed using equation 11 for a 1978–2007 scenario, and 2020 land cover; SD RCP8.5 2071–99 scenario is projected 2071–99 rainfall for a RCP8.5 warming scenario from Elison Timm and others (2015), daily rainfall fragments from Longman and others (2019), ET_o from Giambelluca and others (2014), V ratios computed using equation 11 for a 1978–2007 scenario, and 2020 land cover; HRCM2 RCP4.5 2080–99 scenario is projected 2080–99 rainfall for a RCP4.5 warming scenario from Zhang and Wang (2017), daily rainfall fragments from Longman and others (2019), ET_o from Giambelluca and others (2014), V ratios computed using equation 11 for a 1978–2007 scenario, and 2020 land cover. Components might not balance because of rounding, direct recharge from cesspools and reservoirs, and changes in root zone soil moisture storage. The source of water for direct recharge might be from external sources, including groundwater and imported streamflow. Some recharge might discharge to streams as base flow.]

Aquifer sector	Aquifer system	Area, in square miles	Scenario	Water-budget component, in million gallons per day												
				Rain	Fog	Total precipitation	Irrigation	Septic	Total direct recharge	Storm-drain capture	Runoff	Canopy evaporation	Actual ET	Total ET	Ground-water use, kiawe	Recharge
Island of Kauaʻi	--	554.27	1978–2007	2,220.35	105.25	2,325.60	48.86	3.88	13.72	2.74	550.83	313.40	721.72	1,035.11	0.37	806.77
			SD RCP8.5 2041–71	1,876.37	88.91	1,965.25	52.83	3.88	13.72	1.96	438.64	282.53	678.05	960.58	0.40	639.89
			SD RCP8.5 2071–99	1,702.84	80.30	1,783.13	55.30	3.88	13.72	1.65	376.93	265.98	645.48	911.45	0.41	572.33
			HRCM2 RCP4.5 2080–99	2,261.50	101.84	2,363.31	47.17	3.88	13.72	3.07	557.25	313.58	731.58	1,045.20	0.37	825.45
Hanalei	Hanalei	32.28	1978–2007	217.08	6.76	223.85	4.11	0.14	0.29	0.12	80.91	23.42	49.03	72.46	0.00	74.59
			SD RCP8.5 2041–71	197.29	6.25	203.53	4.15	0.14	0.29	0.10	68.62	22.56	48.49	71.05	0.00	68.20
			SD RCP8.5 2071–99	187.15	5.94	193.09	4.17	0.14	0.29	0.09	61.69	22.07	47.91	69.98	0.00	65.86
			HRCM2 RCP4.5 2080–99	227.65	6.98	234.62	4.06	0.14	0.29	0.15	84.10	23.89	49.75	73.64	0.00	80.77
Hanalei	Kalihiwai	17.84	1978–2007	84.36	1.41	85.77	0.77	0.28	0.43	0.12	17.90	10.97	32.71	43.68	0.00	25.52
			SD RCP8.5 2041–71	76.62	1.29	77.90	0.83	0.28	0.43	0.10	15.13	10.50	32.00	42.51	0.00	21.77
			SD RCP8.5 2071–99	72.69	1.24	73.93	0.87	0.28	0.43	0.09	13.65	10.21	31.37	41.58	0.00	20.31
			HRCM2 RCP4.5 2080–99	92.12	1.47	93.58	0.66	0.28	0.43	0.14	19.73	11.50	34.45	45.95	0.00	28.99
Hanalei	Nāpali	35.37	1978–2007	103.91	14.92	118.82	0.00	0.00	0.03	0.01	19.74	24.94	29.92	54.86	0.00	44.16
			SD RCP8.5 2041–71	77.93	11.99	89.91	0.00	0.00	0.03	0.00	13.61	20.77	27.47	48.24	0.00	28.02
			SD RCP8.5 2071–99	65.44	10.52	75.96	0.00	0.00	0.03	0.00	10.68	18.57	25.44	44.01	0.00	21.24
			HRCM2 RCP4.5 2080–99	94.69	13.59	108.29	0.00	0.00	0.03	0.01	17.39	23.30	28.92	52.23	0.00	38.62
Hanalei	Wainiha	39.05	1978–2007	295.26	21.79	317.05	0.10	0.16	0.31	0.01	104.79	32.95	40.57	73.52	0.00	138.29
			SD RCP8.5 2041–71	263.00	19.81	282.81	0.10	0.16	0.31	0.01	88.48	31.46	40.27	71.73	0.00	122.41
			SD RCP8.5 2071–99	243.42	18.50	261.92	0.10	0.16	0.31	0.01	77.58	30.57	39.95	70.52	0.00	113.75
			HRCM2 RCP4.5 2080–99	287.40	21.29	308.69	0.10	0.16	0.31	0.01	100.09	32.62	40.68	73.30	0.00	134.90
Līhuʻe	Anahola	51.01	1978–2007	178.96	2.85	181.80	2.51	1.05	2.82	0.33	45.28	26.35	79.73	106.08	0.01	36.57
			SD RCP8.5 2041–71	154.90	2.59	157.49	2.67	1.05	2.82	0.23	36.32	24.23	75.29	99.52	0.01	28.14
			SD RCP8.5 2071–99	145.09	2.52	147.61	2.76	1.05	2.82	0.19	32.28	23.27	72.68	95.95	0.01	26.02
			HRCM2 RCP4.5 2080–99	189.35	3.02	192.37	2.43	1.05	2.82	0.33	49.77	27.16	81.83	108.99	0.01	39.70

Table 2-1.—Continued

Aquifer sector	Aquifer system	Area, in square miles	Scenario	Water-budget component, in million gallons per day												
				Rain	Fog	Total precipitation	Irrigation	Septic	Total direct recharge	Storm-drain capture	Runoff	Canopy evaporation	Actual ET	Total ET	Ground-water use, kiawe	Recharge
Līhu‘e	Hanamā‘ulu	54.64	1978–2007	185.88	1.36	187.24	6.47	0.17	1.53	1.33	27.42	24.08	98.28	122.36	0.01	44.92
			SD RCP8.5 2041–71	163.10	1.19	164.29	6.92	0.17	1.53	1.01	21.96	22.34	93.74	116.07	0.01	34.68
			SD RCP8.5 2071–99	152.97	1.13	154.10	7.16	0.17	1.53	0.87	19.36	21.52	91.13	112.65	0.01	30.98
			HRCM2 RCP4.5 2080–99	197.77	1.41	199.17	6.40	0.17	1.53	1.49	29.74	24.90	100.11	125.02	0.01	51.59
Līhu‘e	Kīlauea	19.33	1978–2007	75.46	0.21	75.67	0.46	0.34	0.71	0.05	14.69	13.75	33.66	47.41	0.00	15.15
			SD RCP8.5 2041–71	66.35	0.20	66.54	0.50	0.34	0.71	0.04	11.93	12.85	32.36	45.22	0.00	11.15
			SD RCP8.5 2071–99	61.92	0.19	62.10	0.52	0.34	0.71	0.04	10.48	12.35	31.36	43.71	0.00	9.75
			HRCM2 RCP4.5 2080–99	83.90	0.23	84.13	0.39	0.34	0.71	0.06	16.60	14.60	36.23	50.83	0.00	18.14
Līhu‘e	Kōloa	50.96	1978–2007	149.47	1.36	150.84	22.65	1.03	4.99	0.36	15.49	19.10	101.73	120.83	0.02	44.34
			SD RCP8.5 2041–71	128.13	1.18	129.31	24.47	1.03	4.99	0.26	11.81	17.41	97.66	115.07	0.02	34.69
			SD RCP8.5 2071–99	118.05	1.13	119.18	25.62	1.03	4.99	0.21	10.01	16.58	94.86	111.45	0.02	31.39
			HRCM2 RCP4.5 2080–99	162.06	1.42	163.48	21.41	1.03	4.99	0.43	17.86	19.97	104.13	124.11	0.02	49.79
Līhu‘e	Wailua	50.77	1978–2007	300.55	6.44	306.99	0.18	0.42	1.25	0.16	93.34	36.95	81.06	118.00	0.00	97.52
			SD RCP8.5 2041–71	270.06	5.89	275.95	0.20	0.42	1.25	0.12	78.58	35.11	79.29	114.40	0.00	85.00
			SD RCP8.5 2071–99	255.53	5.55	261.09	0.20	0.42	1.25	0.11	70.59	34.19	78.01	112.20	0.00	80.39
			HRCM2 RCP4.5 2080–99	307.78	6.68	314.45	0.18	0.42	1.25	0.16	97.02	37.16	80.58	117.74	0.00	101.57
Waimea	Hanapēpē	22.49	1978–2007	123.17	5.38	128.55	0.45	0.01	0.06	0.02	27.14	19.22	24.71	43.93	0.00	58.16
			SD RCP8.5 2041–71	105.35	4.70	110.05	0.48	0.01	0.06	0.01	20.98	17.63	23.51	41.13	0.00	48.68
			SD RCP8.5 2071–99	95.72	4.31	100.03	0.50	0.01	0.06	0.00	17.33	16.73	22.59	39.32	0.00	44.17
			HRCM2 RCP4.5 2080–99	124.54	5.38	129.92	0.44	0.01	0.06	0.02	27.80	19.27	24.74	44.01	0.00	58.76
Waimea	Kekaha	60.81	1978–2007	78.05	6.72	84.77	6.11	0.26	0.80	0.18	3.84	13.72	49.60	63.32	0.28	26.31
			SD RCP8.5 2041–71	46.91	4.70	51.61	6.84	0.26	0.80	0.06	1.58	9.97	39.96	49.93	0.30	9.91
			SD RCP8.5 2071–99	30.72	3.64	34.35	7.33	0.26	0.80	0.03	0.60	7.71	31.74	39.44	0.31	4.81
			HRCM2 RCP4.5 2080–99	80.03	6.20	86.23	6.08	0.26	0.80	0.21	3.94	13.43	49.98	63.41	0.28	27.50
Waimea	Makaweli	68.32	1978–2007	210.27	11.85	222.12	4.88	0.00	0.40	0.03	38.69	27.40	65.87	93.26	0.05	95.78
			SD RCP8.5 2041–71	163.64	9.83	173.48	5.49	0.00	0.40	0.01	29.69	23.52	55.81	79.33	0.06	70.80
			SD RCP8.5 2071–99	137.07	8.65	145.72	5.89	0.00	0.40	0.01	22.96	21.25	48.62	69.86	0.06	59.69
			HRCM2 RCP4.5 2080–99	210.44	11.63	222.07	4.85	0.00	0.40	0.04	38.31	27.14	65.74	92.88	0.05	96.46
Waimea	Waimea	51.42	1978–2007	217.93	24.20	242.13	0.17	0.02	0.10	0.02	61.60	40.55	34.85	75.40	0.00	105.46
			SD RCP8.5 2041–71	163.09	19.29	182.38	0.18	0.02	0.10	0.01	39.95	34.18	32.20	66.38	0.00	76.44
			SD RCP8.5 2071–99	137.07	16.98	154.05	0.18	0.02	0.10	0.00	29.72	30.96	29.82	60.78	0.00	63.97
			HRCM2 RCP4.5 2080–99	203.77	22.54	226.31	0.17	0.02	0.10	0.02	54.90	38.64	34.44	73.09	0.00	98.66

Table 2-2. Mean annual water-budget components from this study for aquifer systems and all of O'ahu.

[See figure 2 for locations of aquifer systems; --, not applicable; Water-budget area descriptions: A, entire aquifer-system area; B, entire aquifer-system area excluding areas seaward of the 0-foot altitude contour of the top of the volcanic aquifer (see fig. 2); Total precipitation, sum of Rain and Fog; Septic, septic-system leachate; Total direct recharge, sum of cesspool seepage and constant recharge from unlined reservoirs and inland water bodies; Runoff, direct runoff; ET, evapotranspiration; Total ET, sum of Canopy evaporation and Actual ET; Groundwater use, kiawe, groundwater use by kiawe land cover; Recharge, groundwater recharge which includes Total direct recharge; ET_o, reference evapotranspiration for grass surface; V, ratio of the mean evaporation rate to mean precipitation rate during saturated conditions; SD, statistical downscaling; HRCM2, Hawai'i Regional Climate Model version 2 from Zhang and Wang (2017); RCP8.5, Representative Concentration Pathway with total radiative forcing of 8.5 Watts per square meter by the year 2100 from Intergovernmental Panel on Climate Change (2014); 1978–2007 scenario is 1978–2007 rainfall from Frazier and others (2016), daily rainfall fragments from Longman and others (2019), ET_o from Giambelluca and others (2014), V ratios computed using equation 11 for a 1978–2007 scenario, and 2020 land cover; SD RCP8.5 2041–71 scenario is projected 2041–71 rainfall for a RCP8.5 warming scenario from Elison Timm and others (2015), daily rainfall fragments from Longman and others (2019), ET_o from Giambelluca and others (2014), V ratios computed using equation 11 for a 1978–2007 scenario, and 2020 land cover; SD RCP8.5 2071–99 scenario is projected 2071–99 rainfall for a RCP8.5 warming scenario from Elison Timm and others (2015), daily rainfall fragments from Longman and others (2019), ET_o from Giambelluca and others (2014), V ratios computed using equation 11 for a 1978–2007 scenario, and 2020 land cover; HRCM2 RCP8.5 2080–99 scenario is projected 2080–99 rainfall for a RCP8.5 warming scenario from Zhang and Wang (2017), daily rainfall fragments from Longman and others (2019), ET_o from Giambelluca and others (2014), V ratios computed using equation 11 for a 1978–2007 scenario, and 2020 land cover. Components might not balance because of rounding, direct recharge from cesspools and reservoirs, and changes in root zone soil moisture storage. The source of water for direct recharge might be from external sources, including groundwater and imported streamflow. Some recharge might discharge to streams as base flow. Calculations exclude areas that are unassigned in the Department of Land and Natural Resources aquifer system boundaries from State of Hawai'i (2014).]

Aquifer sector	Aquifer system	Water-budget area description	Area, in square miles	Scenario	Water-budget component, in million gallons per day												
					Rain	Fog	Total precipitation	Irrigation	Septic	Total direct recharge	Storm-drain capture	Runoff	Canopy evaporation	Actual ET	Total ET	Ground-water use, kiawe	Recharge
Island of	--	A	595.48	1978–2007	1,820.46	16.09	1,836.56	102.58	2.43	9.53	33.76	263.69	282.75	783.22	1,065.93	0.73	597.06
O'ahu		A		SD RCP8.5 2041–71	1,532.55	14.38	1,546.91	112.67	2.43	9.53	23.63	215.31	254.37	703.03	957.43	0.79	485.99
		A		SD RCP8.5 2071–99	1,417.62	13.66	1,431.30	117.24	2.43	9.53	20.48	194.84	242.33	663.27	905.65	0.82	450.86
		A		HRCM2 RCP8.5 2080–99	1,893.40	17.19	1,910.60	102.95	2.43	9.53	38.40	288.70	286.81	779.46	1,066.25	0.73	641.11
		B	459.75	1978–2007	1,612.31	16.09	1,628.41	44.68	0.66	4.49	16.02	244.32	272.50	585.46	857.99	0.01	559.91
		B		SD RCP8.5 2041–71	1,375.30	14.38	1,389.69	50.33	0.66	4.49	11.59	201.20	246.39	527.72	774.14	0.01	458.72
		B		SD RCP8.5 2071–99	1,279.41	13.66	1,293.05	53.11	0.66	4.49	10.09	182.70	235.27	497.95	733.25	0.01	425.95
		B		HRCM2 RCP8.5 2080–99	1,671.99	17.19	1,689.16	45.66	0.66	4.49	17.74	266.55	276.23	579.81	856.02	0.01	599.63
Central	Wahiawa	A	68.82	1978–2007	303.67	3.94	307.61	3.70	0.01	1.78	2.04	63.21	46.11	82.98	129.08	0.00	119.27
		A		SD RCP8.5 2041–71	265.39	3.60	268.99	4.32	0.01	1.78	1.35	52.85	42.61	75.87	118.48	0.00	103.14
		A		SD RCP8.5 2071–99	248.57	3.44	252.02	4.62	0.01	1.78	1.11	47.75	40.97	72.24	113.22	0.00	97.15
		A		HRCM2 RCP8.5 2080–99	305.43	4.24	309.67	4.04	0.01	1.78	2.02	66.45	45.91	80.24	126.14	0.00	121.45
Honolulu	Kalihi	A	9.82	1978–2007	34.92	0.09	35.02	1.00	0.00	0.18	2.11	4.88	6.10	13.66	19.76	0.00	9.50
		A		SD RCP8.5 2041–71	33.10	0.09	33.19	1.06	0.00	0.18	1.82	4.54	5.96	13.14	19.10	0.00	9.01
		A		SD RCP8.5 2071–99	32.54	0.09	32.63	1.08	0.00	0.18	1.75	4.41	5.91	12.98	18.89	0.00	8.88
		A		HRCM2 RCP8.5 2080–99	37.09	0.11	37.20	1.01	0.00	0.18	2.39	5.49	6.31	13.55	19.86	0.00	10.70
		B	6.56	1978–2007	28.81	0.09	28.91	0.43	0.00	0.04	0.66	4.31	6.08	9.28	15.36	0.00	9.03
		B		SD RCP8.5 2041–71	27.72	0.09	27.81	0.46	0.00	0.04	0.63	4.04	5.94	9.11	15.06	0.00	8.57
		B		SD RCP8.5 2071–99	27.36	0.09	27.45	0.47	0.00	0.04	0.63	3.94	5.89	9.05	14.94	0.00	8.44
		B		HRCM2 RCP8.5 2080–99	30.75	0.11	30.85	0.44	0.00	0.04	0.75	4.85	6.30	9.25	15.55	0.00	10.17
Honolulu	Moanalua	A	22.93	1978–2007	63.14	0.20	63.34	5.20	0.00	0.15	3.21	6.90	10.43	29.84	40.26	0.00	18.88
		A		SD RCP8.5 2041–71	55.57	0.19	55.77	5.65	0.00	0.15	2.12	5.99	10.00	27.03	37.03	0.00	17.04
		A		SD RCP8.5 2071–99	52.99	0.19	53.18	5.81	0.00	0.15	1.79	5.65	9.84	26.01	35.85	0.00	16.49
		A		HRCM2 RCP8.5 2080–99	66.46	0.22	66.68	5.24	0.00	0.15	3.58	7.69	10.76	29.59	40.35	0.00	21.01
		B	10.94	1978–2007	46.79	0.20	46.99	0.71	0.00	0.00	0.45	5.92	10.32	14.16	24.48	0.00	16.84
		B		SD RCP8.5 2041–71	43.76	0.19	43.95	0.78	0.00	0.00	0.35	5.37	9.92	13.53	23.45	0.00	15.56

Table 2-2.—Continued

Aquifer sector	Aquifer system	Water-budget area description	Area, in square miles	Scenario	Water-budget component, in million gallons per day												
					Rain	Fog	Total precipitation	Irrigation	Septic	Total direct recharge	Storm-drain capture	Runoff	Canopy evaporation	Actual ET	Total ET	Ground-water use, kiawe	Recharge
		B		SD RCP8.5 2071–99	42.70	0.19	42.89	0.81	0.00	0.00	0.32	5.14	9.77	13.29	23.06	0.00	15.16
		B		HRCM2 RCP8.5 2080–99	49.51	0.22	49.73	0.73	0.00	0.00	0.47	6.63	10.65	14.00	24.65	0.00	18.69
Honolulu	Nu‘uanu	A	14.78	1978–2007	55.80	0.26	56.06	1.30	0.06	0.88	2.38	8.21	8.73	21.02	29.75	0.00	17.80
		A		SD RCP8.5 2041–71	54.14	0.25	54.39	1.36	0.06	0.88	2.13	8.01	8.64	20.27	28.91	0.00	17.50
		A		SD RCP8.5 2071–99	53.72	0.25	53.97	1.37	0.06	0.88	2.10	7.89	8.60	20.04	28.64	0.00	17.52
		A		HRCM2 RCP8.5 2080–99	60.40	0.30	60.70	1.30	0.06	0.88	2.76	9.63	9.14	20.95	30.09	0.00	20.25
		B	9.40	1978–2007	46.10	0.26	46.36	0.27	0.06	0.70	0.60	7.37	8.64	13.69	22.33	0.00	16.85
		B		SD RCP8.5 2041–71	45.40	0.25	45.65	0.28	0.06	0.70	0.59	7.25	8.56	13.47	22.03	0.00	16.60
		B		SD RCP8.5 2071–99	45.20	0.25	45.45	0.29	0.06	0.70	0.60	7.16	8.52	13.39	21.91	0.00	16.61
		B		HRCM2 RCP8.5 2080–99	50.20	0.30	50.50	0.27	0.06	0.70	0.68	8.68	9.05	13.72	22.77	0.00	19.10
Honolulu	Pālolo	A	10.10	1978–2007	28.27	0.08	28.36	2.31	0.02	0.22	2.29	3.07	3.82	14.61	18.43	0.00	7.52
		A		SD RCP8.5 2041–71	25.62	0.08	25.70	2.47	0.02	0.22	1.86	2.75	3.67	13.61	17.28	0.00	6.96
		A		SD RCP8.5 2071–99	24.66	0.08	24.74	2.53	0.02	0.22	1.75	2.60	3.60	13.19	16.80	0.00	6.80
		A		HRCM2 RCP8.5 2080–99	30.83	0.09	30.92	2.23	0.02	0.22	2.79	3.55	3.97	14.97	18.94	0.00	8.50
		B	5.09	1978–2007	20.50	0.08	20.59	0.45	0.01	0.06	0.87	2.55	3.76	7.29	11.05	0.00	6.63
		B		SD RCP8.5 2041–71	19.28	0.08	19.36	0.48	0.01	0.06	0.77	2.34	3.62	7.00	10.62	0.00	6.17
		B		SD RCP8.5 2071–99	18.81	0.08	18.89	0.49	0.01	0.06	0.75	2.24	3.56	6.86	10.42	0.00	6.03
		B		HRCM2 RCP8.5 2080–99	22.17	0.09	22.26	0.42	0.01	0.06	1.03	2.93	3.90	7.44	11.35	0.00	7.44
Honolulu	Wai‘alae-East	A	19.98	1978–2007	39.53	0.07	39.60	4.31	0.00	0.16	1.85	3.22	8.39	23.11	31.50	0.00	7.76
		A		SD RCP8.5 2041–71	33.05	0.07	33.11	4.56	0.00	0.16	1.42	2.41	7.33	21.03	28.36	0.00	5.94
		A		SD RCP8.5 2071–99	31.04	0.06	31.10	4.65	0.00	0.16	1.32	2.11	6.97	20.23	27.20	0.00	5.58
		A		HRCM2 RCP8.5 2080–99	45.12	0.08	45.21	4.16	0.00	0.16	2.29	4.02	9.25	24.41	33.66	0.00	9.82
		B	14.11	1978–2007	31.19	0.07	31.27	1.55	0.00	0.09	0.95	2.71	7.88	14.95	22.83	0.00	6.42
		B		SD RCP8.5 2041–71	26.16	0.07	26.23	1.65	0.00	0.09	0.73	2.03	6.91	13.49	20.40	0.00	4.81
		B		SD RCP8.5 2071–99	24.58	0.06	24.64	1.69	0.00	0.09	0.67	1.78	6.57	12.91	19.48	0.00	4.48
		B		HRCM2 RCP8.5 2080–99	35.61	0.08	35.70	1.49	0.00	0.09	1.16	3.37	8.69	15.95	24.64	0.00	8.11
Honolulu	Wai‘alae-West	A	7.29	1978–2007	18.84	0.06	18.90	1.49	0.04	0.17	0.75	1.88	4.29	8.97	13.26	0.00	4.78
		A		SD RCP8.5 2041–71	16.06	0.06	16.12	1.60	0.04	0.17	0.53	1.48	3.91	8.15	12.06	0.00	3.93
		A		SD RCP8.5 2071–99	15.05	0.06	15.11	1.64	0.04	0.17	0.47	1.32	3.76	7.78	11.55	0.00	3.71
		A		HRCM2 RCP8.5 2080–99	21.30	0.07	21.37	1.41	0.04	0.17	1.01	2.27	4.59	9.47	14.06	0.00	5.72
		B	4.71	1978–2007	15.42	0.06	15.49	0.33	0.00	0.03	0.32	1.69	4.18	5.39	9.57	0.00	4.27
		B		SD RCP8.5 2041–71	13.52	0.06	13.58	0.36	0.00	0.03	0.25	1.36	3.83	5.01	8.85	0.00	3.51
		B		SD RCP8.5 2071–99	12.85	0.06	12.91	0.37	0.00	0.03	0.23	1.23	3.70	4.83	8.54	0.00	3.31
		B		HRCM2 RCP8.5 2080–99	17.21	0.07	17.28	0.31	0.00	0.03	0.41	2.02	4.47	5.64	10.10	0.00	5.08
North	Kawailoa	A	37.64	1978–2007	122.42	0.03	122.44	1.34	0.44	0.64	0.11	15.68	25.64	53.64	79.28	0.01	29.95
		A		SD RCP8.5 2041–71	102.05	0.02	102.07	1.49	0.44	0.64	0.08	11.48	23.04	48.97	72.01	0.01	21.28
		A		SD RCP8.5 2071–99	94.16	0.02	94.18	1.57	0.44	0.64	0.06	9.81	21.99	46.65	68.65	0.01	18.54
		A		HRCM2 RCP8.5 2080–99	126.55	0.03	126.58	1.39	0.44	0.64	0.13	16.93	26.10	53.35	79.45	0.01	32.70

Table 2-2—Continued

Aquifer sector	Aquifer system	Water-budget area description	Area, in square miles	Scenario	Water-budget component, in million gallons per day												
					Rain	Fog	Total precipitation	Irrigation	Septic	Total direct recharge	Storm-drain capture	Runoff	Canopy evaporation	Actual ET	Total ET	Ground-water use, kiawe	Recharge
		B	35.12	1978–2007	117.52	0.03	117.54	0.87	0.20	0.24	0.02	15.32	25.22	49.30	74.52	0.00	28.93
		B		SD RCP8.5 2041–71	98.20	0.02	98.22	0.98	0.20	0.24	0.01	11.25	22.69	45.14	67.83	0.00	20.50
		B		SD RCP8.5 2071–99	90.72	0.02	90.75	1.04	0.20	0.24	0.01	9.62	21.67	43.05	64.73	0.00	17.83
		B		HRCM2 RCP8.5 2080–99	121.61	0.03	121.63	0.93	0.20	0.24	0.02	16.56	25.69	49.07	74.75	0.00	31.60
North	Mokulēʻia	A	34.87	1978–2007	73.73	0.92	74.65	2.90	0.10	0.18	0.05	5.98	15.53	38.59	54.12	0.01	17.73
		A		SD RCP8.5 2041–71	55.10	0.68	55.78	3.16	0.10	0.18	0.03	3.63	12.47	33.30	45.78	0.01	9.89
		A		SD RCP8.5 2071–99	48.42	0.59	49.01	3.27	0.10	0.18	0.03	2.77	11.30	30.85	42.16	0.01	7.72
		A		HRCM2 RCP8.5 2080–99	66.60	0.89	67.49	3.08	0.10	0.18	0.05	5.11	14.23	35.71	49.95	0.01	15.83
		B	27.50	1978–2007	61.81	0.92	62.73	0.26	0.00	0.00	0.00	5.19	14.97	27.98	42.95	0.00	14.82
		B		SD RCP8.5 2041–71	46.13	0.68	46.81	0.29	0.00	0.00	0.00	3.15	12.03	23.87	35.90	0.00	8.06
		B		SD RCP8.5 2071–99	40.53	0.59	41.12	0.30	0.00	0.00	0.00	2.41	10.91	21.92	32.83	0.00	6.19
		B		HRCM2 RCP8.5 2080–99	55.99	0.89	56.88	0.28	0.00	0.00	0.00	4.46	13.73	25.78	39.51	0.00	13.17
North	Waialua	A	21.39	1978–2007	40.66	0.00	40.66	6.05	0.30	0.97	0.22	3.17	4.60	30.30	34.90	0.00	10.03
		A		SD RCP8.5 2041–71	32.25	0.00	32.25	6.85	0.30	0.97	0.13	2.19	3.99	27.48	31.47	0.00	6.97
		A		SD RCP8.5 2071–99	29.03	0.00	29.03	7.20	0.30	0.97	0.10	1.78	3.74	26.18	29.92	0.00	6.10
		A		HRCM2 RCP8.5 2080–99	38.98	0.00	38.98	6.55	0.30	0.97	0.21	3.03	4.46	29.11	33.56	0.00	10.35
		B	18.33	1978–2007	36.09	0.00	36.09	4.14	0.03	0.24	0.04	2.91	4.38	24.78	29.16	0.00	8.42
		B		SD RCP8.5 2041–71	28.92	0.00	28.92	4.77	0.03	0.24	0.03	2.04	3.82	22.46	26.29	0.00	5.67
		B		SD RCP8.5 2071–99	26.17	0.00	26.17	5.06	0.03	0.24	0.02	1.68	3.59	21.39	24.98	0.00	4.90
		B		HRCM2 RCP8.5 2080–99	34.70	0.00	34.70	4.59	0.03	0.24	0.04	2.80	4.25	23.76	28.02	0.00	8.74
Pearl Harbor	ʻEwa-Kunia	A	38.63	1978–2007	55.06	0.28	55.35	7.91	0.04	0.26	1.65	2.97	7.07	41.63	48.69	0.23	11.00
		A		SD RCP8.5 2041–71	34.19	0.18	34.36	8.72	0.04	0.26	0.82	1.27	4.79	32.59	37.38	0.25	4.81
		A		SD RCP8.5 2071–99	25.50	0.13	25.63	9.12	0.04	0.26	0.56	0.63	3.74	27.82	31.56	0.26	3.24
		A		HRCM2 RCP8.5 2080–99	54.99	0.29	55.28	7.98	0.04	0.26	1.74	3.07	6.95	41.10	48.05	0.23	11.48
		B	26.13	1978–2007	42.25	0.28	42.53	4.37	0.00	0.07	0.47	2.47	6.18	29.45	35.63	0.00	8.42
		B		SD RCP8.5 2041–71	26.05	0.18	26.22	4.92	0.00	0.07	0.23	1.03	4.20	22.56	26.77	0.00	3.24
		B		SD RCP8.5 2071–99	19.23	0.13	19.35	5.19	0.00	0.07	0.15	0.50	3.28	18.84	22.12	0.00	1.93
		B		HRCM2 RCP8.5 2080–99	41.90	0.29	42.18	4.44	0.00	0.07	0.50	2.52	6.05	28.92	34.97	0.00	8.73
Pearl Harbor	Makaīwa	A	3.07	1978–2007	4.09	0.00	4.09	0.50	0.00	0.01	0.07	0.21	0.30	3.23	3.53	0.00	0.92
		A		SD RCP8.5 2041–71	2.21	0.00	2.21	0.55	0.00	0.01	0.02	0.07	0.18	2.31	2.49	0.00	0.33
		A		SD RCP8.5 2071–99	1.43	0.00	1.43	0.58	0.00	0.01	0.01	0.02	0.12	1.82	1.94	0.00	0.19
		A		HRCM2 RCP8.5 2080–99	4.20	0.00	4.20	0.50	0.00	0.01	0.07	0.23	0.30	3.24	3.54	0.00	0.99
		B	2.31	1978–2007	3.23	0.00	3.23	0.09	0.00	0.01	0.01	0.17	0.29	2.17	2.46	0.00	0.70
		B		SD RCP8.5 2041–71	1.76	0.00	1.76	0.09	0.00	0.01	0.00	0.06	0.17	1.44	1.61	0.00	0.20
		B		SD RCP8.5 2071–99	1.14	0.00	1.14	0.10	0.00	0.01	0.00	0.02	0.12	1.03	1.16	0.00	0.08
		B		HRCM2 RCP8.5 2080–99	3.33	0.00	3.33	0.09	0.00	0.01	0.01	0.18	0.29	2.18	2.47	0.00	0.76

Table 2-2.—Continued

Aquifer sector	Aquifer system	Water-budget area description	Area, in square miles	Scenario	Water-budget component, in million gallons per day												
					Rain	Fog	Total precipitation	Irrigation	Septic	Total direct recharge	Storm-drain capture	Runoff	Canopy evaporation	Actual ET	Total ET	Ground-water use, kiawe	Recharge
Pearl Harbor	Waimalu	A	38.18	1978–2007	153.18	0.86	154.05	4.41	0.02	0.11	3.49	22.68	27.09	46.78	73.87	0.00	58.68
		A		SD RCP8.5 2041–71	135.51	0.82	136.33	4.83	0.02	0.11	2.12	19.11	25.54	43.41	68.95	0.00	51.26
		A		SD RCP8.5 2071–99	128.40	0.81	129.21	5.01	0.02	0.11	1.64	17.55	24.89	41.80	66.69	0.00	48.64
		A		HRCM2 RCP8.5 2080–99	159.49	0.97	160.46	4.48	0.02	0.11	3.72	25.11	27.56	45.82	73.37	0.00	63.00
		B	31.45	1978–2007	144.34	0.86	145.20	2.55	0.02	0.09	1.92	22.15	26.97	38.78	65.76	0.00	57.97
		B		SD RCP8.5 2041–71	129.43	0.82	130.25	2.82	0.02	0.09	1.26	18.81	25.45	36.70	62.15	0.00	50.90
		B		SD RCP8.5 2071–99	123.38	0.81	124.19	2.95	0.02	0.09	1.01	17.33	24.81	35.64	60.46	0.00	48.39
		B		HRCM2 RCP8.5 2080–99	150.61	0.97	151.58	2.61	0.02	0.09	2.05	24.56	27.45	37.99	65.44	0.00	62.20
Pearl Harbor	Waipahu-Waiawa	A	82.86	1978–2007	231.52	1.26	232.78	33.55	0.12	0.88	5.85	32.97	30.00	112.13	142.13	0.38	87.90
		A		SD RCP8.5 2041–71	194.00	1.21	195.21	37.25	0.12	0.88	3.28	27.83	27.60	101.01	128.61	0.42	75.41
		A		SD RCP8.5 2071–99	178.50	1.18	179.69	39.02	0.12	0.88	2.39	25.57	26.59	95.41	122.00	0.43	71.50
		A		HRCM2 RCP8.5 2080–99	233.65	1.38	235.04	33.95	0.12	0.88	6.06	34.93	29.93	110.27	140.20	0.38	90.31
		B	60.37	1978–2007	207.70	1.26	208.96	17.57	0.01	0.10	4.25	32.02	28.14	80.84	108.99	0.00	81.34
		B		SD RCP8.5 2041–71	179.21	1.21	180.42	19.95	0.01	0.10	2.50	27.42	26.40	73.66	100.06	0.00	70.47
		B		SD RCP8.5 2071–99	167.21	1.18	168.39	21.18	0.01	0.10	1.86	25.36	25.65	69.87	95.52	0.00	66.93
		B		HRCM2 RCP8.5 2080–99	208.81	1.38	210.20	17.99	0.01	0.10	4.25	33.87	28.02	78.93	106.95	0.00	83.18
Waianae	Kea‘au	A	12.67	1978–2007	25.89	0.13	26.02	0.01	0.00	0.00	0.00	2.01	2.17	15.56	17.73	0.02	6.49
		A		SD RCP8.5 2041–71	16.53	0.08	16.61	0.01	0.00	0.00	0.00	0.91	1.54	11.79	13.33	0.02	2.62
		A		SD RCP8.5 2071–99	12.85	0.07	12.92	0.01	0.00	0.00	0.00	0.52	1.27	9.79	11.07	0.02	1.60
		A		HRCM2 RCP8.5 2080–99	24.54	0.12	24.66	0.01	0.00	0.00	0.00	1.84	2.04	14.50	16.54	0.02	6.50
		B	11.45	1978–2007	23.98	0.13	24.10	0.00	0.00	0.00	0.00	1.89	2.14	13.96	16.10	0.01	6.17
		B		SD RCP8.5 2041–71	15.39	0.08	15.47	0.00	0.00	0.00	0.00	0.86	1.52	10.63	12.15	0.01	2.53
		B		SD RCP8.5 2071–99	12.03	0.07	12.09	0.00	0.00	0.00	0.00	0.50	1.26	8.86	10.11	0.01	1.56
		B		HRCM2 RCP8.5 2080–99	22.65	0.12	22.77	0.00	0.00	0.00	0.00	1.72	2.00	12.96	14.97	0.01	6.13
Waianae	Lualualei	A	23.97	1978–2007	37.22	0.13	37.35	2.89	0.12	0.28	0.51	2.22	6.41	24.09	30.49	0.02	7.51
		A		SD RCP8.5 2041–71	21.59	0.08	21.67	3.20	0.12	0.28	0.17	0.84	4.27	17.47	21.74	0.02	2.66
		A		SD RCP8.5 2071–99	14.92	0.06	14.99	3.36	0.12	0.28	0.06	0.36	3.28	13.78	17.05	0.02	1.42
		A		HRCM2 RCP8.5 2080–99	35.12	0.13	35.25	2.94	0.12	0.28	0.51	2.07	6.03	22.59	28.62	0.02	7.49
		B	15.31	1978–2007	27.32	0.13	27.45	0.41	0.02	0.06	0.06	1.74	5.80	14.59	20.40	0.00	5.73
		B		SD RCP8.5 2041–71	16.40	0.08	16.48	0.45	0.02	0.06	0.02	0.69	3.91	10.36	14.27	0.00	2.04
		B		SD RCP8.5 2071–99	11.75	0.06	11.81	0.47	0.02	0.06	0.01	0.31	3.03	7.97	11.00	0.00	1.06
		B		HRCM2 RCP8.5 2080–99	25.70	0.13	25.83	0.41	0.02	0.06	0.06	1.61	5.47	13.55	19.01	0.00	5.63
Waianae	Mākaha	A	9.49	1978–2007	20.84	0.34	21.17	1.94	0.06	0.12	0.10	1.71	3.51	10.59	14.10	0.00	7.42
		A		SD RCP8.5 2041–71	12.48	0.23	12.71	2.31	0.06	0.12	0.02	0.73	2.45	8.34	10.79	0.00	3.69
		A		SD RCP8.5 2071–99	9.17	0.19	9.36	2.50	0.06	0.12	0.01	0.41	2.00	6.96	8.95	0.01	2.73
		A		HRCM2 RCP8.5 2080–99	19.90	0.33	20.23	1.97	0.06	0.12	0.10	1.62	3.34	10.15	13.49	0.00	7.21

Table 2-2—Continued

Aquifer sector	Aquifer system	Water-budget area description	Area, in square miles	Scenario	Water-budget component, in million gallons per day												
					Rain	Fog	Total precipitation	Irrigation	Septic	Total direct recharge	Storm-drain capture	Runoff	Canopy evaporation	Actual ET	Total ET	Ground-water use, kiawe	Recharge
		B	7.77	1978–2007	18.76	0.34	19.09	0.95	0.00	0.00	0.02	1.59	3.43	8.33	11.76	0.00	6.68
		B		SD RCP8.5 2041–71	11.60	0.23	11.83	1.15	0.00	0.00	0.01	0.70	2.41	6.65	9.06	0.00	3.22
		B		SD RCP8.5 2071–99	8.80	0.19	8.98	1.25	0.00	0.00	0.00	0.40	1.98	5.60	7.58	0.00	2.28
		B		HRCM2 RCP8.5 2080–99	17.82	0.33	18.15	0.97	0.00	0.00	0.02	1.50	3.26	7.93	11.19	0.00	6.42
Waianae	Nānākuli	A	5.42	1978–2007	9.28	0.05	9.34	0.27	0.02	0.07	0.11	0.58	1.69	5.20	6.89	0.00	2.13
		A		SD RCP8.5 2041–71	5.24	0.03	5.28	0.29	0.02	0.07	0.04	0.20	1.08	3.64	4.72	0.00	0.72
		A		SD RCP8.5 2071–99	3.48	0.02	3.50	0.31	0.02	0.07	0.02	0.07	0.78	2.71	3.49	0.00	0.35
		A		HRCM2 RCP8.5 2080–99	9.15	0.06	9.21	0.27	0.02	0.07	0.11	0.59	1.67	5.04	6.71	0.00	2.17
		B	4.59	1978–2007	8.22	0.05	8.28	0.08	0.00	0.01	0.03	0.53	1.63	4.29	5.92	0.00	1.88
		B		SD RCP8.5 2041–71	4.67	0.03	4.71	0.09	0.00	0.01	0.01	0.18	1.04	2.97	4.01	0.00	0.60
		B		SD RCP8.5 2071–99	3.12	0.02	3.14	0.09	0.00	0.01	0.01	0.06	0.75	2.16	2.91	0.00	0.26
		B		HRCM2 RCP8.5 2080–99	8.11	0.06	8.16	0.08	0.00	0.01	0.03	0.54	1.62	4.16	5.77	0.00	1.91
Waianae	Wai‘anae	A	9.35	1978–2007	18.13	0.22	18.35	1.05	0.06	0.27	0.19	1.44	2.97	9.84	12.81	0.03	5.38
		A		SD RCP8.5 2041–71	10.38	0.15	10.52	1.19	0.06	0.27	0.05	0.56	2.05	7.34	9.39	0.03	2.15
		A		SD RCP8.5 2071–99	7.35	0.12	7.46	1.25	0.06	0.27	0.01	0.28	1.66	5.88	7.54	0.03	1.33
		A		HRCM2 RCP8.5 2080–99	17.69	0.22	17.91	1.07	0.06	0.27	0.20	1.42	2.87	9.33	12.21	0.03	5.57
		B	7.56	1978–2007	16.17	0.22	16.39	0.42	0.03	0.17	0.06	1.34	2.83	7.86	10.70	0.00	4.91
		B		SD RCP8.5 2041–71	9.56	0.15	9.71	0.48	0.03	0.17	0.02	0.54	1.98	5.86	7.85	0.00	1.99
		B		SD RCP8.5 2071–99	7.01	0.12	7.12	0.51	0.03	0.17	0.01	0.28	1.63	4.71	6.34	0.00	1.22
		B		HRCM2 RCP8.5 2080–99	15.75	0.22	15.97	0.43	0.03	0.17	0.07	1.31	2.74	7.42	10.16	0.00	5.06
Windward	Kahana	A	18.43	1978–2007	101.72	0.87	102.59	0.16	0.13	0.22	0.06	21.42	13.47	27.05	40.52	0.00	41.56
		A		SD RCP8.5 2041–71	90.57	0.81	91.38	0.17	0.13	0.22	0.04	18.20	12.64	25.78	38.43	0.00	35.82
		A		SD RCP8.5 2071–99	86.01	0.79	86.80	0.18	0.13	0.22	0.04	16.79	12.30	25.18	37.49	0.00	33.64
		A		HRCM2 RCP8.5 2080–99	114.21	0.93	115.14	0.15	0.13	0.22	0.07	25.35	14.31	28.21	42.52	0.00	48.03
		B	16.38	1978–2007	95.60	0.87	96.46	0.01	0.02	0.05	0.00	20.21	12.92	22.69	35.61	0.00	40.68
		B		SD RCP8.5 2041–71	85.47	0.81	86.28	0.01	0.02	0.05	0.00	17.27	12.15	21.72	33.86	0.00	35.19
		B		SD RCP8.5 2071–99	81.29	0.79	82.07	0.01	0.02	0.05	0.00	15.97	11.83	21.24	33.07	0.00	33.08
		B		HRCM2 RCP8.5 2080–99	107.08	0.93	108.01	0.01	0.02	0.05	0.01	23.86	13.72	23.59	37.31	0.00	46.87
Windward	Ko‘olauloa	A	41.15	1978–2007	173.67	4.33	177.99	7.62	0.36	0.57	0.72	21.23	26.26	67.96	94.22	0.01	71.72
		A		SD RCP8.5 2041–71	149.61	3.91	153.52	8.14	0.36	0.57	0.56	16.99	24.09	64.22	88.32	0.01	58.18
		A		SD RCP8.5 2071–99	140.11	3.72	143.83	8.33	0.36	0.57	0.50	15.33	23.21	62.52	85.73	0.01	53.00
		A		HRCM2 RCP8.5 2080–99	183.59	4.55	188.13	7.31	0.36	0.57	0.83	23.34	26.93	69.14	96.07	0.01	77.39
		B	30.55	1978–2007	148.06	4.33	152.39	1.18	0.01	0.02	0.03	17.73	24.34	44.60	68.94	0.00	66.82
		B		SD RCP8.5 2041–71	127.63	3.91	131.54	1.27	0.01	0.02	0.02	14.21	22.37	41.93	64.29	0.00	54.24
		B		SD RCP8.5 2071–99	119.35	3.72	123.08	1.31	0.01	0.02	0.02	12.81	21.55	40.66	62.20	0.00	49.31
		B		HRCM2 RCP8.5 2080–99	155.57	4.55	160.12	1.15	0.01	0.02	0.03	19.37	24.90	45.13	70.03	0.00	71.78

Table 2-2.—Continued

Aquifer sector	Aquifer system	Water-budget area description	Area, in square miles	Scenario	Water-budget component, in million gallons per day												
					Rain	Fog	Total precipitation	Irrigation	Septic	Total direct recharge	Storm-drain capture	Runoff	Canopy evaporation	Actual ET	Total ET	Ground-water use, kiawe	Recharge
Windward	Koʻolaupoko	A	29.69	1978–2007	125.34	1.51	126.84	3.86	0.31	0.88	2.81	23.74	16.90	49.86	66.76	0.00	38.72
		A		SD RCP8.5 2041–71	114.92	1.42	116.33	4.14	0.31	0.88	2.49	21.26	16.16	47.59	63.75	0.00	34.35
		A		SD RCP8.5 2071–99	110.92	1.38	112.30	4.26	0.31	0.88	2.39	20.22	15.85	46.56	62.41	0.00	32.92
		A		HRCM2 RCP8.5 2080–99	141.34	1.66	143.00	3.64	0.31	0.88	3.58	27.71	17.89	52.15	70.04	0.00	46.58
		B	26.52	1978–2007	115.91	1.51	117.42	2.18	0.19	0.66	2.28	22.06	16.43	43.17	59.60	0.00	36.52
		B		SD RCP8.5 2041–71	106.49	1.42	107.91	2.42	0.19	0.66	2.03	19.79	15.73	41.32	57.05	0.00	32.36
		B		SD RCP8.5 2071–99	102.86	1.38	104.24	2.53	0.19	0.66	1.95	18.84	15.43	40.46	55.89	0.00	30.98
		B		HRCM2 RCP8.5 2080–99	130.37	1.66	132.03	2.00	0.19	0.66	2.88	25.68	17.37	45.01	62.38	0.00	43.94
Windward	Waimānalo	A	34.95	1978–2007	83.54	0.46	84.00	8.81	0.22	0.53	3.19	14.31	11.27	52.58	63.85	0.02	14.41
		A		SD RCP8.5 2041–71	72.99	0.42	73.41	9.35	0.22	0.53	2.55	12.01	10.36	48.69	59.05	0.02	12.33
		A		SD RCP8.5 2071–99	68.80	0.41	69.21	9.57	0.22	0.53	2.37	11.00	9.96	46.89	56.85	0.02	11.81
		A		HRCM2 RCP8.5 2080–99	96.77	0.52	97.29	8.27	0.22	0.53	4.18	17.25	12.27	56.57	68.83	0.02	18.06
		B	18.78	1978–2007	52.87	0.46	53.33	2.16	0.05	0.07	0.94	9.24	9.86	24.93	34.79	0.00	10.61
		B		SD RCP8.5 2041–71	47.16	0.42	47.59	2.31	0.05	0.07	0.78	7.96	9.13	22.97	32.10	0.00	9.15
		B		SD RCP8.5 2071–99	44.75	0.41	45.16	2.38	0.05	0.07	0.73	7.37	8.80	21.98	30.78	0.00	8.77
		B		HRCM2 RCP8.5 2080–99	61.11	0.52	61.63	1.98	0.05	0.07	1.25	11.08	10.70	27.19	37.89	0.00	13.47

Table 2-3. Mean annual water-budget components from this study for aquifer systems and all of Moloka‘i

[See figure 2 for locations of aquifer systems; --, not applicable; Total precipitation, sum of Rain and Fog; Septic, septic-system leachate; Total direct recharge, sum of cesspool seepage and constant recharge from unlined reservoirs and inland water bodies; Runoff, direct runoff; ET, evapotranspiration; Total ET, sum of Canopy evaporation and Actual ET; Groundwater use, kiawe, groundwater use by kiawe land cover; Recharge, groundwater recharge which includes Total direct recharge; ET_o, reference evapotranspiration for grass surface; V, ratio of the mean evaporation rate to mean precipitation rate during saturated conditions; SD, statistical downscaling; HRCM2, Hawai‘i Regional Climate Model version 2 from Zhang and Wang (2017); RCP8.5, Representative Concentration Pathway with total radiative forcing of 8.5 Watts per square meter by the year 2100 from Intergovernmental Panel on Climate Change (2014); 1978–2007 scenario is 1978–2007 rainfall from Frazier and others (2016), daily rainfall fragments from Longman and others (2019), ET_o from Giambelluca and others (2014), V ratios computed using equation 11 for a 1978–2007 scenario, and 2020 land cover; SD RCP8.5 2041–71 scenario is projected 2041–71 rainfall for a RCP8.5 warming scenario from Elison Timm and others (2015), daily rainfall fragments from Longman and others (2019), ET_o from Giambelluca and others (2014), V ratios computed using equation 11 for a 1978–2007 scenario, and 2020 land cover; SD RCP8.5 2071–99 scenario is projected 2071–99 rainfall for a RCP8.5 warming scenario from Elison Timm and others (2015), daily rainfall fragments from Longman and others (2019), ET_o from Giambelluca and others (2014), V ratios computed using equation 11 for a 1978–2007 scenario, and 2020 land cover; HRCM2 RCP8.5 2080–99 scenario is projected 2080–99 rainfall for a RCP8.5 warming scenario from Zhang and Wang (2017), daily rainfall fragments from Longman and others (2019), ET_o from Giambelluca and others (2014), V ratios computed using equation 11 for a 1978–2007 scenario, and 2020 land cover. Components might not balance because of rounding, direct recharge from cesspools and reservoirs, and changes in root zone soil moisture storage. The source of water for direct recharge might be from external sources, including groundwater and imported streamflow. Some recharge might discharge to streams as base flow.]

Aquifer sector	Aquifer system	Area, in square miles	Scenario	Water-budget component, in million gallons per day												
				Rain	Fog	Total precipitation	Irrigation	Septic	Total direct recharge	Storm-drain capture	Runoff	Canopy evaporation	Actual ET	Total ET	Ground-water use, kiawe	Recharge
Island of Moloka‘i	--	260.01	1978–2007	607.59	27.93	635.53	5.64	0.36	1.65	0.06	82.90	92.43	242.56	334.98	0.92	228.22
			SD RCP8.5 2041–71	528.13	25.82	553.94	6.01	0.36	1.65	0.03	76.28	83.27	215.21	298.47	0.96	190.44
			SD RCP8.5 2071–99	482.16	24.11	506.27	6.15	0.36	1.65	0.03	66.42	78.14	200.88	279.00	0.97	172.43
			HRCM2 RCP8.5 2080–99	683.33	31.00	714.34	5.64	0.36	1.65	0.07	109.37	98.61	248.46	347.06	0.91	268.05
Central	Ho‘olehua	13.73	1978–2007	17.83	0.00	17.83	0.37	0.00	0.16	0.00	1.06	2.91	11.80	14.71	0.00	2.62
			SD RCP8.5 2041–71	14.41	0.00	14.41	0.39	0.00	0.16	0.00	0.75	2.44	10.15	12.59	0.00	1.64
			SD RCP8.5 2071–99	13.21	0.00	13.21	0.40	0.00	0.16	0.00	0.64	2.28	9.54	11.82	0.00	1.34
			HRCM2 RCP8.5 2080–99	20.49	0.00	20.49	0.37	0.00	0.16	0.00	1.36	3.22	12.21	15.43	0.00	4.24
Central	Kualapu‘u	17.95	1978–2007	42.80	1.55	44.35	0.77	0.01	0.81	0.01	4.11	12.13	18.03	30.16	0.00	11.95
			SD RCP8.5 2041–71	34.99	1.31	36.29	0.85	0.01	0.81	0.01	2.93	10.42	16.15	26.56	0.00	8.81
			SD RCP8.5 2071–99	32.14	1.22	33.36	0.88	0.01	0.81	0.01	2.44	9.78	15.27	25.05	0.00	7.94
			HRCM2 RCP8.5 2080–99	43.15	1.51	44.66	0.79	0.01	0.81	0.02	4.22	12.05	17.40	29.45	0.00	12.86
Central	Pālā‘au	24.52	1978–2007	24.44	0.00	24.44	3.74	0.05	0.13	0.02	1.22	1.92	20.92	22.85	0.24	4.77
			SD RCP8.5 2041–71	18.91	0.00	18.91	3.98	0.05	0.13	0.01	0.78	1.53	18.03	19.56	0.25	3.24
			SD RCP8.5 2071–99	16.85	0.00	16.85	4.07	0.05	0.13	0.01	0.59	1.37	16.89	18.26	0.25	2.78
			HRCM2 RCP8.5 2080–99	25.69	0.00	25.69	3.75	0.05	0.13	0.02	1.36	1.94	20.75	22.69	0.24	6.04
Northeast	Hālawa	13.54	1978–2007	59.84	4.33	64.17	0.07	0.00	0.00	0.00	13.83	11.69	14.05	25.73	0.00	24.71
			SD RCP8.5 2041–71	53.98	3.96	57.94	0.07	0.00	0.00	0.00	13.99	10.94	13.54	24.49	0.00	19.58
			SD RCP8.5 2071–99	48.42	3.56	51.98	0.07	0.00	0.00	0.00	11.81	10.25	12.90	23.15	0.00	17.13
			HRCM2 RCP8.5 2080–99	66.87	4.73	71.61	0.07	0.00	0.00	0.00	18.73	12.41	14.52	26.93	0.00	26.03
Northeast	Hā‘upu	2.55	1978–2007	11.19	0.72	11.91	0.00	0.00	0.00	0.00	1.81	1.29	3.38	4.67	0.00	5.42
			SD RCP8.5 2041–71	10.55	0.68	11.23	0.00	0.00	0.00	0.00	1.68	1.24	3.28	4.52	0.00	5.01
			SD RCP8.5 2071–99	9.83	0.64	10.47	0.00	0.00	0.00	0.00	1.53	1.18	3.13	4.31	0.00	4.63
			HRCM2 RCP8.5 2080–99	10.66	0.70	11.36	0.00	0.00	0.00	0.00	1.73	1.24	3.20	4.44	0.00	5.18
Northeast	Kahanui	5.41	1978–2007	20.47	1.57	22.04	0.00	0.00	0.00	0.00	2.66	3.75	5.78	9.53	0.00	9.84
			SD RCP8.5 2041–71	17.91	1.40	19.31	0.00	0.00	0.00	0.00	2.22	3.43	5.45	8.88	0.00	8.20
			SD RCP8.5 2071–99	16.63	1.31	17.94	0.00	0.00	0.00	0.00	1.99	3.26	5.21	8.46	0.00	7.47
			HRCM2 RCP8.5 2080–99	19.96	1.54	21.50	0.00	0.00	0.00	0.00	2.64	3.65	5.36	9.01	0.00	9.84

Table 2-3.—Continued

Aquifer sector	Aquifer system	Area, in square miles	Scenario	Water-budget component, in million gallons per day												
				Rain	Fog	Total precipitation	Irrigation	Septic	Total direct recharge	Storm-drain capture	Runoff	Canopy evaporation	Actual ET	Total ET	Ground-water use, kiawe	Recharge
Northeast	Kalaupapa	4.30	1978–2007	9.59	0.00	9.59	0.03	0.00	0.00	0.00	1.70	0.46	6.18	6.64	0.00	1.33
			SD RCP8.5 2041–71	8.10	0.00	8.10	0.03	0.00	0.00	0.00	1.35	0.40	5.54	5.94	0.00	0.90
			SD RCP8.5 2071–99	7.46	0.00	7.46	0.03	0.00	0.00	0.00	1.19	0.37	5.19	5.57	0.00	0.79
			HRCM2 RCP8.5 2080–99	10.51	0.00	10.51	0.03	0.00	0.00	0.00	1.90	0.50	6.27	6.77	0.00	1.92
Northeast	Pelekunu	7.38	1978–2007	38.81	1.64	40.46	0.00	0.00	0.00	0.00	4.56	3.22	10.13	13.35	0.00	22.54
			SD RCP8.5 2041–71	37.27	1.58	38.84	0.00	0.00	0.00	0.00	5.25	3.14	9.96	13.10	0.00	20.48
			SD RCP8.5 2071–99	35.01	1.48	36.49	0.00	0.00	0.00	0.00	4.85	3.03	9.68	12.70	0.00	18.93
			HRCM2 RCP8.5 2080–99	40.46	1.79	42.25	0.00	0.00	0.00	0.00	6.05	3.17	9.89	13.06	0.00	23.12
Northeast	Waikolu	4.44	1978–2007	21.41	1.54	22.95	0.00	0.00	0.00	0.00	4.39	2.28	5.60	7.87	0.00	10.68
			SD RCP8.5 2041–71	20.08	1.47	21.55	0.00	0.00	0.00	0.00	4.16	2.20	5.43	7.63	0.00	9.76
			SD RCP8.5 2071–99	19.04	1.41	20.45	0.00	0.00	0.00	0.00	3.87	2.13	5.24	7.37	0.00	9.20
			HRCM2 RCP8.5 2080–99	21.27	1.60	22.87	0.00	0.00	0.00	0.00	4.60	2.27	5.24	7.51	0.00	10.75
Northeast	Wailau	13.12	1978–2007	75.32	4.11	79.43	0.00	0.00	0.00	0.00	12.76	7.74	15.73	23.47	0.00	43.15
			SD RCP8.5 2041–71	70.57	3.86	74.43	0.00	0.00	0.00	0.00	14.46	7.49	15.52	23.01	0.00	36.92
			SD RCP8.5 2071–99	64.96	3.56	68.52	0.00	0.00	0.00	0.00	12.89	7.17	15.08	22.25	0.00	33.35
			HRCM2 RCP8.5 2080–99	84.08	4.58	88.66	0.00	0.00	0.00	0.00	18.78	7.92	15.66	23.58	0.00	46.26
Southeast	Kamiloloa	17.17	1978–2007	37.23	2.99	40.21	0.13	0.09	0.20	0.02	3.65	7.34	13.67	21.01	0.03	16.02
			SD RCP8.5 2041–71	32.47	2.71	35.19	0.13	0.09	0.20	0.01	2.59	6.60	12.33	18.93	0.03	14.14
			SD RCP8.5 2071–99	30.65	2.60	33.24	0.13	0.09	0.20	0.01	2.29	6.30	11.73	18.03	0.03	13.40
			HRCM2 RCP8.5 2080–99	39.67	3.13	42.80	0.12	0.09	0.20	0.02	3.58	7.56	14.33	21.88	0.03	17.79
Southeast	Kawela	23.52	1978–2007	46.46	4.12	50.58	0.00	0.09	0.07	0.00	8.15	6.69	17.81	24.50	0.13	18.33
			SD RCP8.5 2041–71	43.69	3.92	47.61	0.00	0.09	0.07	0.00	7.68	6.38	16.89	23.26	0.13	17.06
			SD RCP8.5 2071–99	41.97	3.79	45.76	0.00	0.09	0.07	0.00	7.20	6.18	16.33	22.51	0.13	16.45
			HRCM2 RCP8.5 2080–99	55.59	4.73	60.32	0.00	0.09	0.07	0.00	11.02	7.56	20.23	27.79	0.12	21.90
Southeast	ʻUalapuʻe	17.74	1978–2007	69.09	3.90	72.99	0.08	0.05	0.10	0.00	12.38	11.14	22.08	33.22	0.04	28.25
			SD RCP8.5 2041–71	62.26	3.60	65.86	0.09	0.05	0.10	0.00	10.65	10.41	21.10	31.51	0.04	24.63
			SD RCP8.5 2071–99	56.98	3.34	60.32	0.09	0.05	0.10	0.00	9.06	9.82	20.12	29.93	0.04	22.31
			HRCM2 RCP8.5 2080–99	92.13	4.99	97.12	0.07	0.05	0.10	0.00	19.45	13.34	24.11	37.45	0.04	40.86
Southeast	Waialua	14.55	1978–2007	53.33	1.46	54.80	0.03	0.01	0.04	0.00	7.40	10.16	19.68	29.84	0.00	17.73
			SD RCP8.5 2041–71	47.07	1.33	48.40	0.03	0.01	0.04	0.00	6.08	9.39	18.54	27.93	0.00	14.60
			SD RCP8.5 2071–99	42.38	1.20	43.59	0.03	0.01	0.04	0.00	4.98	8.78	17.40	26.19	0.00	12.65
			HRCM2 RCP8.5 2080–99	65.86	1.70	67.56	0.03	0.01	0.04	0.00	10.07	11.47	21.16	32.63	0.00	24.98
West	Kaluakoʻi	44.99	1978–2007	45.48	0.00	45.48	0.01	0.06	0.14	0.01	1.84	6.89	31.96	38.85	0.15	5.37
			SD RCP8.5 2041–71	31.86	0.00	31.86	0.01	0.06	0.14	0.00	0.99	5.12	23.74	28.86	0.16	2.64
			SD RCP8.5 2071–99	26.59	0.00	26.59	0.01	0.06	0.14	0.00	0.64	4.39	20.28	24.67	0.16	1.94
			HRCM2 RCP8.5 2080–99	47.77	0.00	47.77	0.01	0.06	0.14	0.01	2.10	7.20	31.25	38.45	0.15	7.77
West	Punakou	35.10	1978–2007	34.30	0.00	34.30	0.41	0.00	0.00	0.00	1.38	2.82	25.76	28.58	0.33	5.51
			SD RCP8.5 2041–71	24.01	0.00	24.01	0.43	0.00	0.00	0.00	0.72	2.14	19.56	21.70	0.35	2.83
			SD RCP8.5 2071–99	20.04	0.00	20.04	0.44	0.00	0.00	0.00	0.45	1.85	16.89	18.73	0.36	2.12
			HRCM2 RCP8.5 2080–99	39.17	0.00	39.17	0.40	0.00	0.00	0.00	1.78	3.11	26.88	29.99	0.33	8.51

Table 2-4. Mean annual water-budget components from this study for aquifer systems and all of Lānaʻi.

[See figure 2 for locations of aquifer systems; --, not applicable; Total precipitation, sum of Rain and Fog; Septic, septic-system leachate; Total direct recharge, sum of cesspool seepage and constant recharge from unlined reservoirs and inland water bodies; Runoff, direct runoff; ET, evapotranspiration; Total ET, sum of Canopy evaporation and Actual ET; Groundwater use, kiawe, groundwater use by kiawe land cover; Recharge, groundwater recharge which includes Total direct recharge; ET_o, reference evapotranspiration for grass surface; V, ratio of the mean evaporation rate to mean precipitation rate during saturated conditions; SD, statistical downscaling; HRCM2, Hawaiʻi Regional Climate Model version 2 from Zhang and Wang (2017); RCP4.5 and RCP8.5, Representative Concentration Pathway with total radiative forcing of 4.5 and 8.5 Watts per square meter, respectively, by the year 2100 from Intergovernmental Panel on Climate Change (2014); 1978–2007 scenario is 1978–2007 rainfall from Frazier and others (2016), daily rainfall fragments from Longman and others (2019), ET_o from Giambelluca and others (2014), V ratios computed using equation 11 for a 1978–2007 scenario, and 2020 land cover; SD RCP8.5 2041–71 scenario is projected 2041–71 rainfall for a RCP8.5 warming scenario from Elison Timm and others (2015), daily rainfall fragments from Longman and others (2019), ET_o from Giambelluca and others (2014), V ratios computed using equation 11 for a 1978–2007 scenario, and 2020 land cover; SD RCP8.5 2071–99 scenario is projected 2071–99 rainfall for a RCP8.5 warming scenario from Elison Timm and others (2015), daily rainfall fragments from Longman and others (2019), ET_o from Giambelluca and others (2014), V ratios computed using equation 11 for a 1978–2007 scenario, and 2020 land cover; HRCM2 RCP4.5 2080–99 scenario is projected 2080–99 rainfall for a RCP4.5 warming scenario from Zhang and Wang (2017), daily rainfall fragments from Longman and others (2019), ET_o from Giambelluca and others (2014), V ratios computed using equation 11 for a 1978–2007 scenario, and 2020 land cover. Components might not balance because of rounding, direct recharge from cesspools and reservoirs, and changes in root zone soil moisture storage. The source of water for direct recharge might be from external sources, including groundwater and imported streamflow. Some recharge might discharge to streams as base flow.]

Aquifer sector	Aquifer system	Area, in square miles	Scenario	Water-budget component, in million gallons per day												
				Rain	Fog	Total precipitation	Irrigation	Septic	Total direct recharge	Storm-drain capture	Runoff	Canopy evaporation	Actual ET	Total ET	Ground-water use, kiawe	Recharge
Island of Lānaʻi	--	140.87	1978–2007	147.79	1.01	148.80	1.26	0.00	0.00	0.14	6.28	28.48	90.34	118.81	1.21	26.84
			SD RCP8.5 2041–71	110.55	0.72	111.27	1.44	0.00	0.00	0.07	3.90	22.22	72.75	94.97	1.24	15.95
			SD RCP8.5 2071–99	95.57	0.58	96.15	1.54	0.00	0.00	0.05	2.84	19.58	65.25	84.84	1.26	12.20
			HRCM2 RCP4.5 2080–99	158.48	0.97	159.44	1.33	0.00	0.00	0.17	7.28	29.61	87.25	116.87	1.20	38.45
			2008–12	92.26	0.72	92.98	1.47	0.00	0.00	0.07	3.18	18.60	64.24	82.85	1.28	12.25
Central	Leeward	16.37	1978–2007	21.54	0.41	21.95	0.48	0.00	0.00	0.11	1.06	3.73	13.17	16.90	0.00	4.34
			SD RCP8.5 2041–71	14.19	0.28	14.47	0.60	0.00	0.00	0.06	0.52	2.68	9.82	12.50	0.00	2.00
			SD RCP8.5 2071–99	11.11	0.22	11.33	0.66	0.00	0.00	0.04	0.28	2.21	8.16	10.37	0.00	1.32
			HRCM2 RCP4.5 2080–99	22.57	0.40	22.96	0.54	0.00	0.00	0.13	1.18	3.72	12.43	16.15	0.00	6.03
			2008–12	14.32	0.29	14.61	0.62	0.00	0.00	0.06	0.57	2.64	10.02	12.66	0.00	2.21
Central	Windward	12.73	1978–2007	17.17	0.58	17.75	0.08	0.00	0.00	0.00	0.81	3.30	9.25	12.55	0.00	4.45
			SD RCP8.5 2041–71	12.92	0.42	13.34	0.10	0.00	0.00	0.00	0.51	2.57	7.70	10.27	0.00	2.65
			SD RCP8.5 2071–99	11.49	0.35	11.84	0.11	0.00	0.00	0.00	0.41	2.29	7.18	9.48	0.00	2.05
			HRCM2 RCP4.5 2080–99	16.67	0.55	17.22	0.09	0.00	0.00	0.00	0.83	3.12	8.47	11.59	0.00	4.88
			2008–12	11.47	0.41	11.88	0.11	0.00	0.00	0.00	0.43	2.33	6.86	9.19	0.00	2.50
Kaʻā	Honopū	19.38	1978–2007	20.18	0.00	20.18	0.00	0.00	0.00	0.00	0.81	4.09	12.68	16.77	0.01	3.11
			SD RCP8.5 2041–71	15.94	0.00	15.94	0.00	0.00	0.00	0.00	0.56	3.27	10.60	13.87	0.01	2.05
			SD RCP8.5 2071–99	13.94	0.00	13.94	0.00	0.00	0.00	0.00	0.42	2.86	9.60	12.46	0.01	1.62
			HRCM2 RCP4.5 2080–99	21.31	0.00	21.31	0.00	0.00	0.00	0.00	0.94	4.18	12.07	16.25	0.01	4.63
			2008–12	11.55	0.00	11.55	0.00	0.00	0.00	0.00	0.40	2.41	8.27	10.68	0.01	1.28
Kaʻā	Kaumalapau	14.94	1978–2007	14.83	0.00	14.83	0.12	0.00	0.00	0.02	0.58	3.09	9.44	12.53	0.00	1.86
			SD RCP8.5 2041–71	10.25	0.00	10.25	0.13	0.00	0.00	0.01	0.31	2.27	6.91	9.19	0.00	0.94
			SD RCP8.5 2071–99	8.18	0.00	8.18	0.14	0.00	0.00	0.01	0.18	1.89	5.68	7.57	0.00	0.64
			HRCM2 RCP4.5 2080–99	16.21	0.00	16.21	0.12	0.00	0.00	0.02	0.69	3.24	9.06	12.30	0.00	3.35
			2008–12	8.69	0.00	8.69	0.13	0.00	0.00	0.01	0.25	1.94	6.41	8.35	0.00	0.54

Table 2-4.—Continued

Aquifer sector	Aquifer system	Area, in square miles	Scenario	Water-budget component, in million gallons per day												
				Rain	Fog	Total precipitation	Irrigation	Septic	Total direct recharge	Storm-drain capture	Runoff	Canopy evaporation	Actual ET	Total ET	Ground-water use, kiawe	Recharge
Kanao	Keālia	12.72	1978–2007	11.23	0.00	11.23	0.13	0.00	0.00	0.00	0.40	1.47	7.75	9.22	0.01	1.86
			SD RCP8.5 2041–71	7.57	0.00	7.57	0.14	0.00	0.00	0.00	0.21	1.06	5.60	6.65	0.01	0.99
			SD RCP8.5 2071–99	6.19	0.00	6.19	0.15	0.00	0.00	0.00	0.12	0.90	4.73	5.63	0.01	0.73
			HRCM2 RCP4.5 2080–99	12.38	0.00	12.38	0.13	0.00	0.00	0.01	0.49	1.56	7.36	8.92	0.01	3.22
			2008–12	7.47	0.00	7.47	0.14	0.00	0.00	0.00	0.21	1.04	5.91	6.96	0.01	0.73
Kanao	Mānele	14.81	1978–2007	13.37	0.02	13.39	0.45	0.00	0.00	0.01	0.53	3.10	7.52	10.62	0.09	2.87
			SD RCP8.5 2041–71	9.19	0.02	9.20	0.47	0.00	0.00	0.00	0.27	2.20	5.74	7.94	0.09	1.66
			SD RCP8.5 2071–99	7.65	0.01	7.66	0.48	0.00	0.00	0.00	0.16	1.86	5.05	6.91	0.09	1.27
			HRCM2 RCP4.5 2080–99	15.34	0.02	15.36	0.45	0.00	0.00	0.01	0.67	3.37	7.20	10.57	0.09	4.74
			2008–12	9.15	0.02	9.17	0.47	0.00	0.00	0.00	0.30	2.20	5.96	8.16	0.09	1.48
Mahana	Hauola	13.09	1978–2007	12.19	0.00	12.19	0.00	0.00	0.00	0.00	0.51	2.11	7.88	9.98	0.67	2.37
			SD RCP8.5 2041–71	9.36	0.00	9.37	0.00	0.00	0.00	0.00	0.32	1.65	6.51	8.17	0.70	1.57
			SD RCP8.5 2071–99	8.49	0.00	8.49	0.00	0.00	0.00	0.00	0.26	1.52	6.12	7.64	0.71	1.29
			HRCM2 RCP4.5 2080–99	13.12	0.00	13.12	0.00	0.00	0.00	0.00	0.58	2.23	7.79	10.02	0.67	3.18
			2008–12	7.85	0.00	7.85	0.00	0.00	0.00	0.00	0.26	1.41	5.90	7.31	0.71	1.08
Mahana	Maunalei	13.81	1978–2007	13.41	0.00	13.41	0.00	0.00	0.00	0.00	0.58	3.38	7.64	11.02	0.22	2.02
			SD RCP8.5 2041–71	11.35	0.00	11.35	0.00	0.00	0.00	0.00	0.45	2.94	6.82	9.76	0.22	1.37
			SD RCP8.5 2071–99	10.72	0.00	10.72	0.00	0.00	0.00	0.00	0.40	2.79	6.63	9.42	0.22	1.12
			HRCM2 RCP4.5 2080–99	14.83	0.00	14.83	0.00	0.00	0.00	0.00	0.71	3.67	7.89	11.57	0.21	2.76
			2008–12	8.05	0.00	8.05	0.00	0.00	0.00	0.00	0.27	2.14	5.24	7.38	0.23	0.78
Mahana	Pao-ma‘i	23.03	1978–2007	23.87	0.00	23.87	0.00	0.00	0.00	0.00	1.00	4.21	15.01	19.22	0.21	3.96
			SD RCP8.5 2041–71	19.78	0.00	19.78	0.00	0.00	0.00	0.00	0.75	3.58	13.05	16.62	0.21	2.72
			SD RCP8.5 2071–99	17.80	0.00	17.80	0.00	0.00	0.00	0.00	0.61	3.26	12.10	15.36	0.22	2.16
			HRCM2 RCP4.5 2080–99	26.05	0.00	26.05	0.00	0.00	0.00	0.00	1.19	4.52	14.98	19.50	0.21	5.66
			2008–12	13.71	0.00	13.71	0.00	0.00	0.00	0.00	0.49	2.49	9.67	12.16	0.23	1.65

Table 2-5. Mean annual water-budget components from this study for aquifer systems and all of Maui.

[See fig. 2 for locations of aquifer systems; n/a, not applicable; Water-budget area descriptions: A, entire aquifer-system area: B, entire aquifer-system area excluding areas seaward of the 0-ft altitude contour of the top of the volcanic aquifer (see fig. 2); Total precipitation, sum of Rain and Fog; Septic, septic-system leachate; Total direct recharge, sum of cesspool seepage and constant recharge from unlined reservoirs and inland water bodies; Runoff, direct runoff; ET, evapotranspiration; Total ET, sum of Canopy evaporation and Actual ET; Groundwater use, kiawe, groundwater use by kiawe land cover; Recharge, groundwater recharge which includes Total direct recharge; ET_o, reference evapotranspiration for grass surface; V, ratio of the mean evaporation rate to mean precipitation rate during saturated conditions; SD, statistical downscaling; HRCM2, Hawai'i Regional Climate Model from Zhang and Wang (2017); Representative Concentration Pathway with total radiative forcing of 8.5 Watts per square meter by the year 2100 from Intergovernmental Panel on Climate Change (2014); 1978–2007 scenario is 1978–2007 rainfall from Frazier and others (2016), daily rainfall fragments from Longman and others (2019), ET_o from Giambelluca and others (2014), V ratios computed using equation 11 for a 1978–2007 scenario, and 2020 land cover; SD RCP8.5 2041–71 scenario is projected 2041–71 rainfall for a RCP8.5 warming scenario from Elison Timm and others (2015), daily rainfall fragments from Longman and others (2019), ET_o from Giambelluca and others (2014), V ratios computed using equation 11 for a 1978–2007 scenario, and 2020 land cover; SD RCP8.5 2071–99 scenario is projected 2071–99 rainfall for a RCP8.5 warming scenario from Elison Timm and others (2015), daily rainfall fragments from Longman and others (2019), ET_o from Giambelluca and others (2014), V ratios computed using equation 11 for a 1978–2007 scenario, and 2020 land cover; HRCM1 A1B 2080–99 scenario is projected 2080–99 rainfall for an A1B emission scenario from Zhang and others (2016b), daily rainfall fragments from Longman and others (2019), ET_o from Giambelluca and others (2014), V ratios computed using equation 11 for a 1978–2007 scenario, and 2020 land cover. Components might not balance because of rounding, direct recharge from cesspools and reservoirs, and changes in root zone soil moisture storage. The source of water for direct recharge might be from external sources, including groundwater and imported streamflow. Some recharge might discharge to streams as base flow.]

Aquifer sector	Aquifer system	Water-budget area description	Area, in square miles	Scenario	Water-budget component, in million gallons per day												
					Rain	Fog	Total precipitation	Irrigation	Septic	Total direct recharge	Storm-drain capture	Runoff	Canopy evaporation	Actual ET	Total ET	Ground-water use, kiawe	Recharge
Island of Maui	n/a	A	727.78	1978–2007	2,804.99	171.23	2,976.22	29.45	3.74	37.36	4.57	846.83	256.17	750.62	1,006.79	0.04	1,192.34
		A		SD RCP8.5 2041–71	2,571.76	155.09	2,726.84	31.58	3.74	37.36	3.12	811.84	239.84	671.31	911.14	0.04	1,077.76
		A		SD RCP8.5 2071–99	2,446.10	146.50	2,592.58	32.65	3.74	37.36	2.54	781.43	230.89	625.52	856.43	0.04	1,030.62
		A		HRCM1 A1B 2080–99	3,078.87	193.93	3,272.80	30.18	3.74	37.36	3.95	1,009.38	265.79	749.95	1,015.76	0.04	1,318.75
		B	707.98	1978–2007	2,786.30	171.23	2,957.54	24.47	3.53	33.43	3.51	844.07	255.34	730.81	986.16	0.03	1,185.80
		B		SD RCP8.5 2041–71	2,558.24	155.09	2,713.33	26.37	3.53	33.43	2.44	809.92	239.22	653.89	893.11	0.03	1,072.17
		B		SD RCP8.5 2071–99	2,434.85	146.50	2,581.34	27.33	3.53	33.43	2.04	779.86	230.39	609.28	839.66	0.03	1,025.33
		B		HRCM1 A1B 2080–99	3,062.17	193.93	3,256.08	25.09	3.53	33.43	3.07	1,006.83	265.04	731.06	996.09	0.03	1,312.65
Central	Kahului	A	27.50	1978–2007	20.19	0.00	20.19	3.39	0.08	3.55	1.03	0.82	0.72	21.58	22.30	0.00	6.00
		A		SD RCP8.5 2041–71	12.14	0.00	12.14	3.60	0.08	3.55	0.55	0.31	0.47	16.61	17.08	0.00	4.54
		A		SD RCP8.5 2071–99	8.64	0.00	8.64	3.70	0.08	3.55	0.36	0.13	0.35	14.13	14.49	0.00	4.18
		A		HRCM1 A1B 2080–99	17.20	0.00	17.20	3.47	0.08	3.55	0.86	0.68	0.63	19.59	20.22	0.00	5.52
		B	23.06	1978–2007	17.28	0.00	17.28	2.78	0.07	3.53	0.72	0.73	0.52	16.25	16.77	0.00	5.72
		B		SD RCP8.5 2041–71	10.38	0.00	10.38	2.96	0.07	3.53	0.38	0.28	0.35	11.82	12.17	0.00	4.42
		B		SD RCP8.5 2071–99	7.39	0.00	7.39	3.04	0.07	3.53	0.25	0.12	0.27	9.61	9.88	0.00	4.11
		B		HRCM1 A1B 2080–99	14.66	0.00	14.66	2.85	0.07	3.53	0.59	0.60	0.46	14.44	14.90	0.00	5.30
Central	Kama'ole	A	91.36	1978–2007	98.06	8.05	106.11	6.63	0.53	1.56	0.41	3.97	16.65	63.15	79.80	0.01	30.65
		A		SD RCP8.5 2041–71	56.14	4.80	60.94	7.24	0.53	1.56	0.17	1.42	10.48	44.84	55.31	0.01	13.45
		A		SD RCP8.5 2071–99	38.07	3.45	41.52	7.50	0.53	1.56	0.09	0.53	7.47	34.71	42.17	0.01	8.50
		A		HRCM1 A1B 2080–99	83.13	6.63	89.76	6.79	0.53	1.56	0.35	3.05	14.57	57.24	71.81	0.01	23.45
Central	Makawao	A	51.78	1978–2007	107.86	5.70	113.56	2.90	1.24	3.03	1.44	11.50	6.83	56.13	62.96	0.00	44.75
		A		SD RCP8.5 2041–71	85.19	4.22	89.40	3.37	1.24	3.03	1.09	8.62	5.64	49.57	55.21	0.00	32.05
		A		SD RCP8.5 2071–99	74.29	3.45	77.74	3.67	1.24	3.03	0.95	7.00	5.01	44.95	49.97	0.00	27.71
		A		HRCM1 A1B 2080–99	96.05	5.06	101.11	3.11	1.24	3.03	1.24	10.38	6.26	52.90	59.15	0.00	37.62

Table 2-5.—Continued

Aquifer sector	Aquifer system	Water-budget area description	Area, in square miles	Scenario	Water-budget component, in million gallons per day												
					Rain	Fog	Total precipitation	Irrigation	Septic	Total direct recharge	Storm-drain capture	Runoff	Canopy evaporation	Actual ET	Total ET	Ground-water use, kiawe	Recharge
Central	Pā‘ia	A	59.04	1978–2007	70.54	0.00	70.54	1.55	0.09	18.48	0.22	4.98	1.55	58.83	60.39	0.00	26.13
		A		SD RCP8.5 2041–71	53.08	0.00	53.08	1.65	0.09	18.48	0.16	3.54	1.15	47.50	48.65	0.00	22.17
		A		SD RCP8.5 2071–99	45.75	0.00	45.75	1.70	0.09	18.48	0.13	2.95	0.97	41.93	42.91	0.00	21.31
		A		HRCM1 A1B 2080–99	67.63	0.00	67.63	1.57	0.09	18.48	0.21	4.93	1.47	56.58	58.05	0.00	25.62
Hāna	Kawaipapa	A	31.72	1978–2007	255.00	12.34	267.33	0.18	0.12	0.28	0.08	93.45	19.16	49.36	68.53	0.00	105.78
		A		SD RCP8.5 2041–71	261.19	12.55	273.73	0.18	0.12	0.28	0.08	98.37	19.36	49.26	68.62	0.00	107.16
		A		SD RCP8.5 2071–99	259.55	12.44	271.99	0.19	0.12	0.28	0.08	98.30	19.28	48.37	67.64	0.00	106.47
		A		HRCM1 A1B 2080–99	288.29	14.31	302.60	0.16	0.12	0.28	0.08	111.44	20.07	50.59	70.66	0.00	120.89
Hāna	Kīpahulu	A	29.65	1978–2007	237.19	9.97	247.16	0.11	0.01	0.03	0.00	67.74	22.54	34.83	57.38	0.00	122.11
		A		SD RCP8.5 2041–71	222.89	9.31	232.20	0.11	0.01	0.03	0.00	63.14	21.69	33.55	55.24	0.00	113.89
		A		SD RCP8.5 2071–99	214.10	8.93	223.03	0.11	0.01	0.03	0.00	59.24	21.18	32.86	54.04	0.00	109.83
		A		HRCM1 A1B 2080–99	285.44	12.19	297.63	0.10	0.01	0.03	0.00	91.52	24.62	36.56	61.18	0.00	144.99
Hāna	Kūhiwa	A	13.58	1978–2007	175.75	10.26	186.00	0.00	0.01	0.02	0.00	71.35	11.76	15.42	27.17	0.00	87.45
		A		SD RCP8.5 2041–71	178.05	10.20	188.26	0.00	0.01	0.02	0.00	73.67	11.83	15.42	27.25	0.00	87.31
		A		SD RCP8.5 2071–99	174.89	9.99	184.88	0.00	0.01	0.02	0.00	72.72	11.71	15.18	26.89	0.00	85.25
		A		HRCM1 A1B 2080–99	203.76	12.16	215.92	0.00	0.01	0.02	0.00	87.64	12.34	15.60	27.94	0.00	100.32
Hāna	Waiho‘i	A	14.40	1978–2007	132.07	7.90	139.97	0.03	0.02	0.05	0.00	57.88	9.51	19.67	29.18	0.00	52.97
		A		SD RCP8.5 2041–71	128.86	7.76	136.62	0.03	0.02	0.05	0.00	57.20	9.37	19.26	28.63	0.00	50.85
		A		SD RCP8.5 2071–99	124.17	7.50	131.67	0.03	0.02	0.05	0.00	54.76	9.18	18.66	27.85	0.00	49.13
		A		HRCM1 A1B 2080–99	154.86	9.37	164.24	0.02	0.02	0.05	0.00	72.90	10.10	20.19	30.30	0.00	61.09
Kahikinui	Kaupō	A	20.30	1978–2007	58.16	1.80	59.96	0.00	0.00	0.01	0.00	7.16	1.01	25.31	26.32	0.00	26.48
		A		SD RCP8.5 2041–71	49.81	1.51	51.32	0.00	0.00	0.01	0.00	5.66	0.91	22.64	23.55	0.00	22.11
		A		SD RCP8.5 2071–99	46.50	1.39	47.88	0.00	0.00	0.01	0.00	4.93	0.87	21.39	22.25	0.00	20.71
		A		HRCM1 A1B 2080–99	66.91	2.17	69.08	0.00	0.00	0.01	0.00	8.98	1.15	27.29	28.43	0.00	31.67
Kahikinui	Lualaʻilua	A	62.43	1978–2007	75.58	3.04	78.62	0.00	0.00	0.01	0.00	3.93	5.03	47.98	53.02	0.02	21.66
		A		SD RCP8.5 2041–71	46.93	1.67	48.61	0.00	0.00	0.01	0.00	1.56	3.12	33.41	36.53	0.02	10.54
		A		SD RCP8.5 2071–99	34.96	1.08	36.03	0.00	0.00	0.01	0.00	0.64	2.19	26.26	28.45	0.02	6.98
		A		HRCM1 A1B 2080–99	77.41	2.84	80.25	0.00	0.00	0.01	0.00	4.14	5.24	49.93	55.17	0.02	20.94
Kahikinui	Nakula	A	30.21	1978–2007	62.39	1.14	63.54	0.00	0.00	0.00	0.00	5.70	1.12	33.98	35.11	0.00	22.73
		A		SD RCP8.5 2041–71	48.43	0.80	49.23	0.00	0.00	0.00	0.00	3.67	0.89	28.30	29.20	0.00	16.37
		A		SD RCP8.5 2071–99	42.57	0.66	43.24	0.00	0.00	0.00	0.00	2.70	0.79	25.44	26.23	0.00	14.32
		A		HRCM1 A1B 2080–99	69.76	1.20	70.96	0.00	0.00	0.00	0.00	6.74	1.30	36.80	38.09	0.00	26.12

Table 2-5.—Continued

Aquifer sector	Aquifer system	Water-budget area description	Area, in square miles	Scenario	Water-budget component, in million gallons per day												
					Rain	Fog	Total precipitation	Irrigation	Septic	Total direct recharge	Storm-drain capture	Runoff	Canopy evaporation	Actual ET	Total ET	Ground-water use, kiawe	Recharge
Koʻolau	Haʻikū	A	35.19	1978–2007	161.54	12.84	174.38	0.35	0.85	2.66	0.24	43.27	20.21	55.70	75.91	0.00	58.74
		A		SD RCP8.5 2041–71	159.80	11.87	171.67	0.34	0.85	2.66	0.25	42.10	20.00	56.30	76.29	0.00	56.80
		A		SD RCP8.5 2071–99	158.66	11.44	170.10	0.34	0.85	2.66	0.26	41.45	19.83	56.09	75.91	0.00	56.25
		A		HRCM1 A1B 2080–99	178.33	14.68	193.01	0.32	0.85	2.66	0.27	49.36	21.65	58.82	80.48	0.00	66.63
Koʻolau	Honopou	A	17.70	1978–2007	118.36	6.69	125.05	0.01	0.12	0.51	0.01	43.06	16.06	27.33	43.39	0.00	39.16
		A		SD RCP8.5 2041–71	122.41	6.57	128.98	0.01	0.12	0.51	0.01	45.04	16.37	27.61	43.98	0.00	40.51
		A		SD RCP8.5 2071–99	121.88	6.57	128.45	0.01	0.12	0.51	0.01	45.44	16.34	27.03	43.37	0.00	40.19
		A		HRCM1 A1B 2080–99	137.38	7.99	145.37	0.01	0.12	0.51	0.01	52.75	17.52	28.24	45.76	0.00	47.39
Koʻolau	Keʻanae	A	55.22	1978–2007	454.19	40.30	494.49	0.98	0.01	0.03	0.01	188.94	35.14	53.79	88.92	0.00	217.44
		A		SD RCP8.5 2041–71	449.47	38.21	487.68	0.98	0.01	0.03	0.01	188.96	35.07	52.70	87.77	0.00	211.74
		A		SD RCP8.5 2071–99	436.44	36.24	472.68	0.98	0.01	0.03	0.01	182.84	34.53	51.32	85.85	0.00	204.80
		A		HRCM1 A1B 2080–99	521.89	47.87	569.76	0.98	0.01	0.03	0.01	227.23	37.32	54.42	91.75	0.00	251.56
Koʻolau	Waikamoi	A	26.06	1978–2007	210.88	18.25	229.13	0.00	0.00	0.28	0.00	91.99	24.26	25.86	50.12	0.00	87.11
		A		SD RCP8.5 2041–71	218.77	17.71	236.48	0.00	0.00	0.28	0.00	97.97	24.51	25.85	50.36	0.00	88.23
		A		SD RCP8.5 2071–99	221.50	17.43	238.94	0.00	0.00	0.28	0.00	100.01	24.56	25.57	50.13	0.00	88.88
		A		HRCM1 A1B 2080–99	245.90	21.57	267.48	0.00	0.00	0.28	0.00	114.65	26.22	26.17	52.39	0.00	100.50
Lahaina	Honokōhau	A	13.86	1978–2007	73.31	5.62	78.94	0.11	0.00	0.01	0.00	21.29	10.13	15.03	25.16	0.00	32.58
		A		SD RCP8.5 2041–71	69.45	5.26	74.71	0.11	0.00	0.01	0.00	19.78	9.92	14.75	24.67	0.00	30.37
		A		SD RCP8.5 2071–99	67.22	5.07	72.29	0.11	0.00	0.01	0.00	18.80	9.77	14.43	24.20	0.00	29.39
		A		HRCM1 A1B 2080–99	77.99	6.31	84.30	0.11	0.00	0.01	0.00	23.72	10.28	14.87	25.15	0.00	35.53
		B	13.79	1978–2007	73.18	5.62	78.80	0.11	0.00	0.01	0.00	21.28	10.13	14.92	25.05	0.00	32.56
		B		SD RCP8.5 2041–71	69.32	5.26	74.58	0.11	0.00	0.01	0.00	19.77	9.92	14.65	24.57	0.00	30.35
		B		SD RCP8.5 2071–99	67.10	5.07	72.17	0.11	0.00	0.01	0.00	18.79	9.76	14.34	24.10	0.00	29.38
		B		HRCM1 A1B 2080–99	77.86	6.31	84.17	0.11	0.00	0.01	0.00	23.71	10.27	14.77	25.05	0.00	35.51
Lahaina	Honokōwai	A	23.54	1978–2007	70.38	2.36	72.74	5.32	0.05	1.17	0.22	10.86	8.26	27.48	35.73	0.00	32.83
		A		SD RCP8.5 2041–71	58.86	1.99	60.85	5.69	0.05	1.17	0.17	8.13	7.35	24.97	32.32	0.00	27.55
		A		SD RCP8.5 2071–99	54.23	1.83	56.05	5.86	0.05	1.17	0.14	6.91	6.96	23.88	30.85	0.00	25.66
		A		HRCM1 A1B 2080–99	72.41	2.62	75.02	5.47	0.05	1.17	0.18	12.20	8.34	26.89	35.23	0.00	34.49
		B	22.71	1978–2007	69.59	2.36	71.95	4.69	0.05	1.13	0.16	10.83	8.25	26.19	34.44	0.00	32.60
		B		SD RCP8.5 2041–71	58.24	1.99	60.24	5.03	0.05	1.13	0.12	8.11	7.34	23.77	31.12	0.00	27.34
		B		SD RCP8.5 2071–99	53.70	1.83	55.52	5.19	0.05	1.13	0.11	6.89	6.96	22.73	29.69	0.00	25.46
		B		HRCM1 A1B 2080–99	71.72	2.62	74.33	4.82	0.05	1.13	0.13	12.17	8.33	25.65	33.98	0.00	34.27

Table 2-5.—Continued

Aquifer sector	Aquifer system	Water-budget area description	Area, in square miles	Scenario	Water-budget component, in million gallons per day												
					Rain	Fog	Total precipitation	Irrigation	Septic	Total direct recharge	Storm-drain capture	Runoff	Canopy evaporation	Actual ET	Total ET	Ground-water use, kiawe	Recharge
Lahaina	Honolua	A	17.77	1978–2007	67.17	1.25	68.41	1.98	0.05	0.41	0.15	9.93	11.24	25.11	36.35	0.00	24.44
		A		SD RCP8.5 2041–71	62.34	1.13	63.47	2.06	0.05	0.41	0.13	8.82	10.77	24.50	35.27	0.00	21.79
		A		SD RCP8.5 2071–99	60.86	1.06	61.92	2.10	0.05	0.41	0.12	8.36	10.65	24.39	35.03	0.00	20.98
		A		HRCM1 A1B 2080–99	69.41	1.41	70.82	2.03	0.05	0.41	0.13	10.71	11.49	24.97	36.46	0.00	26.01
		B	16.94	1978–2007	65.67	1.25	66.92	1.51	0.05	0.32	0.10	9.82	11.23	23.59	34.81	0.00	24.06
		B		SD RCP8.5 2041–71	60.97	1.13	62.10	1.56	0.05	0.32	0.09	8.72	10.75	23.04	33.79	0.00	21.44
		B		SD RCP8.5 2071–99	59.56	1.06	60.62	1.59	0.05	0.32	0.08	8.27	10.63	22.96	33.60	0.00	20.63
		B		HRCM1 A1B 2080–99	67.97	1.41	69.38	1.54	0.05	0.32	0.09	10.60	11.48	23.47	34.95	0.00	25.65
Lahaina	Launiupoko	A	20.74	1978–2007	64.05	3.19	67.24	0.77	0.24	0.45	0.14	9.23	5.62	15.02	20.64	0.00	38.75
		A		SD RCP8.5 2041–71	50.53	2.53	53.05	0.80	0.24	0.45	0.10	6.12	4.77	12.97	17.74	0.00	30.66
		A		SD RCP8.5 2071–99	46.68	2.39	49.06	0.81	0.24	0.45	0.08	5.16	4.51	12.24	16.76	0.00	28.65
		A		HRCM1 A1B 2080–99	63.55	3.30	66.85	0.79	0.24	0.45	0.10	9.49	5.48	14.25	19.73	0.00	39.09
		B	18.73	1978–2007	62.84	3.19	66.04	0.41	0.24	0.44	0.06	9.17	5.53	13.83	19.36	0.00	38.55
		B		SD RCP8.5 2041–71	49.76	2.53	52.28	0.43	0.24	0.44	0.04	6.09	4.71	12.01	16.73	0.00	30.55
		B		SD RCP8.5 2071–99	46.05	2.39	48.44	0.43	0.24	0.44	0.04	5.13	4.47	11.36	15.83	0.00	28.58
		B		HRCM1 A1B 2080–99	62.56	3.30	65.86	0.42	0.24	0.44	0.04	9.44	5.40	13.16	18.55	0.00	38.95
Lahaina	Olowalu	A	8.11	1978–2007	21.33	0.60	21.93	0.09	0.02	0.01	0.00	2.59	2.08	6.00	8.09	0.00	11.39
		A		SD RCP8.5 2041–71	14.64	0.42	15.06	0.10	0.02	0.01	0.00	1.34	1.62	4.90	6.52	0.00	7.34
		A		SD RCP8.5 2071–99	12.84	0.38	13.22	0.10	0.02	0.01	0.00	1.02	1.49	4.45	5.94	0.00	6.40
		A		HRCM1 A1B 2080–99	21.46	0.63	22.09	0.09	0.02	0.01	0.00	2.69	2.10	5.81	7.91	0.00	11.63
		B	7.12	1978–2007	20.77	0.60	21.37	0.03	0.01	0.00	0.00	2.55	2.07	5.51	7.58	0.00	11.28
		B		SD RCP8.5 2041–71	14.36	0.42	14.78	0.03	0.01	0.00	0.00	1.32	1.61	4.59	6.19	0.00	7.31
		B		SD RCP8.5 2071–99	12.66	0.38	13.04	0.03	0.01	0.00	0.00	1.02	1.49	4.21	5.69	0.00	6.38
		B		HRCM1 A1B 2080–99	20.99	0.63	21.62	0.03	0.01	0.00	0.00	2.66	2.09	5.36	7.45	0.00	11.55
Lahaina	Ukumehame	A	11.82	1978–2007	23.63	0.71	24.34	0.03	0.00	0.09	0.00	2.39	1.54	9.09	10.63	0.01	11.46
		A		SD RCP8.5 2041–71	13.62	0.42	14.04	0.03	0.00	0.09	0.00	0.90	1.05	6.67	7.72	0.01	5.57
		A		SD RCP8.5 2071–99	9.85	0.32	10.17	0.03	0.00	0.09	0.00	0.44	0.85	5.30	6.16	0.01	3.72
		A		HRCM1 A1B 2080–99	22.62	0.72	23.34	0.03	0.00	0.09	0.00	2.33	1.50	8.66	10.17	0.01	10.98
		B	10.79	1978–2007	22.98	0.71	23.70	0.00	0.00	0.00	0.00	2.36	1.50	8.59	10.09	0.00	11.24
		B		SD RCP8.5 2041–71	13.29	0.42	13.71	0.00	0.00	0.00	0.00	0.89	1.03	6.35	7.38	0.00	5.44
		B		SD RCP8.5 2071–99	9.64	0.32	9.96	0.00	0.00	0.00	0.00	0.43	0.84	5.07	5.91	0.00	3.62
		B		HRCM1 A1B 2080–99	22.06	0.72	22.77	0.00	0.00	0.00	0.00	2.30	1.47	8.22	9.69	0.00	10.78

Table 2-5.—Continued

Aquifer sector	Aquifer system	Water-budget area description	Area, in square miles	Scenario	Water-budget component, in million gallons per day												
					Rain	Fog	Total precipitation	Irrigation	Septic	Total direct recharge	Storm-drain capture	Runoff	Canopy evaporation	Actual ET	Total ET	Ground-water use, kiawe	Recharge
Wailuku	ʻĪao	A	25.27	1978–2007	98.50	7.31	105.81	2.88	0.11	4.14	0.60	39.65	8.68	25.36	34.03	0.00	38.87
		A		SD RCP8.5 2041–71	82.76	6.16	88.92	3.00	0.11	4.14	0.39	30.71	7.90	24.21	32.11	0.00	33.20
		A		SD RCP8.5 2071–99	76.74	5.76	82.50	3.06	0.11	4.14	0.30	27.43	7.56	23.41	30.97	0.00	31.36
		A		HRCM1 A1B 2080–99	103.28	8.05	111.33	2.92	0.11	4.14	0.50	42.48	8.91	25.50	34.41	0.00	41.34
		B	18.21	1978–2007	89.94	7.31	97.25	0.33	0.04	0.56	0.06	37.63	8.31	17.91	26.22	0.00	34.25
		B		SD RCP8.5 2041–71	76.05	6.16	82.21	0.34	0.04	0.56	0.04	29.24	7.59	17.35	24.94	0.00	28.93
		B		SD RCP8.5 2071–99	70.90	5.76	76.66	0.35	0.04	0.56	0.03	26.21	7.28	16.88	24.16	0.00	27.20
		B		HRCM1 A1B 2080–99	95.39	8.05	103.43	0.33	0.04	0.56	0.05	40.59	8.57	18.25	26.81	0.00	36.89
Wailuku	Kahakuloa	A	10.77	1978–2007	45.02	3.41	48.43	0.07	0.00	0.02	0.00	15.53	7.91	10.66	18.56	0.00	14.41
		A		SD RCP8.5 2041–71	42.92	3.29	46.21	0.07	0.00	0.02	0.00	14.61	7.69	10.47	18.16	0.00	13.52
		A		SD RCP8.5 2071–99	41.29	3.16	44.45	0.07	0.00	0.02	0.00	13.77	7.51	10.26	17.76	0.00	13.00
		A		HRCM1 A1B 2080–99	47.18	3.77	50.95	0.07	0.00	0.02	0.00	16.75	8.04	10.67	18.71	0.00	15.57
Wailuku	Waiheʻe	A	12.44	1978–2007	61.43	4.26	65.69	0.32	0.08	0.04	0.00	25.37	5.45	13.01	18.45	0.00	22.29
		A		SD RCP8.5 2041–71	56.58	3.98	60.56	0.32	0.08	0.04	0.00	22.76	5.21	12.65	17.86	0.00	20.38
		A		SD RCP8.5 2071–99	53.86	3.80	57.66	0.32	0.08	0.04	0.00	21.10	5.07	12.38	17.45	0.00	19.55
		A		HRCM1 A1B 2080–99	66.42	4.76	71.18	0.32	0.08	0.04	0.00	28.82	5.59	13.12	18.71	0.00	24.09
		B	12.17	1978–2007	61.01	4.26	65.27	0.21	0.07	0.03	0.00	25.21	5.40	12.76	18.16	0.00	22.18
		B		SD RCP8.5 2041–71	56.22	3.98	60.20	0.21	0.07	0.03	0.00	22.63	5.17	12.42	17.59	0.00	20.27
		B		SD RCP8.5 2071–99	53.53	3.80	57.33	0.21	0.07	0.03	0.00	20.98	5.03	12.16	17.20	0.00	19.44
		B		HRCM1 A1B 2080–99	66.03	4.76	70.78	0.21	0.07	0.03	0.00	28.67	5.54	12.89	18.43	0.00	23.97
Wailuku	Waikapū	A	17.31	1978–2007	42.41	4.24	46.66	1.75	0.11	0.52	0.02	14.25	3.71	14.94	18.65	0.00	16.16
		A		SD RCP8.5 2041–71	26.90	2.73	29.63	1.89	0.11	0.52	0.01	7.44	2.70	12.40	15.10	0.00	9.66
		A		SD RCP8.5 2071–99	20.56	2.16	22.72	1.96	0.11	0.52	0.01	4.80	2.26	10.89	13.16	0.00	7.40
		A		HRCM1 A1B 2080–99	40.61	4.32	44.92	1.82	0.11	0.52	0.01	13.80	3.60	14.29	17.90	0.00	15.71
		B	15.06	1978–2007	40.45	4.24	44.69	1.59	0.00	0.44	0.00	14.04	3.66	13.26	16.92	0.00	15.79
		B		SD RCP8.5 2041–71	25.71	2.73	28.44	1.72	0.00	0.44	0.00	7.34	2.67	11.21	13.88	0.00	9.42
		B		SD RCP8.5 2071–99	19.70	2.16	21.86	1.78	0.00	0.44	0.00	4.74	2.24	9.94	12.18	0.00	7.20
		B		HRCM1 A1B 2080–99	39.01	4.32	43.33	1.65	0.00	0.44	0.00	13.63	3.56	12.85	16.41	0.00	15.42

Table 2-6. Mean annual water-budget components from this study for aquifer systems and all of the Island of Hawai‘i.

[See figure 2 for locations of aquifer systems; --, not applicable; Total precipitation, sum of Rain and Fog; Septic, septic-system leachate; Total direct recharge, sum of cesspool seepage and constant recharge from unlined reservoirs and inland water bodies; Runoff, direct runoff; ET, evapotranspiration; Total ET, sum of Canopy evaporation and Actual ET; Groundwater use, kiawe, groundwater use by kiawe land cover; Recharge, groundwater recharge which includes Total direct recharge; ET_o, reference evapotranspiration for grass surface; V, ratio of the mean evaporation rate to mean precipitation rate during saturated conditions; SD, statistical downscaling; HRCM2, Hawai‘i Regional Climate Model version 2 from Zhang and Wang (2017); RCP4.5 and RCP8.5, Representative Concentration Pathway with total radiative forcing of 4.5 and 8.5 Watts per square meter, respectively, by the year 2100 from Intergovernmental Panel on Climate Change (2014); 1978–2007 scenario is 1978–2007 rainfall from Frazier and others (2016), daily rainfall fragments from Longman and others (2019), ET_o from Giambelluca and others (2014), V ratios computed using equation 11 for a 1978–2007 scenario, and 2020 land cover; SD RCP8.5 2041–71 scenario is projected 2041–71 rainfall for a RCP8.5 warming scenario from Elison Timm and others (2015), daily rainfall fragments from Longman and others (2019), ET_o from Giambelluca and others (2014), V ratios computed using equation 11 for a 1978–2007 scenario, and 2020 land cover; SD RCP8.5 2071–99 scenario is projected 2071–99 rainfall for a RCP8.5 warming scenario from Elison Timm and others (2015), daily rainfall fragments from Longman and others (2019), ET_o from Giambelluca and others (2014), V ratios computed using equation 11 for a 1978–2007 scenario, and 2020 land cover; HRCM2 RCP4.5 2080–99 scenario is projected 2080–99 rainfall for a RCP4.5 warming scenario from Zhang and Wang (2017), daily rainfall fragments from Longman and others (2019), ET_o from Giambelluca and others (2014), V ratios computed using equation 11 for a 1978–2007 scenario, and 2020 land cover. Components might not balance because of rounding, direct recharge from cesspools and reservoirs, and changes in root zone soil moisture storage. The source of water for direct recharge might be from external sources, including groundwater and imported streamflow. Some recharge might discharge to streams as base flow. Calculations exclude areas that are unassigned in the Department of Land and Natural Resources aquifer system boundaries from State of Hawai‘i (2014)]

Aquifer sector	Aquifer system	Area, in square miles	Scenario	Water-budget component, in million gallons per day												
				Rain	Fog	Total precipitation	Irrigation	Septic	Total direct recharge	Storm-drain capture	Runoff	Canopy evaporation	Actual ET	Total ET	Ground-water use, kiawe	Recharge
Island of Hawai‘i	--	4,028.82	1978–2007	13,931.86	1,017.77	14,949.63	59.45	6.38	29.54	14.36	2,650.27	1,383.15	4,105.38	5,488.59	0.69	6,885.38
			SD RCP8.5 2041–71	12,410.97	895.47	13,306.45	70.60	6.38	29.54	13.75	2,400.33	1,292.80	3,755.52	5,048.32	0.74	5,945.24
			SD RCP8.5 2071–99	11,657.06	827.90	12,484.97	77.77	6.38	29.54	13.48	2,270.93	1,239.52	3,533.16	4,772.72	0.76	5,536.54
			HRCM2 RCP4.5 2080–99	13,434.17	928.71	14,362.85	63.41	6.38	29.54	14.83	2,569.24	1,353.69	3,946.84	5,300.51	0.69	6,571.27
East Mauna Kea	Hakalau	166.66	1978–2007	1,249.64	149.36	1,399.00	0.63	0.15	0.71	0.24	218.59	123.99	204.31	328.30	0.00	852.39
			SD RCP8.5 2041–71	1,268.34	144.65	1,412.99	0.63	0.15	0.71	0.25	229.54	124.60	202.86	327.45	0.00	856.29
			SD RCP8.5 2071–99	1,271.98	140.88	1,412.86	0.66	0.15	0.71	0.25	233.37	124.44	201.38	325.82	0.00	853.99
			HRCM2 RCP4.5 2080–99	1,222.72	141.17	1,363.89	0.72	0.15	0.71	0.25	214.23	122.04	200.15	322.19	0.00	827.85
East Mauna Kea	Honoka‘a	103.17	1978–2007	314.36	5.20	319.56	1.00	0.06	1.24	0.16	50.78	30.72	143.44	174.15	0.00	96.17
			SD RCP8.5 2041–71	318.45	5.03	323.48	0.87	0.06	1.24	0.18	51.20	31.42	143.00	174.42	0.00	99.28
			SD RCP8.5 2071–99	316.51	4.88	321.40	0.86	0.06	1.24	0.18	50.53	31.41	141.82	173.23	0.00	99.05
			HRCM2 RCP4.5 2080–99	345.39	5.76	351.15	0.85	0.06	1.24	0.19	56.99	33.27	144.48	177.75	0.00	117.78
East Mauna Kea	Onomea	182.77	1978–2007	1,248.88	162.98	1,411.87	1.68	0.03	0.67	0.61	200.00	97.06	220.47	317.54	0.00	895.09
			SD RCP8.5 2041–71	1,233.72	150.42	1,384.14	1.69	0.03	0.67	0.62	207.19	96.16	214.31	310.47	0.00	867.25
			SD RCP8.5 2071–99	1,225.26	143.24	1,368.49	1.79	0.03	0.67	0.62	210.53	95.44	209.73	305.17	0.00	853.66
			HRCM2 RCP4.5 2080–99	1,174.82	147.64	1,322.46	1.94	0.03	0.67	0.64	189.61	94.24	212.64	306.88	0.00	826.99
East Mauna Kea	Pa‘auilo	149.37	1978–2007	541.76	29.06	570.82	0.54	0.01	0.62	0.06	88.49	67.63	183.78	251.41	0.00	231.40
			SD RCP8.5 2041–71	561.42	28.80	590.22	0.48	0.01	0.62	0.07	93.05	69.51	183.33	252.84	0.00	244.74
			SD RCP8.5 2071–99	564.47	28.19	592.66	0.47	0.01	0.62	0.07	93.78	69.81	181.77	251.58	0.00	247.69
			HRCM2 RCP4.5 2080–99	554.59	28.91	583.50	0.52	0.01	0.62	0.07	90.85	69.21	181.05	250.26	0.00	242.86
Hualālai	Keauhou	164.27	1978–2007	287.94	11.81	299.76	6.27	1.25	4.53	0.90	17.71	50.86	142.07	192.93	0.05	101.32
			SD RCP8.5 2041–71	218.59	9.35	227.94	7.75	1.25	4.53	0.50	10.49	44.19	124.68	168.88	0.06	62.78
			SD RCP8.5 2071–99	186.48	8.16	194.63	8.62	1.25	4.53	0.36	7.38	40.59	114.09	154.69	0.06	47.83
			HRCM2 RCP4.5 2080–99	252.61	9.93	262.54	6.87	1.25	4.53	0.95	14.17	46.54	134.96	181.50	0.05	79.60

Table 2-6.—Continued

Aquifer sector	Aquifer system	Area, in square miles	Scenario	Water-budget component, in million gallons per day												
				Rain	Fog	Total precipitation	Irrigation	Septic	Total direct recharge	Storm-drain capture	Runoff	Canopy evapora-tion	Actual ET	Total ET	Ground-water use, kiawe	Recharge
Hualālai	Kīholo	147.20	1978–2007	151.60	5.83	157.43	3.48	0.07	0.04	0.01	5.61	16.10	95.29	111.39	0.13	44.13
			SD RCP8.5 2041–71	109.27	4.45	113.71	3.82	0.07	0.04	0.01	3.14	12.90	77.28	90.18	0.14	24.48
			SD RCP8.5 2071–99	90.19	3.80	93.99	3.99	0.07	0.04	0.00	2.06	11.27	67.19	78.47	0.14	17.73
			HRCM2 RCP4.5 2080–99	121.30	4.49	125.79	3.59	0.07	0.04	0.01	3.79	13.55	83.39	96.94	0.13	28.88
Kīlauea	Hilina	59.32	1978–2007	183.99	1.31	185.30	0.00	0.00	0.00	0.01	23.06	9.39	78.07	87.46	0.00	74.73
			SD RCP8.5 2041–71	130.50	0.95	131.45	0.00	0.00	0.00	0.01	12.88	7.33	63.47	70.80	0.00	47.73
			SD RCP8.5 2071–99	106.03	0.79	106.82	0.00	0.00	0.00	0.00	8.04	6.29	55.23	61.52	0.00	37.22
			HRCM2 RCP4.5 2080–99	174.76	1.23	175.99	0.00	0.00	0.00	0.01	21.35	8.65	71.78	80.43	0.00	74.17
Kīlauea	Kalapana	189.35	1978–2007	948.60	9.75	958.34	1.50	0.18	0.95	0.02	253.21	91.35	281.70	373.05	0.00	334.42
			SD RCP8.5 2041–71	792.29	7.24	799.52	1.78	0.18	0.95	0.02	194.79	83.59	261.67	345.26	0.00	262.10
			SD RCP8.5 2071–99	719.62	6.11	725.74	1.94	0.18	0.95	0.02	167.04	79.51	249.62	329.13	0.00	232.36
			HRCM2 RCP4.5 2080–99	949.19	9.75	958.93	1.81	0.18	0.95	0.03	259.34	89.39	273.44	362.83	0.00	339.41
Kīlauea	Keaīwa	85.14	1978–2007	224.41	0.41	224.82	2.04	0.00	0.00	0.00	22.51	6.19	92.01	98.20	0.01	106.18
			SD RCP8.5 2041–71	155.64	0.31	155.96	2.43	0.00	0.00	0.00	12.30	4.78	75.07	79.85	0.01	66.28
			SD RCP8.5 2071–99	124.62	0.27	124.89	2.61	0.00	0.00	0.00	7.63	4.06	64.98	69.04	0.01	50.89
			HRCM2 RCP4.5 2080–99	237.08	0.36	237.44	2.13	0.00	0.00	0.00	23.32	6.04	85.24	91.27	0.01	125.01
Kīlauea	Pāhoa	223.14	1978–2007	1,601.66	36.30	1,637.96	0.74	1.42	5.94	0.58	478.19	170.02	342.52	512.55	0.00	654.02
			SD RCP8.5 2041–71	1,474.99	28.84	1,503.83	0.82	1.42	5.94	0.53	426.60	164.20	339.03	503.24	0.00	580.91
			SD RCP8.5 2071–99	1,410.68	25.16	1,435.84	0.90	1.42	5.94	0.51	399.81	160.68	335.52	496.20	0.00	546.85
			HRCM2 RCP4.5 2080–99	1,596.79	35.93	1,632.72	0.88	1.42	5.94	0.59	482.40	168.61	337.58	506.19	0.00	651.06
Kohala	Hāwī	47.34	1978–2007	173.53	3.30	176.82	2.67	0.07	1.82	0.06	26.87	27.05	80.02	107.07	0.01	47.17
			SD RCP8.5 2041–71	162.95	3.09	166.04	2.73	0.07	1.82	0.05	24.32	26.11	78.92	105.03	0.01	41.06
			SD RCP8.5 2071–99	156.37	2.96	159.33	2.86	0.07	1.82	0.05	22.86	25.47	77.46	102.93	0.01	38.05
			HRCM2 RCP4.5 2080–99	176.90	3.45	180.34	2.73	0.07	1.82	0.06	27.76	27.30	80.93	108.23	0.01	48.69
Kohala	Māhukona	118.09	1978–2007	235.05	16.53	251.58	0.26	0.47	1.53	0.15	21.30	19.53	138.67	158.21	0.07	74.02
			SD RCP8.5 2041–71	201.24	15.14	216.38	0.28	0.47	1.53	0.11	17.12	18.11	124.48	142.59	0.07	58.82
			SD RCP8.5 2071–99	184.74	14.38	199.12	0.29	0.47	1.53	0.09	14.84	17.34	116.60	133.94	0.07	52.56
			HRCM2 RCP4.5 2080–99	241.36	14.97	256.33	0.26	0.47	1.53	0.15	22.20	19.30	143.43	162.72	0.07	73.35
Kohala	Waimanu	74.89	1978–2007	437.57	43.54	481.11	1.80	0.05	0.23	0.01	193.83	64.90	72.78	137.68	0.01	151.31
			SD RCP8.5 2041–71	439.60	43.54	483.14	1.80	0.05	0.23	0.01	194.55	65.34	73.98	139.32	0.01	151.00
			SD RCP8.5 2071–99	434.35	42.85	477.20	1.80	0.05	0.23	0.01	190.44	64.95	73.95	138.90	0.01	149.57
			HRCM2 RCP4.5 2080–99	446.28	45.58	491.87	1.80	0.05	0.23	0.01	198.69	65.71	73.96	139.67	0.01	155.23

Table 2-6.—Continued

Aquifer sector	Aquifer system	Area, in square miles	Scenario	Water-budget component, in million gallons per day												
				Rain	Fog	Total precipitation	Irrigation	Septic	Total direct recharge	Storm-drain capture	Runoff	Canopy evapora-tion	Actual ET	Total ET	Ground-water use, kiawe	Recharge
Northeast Mauna Loa	Hilo	187.77	1978–2007	1,104.34	142.34	1,246.68	0.64	0.85	4.92	9.82	367.54	86.46	187.58	274.04	0.00	601.18
			SD RCP8.5 2041–71	1,078.23	127.58	1,205.81	0.65	0.85	4.92	9.84	369.04	85.68	181.88	267.56	0.00	565.29
			SD RCP8.5 2071–99	1,060.72	118.46	1,179.18	0.69	0.85	4.92	9.87	367.50	84.78	176.79	261.57	0.00	546.19
			HRCM2 RCP4.5 2080–99	1,019.00	125.56	1,144.56	0.74	0.85	4.92	10.05	336.41	84.01	179.01	263.02	0.00	541.09
Northeast Mauna Loa	Kea‘au	200.45	1978–2007	1,381.90	140.14	1,522.04	3.83	0.63	2.38	1.27	311.15	128.29	239.11	367.40	0.00	848.47
			SD RCP8.5 2041–71	1,348.28	125.22	1,473.50	4.03	0.63	2.38	1.26	306.96	126.84	235.70	362.54	0.00	809.20
			SD RCP8.5 2071–99	1,327.19	115.84	1,443.04	4.30	0.63	2.38	1.25	303.17	125.55	232.67	358.22	0.00	787.12
			HRCM2 RCP4.5 2080–99	1,321.33	123.59	1,444.92	4.40	0.63	2.38	1.35	292.47	125.13	234.11	359.24	0.00	798.68
Northwest Mauna Loa	‘Anaeho‘omalu	318.78	1978–2007	312.12	8.38	320.49	5.01	0.11	0.12	0.04	10.80	11.90	186.25	198.16	0.23	116.94
			SD RCP8.5 2041–71	240.37	6.43	246.80	5.21	0.11	0.12	0.02	7.00	9.36	154.89	164.25	0.24	81.22
			SD RCP8.5 2071–99	205.99	5.49	211.48	5.30	0.11	0.12	0.01	5.15	8.11	136.60	144.71	0.24	67.41
			HRCM2 RCP4.5 2080–99	241.93	6.35	248.27	5.03	0.11	0.12	0.04	6.97	9.74	158.48	168.23	0.23	78.52
Southeast Mauna Loa	Ka Lae	133.60	1978–2007	294.00	8.28	302.28	0.51	0.02	0.42	0.00	22.67	36.86	131.35	168.21	0.04	112.32
			SD RCP8.5 2041–71	198.85	5.51	204.37	0.79	0.02	0.42	0.00	10.64	27.59	107.15	134.73	0.04	60.25
			SD RCP8.5 2071–99	153.38	4.22	157.60	0.98	0.02	0.42	0.00	5.66	22.60	91.42	114.02	0.04	39.40
			HRCM2 RCP4.5 2080–99	328.27	7.56	335.83	0.48	0.02	0.42	0.00	25.92	41.54	138.67	180.21	0.04	130.58
Southeast Mauna Loa	Kapāpala	76.41	1978–2007	196.91	13.29	210.19	0.00	0.00	0.00	0.00	17.35	12.66	79.07	91.74	0.00	101.00
			SD RCP8.5 2041–71	149.03	10.06	159.09	0.00	0.00	0.00	0.00	10.79	10.56	70.14	80.71	0.00	67.50
			SD RCP8.5 2071–99	126.37	8.55	134.92	0.00	0.00	0.00	0.00	7.63	9.38	63.48	72.86	0.00	54.33
			HRCM2 RCP4.5 2080–99	168.40	11.37	179.78	0.00	0.00	0.00	0.00	13.15	11.19	71.77	82.96	0.00	83.57
Southeast Mauna Loa	Nā‘ālehu	347.66	1978–2007	1,121.44	79.48	1,200.92	11.97	0.10	0.38	0.05	122.89	111.62	334.88	446.51	0.01	643.13
			SD RCP8.5 2041–71	787.86	56.19	844.05	15.67	0.10	0.38	0.03	64.29	91.40	299.58	390.97	0.02	404.17
			SD RCP8.5 2071–99	636.21	45.52	681.73	17.66	0.10	0.38	0.02	39.75	80.44	275.04	355.48	0.02	303.90
			HRCM2 RCP4.5 2080–99	1,047.44	69.48	1,116.91	13.36	0.10	0.38	0.07	108.26	105.71	322.30	428.01	0.01	593.67
Southeast Mauna Loa	‘Ōla‘a	132.17	1978–2007	676.35	95.82	772.18	0.02	0.11	0.43	0.03	133.35	75.11	106.79	181.90	0.00	456.94
			SD RCP8.5 2041–71	607.39	80.63	688.02	0.03	0.11	0.43	0.03	117.27	71.07	101.33	172.40	0.00	398.38
			SD RCP8.5 2071–99	571.88	72.21	644.08	0.04	0.11	0.43	0.02	108.77	68.55	97.25	165.80	0.00	369.56
			HRCM2 RCP4.5 2080–99	617.04	87.05	704.09	0.03	0.11	0.43	0.03	117.89	71.33	101.01	172.33	0.00	413.90
Southwest Mauna Loa	Ka‘apuna	223.31	1978–2007	379.03	18.64	397.67	1.58	0.08	0.43	0.00	21.51	60.30	179.16	239.47	0.00	138.67
			SD RCP8.5 2041–71	276.20	13.71	289.91	2.29	0.08	0.43	0.00	11.36	50.25	154.36	204.61	0.01	76.65
			SD RCP8.5 2071–99	228.49	11.40	239.90	2.72	0.08	0.43	0.00	7.16	44.69	137.54	182.24	0.01	53.66
			HRCM2 RCP4.5 2080–99	368.68	17.69	386.37	1.38	0.08	0.43	0.00	21.45	60.43	170.21	230.64	0.00	136.05
Southwest Mauna Loa	Kealakekua	230.44	1978–2007	323.79	21.62	345.42	4.62	0.26	1.34	0.21	17.79	49.21	177.71	226.92	0.01	106.49
			SD RCP8.5 2041–71	253.93	17.18	271.11	6.19	0.26	1.34	0.14	10.74	42.72	158.62	201.34	0.01	66.49
			SD RCP8.5 2071–99	220.47	15.00	235.47	7.18	0.26	1.34	0.11	7.58	39.12	146.32	185.44	0.02	50.96
			HRCM2 RCP4.5 2080–99	300.97	19.36	320.32	4.58	0.26	1.34	0.20	17.06	46.84	166.97	213.81	0.01	95.20

Table 2-6.—Continued

Aquifer sector	Aquifer system	Area, in square miles	Scenario	Water-budget component, in million gallons per day											
				Rain	Fog	Total precipitation	Irrigation	Septic	Total direct recharge	Storm-drain capture	Runoff	Canopy evapora-tion	Actual ET	Total ET	Ground-water use, kiawe	Recharge
Southwest Mauna Loa	Manukā	167.96	1978–2007	248.40	4.57	252.97	5.80	0.40	0.48	0.01	12.63	28.91	146.12	175.03	0.06	72.14
			SD RCP8.5 2041–71	177.56	3.24	180.81	7.62	0.40	0.48	0.00	6.75	23.37	120.64	144.01	0.06	38.73
			SD RCP8.5 2071–99	139.59	2.56	142.15	8.99	0.40	0.48	0.00	3.82	19.99	103.53	123.53	0.07	24.90
			HRCM2 RCP4.5 2080–99	266.33	4.06	270.39	6.47	0.40	0.48	0.01	14.42	27.91	143.60	171.51	0.06	91.96
West Mauna Kea	Waimea	299.58	1978–2007	294.59	9.83	304.42	2.86	0.06	0.36	0.12	12.44	7.04	262.23	269.27	0.06	25.75
			SD RCP8.5 2041–71	226.27	7.91	234.18	3.04	0.06	0.36	0.07	8.32	5.72	209.15	214.87	0.06	14.64
			SD RCP8.5 2071–99	195.47	6.98	202.45	3.12	0.06	0.36	0.04	6.43	5.05	183.18	188.23	0.06	11.66
			HRCM2 RCP4.5 2080–99	260.99	7.47	268.46	2.84	0.06	0.36	0.12	10.54	6.01	237.68	243.69	0.06	17.17

References Cited

Elison Timm, O., Giambelluca, T.W., and Diaz, H.F., 2015, Statistical downscaling of rainfall changes in Hawai‘i based on the CMIP5 global model projections: Journal of Geophysical Research, Atmospheres, v. 120, no. 1, p. 92–112, https://doi.org/10.1002/2014JD022059.

Frazier, A.G., Giambelluca, T.W., Diaz, H.F., and Needham, H.L., 2016, Comparison of geostatistical approaches to spatially interpolate month-year rainfall for the Hawaiian Islands: International Journal of Climatology, v. 36, no. 3, p. 1459–1470, https://doi.org/10.1002/joc.4437.

Giambelluca, T.W., Shuai, X., Barnes, M.L., Alliss, R.J., Longman, R.J., Miura, T., Chen, Q., Frazier, A.G., Mudd, R.G., Cuo, L., and Businger, A.D., 2014, Evapotranspiration of Hawai‘i: University of Hawai‘i at Mānoa, accessed on November 12, 2021, at http://evapotranspiration.geography.hawaii.edu/.

Intergovernmental Panel on Climate Change, 2014, Climate change 2014—Synthesis report: Contribution of Working Groups I, II and III to the Fifth Assessment Report of the Intergovernmental Panel on Climate Change (Core Writing Team, Pachauri, R.K., and Meyer, L.A., eds.), IPCC, Geneva, Switzerland, 151 p.

Longman, R.J., Frazier, A.G., Newman, A.J., Giambelluca, T.W., Schanzenbach, D., Kagawa-Viviani, A., Needham, H., Arnold, J.R., and Clark, M.P., 2019, High-resolution gridded daily rainfall and temperature for the Hawaiian Islands (1990–2014): Journal of Hydrometeorology, v. 20, no. 3, p. 489–508, https://doi.org/10.1175/JHM-D-18-0112.1.

State of Hawai‘i, 2014, Aquifer sector and aquifer system area boundaries for the Islands of Hawaii, Kauai, Lanai, Maui, Molokai, and Oahu: Department of Land and Natural Resources, accessed August 6, 2020, at http://planning.hawaii.gov/gis/download-gis-data/.

Zhang, C., Wang, Y., Hamilton, K., and Lauer, A., 2016b, Dynamical downscaling of the climate for the Hawaiian Islands Part II—Projection for the late twenty-first century: Journal of Climate, v. 29, no. 23, p. 8333–8354, https://doi.org/10.1175/JCLI-D-16-0038.1.

Zhang, C., and Wang, Y., 2017, Projected future changes of tropical cyclone activity over the western north and south Pacific in a 20-km-mesh regional climate model: Journal of Climate, v. 30, no. 15, p. 5923–5941, https://doi.org/10.1175/JCLI-D-16-0597.1.

Appendix 3. Development of Daily Rainfall Fragment Sets, Forest-Canopy Evaporation Parameters, and Reference-Evapotranspiration Rates for HRCM1 and HRCM2 Scenarios

A separate set of model inputs to the Water-budget Accounting for Tropical Regions Model (WATRMod; Oki, 2022) was developed for the Hawai‘i Regional Climate Model version 1 (HRCM1) A1B 2080–99 scenario for Maui (Zhang and others, 2016a,b), the HRCM version 2 (HRCM2) Representative Concentration Pathway (RCP) 4.5 2080–99 scenario for Kaua‘i, Lāna‘i, and the Island of Hawai‘i (Zhang and Wang, 2017), and the HRCM2 RCP8.5 2080–99 scenario for O‘ahu and Moloka‘i (Zhang and Wang, 2017). The HRCM1 and HRCM2 retrospective simulations during 1990–2009 (Zhang and others, 2016a,b; Zhang and Wang, 2017), hereinafter HRCM1 and HRCM2 1990–2009 simulations, were used to help develop the set of model inputs for WATRMod. Simulations using WATRMod were then used to assess the effect of projected changes in daily rainfall frequency, and estimated changes in forest-canopy evaporation and reference evapotranspiration (ET) on water-budget results. This appendix describes the methods used to develop these model inputs for the selected HRCM1 and HRCM2 future scenarios. The selection of the HRCM1 and HRCM2 future-climate scenarios is described in the “Selected Future-Climate Scenarios” section of the associated report. The effect of changes in these model inputs on water-budget results is examined in the “Effects of Selected Climate Inputs on Estimated Recharge” section of the associated report.

Daily Rainfall Fragments

Separate datasets of daily rainfall fragments were developed for WATRMod using the distributions of simulated daily rainfall from the HRCM1 and HRCM2 retrospective simulations and future-climate scenarios (Zhang and others, 2016a,b; Zhang and Wang, 2017). The datasets of daily rainfall fragments were used to assess the effects of potential changes in daily rainfall frequency on estimated recharge. For each grid cell in the HRCM1 and HRCM2 1990–2009 simulations and future-climate scenarios, daily rainfall fragments were computed by dividing each simulated daily rainfall value for a particular month by the total rainfall simulated at the grid cell for that month. Fragment sets were then grouped by month of the year and by grid cell for each island. The 20-year simulation period used by the HRCM1 and HRCM2 1990–2009 simulations and future-climate scenarios produced a total of 240 fragment sets for each grid cell (20 fragment sets for each calendar month), except for grid cells where the HRCM1 and HRCM2 1990–2009 simulations and future-climate scenarios resulted in no rainfall over an entire calendar month. In cases with no monthly rainfall, the number of fragment sets at a particular grid cell was reduced by the number of months in the HRCM1 and HRCM2 1990–2009 simulations and future-climate scenarios with no rainfall. The daily rainfall pattern for all model subareas within each grid cell of the HRCM1 or HRCM2 1990–2009 simulations or future-climate scenarios was assumed to be the same. A description of how fragment sets are used to disaggregate monthly rainfall in the WATRMod simulations is presented in the “1978–2007 Daily Rainfall” section of the associated report.

Forest-Canopy Evaporation

In WATRMod, the method derived from the Gash model (Gash and others, 1995) was used to estimate forest-canopy evaporation. The Gash model requires an estimate of the mean evaporation rate to mean precipitation rate ratio during saturated conditions, V. A common bias-correction approach is the change-factor method (Hay and others, 2000), which applies relative change factors, derived by comparing simulated reference climate and projected future-climate data to a reference climate series. The change-factor approach was used in this study to develop the spatial distribution of V by adjusting the ratio of mean annual wind speed to mean annual rainfall (w, equation 11 of main report) with relative change factors for mean annual wind speed and mean annual rainfall determined from the HRCM1 and HRCM2 1990–2009 simulations and future-climate scenarios (Zhang and others, 2016a, b; Zhang and Wang, 2017). Relative change factors for mean annual wind speed and mean annual rainfall were calculated from the differences in annual means between the retrospective simulations and future scenarios. Mean annual wind speed for the HRCM1 and HRCM2 1990–2009 simulations and future-climate scenarios was computed using simulated hourly zonal and meridional wind speed at a height of 10 meters (m; Zhang and others, 2016a, b; Zhang and Wang, 2017).

An adjusted mean annual wind speed to mean annual rainfall ratio, w_{Δ}, was computed for each subarea and input to equation 11 to determine the spatial distribution of V for the HRCM1 and HRCM2 future-climate scenarios. The adjusted ratio at each subarea was determined from the equation:

$$w_{\Delta} = (F_{\bar{u}_{ann}} \times \bar{u}_{ann}) \div (F_{\bar{P}_{ann}} \times \bar{P}_{ann}) \qquad (3\text{-}1)$$

where

w_{Δ} is the adjusted ratio of mean annual wind speed to mean annual rainfall for the HRCM1 or HRCM2 future-climate scenario ([meter/second]/inch),

$F_{\bar{u}_{ann}}$ is the relative change factor for mean annual wind speed (dimensionless),

$\bar{u}_{ann}$ is the mean annual wind speed from AWS Truewind (2004; meters per second [m/s]),

$F_{\bar{P}_{ann}}$ is the relative change factor for mean annual rainfall (dimensionless),

$\bar{P}_{ann}$ is the mean annual rainfall during 1978–2007 from Giambelluca and others (2013; inches).

The relative change factors, $F_{\bar{u}\,ann}$ and $F_{\bar{P}\,ann}$, for each subarea were calculated for the retrospective simulations and the future-climate scenarios as follows:

$$F_{\bar{u}\,ann} = \frac{u_{fut}}{\bar{u}_{ret}}$$
$$F_{\bar{P}\,ann} = \frac{P_{fut}}{\bar{P}_{ret}} \quad (3\text{-}2)$$

where

$\bar{u}_{fut}$ is the mean annual wind speed during the HRCM1 or HRCM2 future-climate scenario (m/s),

$\bar{u}_{ret}$ is the mean annual wind speed during the HRCM1 or HRCM2 1990–2009 simulation (m/s),

$\bar{P}_{fut}$ is the mean annual rainfall during the HRCM1 or HRCM2 future-climate scenario (millimeters [mm]), and

$\bar{P}_{ret}$ is the mean annual rainfall during the HRCM1 or HRCM2 1990–2009 simulation (mm).

Projected anomalies in V for the selected HRCM1 and HRCM2 future-climate scenarios range from −0.14 to 0.12 (fig. 3-1). In the Gash model, increases in V enhance the potential for greater canopy-evaporation rates relative to the 1978–2007 scenario. Greater canopy-evaporation rates can affect WATRMod computations for forested subareas by increasing evapotranspiration (ET), reducing net precipitation, and reducing the amount of water available for groundwater recharge. Conversely, decreases in V can affect model computations for forested subareas by decreasing canopy evaporation, increasing net precipitation, and increasing the amount water available for groundwater recharge.

Reference Evapotranspiration

Recent water-budget studies on Maui (Brewington and others, 2019; Mair and others, 2019) used gridded hourly meteorological data from the HRCM1 1990–2009 simulation and HRCM1 A1B 2080–99 scenario (Zhang and others, 2016a, b) to estimate changes in reference-ET rates using the "reduced form" Food and Agriculture Organization of the United Nations Penman-Monteith (FAO-PM) equation described by Allen and others (2006) and assess the effect of projected warming changes in reference-evapotranspiration rates on groundwater recharge. However, the FAO-PM equation by Allen and others (2006) does not consider the potential effects of projected increases in atmospheric carbon dioxide (CO_2) concentrations on reference-ET rates. Increases in CO_2 concentrations might act to suppress ET by decreasing plant leaf stomatal conductance, which can decrease transpiration and increase soil moisture available for recharge. Studies of the combined effect of increased temperature and atmospheric CO_2 concentrations on reference-ET rates using General Circulation Model projections indicate that increases in reference ET owing to projected warming might be partially or completely offset by decreases in reference ET owing to projected increases in atmospheric CO_2 concentrations (Islam and others, 2012; Yang and others, 2019). However, the extent to which increased soil moisture could enhance recharge estimates on Maui owing to increased CO_2 concentrations was not evaluated by Brewington and others (2019) or Mair and others (2019).

Yang and others (2019) developed a modified form of the FAO-PM formula to account for the effect of changing CO_2 concentrations on reference-ET rates. Their modified form of the FAO-PM formula was developed to estimate reference-ET rates for increases in CO_2 concentrations of up to 600 parts per million (ppm) above the reference CO_2 concentration of 300 ppm during the 1861–1960 baseline period used by phase 5 of the Coupled Model Intercomparison Project (CMIP5) model projections (Y. Yang, Tsinghua University, written commun. 2020). Yang and others (2019) found that increases in ET in non-water-limited conditions owing to warming-induced increases in the vapor pressure deficit (or evaporative demand) in 16 CMIP5 models were almost entirely offset by a decrease in ET caused by CO_2-induced increases in surface resistance, which describes the resistance of vapor flow through plant leaf stomata, total leaf area, and the soil surface (Allen and others, 1998).

In this study, the change-factor approach was used for the HRCM1 and HRCM2 future-climate scenarios to develop estimates of future mean monthly reference ET by adjusting maps of mean monthly grass reference-surface ET (Giambelluca and others, 2014). Relative change factors for monthly reference ET were calculated from differences in monthly means of reference ET that were computed using hourly gridded meteorological output from the HRCM1 and HRCM2 retrospective simulations and future-climate scenarios. Monthly means of reference ET were determined from hourly reference ET for each of two hypothetical conditions: (1) no increase in atmospheric CO_2 concentrations, and (2) increased atmospheric CO_2 concentrations reflective of the selected HRCM1 and HRCM2 future-climate scenarios. The "reduced form" FAO-PM equation (Allen and others, 2006) was used to estimate reference-ET rates for the first hypothetical condition assuming no increase in atmospheric CO_2 concentrations. Next, a modified form of the FAO-PM equation (Yang and others, 2019) was used to estimate reference-ET rates for the second condition assuming increased CO_2 concentrations reflective of the selected HRCM1 and HRCM2 future-climate scenarios. The HRCM output was used as input to both forms of the FAO-PM formulas to compute hourly reference ET for each model subarea.

Condition 1—Reference ET Adjusted for Projected Warming

Hourly reference ET for each subarea was determined from the "reduced form" FAO-PM equation (Allen and others, 2006):

$$ET_{series} = \frac{0.408\Delta(R_n - G) + \gamma \frac{37}{T_2 + 273} u_2 (e_s - e_a)}{\Delta + \gamma(1 + C_d u_2)} \quad (3\text{-}3)$$

where

ET_{series} is the hourly grass reference ET [mm/hour] and series denotes either the 1990–2009 simulation or future-climate scenario,
Δ is the slope of the saturation vapor pressure versus temperature curve in kilopascals/temperature in degrees Celsius (kPa/°C),
R_n is the net radiation at the surface in megajoule per square meter per hour (MJ/m^{2}h),
G is the sensible heat flux from the surface to the soil (MJ/m^{2}h),
γ is the psychometric constant (kPa/°C),
T_2 is the air temperature at 2-m height (°C),
u_2 is the wind speed at 2-m height (m/s),
$e_s - e_a$ is the saturation vapor pressure deficit (kPa), and
C_d is 0.24 when $R_n > 0$, 0.96 when $R_n \leq 0$ (s/m).

Net radiation, R_n, was calculated from HRCM data describing incoming and outgoing short- and long-wave radiation from the following equation (Brutsaert, 2005):

$$R_n = R_{ns} + \varepsilon_s R_{dlw} - R_{ulw} \quad (3\text{-}4)$$

where

R_{ns} is the net short-wave radiation (MJ/m^{2}h),
ε_s is the surface emissivity (unitless),
R_{dlw} is the downward long-wave radiation (MJ/m^{2}h), and
R_{ulw} is the upward long-wave radiation (MJ/m^{2}h).

Net short-wave radiation, R_{ns}, and downward long-wave radiation, R_{dlw}, were provided by HRCM1 and HRCM2 data. Upward long-wave radiation, R_{ulw}, was calculated from HRCM1 and HRCM2 data assuming that the ground, canopy, or water surface under consideration was equivalent to an infinitely deep gray body of uniform temperature, with constant wavelengths for transmission, absorption, and reflection, and surface emissivity, ε_s (Brutsaert, 2005). In a separate study, a value of one for surface emissivity, ε_s, was used to estimate the spatial patterns of ET for the main Hawaiian Islands (Giambelluca and others, 2014). For this study, the value for surface emissivity, ε_s, also was assumed to be one ($\varepsilon_s = 1$). Upward long-wave radiation, R_{ulw}, was calculated as follows:

$$R_{ulw} = \varepsilon_s \sigma T_s^4 \quad (3\text{-}5)$$

where

σ is the Stefan-Boltzmann constant (MJ/m^{2}h) and
T_s is the surface skin temperature (Kelvin).

Input values for sensible heat flux, G, and air temperature at 2-m height, T_2, in equation 3-3, and surface skin temperature, T_s, in equation 3-5 were provided by HRCM1 and HRCM2 data. The remaining input parameters in equation 3-3 were computed using procedures described in Allen and others (1998) for estimating hourly reference ET and data from HRCM output.

The mean monthly reference ET at each model subarea for the HRCM1 and HRCM2 future-climate scenarios was determined from the equation:

$$ET_{\delta}(i) = F_{\overline{ET}}(i) \times ET_{ref}(i), \quad (3\text{-}6)$$

where

$ET_{\delta}(i)$ is the mean monthly reference ET for month i during the HRCM1 or HRCM2 future-climate scenario (inches),
$F_{\overline{ET}}(i)$ is the relative change factor for mean monthly reference ET for month i (dimensionless),
$ET_{ref}(i)$ is the mean monthly grass reference-surface ET for month i from Giambelluca and others (2014; inches), and
i is the variable designating month in a calendar year.

The relative change factor, $F_{\overline{ET}}(i)$, for mean monthly reference ET at each model subarea was calculated as follows:

$$F_{\overline{ET}}(i) = \frac{\overline{ET}_{fut}(i)}{\overline{ET}_{pre}(i)}, \quad (3\text{-}7)$$

where

$\overline{ET}_{fut}(i)$ is the mean monthly reference ET for month i during the HRCM1 or HRCM2 future scenario (mm) and
$\overline{ET}_{pre}(i)$ is the mean monthly reference ET for month i during the HRCM1 or HRCM2 1990–2009 simulation (mm).

The estimates of mean monthly reference ET for the HRCM1 and HRCM2 1990–2009 simulations and the selected HRCM1 and HRCM2 future scenarios were transformed into mean annual reference-ET anomalies for each island. First, mean annual reference-ET distribution grids were computed for the HRCM1 and HRCM2 1990–2009 simulations and the HRCM1 and HRCM2

future scenarios. Next, absolute mean annual reference-ET anomalies, expressed in mm, were computed as the difference between the HRCM1 and HRCM2 future scenario means and the HRCM1 and HRCM2 1990–2009 simulation means, respectively, for each grid cell. The relative mean annual reference-ET anomalies, expressed as a fraction of the 1990–2009 simulation mean, were then computed by dividing the absolute mean reference-ET anomalies by the HRCM1 and HRCM2 1990–2009 simulation means for each grid cell. Estimated changes in mean annual reference ET for the selected HRCM1 and HRCM2 scenarios range from increases of up to 23 percent across parts of Moloka‘i, Maui, and the Island of Hawai‘i to decreases of more than 12 percent across a small part of eastern Maui (fig. 3-2).

Condition 2—Reference Evapotranspiration Adjusted for Projected Warming and Increased Atmospheric CO_2 Concentrations

Atmospheric CO_2 concentrations measured at the Mauna Loa Observatory, Hawai‘i, have steadily increased from about 315 ppm in 1958 to about 415 ppm in 2021 (Keeling and Keeling, 2017). Further increases in atmospheric CO_2 concentrations are projected for each of the HRCM1 and HRCM2 future scenarios considered in this study (Intergovernmental Panel on Climate Change, 2007; 2014). For the A1B emission scenario, CO_2 concentrations are projected to steadily increase to about 700 ppm by 2100 (Intergovernmental Panel on Climate Change, 2007; 2021a), which represents a net increase of 285 to 385 ppm above CO_2 concentrations measured during 1958–2021. For the RCP4.5 warming scenario, CO_2 concentrations are projected to increase and stabilize at about 600 ppm by 2080 (Clarke and others, 2014; Intergovernmental Panel on Climate Change, 2021b), which represents a net increase of about 185 to 285 ppm above CO_2 concentrations measured during 1958–2021. For the RCP8.5 warming scenario, CO_2 concentrations are projected to steadily increase and reach more than 1,200 ppm by 2100, which represents a net increase of about 785 to 885 ppm above CO_2 concentrations measured during 1958–2021.

Zhang and others (2016a, b) gradually increased the prescribed atmospheric CO_2 concentrations from 360 to 390 ppm during the HRCM1 1990–2009 simulation and from 660 to 720 ppm during the HRCM1 A1B 2080–99 scenario. Their adjustments to the HRCM1 A1B 2080–99 scenario represent a net increase of 300 to 330 ppm in atmospheric CO_2 concentrations above the HRCM1 1990–2009 simulation values. Zhang and Wang (2017) gradually increased the prescribed atmospheric CO_2 concentrations from 354 to 387 ppm during the HRCM2 1990–2009 simulation, from 531 to 538 ppm during the HRCM2 RCP4.5 2080–99 scenario, and from 758 to 927 ppm during the HRCM2 RCP8.5 2080–99 scenario (C. Zhang, I.M. Systems Group, Inc., written commun. 2021). Their adjustments represent net increases in atmospheric CO_2 concentrations of 151 to 177 ppm for the HRCM2 RCP4.5 2080–99 scenario and 404 to 540 ppm for the HRCM2 RCP8.5 2080–99 scenario.

For this analysis, the mean value from the range of net increases in atmospheric CO_2 concentrations prescribed for each HRCM1 and HRCM2 future scenario was used to assess the effect on reference-ET rates. The mean increase in atmospheric CO_2 concentrations was computed as 315 ppm for the HRCM1 A1B 2080–99 scenario, 164 ppm for the HRCM2 RCP4.5 2080–99 scenario, and 472 ppm for the HRCM2 RCP8.5 2080–99 scenario. An adjusted atmospheric CO_2 concentration, $[CO_2]_{fut}$, was then computed as the sum of the reference atmospheric CO_2 concentration from Yang and others (2019), $[CO_2]_{base}$ = 300 ppm, and the mean increase in atmospheric CO_2 concentrations for each HRCM1 and HRCM2 future scenario. The value for $[CO_2]_{fut}$ was computed as 615 ppm for the HRCM1 A1B 2080–99 scenario, 464 ppm for the HRCM2 RCP4.5 2080–99 scenario, and 772 ppm for the HRCM2 RCP8.5 2080–99 scenario.

Hourly reference ET for each subarea was then determined from the following equation (modified from equation 14 in Yang and others [2019]):

$$ET_{series} = \frac{0.408\Delta(R_n - G) + \gamma \frac{37}{T_2 + 273} u_2 (e_s - e_a)}{\Delta + \gamma(1 + u_2(C_d + 2.4 \times 10^{-4}([CO_2]_{fut} - [CO_2]_{base})))} \quad (3\text{-}8)$$

where

$[CO_2]_{fut}$ is the adjusted atmospheric CO_2 concentration [ppm] and computed as 615 ppm for the HRCM1 A1B 2080–99 scenario, 464 ppm for the HRCM2 RCP4.5 2080–99 scenarios, and 772 ppm for the RCP8.5 2080–99 scenario, and

$[CO_2]_{base}$ is the reference atmospheric CO_2 concentration (300 ppm; Yang and others, 2019).

All other input variables in equation 3-8 were computed as described earlier. The mean monthly reference ET at each model subarea for the HRCM1 and HRCM2 future scenarios was determined using equations 3-6 and 3-7 and then used to determine relative change factors. Estimates of mean monthly reference ET for the HRCM1 and HRCM2 1990–2009 simulations and HRCM1 and HRCM2 future scenarios were transformed into mean annual reference-ET anomalies for each island as described earlier. Estimated changes in mean annual reference ET for the selected HRCM1 and HRCM2 scenarios range from increases of up to 23 percent across high-altitude areas of Maui and the Island of Hawai‘i to decreases of up to 18 percent across parts of western and eastern Maui (fig. 3-3).

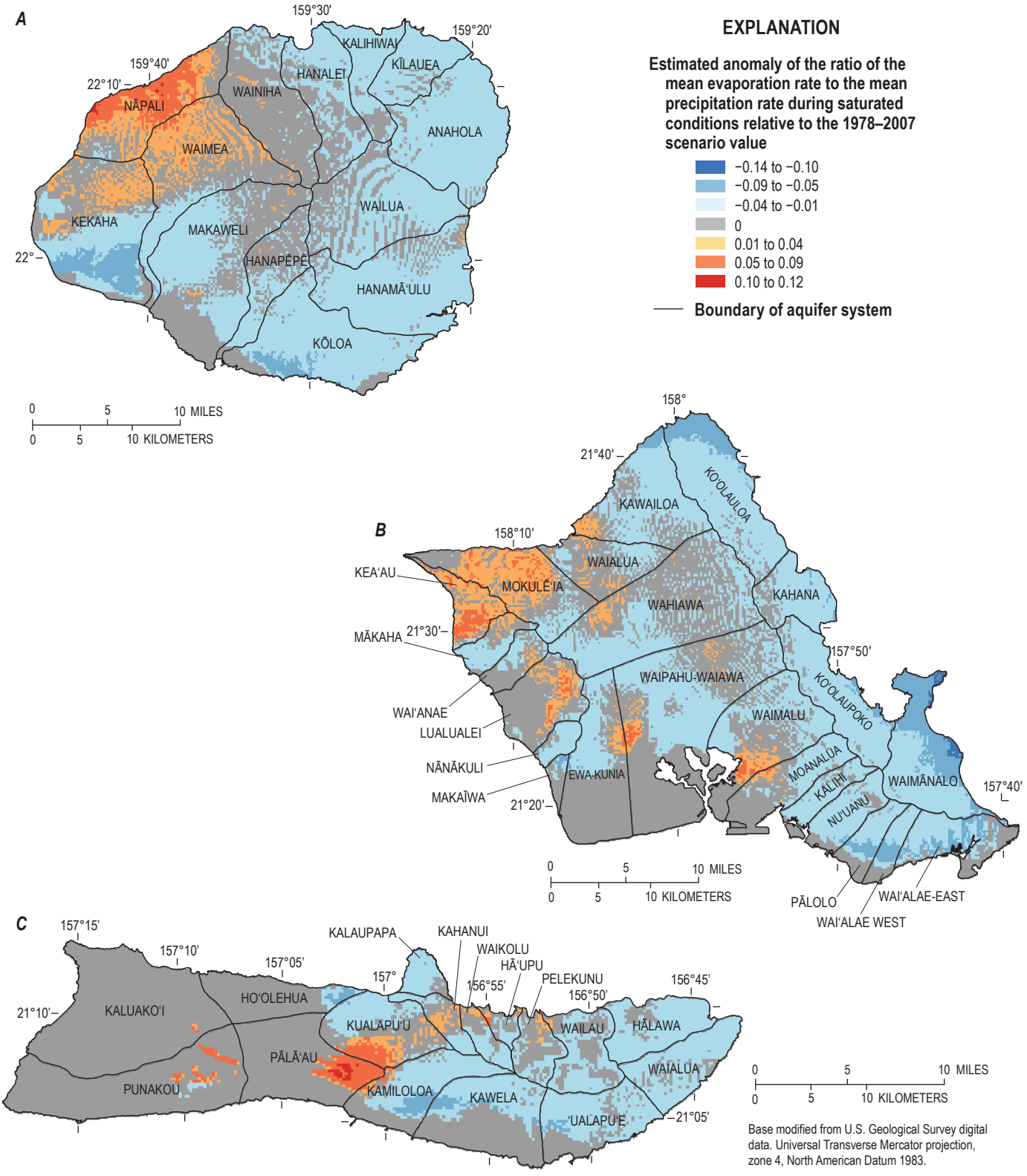
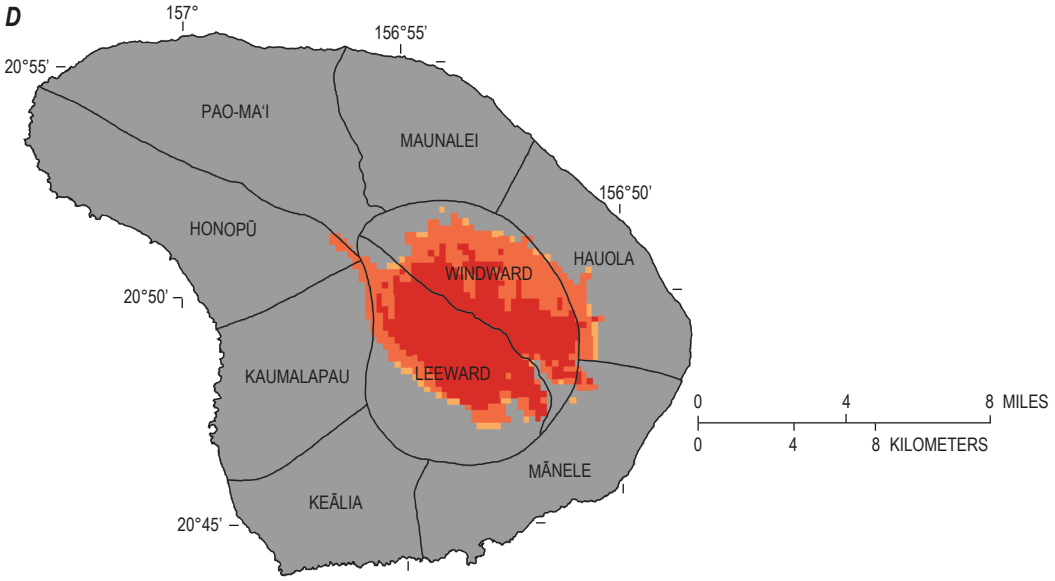
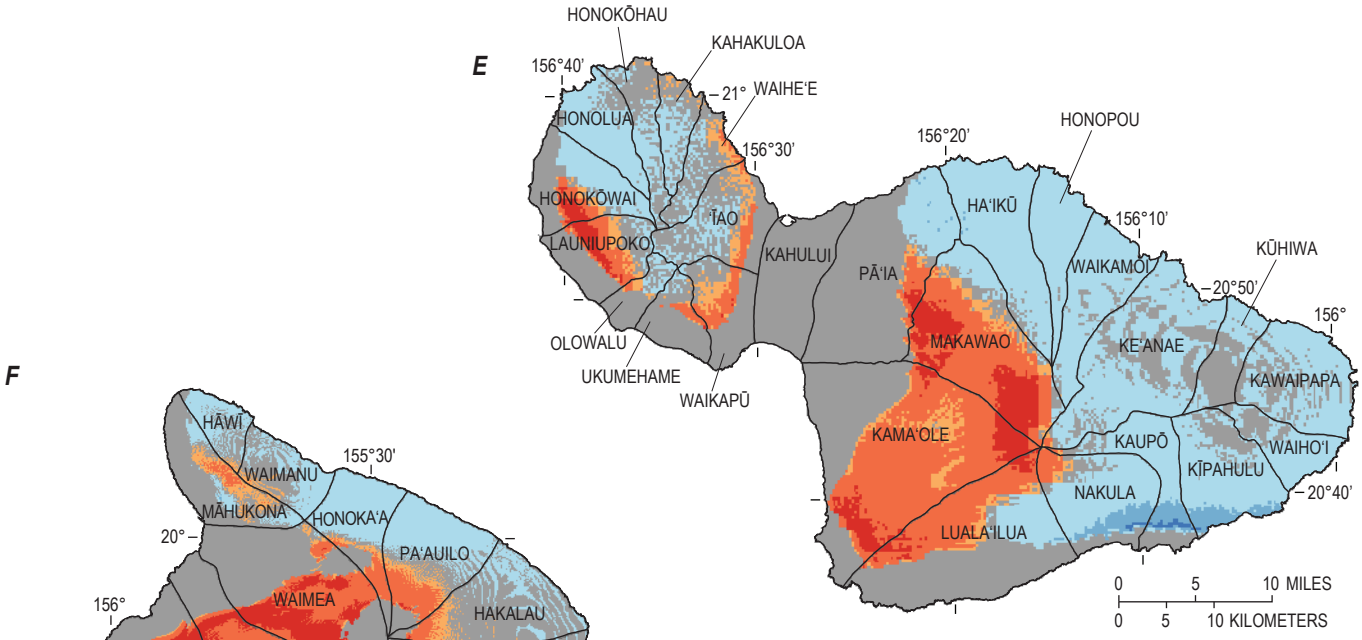
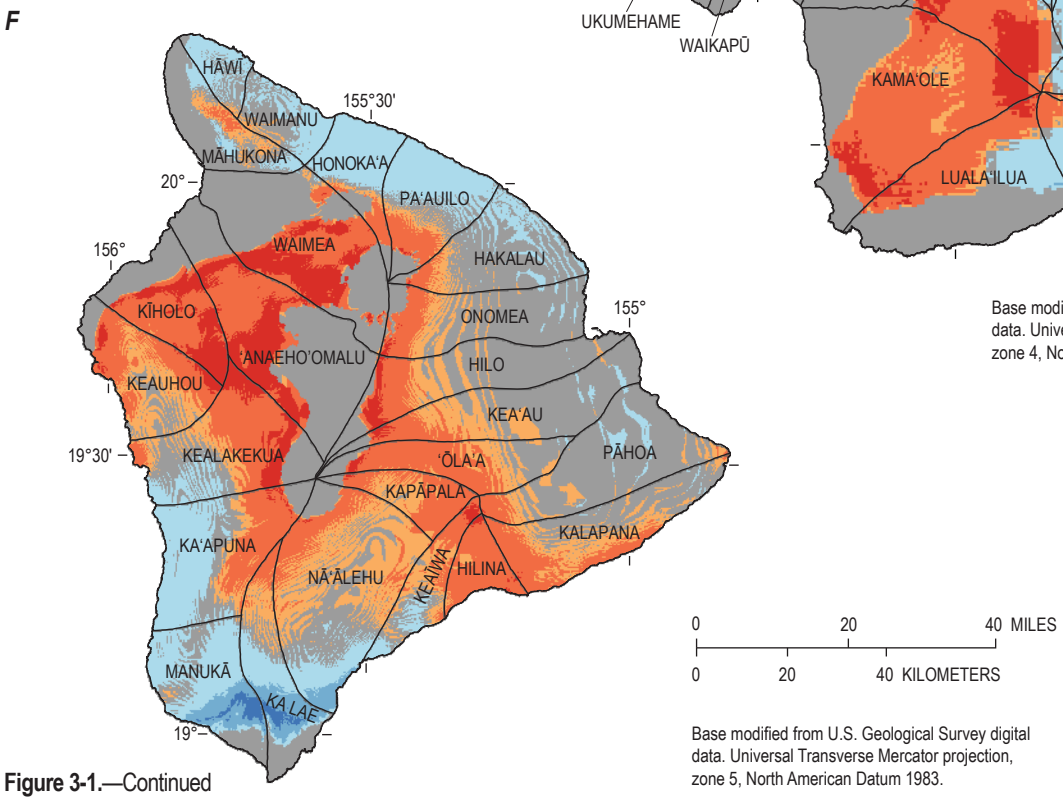


Figure 3-1. Estimated anomaly map of the ratio of the mean evaporation rate to the mean precipitation rate during saturated conditions for selected future-climate scenarios for Kaua'i, O'ahu, Moloka'i, Lāna'i, Maui, and the Island of Hawai'i. (*A*) HRCM2 RCP4.5 2080–99 scenario for Kaua'i, (*B*) HRCM2 RCP8.5 2080–99 scenario for O'ahu, (*C*) HRCM2 RCP8.5 2080–99 scenario for Moloka'i, (*D*) HRCM2 RCP4.5 2080–99 scenario for Lāna'i, (*E*) HRCM1 A1B 2080–99 scenario for Maui, and (*F*) HRCM2 RCP4.5 2080–99 scenario for the Island of Hawai'i. Estimated changes are relative to a 1978–2007 scenario. Aquifer system boundaries from State of Hawai'i (2014).





Base modified from U.S. Geological Survey digital data. Universal Transverse Mercator projection, zone 4, North American Datum 1983.



Base modified from U.S. Geological Survey digital data. Universal Transverse Mercator projection, zone 5, North American Datum 1983.

Figure 3-1.—Continued

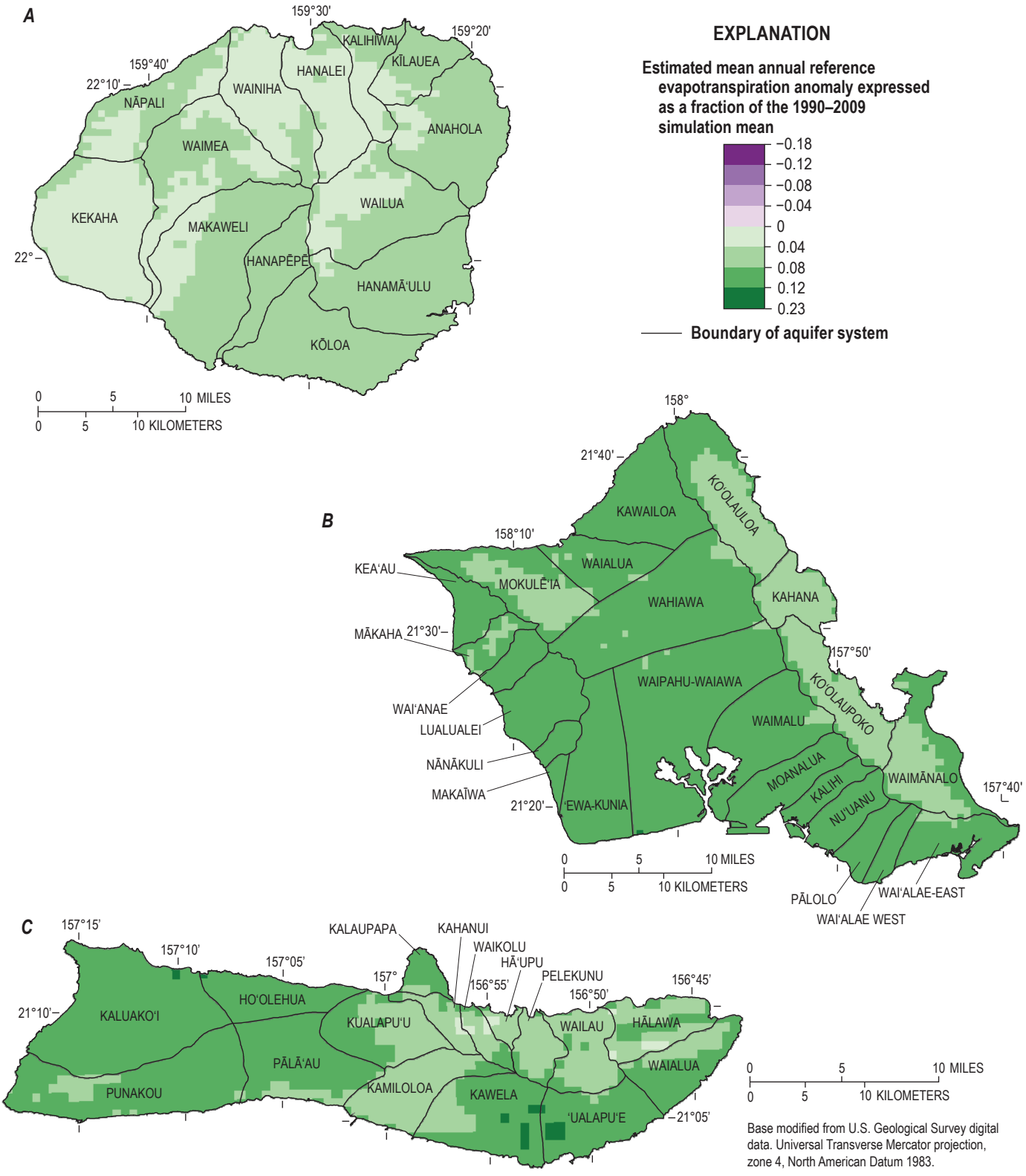


Figure 3-2. Map of anomalies of mean annual reference evapotranspiration adjusted for projected warming using a modified form of the Food and Agriculture Organization of the United Nations Penman-Monteith formula by Allen and others (2006) for selected future-climate scenarios for Kauaʻi, Oʻahu, Molokaʻi, Lānaʻi, Maui, and the Island of Hawaiʻi. (*A*) HRCM2 RCP4.5 2080–99 scenario for Kauaʻi, (*B*) HRCM2 RCP8.5 2080–99 scenario for Oʻahu, (*C*) HRCM2 RCP8.5 2080–99 scenario for Molokaʻi, (*D*) HRCM2 RCP4.5 2080–99 scenario for Lānaʻi, (*E*) HRCM1 A1B 2080–99 scenario for Maui, and (*F*) HRCM2 RCP4.5 2080–99 scenario for the Island of Hawaiʻi. Estimated changes are relative to a 1990–2009 simulation mean. Aquifer system boundaries from State of Hawaiʻi (2014).

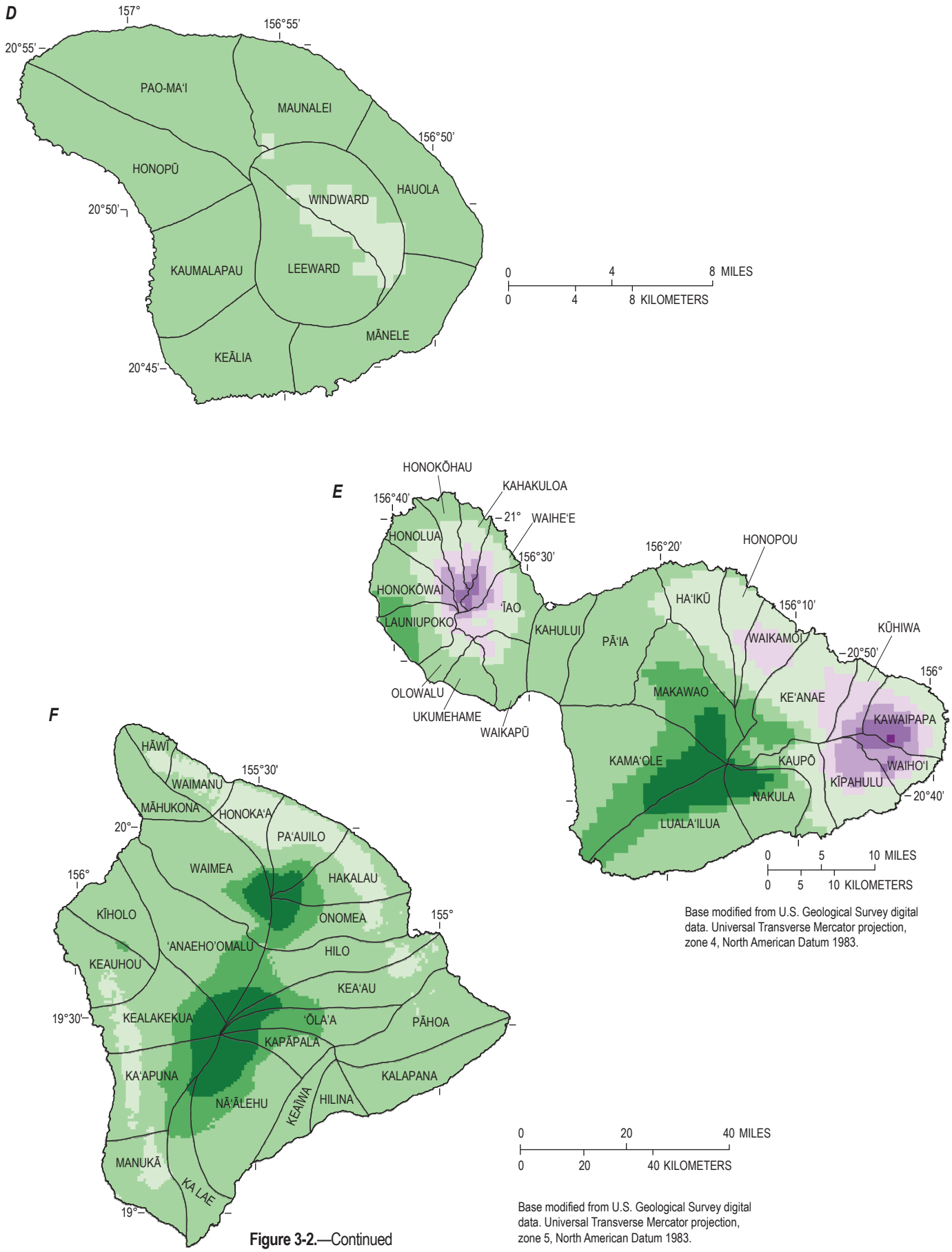


Figure 3-2.—Continued

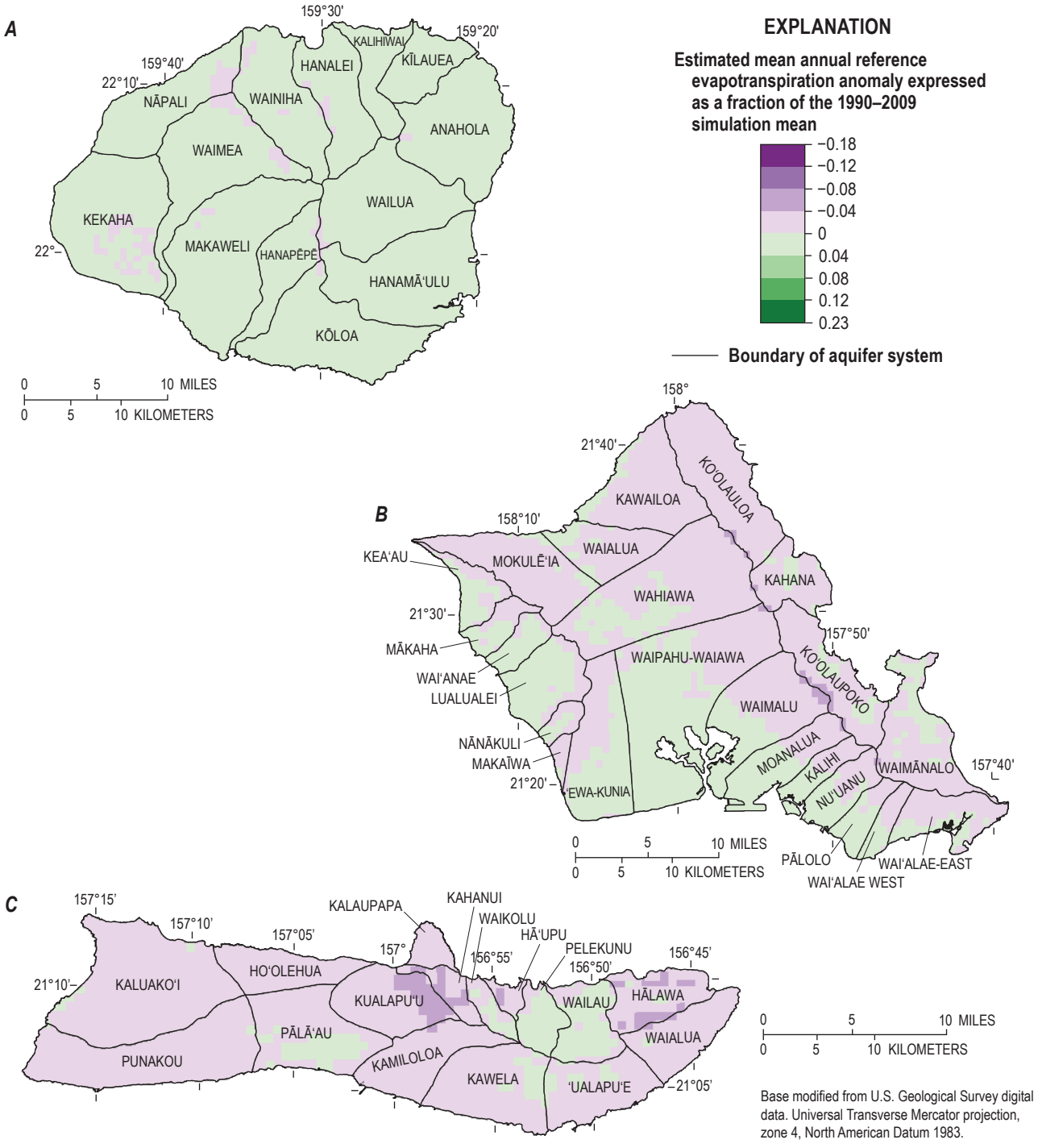


Figure 3-3. Map of anomalies of mean annual reference evapotranspiration adjusted for projected warming and increased atmospheric carbon dioxide (CO_2) concentration computed using a modified form of the Food and Agriculture Organization of the United Nations Penman-Monteith formula by Yang and others (2019) for selected future-climate scenarios for Kaua'i, O'ahu, Moloka'i, Lāna'i, Maui, and the Island of Hawai'i. (*A*) HRCM2 RCP4.5 2080–99 scenario and increase of 164 parts per million (ppm) in atmospheric CO_2 concentration for Kaua'i, (*B*) HRCM2 RCP8.5 2080–99 scenario and increase of 472 ppm in atmospheric CO_2 concentration for O'ahu, (*C*) HRCM2 RCP8.5 2080–99 scenario and increase of 472 ppm in atmospheric CO_2 concentration for Moloka'i, (*D*) HRCM2 RCP4.5 2080–99 scenario and increase of 164 ppm in atmospheric CO_2 concentration for Lāna'i, (*E*) HRCM1 A1B 2080–99 scenario and increase of 315 ppm in atmospheric CO_2 concentration for Maui, and (*F*) HRCM2 RCP4.5 2080–99 scenario and increase of 164 ppm in atmospheric CO_2 concentration for the Island of Hawai'i. Estimated changes are relative to a 1990–2009 simulation mean. Aquifer system boundaries from State of Hawai'i (2014).

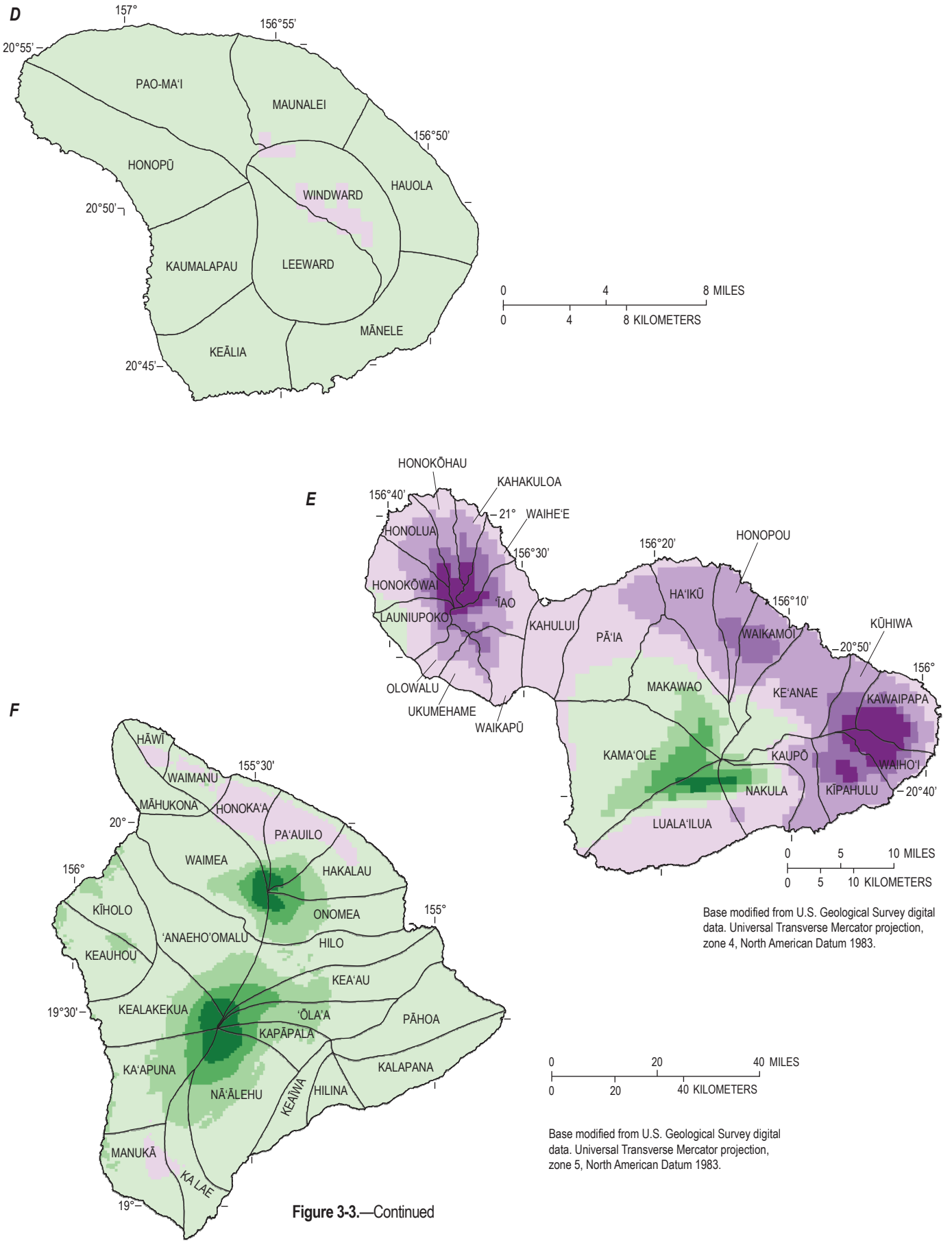


Figure 3-3.—Continued

References Cited

Allen, R.G., Pereira, L.S., Raes, D., and Smith, M., 1998, Crop evapotranspiration—Guidelines for computing crop water requirements: Food and Agriculture Organization of the United Nations, Irrigation and Drainage Paper 56, 300 p.

Allen, R.G., Pruitt, W.O., Wright, J.L., Howell, T.A., Ventura, F., Snyder, R., Itenfisu, D., Steduto, P., Berengena, J., Yrisarry, J.B., Smith, M., Pereira, L.S., Raes, D., Perrier, A., Alves, I., Walter, I., and Elliott, R., 2006, A recommendation on standardized surface resistance for hourly calculation of reference ETo by the FAO56 Penman-Monteith method: Agricultural Water Management, v. 81, no. 1–2, p. 1–22, https://doi.org/10.1016/j.agwat.2005.03.007.

AWS Truewind LLC, 2004, Wind energy resource maps of Hawaii: Prepared for Hawai‘i Department of Business, Economic Development, and Tourism Strategic Industries Division.

Brewington, L., Keener, V., and Mair, A., 2019, Simulating land cover change impacts on groundwater recharge under selected climate projections, Maui, Hawai‘i: Remote Sensing, v. 11, no. 24, https://doi.org/10.3390/rs11243048.

Brutsaert, W., 2005, Hydrology—An introduction: Cambridge University Press, Cambridge, UK, 605 p.

Clarke L., K. Jiang, K. Akimoto, M. Babiker, G. Blanford, K. Fisher-Vanden, J.-C. Hourcade, V. Krey, E. Kriegler, A. Löschel, D. McCollum, S. Paltsev, S. Rose, P.R. Shukla, M. Tavoni, B.C.C. van der Zwaan, and D.P. van Vuuren, 2014, Assessing transformation pathways, in climate change 2014—Mitigation of climate change: Contribution of Working Group III to the Fifth Assessment Report of the Intergovernmental Panel on Climate Change [Edenhofer, O., Pichs-Madruga, R., Sokona, Y., Farahani, E., Kadner, S., Seyboth, K., Adler, A., Baum, I., Brunner, S., Eickemeier, P., Kriemann, B., Savolainen, J., Schlömer, S., von Stechow, C., Zwickel T., and Minx J.C., eds.]. Cambridge University Press, Cambridge, United Kingdom and New York, N.Y., USA, https://www.ipcc.ch/site/assets/uploads/2018/02/ipcc_wg3_ar5_chapter6.pdf.

Gash, J.H.C., Lloyd, C.R., and Lachaud, G., 1995, Estimating sparse forest rainfall interception with an analytical model: Journal of Hydrology, v. 170, no. 1–4, p. 79–86, https://doi.org/10.1016/0022-1694(95)02697-N.

Giambelluca, T.W., Chen, Q., Frazier, A.G., Price, J.P., Chen, Y.-L., Chu, P.-S., Eischeid, J.K., and Delparte, D.M., 2013, Online rainfall atlas of Hawai‘i: Bulletin of the American Meteorological Society, v. 94, no. 3, p. 313–316, https://doi.org/10.1175/BAMS-D-11-00228.1.

Giambelluca, T.W., Shuai, X., Barnes, M.L., Alliss, R.J., Longman, R.J., Miura, T., Chen, Q., Frazier, A.G., Mudd, R.G., Cuo, L., and Businger, A.D., 2014, Evapotranspiration of Hawai‘i: University of Hawai‘i at Mānoa, accessed on November 12, 2021, at http://evapotranspiration.geography.hawaii.edu/.

Hay, L.E., Wilby, R.L., and Leavesley, G.H., 2000, A comparison of delta change and downscaled GCM scenarios for three mountainous basins in the United States: Journal of the American Water Resources Association, v. 36, no. 2, p. 387–397, https://doi.org/10.1111/j.1752-1688.2000.tb04276.x.

Intergovernmental Panel on Climate Change, 2007, Climate Change 2007—The Physical Science Basis: Working Group I Contribution to the Fourth Assessment Report of the Intergovernmental Panel on Climate Change (Solomon, S., Qin, D., Manning, M., Chen, Z., Marquis, M., Averyt, K.B., Tignor, M., and Miller, H.L., eds.), Cambridge University Press, Cambridge, United Kingdom and New York, NY, USA, 996 p.

Intergovernmental Panel on Climate Change, 2014, Climate Change 2014—Synthesis Report: Contribution of Working Groups I, II and III to the Fifth Assessment Report of the Intergovernmental Panel on Climate Change (Core Writing Team, Pachauri, R.K., and Meyer, L.A., eds.), IPCC, Geneva, Switzerland, 151 p.

Intergovernmental Panel on Climate Change, 2021a, Carbon Dioxide—Projected emissions and concentrations: Figure 1, accessed October 1, 2021, at https://www.ipcc-data.org/observ/ddc_co2.html.

Intergovernmental Panel on Climate Change, 2021b, Figure 6.6, plot of projected CO_2 concentrations for Representative Concentration Pathway scenarios, accessed October 1, 2021, at https://www.ipcc.ch/report/ar5/wg3/assessing-transformation-pathways/06_figure_6-6/.

Islam, A., Ahuja, R.L., Garcia, L.A., Ma, L., and Saseendran, A.S., 2012, Modeling the effect of elevated CO2 and climate change on reference evapotranspiration in the semi-arid central Great Plains: Transactions of the American Society of Agricultural and Biological Engineers, v. 55, no. 6, p. 2135–2146, https://doi.org/10.13031/2013.42505.

Keeling, R.F., and Keeling, C.D., 2017, Atmospheric monthly in situ CO2 data: Mauna Loa Observatory, Hawaii, Scripps CO2 program data: University of California San Diego Library Digital Collections, at https://doi.org/10.6075/J08W3BHW.

Mair, A., Johnson, A.G., Rotzoll, K., and Oki, D.S., 2019, Estimated groundwater recharge from a water-budget model incorporating selected climate projections, Island of Maui, Hawai‘i: U.S. Geological Survey Scientific Investigations Report 2019–5064, 46 p., https://doi.org/10.3133/sir20195064.

Yang, Y., Roderick, M.L., Zhang, S., McVicar, T.R., and Donohue, R.J., 2019, Hydrologic implications of vegetation response to elevated CO_2 in climate projections: Nature Climate Change, v. 9, no. 1, p. 44–48, https://doi.org/10.1038/s41558-018-0361-0.

Zhang, C., and Wang, Y., 2017, Projected future changes of tropical cyclone activity over the western north and south Pacific in a 20-km-mesh regional climate model: Journal of Climate, v. 30, no. 15, p. 5923–5941, https://doi.org/10.1175/JCLI-D-16-0597.1.

Zhang, C., Wang, Y., Hamilton, K., and Lauer, A., 2016a, Dynamical downscaling of the climate for the Hawaiian Islands Part I—Present day: Journal of Climate, v. 29, no. 8, p. 3027–3048, https://doi.org/10.1175/JCLI-D-15-0432.1.

Zhang, C., Wang, Y., Hamilton, K., and Lauer, A., 2016b, Dynamical downscaling of the climate for the Hawaiian Islands Part II—Projection for the late twenty-first century: Journal of Climate, v. 29, no. 23, p. 8333–8354, https://doi.org/10.1175/JCLI-D-16-0038.1.

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